PROBEMATRIX

HUMAN FACTORS MULTI-LEVEL ANALYSIS PROTOCOL

OFFICIAL INVESTIGATION RECORD

INC-2026-031

Date: 2020-08-24

Status: IN PROGRESS

1. INCIDENT OVERVIEW

Title: Hazira 2020
Type: aviation
Severity: major
Location: ONGC Hazira

Narrative:

INITIAL INCIDENT REPORT

To: Plant Manager (Hazira); Asset HSE Lead; Statutory authorities under ERDMP notification requirements
From: Health & Safety Manager, Hazira Gas Processing Complex
Date / time of report: 24 September 2020, mid-day IST (filed within 6 hours of all-clear)
Incident type: Hydrocarbon vapour release with subsequent fire — Phase I, Gas Terminal area
Status: Plant in emergency shutdown; affected area cordoned and under guard; site investigation begun. **Cause and root cause not yet established.**

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1. Summary of event

In the early hours of this morning a hydrocarbon vapour release occurred from the 36" condensate metering skid in the Gas Terminal area (Phase I). The release was first detected by perimeter security at approximately 02:15 hrs. The vapour cloud accumulated for approximately 56 minutes during which plant personnel attempted, but were physically unable, to approach and locate the source on foot due to gas density. At **03:08:15 hrs (timestamp from plant CCTV)** the cloud ignited and developed into a major fire affecting the metering skid and the immediately surrounding piping. The fire was contained by the on-site fire team and brought under control by approximately 06:00 hrs. The Plant Manager, acting as Commander-in-Charge, declared all-clear at **07:30 hrs** following confirmation of zero LEL and a complete personnel head-count.

Casualties / injuries (as of this report): Nil to plant personnel and on-shift contractors. CISF have confirmed no injuries inside the plant boundary. Status of the surrounding shanty/civilian area outside the plant perimeter is being verified with the district authorities and CISF; this report will be updated.

Cause: Under investigation. Working hypotheses are stated in §10 below; they are hypotheses only.

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2. Plant and facility context

Hazira Gas Processing Complex receives gas and associated condensate from Mumbai Offshore production via two

submarine pipelines: 36" (commissioned 1986, ~230 km) and 42" (commissioned 1996, ~244 km). Plant throughput capacity ~46.9 mmscmd. Major downstream consumers include the HVJ pipeline network, KRIBHCO fertiliser plant, and NTPC power generation.

The Gas Terminal area (Phase I) is the inlet receiving section where the offshore stream is separated and routed downstream. The condensate metering skid at issue contains two parallel 6" meters in service — **310FT01** and **310FE2** — measuring condensate flow downstream of the slug catcher. Both meters are part of the original 1986 plant inventory.

Relevant recent maintenance activity in the affected area:

- **17 September 2020:** SAP Notification 3800419840 raised by operations for malfunction of meter 310FT01.
- **18 September 2020:** Retaining-plate gasket of meter 310FT01 replaced. Per the maintenance team's verbal account this morning, the original OEM gasket specification (a 1986 rectangular-slot design) is not currently held on file at site, and the replacement was effected using a substitute O-ring procured locally. After installation, the system was pressurised to operating pressure with both meters in line and isolated; **no separate hydrostatic test to design pressure was performed.**
- **19 September 2020:** Meter 310FT01 returned to service, ~46 hours after the gasket replacement.
- This gasket replacement is the **third on this meter in the past 22 months** (previous replacements: November 2018, February 2019). Investigation should determine whether the recurrence pattern was treated as a reliability concern.

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3. Detailed timeline of events

Times below are reconciled from: the Gas Terminal Control Room operator log; the DCS annunciator history; CCTV recording for the metering area (camera: PELCO PTZ D5230D, single coverage, ~90-second pan turn-around); CISF watch-tower and gate logs; personnel statements taken between 09:00 and 11:30 hrs this morning. Where DCS and CCTV disagree the CCTV timestamp is taken as canonical; we have observed that DCS clocks across the multiple control rooms drift between approximately 2 and 12 minutes against each other and this is being corrected.

Time (IST)	Event	Source
17 Sep 2020	SAP Notification 3800419840 — malfunction of meter 310FT01.	SAP
18 Sep 2020	310FT01 retaining-plate gasket replaced. Pressurised to operating pressure only; isolated.	Maintenance log
19 Sep 2020	310FT01 returned to service.	SAP / operator log
23 Sep 20:00	Night shift assumes duty in Gas Terminal Control Room: Chief Engineer (Production) and Junior Engineer (Mechanical).	Roster
02:11:43	CCTV baseline frame — facilities visibly intact.	CCTV
02:12:50	**CCTV records first visible vapour leakage from the metering skid.** No DCS alarm yet.	CCTV
~02:15	**CISF Constable A** (Watch Tower 3, NW periphery) hears noise and sees vapour cloud emerging behind Gas Terminal. Independently, **CISF Head Constable B** (Gas Terminal Gate, East) hears noise and sees vapour cloud. Both notify internal channels (CISF Main Gate, Fire Station, Main LPG Control Room, CCTV Control Room).	Witness statements
~02:15	**CISF Sub-Inspector C** (CCTV Control Room) sees heavy leakage on monitor.	Witness statement
02:18	CISF Sub-Inspector C calls Gas Terminal Control Room.	Operator log
~02:20	DCS annunciator panel registers first alarm in Gas Terminal Control Room — **5 minutes after human detection.** Chief Engineer (Production) and Junior Engineer (Mechanical) hear alarm and answer CCTV phone call simultaneously.	DCS / witness
~02:20	Both engineers attempt to walk to the affected area to locate the source. **Density of the vapour cloud forces them back.** Chief Engineer (Production) telephones the GSP Shift In-charge (DGM), the LPG Shift In-charge (DGM), and the CFU Panel Engineer in Main Control Room.	Witness
~02:35	GSP Shift In-charge and Overall Shift In-charge (GM) reach the Gas Terminal Gate. They too cannot enter the plant area because of gas density. Overall Shift In-charge directs the Gas Terminal CR to close the 36" pressure control valve; informs Fire Station, Resident Engineer, Head Operations, and Plant Manager.	Witness
~02:40	CISF barricade road to Slug Catcher area on Overall Shift In-charge's instruction.	Operator log
~02:45	Resident Engineer arrives. Head Operations, Plant Manager, Area In-charge informed.	Witness
~03:00	Head Operations (CGM) assumes Site Incident Controller (SIC) role.	Witness

03:01	**First isolation valve closed: MOV 54** (slug catcher to 36" metering) by GSP Shift In-charge. **46 minutes after first detection.**	Operator log
03:04	36" Pressure Reducing Skid PCV closed from Gas Terminal CR.	Operator log
03:04	SIC instructs Main Control Room (walkie-talkie) to inform Offshore to halt gas/condensate supply through 36" and 42" trunk lines.	Operator log
~03:08	Mobile teams dispatched to close trunk-line sectionalisation valves at Gavier (36") and Abhwa (42").	Operator log
03:08:15	**Vapour cloud ignites.** First flame visible on CCTV. **56 minutes after leak commenced.**	CCTV
03:08:22	CCTV cable burns through; recording ends.	CCTV
~03:10	**Level-III Off-site Emergency declared** by SIC in consultation with Plant Manager. Mutual-aid call placed.	Operator log
~03:10	Offshore (VCR) instructed to stop gas supply in 36" and 42" SBHTs.	Operator log
~03:10	LPG, GSU, GDU, DPD, SRU, CFU, KRU trains shutdown. Cogen GTs and Boilers shutdown. Cooling towers kept functional. Steam unavailable; grid power available.	Operator log
~03:15	Plant Manager assumes CIC role; reaches Emergency Control Room.	Operator log
~03:15	Isolation and depressurisation of connected process units commenced.	Operator log
~03:30	Sectionalisation valves at Gavier (36") and Abhwa (42") confirmed closed by mobile teams.	Field confirmation
~03:30	Depressurisation from GSU Trains 31–39 begins.	Operator log
~03:30	Initial condensate evacuation via CFU Trains 71–77 begins.	Operator log
~03:50	Active fire-cooling operations begin.	Fire team
~04:30	8" CFU header inter-connection valve between Phase-I and Phase-II closed; depressurisation continues from CFU Trains 71 & 72.	Operator log
~05:30	MOV #50 and #52 closed — bulk hydrocarbons isolated from Slug Catcher Phase-I.	Operator log
~06:00	Active fire extinguished; cooling continues.	Fire team
~07:00	MOV #03 (36" main trunk-line inlet) closed once area is approachable; zero LEL verified.	Field
07:30	**All-clear declared** by Plant Manager (CIC).	Operator log

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4. Personnel accounts (de-identified by role)

Statements were taken this morning between 09:00 and 11:30 hrs from the seven personnel directly involved in the detection-to-isolation phase. Original signed statements are held in the H&S office; summaries below.

- **CISF Constable A — Watch Tower 3 (NW periphery):** "Heard a sharp hissing noise around 02:15. Saw a vapour cloud rising behind the Gas Terminal area. Informed CISF Main Gate, Fire Station and LPG Control Room by intercom."
- **CISF Head Constable B — Gas Terminal Gate (East):** "Heard noise at the same time, saw vapour. Informed CCTV Control Room and CISF Main Gate."
- **CISF Sub-Inspector C — CCTV Control Room:** "Took the call from Head Constable B at 02:15. On the monitor I could see heavy leakage near the metering skid. I called the Gas Terminal Control Room at 02:18."
- **Chief Engineer (Production) — Gas Terminal CR:** "On night duty from 20:00 hrs. Around 02:20 the annunciator alarmed and the CCTV operator phoned. We saw a heavy cloud out of the window. The Junior Engineer and I went out together to try and reach the metering area. The cloud was so dense and the smell so strong that we could not get close. We came back and I phoned the GSP Shift In-charge, LPG Shift In-charge and the CFU Panel Engineer."
- **Junior Engineer (Mechanical) — Gas Terminal CR:** Confirms the Chief Engineer's account. Adds that visibility outside the control room was extremely poor.
- **DGM (Production) — GSP Shift In-charge:** "Got the call at about 02:35. Reached the Gas Terminal Gate with the Overall Shift In-charge. We couldn't enter — the smell was overpowering. I closed MOV 54 at 03:01."
- **GM (Production) — Overall Shift In-charge / SIC:** "Got the call at 02:35. Reached the Gas Terminal Gate. Instructed closure of the 36" PCV from the Gas Terminal CR. Informed Fire Station, Resident Engineer, Head Operations and the Plant Manager. By the time the senior team arrived the fire had started."

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5. Initial site observations (post-extinguishment walk-down, 08:30–11:00 hrs)

The site was made safe by the fire team after 07:30. I conducted a controlled walk-down of the affected area beginning 08:30 hrs accompanied by the Resident Engineer and the Mechanical Maintenance lead. Photographs have

been taken; an annotated photo set will follow this report. **All damaged equipment and the surrounding debris field has been cordoned and is under guard; nothing has been moved or disturbed pending forensic inspection.**

Visible findings at this stage (descriptive, not yet causal):

- **Meter 310FT01:** retaining plate extensively damaged; all bolts of the flange arrangement are out. The meter assembly appears to be the most severely damaged single item in the area.
- **Meter 310FE2 (the parallel 6" meter):** comparatively much less damaged — only one bolt of the flange arrangement is damaged.
- **Surrounding condensate piping:** multiple lines around the metering skid are buckled, ruptured or displaced. This is fire-induced damage; it does not, on visual inspection, indicate a separate primary failure.
- **36" gas pipeline at the Gas Terminal:** visibly intact and undamaged. The loss of containment was localised — it did not occur on the main trunk pipeline.
- **A flame-proof HPMV lamp post in the immediate vicinity** has visible damage to its enclosure: three of the bolts on the fitting are missing and the glass case is detached and standing open by approximately 5–6 cm. Whether this condition pre-existed the fire or was caused by the fire is not yet established and will be assessed by the forensic team. Until determined otherwise it is being treated as a possible ignition source candidate and has been preserved in place.
- Eight other HPMV FLP lamps in the area show varying degrees of fire damage to cables and fittings.
- Fire-water system performed during the incident: tank levels and header pressure remained in their normal bands throughout. Three of the four electrical fire-water pumps (47PE003 A, C, D) ran during cooling. (Diesel-pump availability at the time of the event was already compromised — see §10.)

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6. Initial damage assessment

- **Total loss / write-off (preliminary):** condensate metering skid 310 area — meters, instrumentation, immediate piping, instrument cabling.
- **Significant damage:** electrical fittings within ~15 m radius (FLP lamps, conduit, junction boxes, the CCTV camera and its cable).
- **Fire-induced secondary damage:** to surrounding insulation, paint coatings, and minor structural members.
- **No damage:** main 36" and 42" trunk pipelines; Slug Catcher Phase-I body (isolated in time); main control rooms and Emergency Control Room.

2. EVIDENCE SUMMARY

EVD-2130 - log record

Monthly fire drill records in SAP — lacking action/observation data. Source records: "Fire drill reports are available in SAP but without any data on actions ...

EVD-4286 - witness statement

Witness statement of CISF Constable A on duty at Watch Tower 3 (NW periphery) since 21:00 hrs. Source records: "At 0215 hrs, he observed noise followed by vapour cloud from Gas Terminal Area" (p.16); he then informed CISF Main Gate, Fire Station, and ...

EVD-6234 - witness statement

Witness statement of CISF Head Constable B on night-shift duty at Gas Terminal Gate (east of Gas Terminal). Source records: "At 0215 hrs, he heard the noise and observed vapour cloud behind Gas terminal building. At 0216. He informed CCTV Control ...

EVD-8871 - witness statement

Witness statement of CISF Sub-Inspector C on night-shift duty at CCTV Control Room. Source records that at 02:15 he received a phone call from the Gas Terminal Gate Head Constable regarding noise and vapour cloud; at 02:18 he visually verified "heavy ...

EVD-1460 - witness statement

Witness statement of Chief Engineer (Production), Gas Terminal CR, on night shift from 20:00 hrs. Source records: "At around 0220 hrs Mr. [CE] heard alarm on the annunciator panel and the same time he received a phone call from CISF SI [CCTV CR] ...

EVD-9220 - witness statement

Witness statement of Junior Engineer (Mechanical), Gas Terminal CR, on night shift from 20:00 hrs. Source records: "At 0220 hrs Mr. [JEA] heard alarm and phone call simultaneously. He went out of the control room and observed leakage and intense ...

EVD-9612 - witness statement

Witness statement of DGM (Production) - GSP Shift In-charge, called from quarters at ~02:35 hrs. Source records: "At around 0235 hrs, [DGM-GSP] received phone call from LPG Control Room and Gas Terminal control room, it was informed that heavy ...

EVD-7340 - witness statement

Witness statement of GM (Production) - Overall Shift In-charge / SIC, called at ~02:35 hrs. Source records that the SIC phoned Gas Terminal Control Room and "instructed to close the 36 line pressure control valve and get out of the control room. He ...

EVD-9587 - recording

CCTV frame 02:11:43 (PELCO PTZ D5230D, view of metering skid + Gas Terminal Control Room area). Source describes the frame as: "Clear picture of facilities" (p.18) — all metering-skid piping visible intact, no vapour, no smoke. Establishes the ...

EVD-1284 - recording

CCTV frame 02:12:50 — the earliest visible vapour leakage commencement at the condensate metering skid. Source describes the frame as: "Commencement of Vapour leakage" (p.19). This timestamp defines clock-zero for the 46-minute response-delay ...

EVD-9941 - recording

CCTV sequence 02:15:35 -> 02:16:13 (5 frames over 38 seconds): vapour cloud expansion captured progressively as "vapour cloud near GT control room" (02:15:35) -> "vapour cloud on road near GT control room" (02:15:41) -> "vapour cloud in piping ...

EVD-8213 - recording

CCTV frame 03:00:44 — source describes as "vapour all around even at height" (p.22), indicating the cloud now occupies a vertical column above the metering skid area. This frame is captured 48 minutes after CCTV-confirmed leak onset (02:12:50) and ...

EVD-4507 - recording

CCTV frame 03:08:15 — source describes as "flash started" (p.23), the first visible flame. Anchors the absolute ignition timestamp and the incident's classification as fire (vs. release). Critical observation: ignition occurred 4 minutes 15 seconds ...

EVD-2752 - recording

CCTV frame 03:08:22 — source describes as "major fire continued, CCTV capturing discontinued due to burnt of CCTV cable" (p.25). Hard cutoff for the video evidence timeline, 7 seconds after ignition. The cable's exposure to fire-temperature was ...

EVD-0064 - physical item

Damaged retaining plate of meter 310FE1 / 310FT01 (1986-vintage 6" condensate meter, rectangular slot gasket design). Source records: "6 meter with ID 310FE1 was extensively damaged with all bolts out of the flange arrangement" (p.31) and "The ...

EVD-4517 - physical item

Comparator physical evidence: meter 310FE2 retaining plate. Source records: "only one bolt damaged for 310FE2 (6 size)" (p.33). The asymmetry of damage between 310FE1 (all bolts displaced) and 310FE2 (single bolt damaged from the flange ...

EVD-6916 - physical item

HPMV (high-pressure mercury vapour) flame-proof lamp post with compromised integrity, identified by the committee as the probable ignition source. Source records: "three bolts on one of the flame proof lamps were missing resulting in it being open ...

EVD-7561 - document

S&P Global Platts trade-press analysis (25-Sep-2020): "Fire incident at ONGC's Hazira, India plant hits gas supplies to end-users". Day-after coverage characterising the supply-chain disruption: HVJ pipeline impact, GAIL force majeure context (row ...

EVD-3489 - physical item

Damaged condensate pipelines surrounding the metering skid: photographs (c), (g), (i) on pp.13-15 show buckled, ruptured, and displaced pipe segments. Source records: "various pipelines have got damaged/shifted" (p.31). Documents fire-induced ...

EVD-5780 - physical item

36" gas pipeline at the Gas Terminal area, photographed and inspected post-fire. Source records: "Viewing at the footages from CCTV, it clearly indicated huge vapour cloud and condensate lines would have leaked as gas pipelines were intact & not ...

EVD-1394 - photograph

Post-fire wide-angle photograph of the condensate metering skid area (1986-vintage installation, ground level, NW of Gas Terminal Control Room). Source describes the visible state at site inspection (5-8 Oct 2020, ~2 weeks post-fire): "the condensate ...

EVD-0210 - photograph

Photograph of the watch tower and CCTV camera mounting position relative to the damaged area. Source identifies the camera: "PELCO make PTZ cameras (Model D5230D) are provided at few places and has turn around (360 degree) at the time of about ...

EVD-0496 - photograph

Post-incident electrical PM audit photographs of 9 HPMV FLP lamps (125 W each) in the metering-skid blast radius. Source describes the audit report itself: "the report has clubbed all nine 125 W HPMV FLP lamps together and reported 'cable burnt and ...

EVD-8657 - log record

DCS annunciator panel record showing first alarm activation at the Gas Terminal Control Room at ~02:20 hrs. Source records: "At around 0220 hrs Mr. [CE] heard alarm on the annunciator panel" (p.17). Quantitative comparison: leak onset 02:12:50 ...

EVD-5844 - log record

GSU shutdown sequence record. Source records: "Depressurization of Gas Section was initially carried out from all the GSU trains. (Train #31 to #39)... Subsequently after segregation of Gas Sweetening Phase-I & II from Phase-III & IV, Gas Section ...

EVD-0059 - log record

CFU shutdown sequence record. Source records: "Initially, liquid condensate of Slug-Catcher I & II was evacuated to reduce inventory through all Condensate Fractionation Units i.e. Train #71-77. CFU header 8\"-interconnection valve between CFU ...

EVD-2432 - log record

Fire Water System monitoring record during incident response. Source records: "Both the Tank level and the Fire water header pressure were healthy throughout the incident. Three electrical Fire water pumps namely 47PE003 A, C, D were operated during ...

EVD-6470 - log record

Pressure Transmitter Data MISSING for 310PT4, 310PT5, 310PT6. Source records: "Three Pressure transmitters 310PT4, 310PT5 & 310PT6 are shown in the drawing. It was informed that no pressure transmitter data are available in the system. No ...

EVD-5532 - log record

DCS time-stamp drift across multiple control rooms. Source records: "Time stamping in DCS systems provided in various control rooms is varying from 2 minutes to 12 minutes and data matching is difficult" (p.26). 2-12 minute clock drift across CR DCSes. ...

EVD-0591 - log record

SAP Notification 3800419840 raised 17-Sep-2020 for malfunction of meter 310FE1 / 310FT01. Source records: "Notification no. 3800419840 was raised on 17/09/2020 for malfunction of meter no.1" (p.27); confirmed p.33: "SAP Notification 3800419840 dt. ...

EVD-1494 - document

Meter 310FE1 gasket replacement history. Source records: "Cleaning/servicing/calibration of meters 1 & 2 was carried out on 12th & 13th Nov 2018. Afterwards, meter no.1 was cleaned on 25/02/2019 & gasket was changed. Again on 18/09/2019, gasket ...

EVD-8645 - document

Lost OEM gasket specification; market-sourced O-rings used as substitutes without QC. Source records: "The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not ...

EVD-2506 - log record

Calibration record showing undocumented setting change on meter 310FT01 in Dec 2019. Source records: "During calibration from 17/12/2019 to 18/12/2019, calibration was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of ...

EVD-2894 - document

Pressure boundary integrity record: NO hydro test performed after 18-Sep-2020 gasket replacement. Source records: "After calibration of the meter 310FE01, the gasket was replaced on 18.09.2020. No hydro test was done after replacement of the ...

EVD-0185 - log record

Meter 310FE1 returned to service ~46 hours after the 18-Sep-2020 gasket replacement. Source records: "Next day meter 310FE01 was taken on online and was in use at the time of Incident" (p.31). Replacement on 18-Sep; back online 19-Sep; in ...

EVD-8307 - document

Ultrasonic Thickness (UT) pipeline measurement records for the gas-terminal area. Source records: "The laid down schedule for thickness measurement is every four years... As per the records, the last two thickness measurement exercise was carried out in ...

EVD-5527 - document

2017 thickness analysis report identifying pipes with significant thickness loss. Source records: "As per the report of 2017, several pipes have been identified with significant thickness loss and need for corrective action. However, even after three years ...

EVD-2397 - document

July 2019 spot UT measurement on the 36" pipeline at the Gas Terminal area. Source records: "a reduction of thickness of 1.03mm along with heavy pitting and painting peeling off was observed in one of the segments of 36 pipeline in gas terminal in ...

EVD-8287 - document

Pipeline baseline thickness data record — critical absence. Source records: "It is observed that for a large number of pipes, especially in gas terminal area baseline data of pipe thickness in not available. The original drawings and the schedules of the ...

EVD-9692 - document

Self Safety Audit (SSA) Report — generic checklist not tailored to plant specifics. Source records: "Safety self-audit is conducted by a multi-disciplinary team based on the checklist prepared by the location. The checklist is comprehensive but not related to ...

EVD-7681 - document

Internal Safety Audit reports 2018 and 2019. Source records: "Internal Safety Audit was conducted in 2018 & 2019 for 4 & 3 days respectively by a 3-member committee and the observations are given without the filled checklist. The checklist format is not ...

EVD-6852 - document

External Safety Audit Report Feb 2018 — 12 of 308 recommendations overdue without documented authority. Source records: "External safety audit was conducted in February 2018, and 296 out of 308 recommendations are complied with. Balance are in ...

EVD-9999 - document

Post-incident electrical preventive maintenance audit report on the 9 HPMV FLP lamps in the affected area. Source records: "Post the incident, an electrical audit of the affected area was carried out by the plant maintenance team. Considering that the ...

EVD-0933 - document

P&ID Drawing 02-310-AO-101 Rev.1 covering the 2x6" lines and 1x2" line metering system. Source records: "Reference drawings no. 02-310-AO-101 Rev.1, 2x6 lines & 1x2 lines for metering system are shown but valves & meters configuration are ...

EVD-3452 - document

Plant layout drawings reviewed by the committee, including main plant layout and Flame & Gas detector overlay. Source records the document set: "Layout of the Plant / Layout of Plant along with F&G detectors / Layout of MCP-Gas Terminal" (p.18). F&G ...

EVD-9360 - document

36" pipeline route map: offshore (Mumbai) to Hazira via valve stations. Source records the asset context: "One 36, 230 Km long pipeline commissioned in 1986... The 36 pipeline has one valve station at its landfall point at Umbharat followed by three more ...

EVD-4052 - document

On-site Disaster Management Plan (DMP) — hydrocarbon-leakage section omits isolation/stoppage decision criteria. Source records: "As per DMP On site Hazira Plant, guidelines to be followed in case of hydrocarbon leakage are included but does not ...

EVD-2986 - document

Management of Change (MOC) SOP published Aug 2016 — not implemented since 2018. Source records: "Management of Change (MOC) philosophy/SOP was published in August 2016. However, the procedure indicated in the document is not
...

EVD-5105 - document

Work Permit / Job Safety Plan (JSA) system — missing permits without tracking. Source records: "Work Permits are issued in SAP and hard copy of work permits are at sites during execution of work. After completion of work, the originals are filed. On
...

EVD-5634 - document

Control Room Training Program — 4-day general training, no defined qualification, no simulator. Source records: "control room operations are carried out by persons inadequately trained in safety, operations, control room handling, logbook records and
...

EVD-3932 - document

Plant organogram — night-shift authority chain reconstructed from witness statements (rows 1-7) and the committee's document review (p.18). Authority chain: perimeter (CISF Constable A) -> CCTV CR (CISF SI C) -> Gas Terminal CR (CE-P) -> GSP/LPG
...

EVD-6414 - log record

Reconstructed emergency response timeline — composite of CCTV, DCS, and witness statements. Source records the timeline (Section 5, pp.9-12; Section 9, pp.18-25): leak onset 02:12:50 (CCTV) -> first physical detection 02:15 (CISF perimeter) ->
...

EVD-0368 - log record

All-clear declaration log. Source records: "After ensuring zero LEL and head count, SIC - [Head Operations] recommended all clear to CIC. All clear declared by CIC at 07:30 hrs" (p.12). Two procedural gates met: zero LEL (lower-explosive-limit)
...

EVD-8531 - document

PNGRB Enquiry Committee Report (15-Jan-2021) — master upstream source for this evidence pack. Source: cover, pp.3-4. 36-page investigation report by a 3-member committee (1 ex-BPCL Executive Director, 1 ex-ONGC Executive Director, 1
...

EVD-7337 - document

PNGRB Incident Analysis: "Major Fire at ONGC Hazira Plant - 24.09.2020 Incident Analysis", published 8-Feb-2021 (~24 days after the enquiry committee's final report). Public-facing companion regulatory analysis with timeline summary and operational
...

EVD-9548 - document

GPCB (Gujarat Pollution Control Board) notice issued under Air (Prevention & Control of Pollution) Act 1981, Section 31A, ordering interim environmental damage compensation of approximately Rs 1 crore (~Rs 10,000,000), issued December 2020.
...

EVD-1748 - document

ONGC operator press release dated 25-Sep-2020 confirming fire extinguished, no casualties at the plant, and operations resumption in approximately 37 hours. The 37-hour figure is the operator's own characterisation of the recovery timeline;
...

EVD-8209 - document

GAIL India force majeure declaration dated 24-Sep-2020. GAIL formally notified force majeure on gas supplies due to the ONGC Hazira incident — the downstream gas receiver could not perform supply contracts. Force majeure issued the same day as the
...

EVD-2742 - document

Oil & Gas Journal (24-Sep-2020): same-day technical trade-press reporting. Captures the pipeline-rupture cause framing, facility significance (46.9 mmscmd capacity — cited in the article as the Hazira gas-processing capacity), and supply impact to ...

EVD-8602 - document

Business Standard (24-Sep-2020): mainstream-press breaking coverage. Source reports three blasts at approximately 03:05 IST, fire extinguished by 07:35, gas-pipeline-related cause. Cross-reference with PNGRB primary findings: the 03:05 "three ...

3. HFMAP ANALYSIS FINDINGS

Level 1: Unsafe Acts

AJDE 2.5 **Ignored a Caution/Warning** SUBSTANTIATED

2017 pipeline thickness analysis identified pipes requiring corrective action; findings acknowledged but ignored for 3+ years with no action plan developed.

JUSTIFICATION

The 2017 ultrasonic thickness measurement identified several pipes with significant thickness loss requiring corrective action [EVD-5527]. The committee found that 'even after three years no plan has been developed for the corrective actions to be taken for those lines. Mere measurement of thickness without corrective action does not complete the process of maintaining the integrity of pipes' [EVD-5527]. This constitutes ignoring a documented integrity warning. The 2017 report functioned as a caution/warning that certain piping had degraded to a level requiring engineering intervention. By 2020 (3+ years later), no corrective action plan had been developed, indicating the warning was perceived (the report was received and acknowledged) but ignored in practice (no action taken). While the committee's investigation did not directly link the 2017 thickness findings to the specific meter 310FT01 that failed, the systemic pattern of ignoring integrity warnings is relevant to the overall incident causation as a contributing organizational factor that allowed degraded equipment to remain in service.

CAUSAL EXPLANATION

The 2017 thickness measurement report functioned as an integrity warning system - it identified piping with degradation requiring engineering action. By acknowledging the report but failing to develop or implement corrective action plans over 3+ years, operations systematically ignored this warning mechanism. This pattern indicates a broader organizational tendency to perceive but not act on integrity warnings, creating latent conditions where degraded equipment remains in service. While not the direct cause of the meter 310FT01 gasket failure, this systemic ignoring of integrity warnings contributed to the overall degraded-asset environment in which the incident occurred.

Linked Evidence: EVD-5527, EVD-8307, EVD-2397

AJDE 2.5 **Ignored a Caution/Warning** SUBSTANTIATED

Maintenance team skipped mandatory hydrostatic test after gasket replacement on 18-Sep-2020, ignoring fundamental pressure boundary integrity verification requirement.

JUSTIFICATION

After the gasket replacement on September 18, 2020, the maintenance team pressurized the system to operating pressure only and did not perform a hydrostatic test to design pressure [EVD-2894]. Standard pressure boundary integrity practice requires hydrostatic testing after any pressure-retaining component replacement to verify joint integrity before return to service. The absence of hydro-test after gasket replacement is explicitly identified by the committee: 'No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated' [EVD-2894]. The lack of hydro-test constitutes ignoring a fundamental pressure-vessel safety warning/requirement. Industry-standard codes (ASME, API) mandate hydrostatic testing after pressure boundary work. The maintenance team either knew this requirement and chose not to apply it (active ignoring), or the site had no SOP mandating it (systemic ignoring). Either way, a critical verification step that would have revealed the gasket installation deficiency was skipped. The meter was returned to service on September 19 [EVD-0185] and failed

CAUSAL EXPLANATION

The hydro-test requirement after pressure boundary work is a fundamental integrity verification step designed to detect installation defects before pressurization with process fluids. By ignoring this requirement (whether through deliberate omission or absence of SOP), the maintenance team allowed a potentially defective gasket installation to enter service without verification. The meter failed 5 days later, suggesting the gasket joint was compromised from installation. A proper hydro-test would have pressurized the joint to 1.5x design pressure with water, likely revealing the installation deficiency in a controlled manner before the catastrophic failure in live hydrocarbon service.

Linked Evidence: EVD-2894, EVD-0185, EVD-0064

AJDE 2.5 Ignored a Caution/Warning

SUBSTANTIATED

Three gasket failures in 22 months on meter 310FT01 constituted clear warning of chronic integrity issue; pattern ignored, routine replacement continued without root-cause investigation.

JUSTIFICATION

Meter 310FT01 underwent three gasket replacements in 22 months: November 2018, February 2019, and September 18, 2020 [EVD-1494]. The recurring failure pattern constituted a clear warning signal that the meter assembly had a chronic integrity issue requiring root-cause investigation and corrective action beyond repeated gasket replacement. The committee explicitly identified this as a missed signal: 'Investigation should determine whether the recurrence pattern was treated as a reliability concern' (Initial Report §2). Instead of treating the pattern as a warning requiring engineering analysis, operations continued the practice of replacing gaskets with locally-sourced O-rings that did not match the original OEM rectangular-slot specification [EVD-8645]. The warning was present (three failures in 22 months on the same component), it was perceivable (documented in maintenance records), but it was ignored - operations did not escalate to reliability engineering, did not seek OEM consultation, and did not implement enhanced integrity verification (e.g., mandatory hydro-test after each replacement). The decision to continue routine gasket replacement without

CAUSAL EXPLANATION

The recurring gasket failure pattern was a reliability warning signal indicating a chronic integrity issue with meter 310FT01. By ignoring this warning and continuing routine gasket replacement without root-cause analysis or corrective engineering, operations allowed the underlying failure mechanism to persist. The fourth failure on September 24, 2020 resulted in the catastrophic release that initiated the incident. Had the warning been heeded after the second or third failure, engineering intervention (OEM gasket procurement, design review, or meter replacement) would likely have prevented the incident.

Linked Evidence: EVD-1494, EVD-8645, EVD-0591

AJDE 2.7 Delayed a Necessary Action

SUBSTANTIATED

GSP Shift In-charge delayed MOV 54 closure for 46 minutes after leak detection, allowing massive vapour accumulation before first isolation at 03:01.

JUSTIFICATION

The GSP Shift In-charge (DGM Production) delayed the execution of the first isolation action (MOV 54 closure) for 46 minutes after the leak was first detected. Timeline from [EVD-6414]: leak onset 02:12:50 (CCTV), first physical detection 02:15 (CISF perimeter), shift-in-charge notified ~02:35, MOV 54 closed 03:01. The 26-minute gap between notification (02:35) and action (03:01) represents the shift-in-charge's personal response delay. [EVD-9612] documents that the shift-in-charge 'tried to enter the Gas Control room area but could not enter due to heavy gas smell' and then 'closed the MOV 54 inlet condensate valve to metering skid at 03:01 hrs' - a necessary action was available but delayed by the decision to first attempt physical approach. The total 46-minute delay from leak onset to first isolation allowed massive inventory release: [EVD-8213] shows at 03:00:44 'vapour all around even at height' - a substantial flammable cloud had accumulated before any isolation occurred. The committee identifies this delay as a root cause factor at p.32: the DMP 'does not include isolation/stoppage of facilities/pipelines' [EVD-4052], leaving the shift-in-charge without clear decision

CAUSAL EXPLANATION

The 46-minute delay between leak onset and first isolation action was a primary causal factor in the incident severity. Each minute of delay allowed continuous inventory release from the failed gasket under operating pressure. The CCTV sequence [EVD-9941] shows the cloud spreading across the Gas Terminal CR, road, and piping area within 40 seconds of detection - rapid expansion indicating high release rate. By delaying isolation until 03:01, the shift-in-charge allowed 46 minutes of uncontrolled release to accumulate a flammable envelope sufficient to sustain a major fire even after both isolation valves (MOV 54 at 03:01, 36" PCV at 03:04) were closed. Ignition at 03:08:15 occurred post-isolation, proving the already-released inventory was the fire fuel. Had MOV 54 been closed within 5-10 minutes of detection (a reasonable response time for a remote-operated valve), the accumulated inventory would have been substantially smaller, potentially preventing ignition or reducing fire severity.

Linked Evidence: EVD-9612, EVD-6414, EVD-8213, EVD-4052

AJDE 2.8 **Wrong Choice of Action** **SUBSTANTIATED**

Maintenance team wrongly chose to use locally-sourced substitute O-rings without specification control instead of OEM rectangular-slot gaskets for meter 310FT01.

JUSTIFICATION

The maintenance team selected the wrong course of action when they chose to replace the failed OEM rectangular-slot gasket with a locally-sourced substitute O-ring. [EVD-8645] documents: 'The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings.' This represents faulty logic: the maintenance team diagnosed the problem correctly (gasket failure on meter 310FT01) but selected an inappropriate solution (non-OEM substitute without specification match or quality control). The rectangular-slot design from 1986 has specific geometry and sealing-mechanism requirements that a generic O-ring cannot meet. The wrong choice was compounded by: (1) no specification control on the O-ring procurement, (2) no quality inspection of the substitute, and (3) no pressure test to validate the substitute's performance [EVD-2894]. This was 'regular practice' per the source, indicating the wrong choice

CAUSAL EXPLANATION

The wrong choice of substitute O-ring gasket was a primary causal factor. The 1986 OEM rectangular-slot design has specific dimensional and material requirements to maintain pressure-boundary integrity under operating conditions (likely 50-60 bar condensate pressure). A generic locally-sourced O-ring lacks: (1) geometric compatibility with the rectangular slot, (2) material specification for hydrocarbon service, (3) pressure/temperature rating validation. The wrong choice meant the gasket could not maintain proper seating and sealing under operating loads. The recurrence pattern [EVD-1494] - three replacements in 22 months - proves the substitute was inherently inadequate, yet the wrong choice was repeated each time rather than escalating to source OEM-specification gaskets. The final substitute installed on 18-Sep-2020 failed catastrophically within 5 days, initiating the loss of containment at 02:12:50 on 24-Sep.

Linked Evidence: EVD-8645, EVD-2894, EVD-1494, EVD-0064

AJDE 2.8 **Wrong Choice of Action** **SUBSTANTIATED**

Engineering/maintenance wrongly chose painting as corrective action for 1.03mm thickness loss on 36" pipeline with zero corrosion allowance, instead of replacement/analysis.

JUSTIFICATION

[EVD-2397] documents that in July 2019 a spot UT measurement on the 36" pipeline at Gas Terminal revealed '1.03mm thickness reduction along with heavy pitting and painting peeling off... The corrective action of painting was completed in Dec. 2019. However, as per the available data, the design thickness of the pipeline is 23.83mm with zero corrosion allowance. It needs to be examined whether painting as a corrective action was adequate.' This represents wrong choice of action based on incorrect damage assessment. The decision maker selected painting (a cosmetic/protective action) as the corrective measure for a pipe segment that had: (1) measurable thickness loss (1.03mm from 23.83mm design), (2) zero corrosion allowance in the design, and (3) heavy pitting visible. The faulty logic: treating a structural-integrity issue (metal loss beyond allowance) as a surface-protection issue (coating failure). The correct action would be pipe-segment replacement or engineering fitness-for-service analysis, not painting. The committee flags this at p.30: 'It needs to be examined whether painting as a corrective action was adequate for a pipeline with zero corrosion allowance

CAUSAL EXPLANATION

The wrong choice of painting (rather than replacement or fitness-for-service analysis) for a pipe segment with thickness loss beyond design allowance is a secondary contributing factor. It demonstrates organizational tolerance for selecting inadequate corrective actions following inspection findings. While this specific wrong choice occurred on the 36" gas pipeline (which remained intact during the incident [EVD-5780]), the decision-making pattern - treating structural-integrity deficiencies with cosmetic/protective measures - is the same pattern that allowed substitute O-rings to be used repeatedly for pressure-boundary gaskets without engineering evaluation. The organizational culture that accepts painting as adequate for metal-loss beyond allowance is the same culture that accepts locally-sourced O-rings without specification control.

Linked Evidence: EVD-2397, EVD-5527

AJDE 2.9 **Decision Making - During Operation** SUBSTANTIATED

Overall Shift In-charge directed insufficient isolation response - only 36" PCV closure, lacking comprehensive shutdown protocol for major leak scenario.

JUSTIFICATION

The Overall Shift In-charge (GM Production, acting as SIC) made an insufficient emergency response decision by directing closure of only the 36" pressure control valve from the Gas Terminal CR and not immediately ordering closure of MOV 54 or other isolation points. [EVD-7340] documents: the SIC 'instructed to close the 36 line pressure control valve and get out of the control room.' Timeline shows the 36" PCV was closed at 03:04 [EVD-6414], 3 minutes AFTER MOV 54 (which was closed by the GSP Shift In-charge at 03:01 [EVD-9612]). The SIC's response was insufficient because: (1) only one isolation point was directed initially, (2) no comprehensive shutdown sequence was ordered until after ignition at 03:08:15, and (3) the offshore supply stoppage was not ordered until ~03:10 post-ignition. This represents faulty emergency decision-making in a time-constrained environment - the SIC did not select a comprehensive isolation strategy despite being the senior on-site authority. The committee identifies the root cause: the DMP lacks decision criteria for 'isolation/stoppage of facilities/pipelines' [EVD-4052], meaning even the senior shift

CAUSAL EXPLANATION

The SIC's insufficient isolation response contributed to the emergency response delay as a secondary factor. The lack of a comprehensive isolation protocol meant that valve closures were executed sequentially and reactively (MOV 54 at 03:01, 36" PCV at 03:04, offshore stoppage ~03:10) rather than simultaneously per a pre-planned emergency sequence. This serial decision-making extended the time to full isolation and allowed continued inventory feed from upstream sources. However, the GSP Shift In-charge's independent execution of MOV 54 closure at 03:01 demonstrates that isolation would have occurred even without comprehensive SIC direction, which is why this is rated 3 (secondary) rather than 4 (primary). The root issue is organizational: the absence of DMP isolation criteria [EVD-4052] meant no shift authority - including the SIC - had decision tools to execute rapid, comprehensive isolation in response to a major leak.

Linked Evidence: EVD-7340, EVD-6414, EVD-4052, EVD-9612

AJDE 2.10 **Knowledge/Experience inadequate to Make Decision** SUBSTANTIATED

Control room operators and shift supervisors lacked knowledge of isolation decision criteria for major vapor releases, delaying first valve closure 46 minutes after leak onset.

JUSTIFICATION

The control room operators (Chief Engineer Production and Junior Engineer Mechanical) and the shift supervisory chain (GSP Shift In-charge DGM, Overall Shift In-charge GM) lacked the knowledge and experience to make rapid isolation decisions during a major hydrocarbon vapor release. [EVD-6414] documents the 46-minute delay from leak onset (02:12:50) to first isolation valve closure (MOV 54 at 03:01). [EVD-1460] and [EVD-9220] show that both control room engineers attempted to physically locate the leak source on foot rather than immediately initiating isolation based on CISF reports of vapor cloud. [EVD-9612] shows the GSP Shift In-charge also could not enter the area due to gas density when he arrived at 02:35. The critical knowledge gap is documented in [EVD-4052]: 'As per DMP On site Hazira Plant, guidelines to be followed in case of hydrocarbon leakage are included but does not include isolation/stoppage of facilities/pipelines' and 'In ERDMP, scenario for Level 3 emergencies are taken from the plant failures. Heavy product/gas leakages from pipelines & tanks are considered but decision on isolation/closure is not included.' The operators lacked:

CAUSAL EXPLANATION

The operators' and supervisors' inadequate knowledge of isolation decision criteria caused a 46-minute delay in stopping the inventory release. This delay transformed a containable leak into a major vapor cloud that occupied vertical columns around the metering area. The delayed isolation directly enabled the ignition at 03:08:15 - had MOV 54 been closed within 5-10 minutes of detection at 02:15, the flammable inventory would have been substantially reduced and ignition may have been prevented or the fire magnitude minimized.

Linked Evidence: EVD-6414, EVD-1460, EVD-9612, EVD-4052, EVD-3932, EVD-8213

AV 1.1 **Violation - Routine/Widespread** SUBSTANTIATED

Locally-sourced O-rings routinely substituted for lost OEM gasket specs without quality control - normalized practice over 22+ months with three applications on failed meter.

JUSTIFICATION

The use of locally-purchased O-ring gaskets as substitutes for the original OEM rectangular-slot gasket specification constitutes a routine/widespread violation with direct causal connection to the incident. [EVD-8645] documents that 'Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings.' This practice was systemic - not isolated to a single instance - as evidenced by [EVD-1494] showing three gasket replacements in 22 months (Nov 2018, Feb 2019, Sep 2020) on meter 310FT01, all using this substitute approach. The violation was condoned by supervision: [EVD-8645] states 'There is no record of date or authority of the decision to change from rectangular slot gasket to O ring' - indicating no formal challenge or corrective action despite repeated applications. This was the normal way of working for maintaining 1986-vintage meters whose OEM specifications had been lost. The practice directly contributed to the catastrophic flange-joint failure: [EVD-0064]

CAUSAL EXPLANATION

The routine violation of using non-specification substitute gaskets created the material precondition for the flange-joint failure. Without proper OEM gasket geometry (rectangular slot vs O-ring) and material specification, the retaining-plate bolts could not maintain pressure-boundary integrity under operating conditions. The third failure in 22 months indicates the substitute approach was fundamentally inadequate, yet the practice continued without intervention. This violation directly enabled the leak initiation at 02:12:50 on 24-Sep-2020.

Linked Evidence: EVD-8645, EVD-1494, EVD-0064

AJDE 2.4 Decision-Making based on Misinterpretation

SUBSTANTIATED

Chief Engineer misinterpreted DCS alarm severity timing, treating delayed alarm as recent event requiring field verification rather than 7-minute-old major release needing immediate isolation.

JUSTIFICATION

The Chief Engineer (Production) in the Gas Terminal Control Room received correct information via the DCS annunciator alarm at ~02:20 hrs and the simultaneous CISF phone call [EVD-1460]. However, the DCS alarm appeared approximately 7 minutes AFTER the actual leak onset (02:12:50 per CCTV [EVD-1284]) and 5 minutes AFTER human perimeter detection (02:15 [EVD-4286]). The Chief Engineer's immediate response was to physically approach the affected area to locate the source [EVD-1460], rather than immediately initiating isolation procedures from the control room. This suggests he may have misinterpreted the severity or nature of the leak based on the delayed alarm presentation - treating it as a localized issue requiring field verification rather than an ongoing major release requiring immediate remote isolation. The decision to walk into the vapor cloud area (which forced both engineers back due to density) consumed critical time that could have been used for isolation actions. The annunciator alarm was correct information about a leak, but the 7-minute delay between leak onset and alarm may have created a temporal-severity

CAUSAL EXPLANATION

The 7-minute delay between actual leak onset and DCS alarm presentation created an information-severity mismatch. The Chief Engineer received accurate information (leak alarm) but the temporal delay may have caused him to interpret it as a new/recent event rather than an ongoing major release, leading to the decision to investigate on foot rather than immediately isolate remotely. This contributed to the 46-minute response delay before first valve closure at 03:01.

Linked Evidence: EVD-1460, EVD-8657, EVD-1284

AJDE 2.7 Delayed a Necessary Action

SUBSTANTIATED

Maintenance team delayed/omitted mandatory hydrostatic test after gasket replacement on 18-Sep-2020, returning meter to service without pressure boundary validation.

JUSTIFICATION

The maintenance team replaced the retaining-plate gasket on meter 310FT01 on 18-Sep-2020 but deliberately chose to delay/omit the standard hydrostatic pressure test that should have been performed immediately after the gasket replacement. [EVD-2894] documents: 'No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated.' The system was pressurized only to operating pressure rather than the higher design pressure that a hydrostatic test would require. This is a necessary action (pressure boundary integrity validation after maintenance on a critical flange joint) that was delayed indefinitely - the meter was returned to service on 19-Sep without the test ever being performed [EVD-0185]. The decision to delay/omit this validation step allowed an improperly seated gasket to remain in service under operating pressure for 5 days until catastrophic failure on 24-Sep at 02:12:50 [EVD-1284]. The committee explicitly identifies this as a root cause at p.31 and recommends at p.35 (D.2.iv): 'All pressure equipment shall be subject to pressure testing as per approved

CAUSAL EXPLANATION

The decision to delay/omit the hydrostatic pressure test after gasket replacement created a direct causal pathway to the incident. The test would have revealed whether the substitute O-ring gasket was properly seated and could maintain pressure boundary integrity under design conditions. Without this validation, an improperly installed gasket remained in service under operating pressure (likely in the 50-60 bar range typical for condensate metering) for 5 days. The delayed action allowed the gasket to progressively fail until catastrophic loss of containment at 02:12:50 on 24-Sep. Had the test been performed immediately after installation as required, the gasket failure would have been detected in a controlled test environment rather than during live operation, preventing the 56-minute vapour accumulation and subsequent ignition.

Linked Evidence: EVD-2894, EVD-0185, EVD-0591

AJDE 2.7 Delayed a Necessary Action

SUBSTANTIATED

Maintenance/engineering delayed corrective actions on pipes with significant thickness loss identified in 2017 for 3+ years without developing a remediation plan.

JUSTIFICATION

[EVD-5527] documents that the 2017 ultrasonic thickness measurement identified 'several pipes with significant thickness loss and need for corrective action. However, even after three years no plan has been developed for the corrective actions to be taken for those lines.' This is a necessary action (corrective maintenance on integrity-compromised piping) that was delayed for 3+ years from identification (2017) to the incident date (24-Sep-2020). The committee explicitly states: 'Mere measurement of thickness without corrective action does not complete the process of maintaining the integrity of pipes' (p.30). While the 2017 findings did not directly identify the condensate metering skid as compromised (the leak source was a gasket, not pipe-wall failure), the systemic delay in addressing identified integrity deficiencies is a contributing organizational factor. The delay indicates a pattern where necessary corrective actions following inspections are postponed indefinitely without documented authority or risk assessment. This pattern is reinforced by [EVD-8307]: the 4-year UT cycle itself was not adhered to (2011->2017 = 6 years, breaking the

CAUSAL EXPLANATION

The 3+ year delay in addressing 2017 thickness-loss findings is a secondary contributing factor representing organizational tolerance for deferred corrective action. While the delay did not directly cause the gasket failure (the leak source was a flange joint, not pipe-wall corrosion), it reveals a systemic pattern where necessary maintenance actions identified through inspection programs are delayed indefinitely without risk-based prioritization. This organizational pattern likely influenced the broader maintenance culture in which the gasket-replacement procedure was executed without hydrostatic testing (instance 1) and recurrent gasket failures were not escalated as reliability concerns [EVD-1494]. The delay normalizes deviation from integrity-management best practices, creating an environment where pressure-boundary maintenance can be executed with procedural shortcuts.

Linked Evidence: EVD-5527, EVD-8307

AJDE 2.9 Decision Making - During Operation

SUBSTANTIATED

Gas Terminal CR operators unable to execute immediate isolation due to dense vapour cloud preventing physical approach and lack of remote-isolation protocol.

JUSTIFICATION

The Gas Terminal Control Room operators (Chief Engineer Production and Junior Engineer Mechanical) were unable to execute the correct emergency response (immediate isolation of the leaking system) due to the time-critical, hazardous environment created by the dense vapour cloud. [EVD-1460] and [EVD-9220] document that at ~02:20 both engineers 'went out together to try and reach the metering area. The cloud was so dense and the smell so strong that they could not get close.' This represents insufficient response to an urgent situation - the operators recognized the need for action (they attempted to approach the leak source) but could not execute the necessary isolation because the vapour density physically prevented access. The decision-making environment was severely constrained: (1) time-critical (leak ongoing with accumulating flammable cloud), (2) information-limited (leak source location unknown, only 'heavy cloud' visible), (3) physically hazardous (LEL concentrations preventing approach). The operators' response was to retreat and escalate via telephone to the GSP Shift In-charge [EVD-1460], which added 41 minutes of latency before MOV 54 was closed at

CAUSAL EXPLANATION

The operators' inability to execute immediate remote isolation represents a failure of emergency decision-making under time constraint. The correct response to a major hydrocarbon leak with visible vapour cloud accumulation is immediate isolation of the source via remote-operated valves (MOV 54, 36" PCV) from the control room, without requiring field confirmation of the exact leak location. However, the operators chose to attempt physical approach first, which consumed critical minutes and failed due to environmental hazards. When forced to retreat, they escalated via telephone rather than executing remote isolation themselves. This decision-making failure was enabled by the absence of clear DMP protocols for remote isolation [EVD-4052] - the operators lacked training or procedures to guide the urgent response. The 41-minute delay from operator detection (02:20) to first isolation (03:01) allowed massive inventory accumulation [EVD-8213] that sustained the subsequent fire even after isolation was finally achieved.

AJDE 2.10 **Knowledge/Experience inadequate to Make Decision** **SUBSTANTIATED**

Maintenance team lacked technical knowledge to recognize that substituting unspecified O-rings for OEM rectangular-slot gaskets without hydrostatic testing created unacceptable pressure boundary risk.

JUSTIFICATION

The maintenance team that replaced the gasket on meter 310FT01 on 18-Sep-2020 lacked the technical knowledge and experience required to make appropriate decisions regarding pressure boundary integrity without OEM specifications. [EVD-8645] documents that 'Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings.' The team proceeded with a locally-sourced O-ring substitute for a 1986 rectangular-slot gasket design without documented specifications or quality control. This decision required specialized knowledge of: (1) pressure boundary integrity requirements for hydrocarbon service, (2) gasket material compatibility and sealing performance for the specific meter design, (3) the criticality of proper gasket selection for high-pressure condensate metering. The team's knowledge was demonstrably inadequate - they did not recognize that using an unspecified substitute gasket without hydrostatic testing ([EVD-2894]: 'No hydro test was done after

CAUSAL EXPLANATION

The maintenance team's inadequate technical knowledge created the primary failure mechanism. Without understanding of gasket material specifications, pressure boundary integrity requirements, or the need for hydrostatic validation, they installed an unsuitable gasket that failed under operating pressure 5 days later, initiating the 56-minute vapor release that led to ignition and major fire.

Linked Evidence: EVD-8645, EVD-2894, EVD-1494, EVD-0064

ASBE 1.3 **Procedure Not Followed/Not Followed Correctly** **SUBSTANTIATED**

Maintenance personnel used wrong gasket type (O-ring vs rectangular-slot) without specification or quality control, deviating fundamentally from correct replacement procedure.

JUSTIFICATION

This is a primary cause of the incident. [EVD-8645] documents that the correct procedure for gasket replacement was not followed: 'The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings.' The correct procedure would have been: (1) identify OEM gasket specification, (2) procure to specification, (3) verify quality, (4) install, (5) hydro-test. Instead, the procedure followed was: (1) purchase local O-ring without specification, (2) install, (3) pressurize to operating pressure only. The committee identifies this as root cause #2: 'Use of nonstandard O ring' and root cause #3: 'Lack of control over material sourcing' (p.33). The wrong technique (O-ring substitution for rectangular-slot gasket) combined with omission of critical steps (specification verification, QC, hydro-test) directly caused the catastrophic flange failure.

CAUSAL EXPLANATION

The use of wrong gasket material and wrong technique was the direct mechanical cause of the loss of containment. A rectangular-slot gasket design requires a specific geometry and material specification to seal under design pressure; substituting a generic O-ring without specification, quality verification, or hydro-testing meant the pressure boundary was inherently compromised. When the meter operated at normal condensate pressure for 6 days (19-24 Sep), the inadequate seal failed catastrophically, releasing the vapour cloud that ignited 56 minutes later.

Linked Evidence: EVD-8645, EVD-1494

ASBE 1.3 **Procedure Not Followed/Not Followed Correctly** **SUBSTANTIATED**

Emergency isolation executed 46 minutes after leak detection due to procedural sequence requiring escalation before action, allowing massive vapour cloud accumulation.

JUSTIFICATION

The emergency isolation procedure was not executed with correct timing, contributing directly to the severity of the incident. [EVD-9612] documents that the GSP Shift In-charge 'closed the MOV 54 inlet condensate line from slug catcher to 36 metering area at about 0301 hrs' - this is 46 minutes after first physical detection at 02:15 and 48 minutes after CCTV-confirmed leak onset at 02:12:50 [EVD-1284]. [EVD-4052] identifies the procedural gap: 'As per DMP On site Hazira Plant, guidelines to be followed in case of hydrocarbon leakage are included but does not include isolation/stoppage of facilities/pipelines... decision on isolation/closure of plant/pipelines is not mentioned.' The correct procedure for a confirmed hydrocarbon release in a Zone 1 area would be immediate isolation of the source; instead, the actual sequence was: detection -> verification -> escalation -> authority request -> physical travel to valve -> closure. The 46-minute delay allowed massive inventory release [EVD-8213] shows 'vapour all around even at height' at 03:00:44, just before closure. The committee identifies this as root cause #6: 'Non isolation/stoppage of 36 pipeline

CAUSAL EXPLANATION

The 46-minute delay in executing the isolation procedure was the primary factor that allowed the release to transition from a small leak to a major fire. Had MOV 54 been closed within minutes of detection at 02:15 (as would occur with a correct immediate-isolation procedure), the vapour cloud would have been orders of magnitude smaller. Instead, the escalate-first procedure consumed 46 minutes during which condensate vapour continuously released and spread across the metering area. By the time isolation occurred at 03:01, the flammable envelope was already massive enough to find the compromised FLP lamp and ignite at 03:08:15, producing the major fire.

Linked Evidence: EVD-9612, EVD-4052, EVD-8213, EVD-6414

ASBE 1.6 Breakdown in Visual Scan

SUBSTANTIATED

GT Control Room operators failed to detect vapour cloud via external visual scan despite proximity to source, relying instead on alarm and phone call 5+ minutes after leak onset.

JUSTIFICATION

[EVD-1460] and [EVD-9220] document that the Gas Terminal Control Room operators (Chief Engineer Production and Junior Engineer Mechanical) did not detect the leak until 02:20 - approximately 5 minutes after CISF perimeter personnel detected it at 02:15 [EVD-4286], despite being physically closer to the source. [EVD-1460] states 'At around 0220 hrs Mr. [CE] heard alarm on the annunciator panel and the same time he received a phone call from CISF SI [CCTV CR].' The operators' detection came from the phone call and alarm, not from their own visual scan of their operating area. Given that the GT Control Room has windows overlooking the metering skid area and the vapour cloud was visible on CCTV from 02:12:50 [EVD-1284], a proper external visual scan by the on-shift operators should have detected the release before perimeter security 200+ meters away. [EVD-9941] shows the cloud was spreading rapidly from 02:15:35 onward and was 'intense' by 02:16:13 - well within visual range of the GT CR windows. The breakdown in visual scan is compounded by [EVD-5634]: operators received only 'four days training programme... which is general in nature' with no

CAUSAL EXPLANATION

The failure to maintain effective external visual scan contributed to the 5-8 minute detection delay. Had the GT CR operators detected the vapour cloud at 02:12-02:13 (when it first became visible) rather than waiting for the alarm at 02:20, the emergency response could have begun 5-8 minutes earlier, potentially reducing the pre-ignition accumulation period and enabling faster isolation.

Linked Evidence: EVD-1460, EVD-9220, EVD-4286, EVD-9941, EVD-5634

ASBE 1.2 Checklist Not Followed/Not Followed Correctly

SUBSTANTIATED

Calibration team failed to follow checklist during Dec 2019 work on meter 310FT01, leaving undocumented setting changes from 16000 to 2540 mmwc.

JUSTIFICATION

The evidence demonstrates a clear failure to follow calibration checklists. [EVD-2506] documents that during calibration from 17/12/2019 to 18/12/2019, the calibration setting was changed from 16000 mmwc to 2540 mmwc (~6.3x reduction) 'without any mention of this change in the calibration documents signed by the team.' The document further states 'No checklist is followed while carrying such critical jobs of high pressure/risk.' This is a direct instance of checklist not being followed during a critical maintenance activity on the same meter that later failed. The committee's recommendation D.1.i (p.35) explicitly addresses this gap by requiring 'details of Master calibrator/action taken/replacement/deviation from procedure if any, is to be documented in the calibration certificate.' This checklist failure contributed to a pattern of undocumented changes and inadequate supervision of critical pressure-boundary work on meter 310FT01.

CAUSAL EXPLANATION

The absence of checklist discipline during calibration work on meter 310FT01 in Dec 2019 established a pattern of undocumented modifications to critical pressure-boundary equipment. This contributed to the broader maintenance-governance failures that culminated in the Sep 2020 gasket failure, as it normalized deviation from standard procedures on the same meter that later became the leak source.

Linked Evidence: EVD-2506

ASBE 1.2 **Checklist Not Followed/Not Followed Correctly** **SUBSTANTIATED**

Maintenance team failed to perform mandatory hydrostatic pressure test after 18-Sep-2020 gasket replacement, returning meter to service with unvalidated pressure boundary.

JUSTIFICATION

This is a primary contributing factor to the incident. [EVD-2894] documents that after the gasket replacement on 18.09.2020, 'No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated.' The committee directly identifies this as a probable cause: 'Probable cause of leak is due to use of nonstandard gasket and not testing the system to designed pressure. No standard operating procedure is in place for calibration and taking back in service of these meters which are operating at high pressures' (p.32). Standard pressure-boundary integrity procedures universally require hydrostatic testing to design pressure after gasket replacement on high-pressure systems. The failure to execute this critical checklist step meant the non-standard O-ring gasket was returned to service without verification of its ability to hold design pressure, directly enabling the catastrophic flange failure 6 days later (24-Sep-2020). This is not a case of incorrect execution but complete omission of a mandatory test step.

CAUSAL EXPLANATION

The failure to follow the post-maintenance testing checklist (omission of hydrostatic test to design pressure) directly enabled the incident. The non-standard O-ring gasket installed on 18-Sep-2020 was returned to service without verification that it could withstand design pressure. When the meter operated under normal conditions for 6 days, the inadequate gasket seal failed catastrophically at 02:12:50 on 24-Sep-2020, releasing condensate vapour that accumulated for 56 minutes before igniting. This checklist omission is a primary cause in the incident chain.

Linked Evidence: EVD-2894, EVD-0591

AJDE 2.11 **Misapplied or unaware of existing Policy/Procedure** **SUBSTANTIATED**

Maintenance team unaware of or misapplied hydrostatic test requirement after pressure boundary work - pressurized to operating pressure only instead of design pressure validation.

JUSTIFICATION

The maintenance team that performed the 18-Sep-2020 gasket replacement on meter 310FT01 was either unaware of or misapplied the policy/procedure requiring hydrostatic testing of pressure boundaries after maintenance. [EVD-2894] documents: 'After calibration of the meter 310FE01, the gasket was replaced on 18.09.2020. No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated.' Standard pressure equipment integrity procedures require hydrostatic testing to design pressure (typically 1.5x operating pressure) after any work that breaks a pressure boundary seal. The team instead only pressurized to operating pressure with both meters in line - an inadequate validation that does not prove seal integrity under worst-case conditions. The committee's finding at p.31 explicitly identifies this as a procedural failure: the gasket was replaced and the system returned to service without pressure validation. [EVD-0185] documents the meter was returned to service ~46 hours later and remained in operation until catastrophic failure at 02:12:50 on 24-Sep. This represents either: (1)

CAUSAL EXPLANATION

The maintenance team's failure to apply hydrostatic testing policy created the direct mechanism for catastrophic failure. The unvalidated gasket seal was installed with an unspecified O-ring substitute and never tested to design pressure. This allowed a defective pressure boundary to enter service on 19-Sep and fail under operating conditions at 02:12:50 on 24-Sep, initiating the entire incident chain. Had hydrostatic testing been performed per procedure, the inadequate seal would have been detected and corrected before the meter returned to service, preventing the incident.

Linked Evidence: EVD-2894, EVD-0185, EVD-8645

AJDE 2.11 **Misapplied or unaware of existing Policy/Procedure** **SUBSTANTIATED**

Maintenance supervisors unaware of or misapplied SAP notification closure procedure requiring review before return to service, allowing defective gasket work to proceed.

JUSTIFICATION

Maintenance supervisors were either unaware of or misapplied the SAP notification closure procedure that would have required review of the 18-Sep-2020 gasket replacement work quality before return to service. [EVD-0591] documents that SAP Notification 3800419840 was raised on 17-Sep-2020 for meter 310FT01 malfunction. The gasket was replaced on 18-Sep and the meter returned to service on 19-Sep ([EVD-0185]), but the SAP notification closure process was not properly executed. The committee's recommendation D.2.ii at p.35 explicitly addresses this gap: 'Sample check of SAP notifications opened & closed, Permits issued & closed should be carried out by DGM/GM (Ops) monthly basis. The review should include all pending SAP notifications, permits, work orders etc.' This recommendation was made because the investigation found that notification closure procedures were not being followed - if they had been, the supervisory review would have identified: (1) the missing hydrostatic test ([EVD-2894]), (2) the use of unspecified substitute gasket ([EVD-8645]), and (3) the recurrence pattern of gasket failures requiring investigation ([EVD-1494]: third replacement in

CAUSAL EXPLANATION

The supervisors' failure to apply SAP notification closure policy removed a critical quality gate that would have detected the defective gasket work. Proper closure review would have identified the missing hydrostatic test, unspecified materials, and recurrence pattern, preventing the defective work from entering service. The policy misapplication allowed the flawed gasket installation to proceed to operation and fail catastrophically, contributing to the incident.

Linked Evidence: EVD-0591, EVD-0185, EVD-2894, EVD-8645

AJDE 2.11 **Misapplied or unaware of existing Policy/Procedure** **SUBSTANTIATED**

Operators unaware that DMP lacked isolation decision criteria for major leaks, causing them to default to physical verification and multi-layer escalation instead of immediate isolation.

JUSTIFICATION

Control room operators and shift supervisors were unaware that the Disaster Management Plan (DMP) was missing critical guidance on isolation/stoppage decision criteria for major hydrocarbon leakages, leading them to default to physical verification approaches that caused fatal delay. [EVD-4052] explicitly documents the policy gap: 'As per DMP On site Hazira Plant, guidelines to be followed in case of hydrocarbon leakage are included but does not include isolation/stoppage of facilities/pipelines' and 'In ERDMP, scenario for Level 3 emergencies are taken from the plant failures. Heavy product/gas leakages from pipelines & tanks are considered but decision on isolation/closure is not included.' The operators' response pattern ([EVD-1460], [EVD-9220], [EVD-9612]) - attempting physical approach to locate the source before initiating isolation - was consistent with personnel who believed they were following procedure but were actually operating in a policy void. They were unaware that the DMP guidance they thought existed (isolation criteria for confirmed major leaks) was in fact absent from the plan. The 46-minute delay ([EVD-6414]) from leak onset to

CAUSAL EXPLANATION

The operators' unawareness of the DMP policy gap for isolation criteria was a primary cause of the 46-minute delay. Believing they were following procedure by seeking situational awareness and senior authorization, they did not recognize that immediate isolation upon confirmed vapor cloud reports was the appropriate action in the absence of specific guidance. This policy void directly caused the massive inventory release documented in EVD-8213 ('vapour all around even at height' at 03:00:44, just before isolation), which enabled the ignition at 03:08:15 and subsequent major fire. Had the DMP contained clear isolation criteria and the operators been aware of and trained on them, MOV 54 would likely have been closed within 10-15 minutes of detection, preventing or minimizing the fire.

Linked Evidence: EVD-4052, EVD-6414, EVD-1460, EVD-3932

AJDE 2.11 **Misapplied or unaware of existing Policy/Procedure** **SUBSTANTIATED**

Maintenance planners unaware of or misapplied MOC policy when substituting O-rings for OEM gaskets - significant materials change implemented as 'regular practice' without MOC review.

JUSTIFICATION

Maintenance planners were either unaware of or misapplied the Management of Change (MOC) policy when substituting O-rings for OEM rectangular-slot gaskets on meter 310FT01. [EVD-2986] documents that 'Management of Change (MOC) philosophy/SOP was published in August 2016. However, the procedure indicated in the document is not implemented while carrying out change of Low-Pressure Gas to Kribhico from Medium Pressure Gas headers in LPG Plant in 2013 & 2018. It is informed no more MOCs are implemented in the plant since 2018.' The MOC SOP existed since Aug 2016 but was not being applied to plant changes after 2018. [EVD-8645] documents the gasket substitution: 'The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch from rectangular slot gasket to O ring.' The decision to switch from

CAUSAL EXPLANATION

The planners' failure to apply MOC policy to the gasket substitution decision removed a critical engineering review gate that would have: (1) challenged the appropriateness of O-ring substitutes for rectangular-slot gaskets, (2) required material specifications and quality control, (3) mandated engineering approval and documentation. The MOC policy misapplication allowed an undocumented, unreviewed materials change to become 'regular practice', creating the conditions for the inadequate gasket installation that failed catastrophically. While not the immediate trigger, the MOC bypass was a foundational governance failure that enabled the direct causes.

Linked Evidence: EVD-2986, EVD-8645

AV 1.1 **Violation - Routine/Widespread** **SUBSTANTIATED**

SAP notification 3800419840 for meter malfunction not closed after gasket replacement work - systematic non-closure pattern requiring committee recommendation for sample checks.

JUSTIFICATION

[EVD-0591] documents that 'SAP Notification 3800419840 was raised on 17/09/2020 for malfunction of meter no.1' and the committee recommendation D.2.ii (referenced in [EVD-0185]) explicitly addresses 'Sample checks should be done to ensure... SAP notifications are closed immediately with all details of the job completed.' The fact that this recommendation was made indicates the committee found systematic non-closure of SAP notifications to be a routine practice. The sequence confirms non-closure in this instance: notification raised 17-Sep -> gasket replaced 18-Sep -> meter returned to service 19-Sep -> incident 24-Sep, with no documented notification closure. This constitutes a routine violation of SAP work-management discipline. The practice appears widespread - the committee's recommendation uses 'sample checks' language, implying the problem extends beyond this single notification. Leadership sanction is implicit through the absence of documented disciplinary action for incomplete SAP work records. While this violation did not directly cause the physical failure, it prevented traceability of the maintenance quality issue and pattern recognition of the

CAUSAL EXPLANATION

The non-closure of SAP notifications prevented systematic tracking of the meter's recurring gasket-replacement pattern (3 times in 22 months). Had notifications been properly closed with full job details, reliability analysis tools could have flagged the repeat-failure pattern and triggered root-cause investigation before the catastrophic failure. This violation degraded the organization's ability to learn from maintenance history.

Linked Evidence: EVD-0591, EVD-0185

AV 1.1 **Violation - Routine/Widespread** **SUBSTANTIATED**

P&ID 02-310-AO-101 Rev.1 for metering system: original lost, vendor replacement unapproved, field configuration mismatched - routine configuration-management breakdown.

JUSTIFICATION

[EVD-0933] documents that 'Reference drawings no. 02-310-AO-101 Rev.1, 2x6 lines & 1x2 lines for metering system are shown but valves & meters configuration are not matching. As old drawings were lost, new drawings are made by a vendor but the same has not been checked & approved by ONGC.' Three compounding violations documented: (1) Original drawings lost (configuration-management failure); (2) Vendor-prepared replacement drawings not checked/approved by operator; (3) Field configuration does not match drawings. The routine nature is indicated by the matter-of-fact tone - 'old drawings were lost' suggests this is not unique to this P&ID. This violation had limited direct impact on the incident itself (the metering equipment existed and was operated regardless of drawing accuracy), but it reflects the broader erosion of configuration-management discipline and may have complicated the emergency response (responders could not rely on drawings for valve/equipment locations).

CAUSAL EXPLANATION

While not directly causing the gasket failure, the unapproved and inaccurate P&IDs reflect the systematic erosion of configuration management that enabled other violations (lost OEM gasket specs, missing baseline thickness data) to persist. Accurate as-built drawings are a foundational control for safe operation and maintenance of aging assets.

Linked Evidence: EVD-0933

ASBE 1.2 **Checklist Not Followed/Not Followed Correctly** **SUBSTANTIATED**

Audit teams failed to properly complete safety audit checklists, missing critical equipment integrity issues including the FLP lamp that became the ignition source.

JUSTIFICATION

[EVD-9692] documents systematic checklist failures in the self-safety audit process: 'Safety self-audit is conducted by a multi-disciplinary team based on the checklist prepared by the location. The checklist is comprehensive but not related to specific plants and in the process, lot of observations are missing... during our inspection, the data is found to be incomplete & not in line with OISD standards/manuals.' [EVD-7681] reinforces this: 'Internal Safety Audit was conducted in 2018 & 2019... the observations are given without the filled checklist. The checklist format is not available to check completeness of audit areas.' The audit teams were not following their own checklists systematically, resulting in incomplete coverage. The committee recommendation B.1 (p.34) explicitly calls for 'Checklist for Self-Safety Audit for individual plants are to be developed,' indicating the existing checklists were not fit-for-purpose and were not being completed properly. This contributed to the failure to detect preconditions such as the compromised HPMV FLP lamp integrity and the degraded electrical maintenance practices.

CAUSAL EXPLANATION

The failure to properly execute audit checklists allowed latent preconditions to persist undetected, including the compromised HPMV FLP lamp with three missing bolts and 5-6 cm glass-case opening that became the probable ignition source. Had the audit checklists been followed systematically, these Zone 1 electrical-integrity failures would likely have been identified and corrected before the incident.

Linked Evidence: EVD-9692, EVD-7681

ASBE 1.5 **Over-controlled or Under-controlled Vessel/Aircraft/Vehicle or System** **SUBSTANTIATED**

Calibration setting was under-controlled, reducing measurement range from 16000 to 2540 mmwc without documented justification.

JUSTIFICATION

EVD-2506 documents that the calibration of meter 310FT01 was changed from 16000 mmwc to 2540 mmwc. This represents a significant reduction in measurement range (from 16000 to 2540 mmwc), effectively under-controlling the meter's ability to read the process variable. Such an inappropriate control input constitutes a skill-based error (either a slip in data entry or a mistake in selecting the wrong range) that directly affected the instrument's function and reliability. The error went undetected due to the lack of documentation, compounding its negative effect.

CAUSAL EXPLANATION

Setting the calibration too low may have caused the meter to operate out of range or provide false readings, preventing early detection of performance degradation and contributing to the recurrence of gasket failures that ultimately led to the loss of containment.

Linked Evidence: EVD-2506

AJDE 2.10 **Knowledge/Experience inadequate to Make Decision** **SUBSTANTIATED**

Calibration team lacked knowledge of calibration management principles - made major undocumented setting change (16000->2540 mmwc) without engineering justification or approval.

JUSTIFICATION

The calibration team that worked on meter 310FT01 in December 2019 lacked the knowledge and experience to recognize that changing a critical calibration setting from 16000 mmwc to 2540 mmwc without documentation violated fundamental calibration management principles. [EVD-2506] documents: 'During calibration from 17/12/2019 to 18/12/2019, calibration was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of this change in the calibration documents signed by the team. During next calibration for meter 310FT01 was again changed to 16000 MMWC. No checklist is followed while carrying such critical jobs of high pressure/risk.' This represents a major calibration setting change (84% reduction) on a critical flow meter that was not documented in signed records and was later reversed without explanation. The team's knowledge was inadequate in several dimensions: (1) they did not understand that undocumented calibration changes create undetectable measurement errors and safety risks, (2) they did not recognize the need for engineering justification and approval for such a significant setting change, (3) they lacked

CAUSAL EXPLANATION

The calibration team's inadequate knowledge of calibration management contributed to unreliable meter performance and demonstrated the systemic technical knowledge gap in the maintenance organization. The undocumented setting changes created measurement uncertainty that may have contributed to abnormal meter behavior and accelerated gasket degradation, and the same knowledge deficiency was present in the team that performed the fatal gasket replacement 9 months later.

Linked Evidence: EVD-2506

APE 3.2 **Errors due to Tactile/Sensory Misperception** **SUBSTANTIATED**

Dense vapour cloud prevented Gas Terminal CR personnel from accurately perceiving distances and safe approach paths to leak source

JUSTIFICATION

Both the Chief Engineer (Production) and Junior Engineer (Mechanical) at the Gas Terminal Control Room experienced tactile/sensory misperception when attempting to locate the leak source. [EVD-1460] records the Chief Engineer stating that upon going outside 'the cloud was so dense and the smell so strong that we could not get close' and [EVD-9220] documents the Junior Engineer observing 'Visibility was very poor'. Similarly, [EVD-9612] records the DGM (GSP Shift In-charge) and Overall Shift In-charge 'could not enter due to heavy gas smell' when they arrived at approximately 02:35. The dense vapour cloud created a sensory environment where personnel could not accurately perceive distances, boundaries, or safe approach paths. [EVD-9941] provides CCTV evidence of the rapid cloud expansion between 02:15:35 and 02:16:13, documenting how the vapour 'spread from the leak source across the GT CR, road, and piping area within ~40 seconds'. This environmental condition prevented direct visual and spatial assessment of the leak source location, forcing personnel to rely on remote valve operations rather than direct intervention. The misperception was

CAUSAL EXPLANATION

The sensory misperception caused by the dense vapour cloud was a secondary contributing factor to the 46-minute response delay. While personnel correctly recognized they could not safely approach the leak source (appropriate safety behavior), the inability to visually locate and assess the leak forced reliance on remote operations and delayed the first isolation valve closure (MOV 54) until 03:01. The environmental conditions prevented direct sensory assessment that might have enabled faster source identification and isolation decision-making.

Linked Evidence: EVD-1460, EVD-9220, EVD-9612, EVD-9941

AV 1.1 **Violation - Routine/Widespread** **SUBSTANTIATED**

Meter 310FT01 returned to service after gasket replacement with only operating-pressure test, no hydrostatic test to design pressure - routine practice across multiple maintenance events.

JUSTIFICATION

[EVD-2894] documents that 'After calibration of the meter 310FE01, the gasket was replaced on 18.09.2020. No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated.' The omission of hydrostatic testing after pressure-boundary maintenance constitutes a routine violation if this practice is systemic. The evidence strongly suggests this is routine rather than isolated: (1) The incident report (Section 2) notes 'no separate hydrostatic test to design pressure was performed' without expressing surprise, suggesting this was expected practice; (2) [EVD-1494] documents two prior gasket replacements (Nov 2018, Feb 2019) with no mention of hydrostatic testing, implying the same omission occurred previously; (3) No documented standard operating procedure requiring hydrostatic testing is referenced, suggesting the absence of such testing was the norm. The practice was condoned by supervision - [EVD-0185] confirms the meter was returned to service 'Next day meter 310FE01 was taken on online' with no documented hold-point for pressure validation. This violation directly contributed

CAUSAL EXPLANATION

The absence of hydrostatic testing meant the inadequate substitute O-ring gasket's inability to maintain pressure-boundary integrity was never discovered before operational pressurization. Had a proper hydrostatic test to design pressure been conducted, the gasket installation would likely have failed under test conditions in a controlled environment, preventing the operational failure. This violation removed a critical verification barrier between maintenance intervention and operational risk exposure.

Linked Evidence: EVD-2894, EVD-0185, EVD-1494

AV 1.1 **Violation - Routine/Widespread** **SUBSTANTIATED**

Calibration setting changed 16000->2540->16000 mmwc without documentation in signed calibration records - no checklist followed for critical high-pressure work.

JUSTIFICATION

[EVD-2506] documents that 'During calibration from 17/12/2019 to 18/12/2019, calibration was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of this change in the calibration documents signed by the team. During next calibration for meter 310FT01 was again changed to 16000 MMWC.' This constitutes a routine violation of calibration documentation discipline. The evidence indicates this is systemic rather than isolated: (1) The setting was changed twice (16000->2540->16000) without documentation either time; (2) [EVD-2506] concludes 'No checklist is followed while carrying such critical jobs of high pressure/risk' - using present-tense language indicating ongoing practice; (3) The committee's concern about 'high pressure/risk' jobs lacking checklists suggests this documentation gap extends beyond calibration to other critical work. The violation was condoned by supervision - the calibration documents were 'signed by the team' despite the missing change documentation, indicating supervisory review did not catch or correct the omission. While this specific violation did not directly cause the gasket failure, it reflects the broader pattern of

CAUSAL EXPLANATION

The undocumented calibration changes reflect the weak procedural discipline and documentation culture that also enabled the substitute-gasket and no-hydro-test violations to persist. While not directly causal to the gasket failure, this violation is part of the constellation of routine procedural shortcuts that collectively degraded the integrity-management system for the 1986-vintage metering equipment.

Linked Evidence: EVD-2506

AV 1.1 Violation - Routine/Widespread

SUBSTANTIATED

F&G detection system configured for annunciator alarm only, no auto-shutdown or suppression activation - 5-minute human-detection delay, 46-minute isolation delay.

JUSTIFICATION

[EVD-8657] documents that 'At around 0220 hrs Mr. [CE] heard alarm on the annunciator panel' - approximately 7 minutes after CCTV-confirmed leak onset (02:12:50). However, [EVD-3452] reveals the critical design gap: 'F&G detection system does not have provision for auto shut down of plant or activation of fire suppression system.' The F&G system detected the leak (eventually triggering the annunciator alarm at 02:20) but took no automatic protective action. [EVD-8657] further notes: 'Hydrocarbon/H2S detectors are configured to Annunciator panel rather than shutdown hence there was 5 minute delay from leak to alarm.' This constitutes a routine violation of alarm-management philosophy if the organization's procedures or industry standards require safety-critical alarms to be configured for automatic protective action. The 'configured to Annunciator panel rather than shutdown' design choice reflects a policy decision to rely on operator response rather than automatic interlock - a violation if the risk assessment requires automatic action for hydrocarbon releases. The violation is organization-wide (system design level) and has leadership sanction (no evidence

CAUSAL EXPLANATION

The F&G system's alarm-only configuration (no automatic shutdown/suppression) required the full human detection-assessment-decision-action chain to execute before any isolation occurred. This design choice converted what could have been a sub-second automatic response (machine detection at 02:12:50 -> automatic MOV closure) into a 46-minute manual response (machine detection 02:12:50 -> human alarm 02:20 -> authority on-site -> MOV 54 closure 03:01). The 46-minute inventory release directly enabled the massive flammable cloud that found the ignition source at 03:08:15.

Linked Evidence: EVD-3452, EVD-8657, EVD-1284

ASBE 1.2 Checklist Not Followed/Not Followed Correctly

SUBSTANTIATED

Maintenance team omitted mandatory hydrostatic test after gasket replacement on 18-Sep-2020, returning untested pressure boundary to service.

JUSTIFICATION

The maintenance team completed gasket replacement on meter 310FT01 on 18-Sep-2020 but critically omitted the mandatory hydrostatic test step from the pressure boundary integrity checklist. [EVD-2894] directly documents: 'After calibration of the meter 310FE01, the gasket was replaced on 18.09.2020. No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated.' The absence of hydrostatic testing meant the gasket joint was never validated to design pressure before return to service. [EVD-2506] further documents that no checklist was followed during these critical high-pressure jobs: 'No checklist is followed while carrying such critical jobs of high pressure/risk.' The meter was returned to service on 19-Sep [EVD-0185] and failed catastrophically five days later on 24-Sep [EVD-0064], with all retaining-plate bolts displaced. The committee identified this testing omission as a direct causal factor (p.31). This is a primary procedural failure in pressure boundary integrity management - the checklist step designed to detect gasket installation defects was not performed.

CAUSAL EXPLANATION

The omission of the hydrostatic test checklist step after gasket replacement created an undetected defect in the pressure boundary. The substitute O-ring gasket [EVD-8645] was installed without OEM specification and without the pressure validation designed to reveal installation errors or material inadequacy. When the meter returned to service on 19-Sep under operating pressure, the defective gasket joint began degrading. By 24-Sep at 02:12:50 [EVD-1284], the joint failed completely, initiating the hydrocarbon release. The checklist omission eliminated the last technical barrier to prevent an improperly installed gasket from entering service.

Linked Evidence: EVD-2894, EVD-2506, EVD-0185, EVD-0064

ASBE 1.2 **Checklist Not Followed/Not Followed Correctly** **SUBSTANTIATED**

Calibration team changed meter setting from 16000 to 2540 mmwc in Dec-2019 without documenting the change in signed calibration records.

JUSTIFICATION

[EVD-2506] documents that during calibration work on meter 310FT01 from 17-Dec-2019 to 18-Dec-2019, the calibration setting was changed from 16000 mmwc to 2540 mmwc 'without any mention of this change in the calibration documents signed by the team.' The next calibration reversed the setting back to 16000 mmwc. This represents a checklist compliance failure - the calibration documentation checklist should have captured all setting changes for traceability and review. The document further notes: 'No checklist is followed while carrying such critical jobs of high pressure/risk.' While this calibration event occurred 9 months before the incident and the setting was restored, it demonstrates a pattern of checklist non-compliance on the same equipment that later failed. The undocumented setting change indicates that calibration work was being performed without structured procedural documentation, creating unknown configuration states and eliminating management visibility into maintenance activities on critical pressure boundary equipment.

CAUSAL EXPLANATION

The checklist failure during calibration work demonstrates systematic procedural non-compliance on meter 310FT01. While the Dec-2019 calibration event did not directly cause the Sep-2020 gasket failure, it established a pattern: critical maintenance work on high-pressure equipment was routinely performed without checklist discipline. This pattern culminated in the Sep-2020 gasket replacement where the hydrostatic test checklist step was similarly omitted [EVD-2894, instance 1]. The calibration checklist failure is a secondary contributing factor - it reveals organizational tolerance for checklist non-compliance that enabled the later, directly causal testing omission.

Linked Evidence: EVD-2506

ASBE 1.2 **Checklist Not Followed/Not Followed Correctly** **SUBSTANTIATED**

Emergency response checklist omitted isolation/shutdown decision criteria, causing 46-minute delay in source isolation during active HC release.

JUSTIFICATION

The emergency response checklist/procedure for hydrocarbon leakage scenarios omitted critical decision criteria for isolation and source shutdown, resulting in a 46-minute delay from leak detection to first valve closure. [EVD-4052] documents the procedural gap: 'As per DMP On site Hazira Plant, guidelines to be followed in case of hydrocarbon leakage are included but does not include isolation/stoppage of facilities/pipelines' and 'In ERDMP, scenario for Level 3 emergencies are taken from the plant failures. Heavy product/gas leakages from pipelines & tanks are considered but decision on isolation/closure of pipelines & facilities are not documented.' The operational timeline [EVD-6414] shows: leak onset 02:12:50, first detection 02:15, but MOV 54 closure not executed until 03:01 - a 46-minute gap. During this period, multiple shift personnel responded [EVD-1460, EVD-9220, EVD-9612, EVD-7340] but none had standing checklist authority to order immediate source isolation. The GSP Shift In-charge closed MOV 54 only after physically reaching the gate at ~02:35 and consulting with the Overall Shift In-charge [EVD-9612]. The missing checklist step - 'IF

CAUSAL EXPLANATION

The absence of isolation decision criteria in the emergency checklist created an authority vacuum during the critical first 46 minutes of the release. Operating personnel detected the leak via annunciator at 02:20 but had no procedural authority to close the source valve. The checklist gap forced a serial escalation chain (GT CR -> Shift In-charge -> Site) that consumed 41 minutes while hydrocarbon continued flowing. By the time MOV 54 closed at 03:01, sufficient inventory had released [EVD-8213: 'vapour all around even at height' at 03:00:44] to create the ignitable cloud. The missing checklist step converted what should have been an immediate operator-level isolation into a management-approval-gated decision, directly extending the release duration and inventory magnitude.

Linked Evidence: EVD-4052, EVD-6414, EVD-9612, EVD-1460, EVD-8657

ASBE 1.3 **Procedure Not Followed/Not Followed Correctly** **SUBSTANTIATED**

Maintenance team systematically substituted locally-sourced O-rings for OEM rectangular-slot gaskets without specification, QC, or engineering approval.

JUSTIFICATION

[EVD-8645] documents a systematic procedural violation in the gasket replacement procedure: 'The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings.' The correct procedure for a 1986 OEM rectangular-slot gasket design requires either: (1) procure OEM-specification gasket, or (2) perform engineering substitution analysis with documented approval and validation testing. The maintenance team instead substituted a locally-sourced O-ring without specification, without QC, and without documented authority for the design change. [EVD-1494] shows this incorrect procedure was repeated three times in 22 months (Nov-2018, Feb-2019, Sep-2020). The Sep-2020 instance used this wrong technique and resulted in catastrophic failure [EVD-0064: all bolts displaced]. The committee's root-cause finding (p.33) directly identifies this procedural error: the gaskets 'were not being

CAUSAL EXPLANATION

The wrong-technique gasket substitution procedure created a material incompatibility that the pressure boundary could not sustain. The OEM rectangular-slot gasket design (1986) requires a specific gasket geometry and material specification; substituting a generic O-ring changes the load distribution, sealing mechanism, and pressure capacity. Without QC or specification control, the locally-sourced O-rings had unknown material properties and dimensional tolerances. When installed on 18-Sep-2020 and pressurized to operating pressure (but never validated to design pressure [EVD-2894]), the incompatible gasket began degrading under cyclic loading. By 24-Sep at 02:12:50, the joint failed completely. The wrong procedure (O-ring substitution without engineering analysis) combined with the wrong technique (no hydrostatic validation) produced the initiating equipment failure.

Linked Evidence: EVD-8645, EVD-1494, EVD-0064

Level 2: Preconditions

PTE 2.3 Visibility Restrictions (Not Weather Related)

SUBSTANTIATED

Dense hydrocarbon vapour cloud physically prevented operating personnel from approaching the metering area to locate and isolate the leak source, introducing a 41-minute delay before remote isolation.

JUSTIFICATION

The condensate vapour cloud created absolute visibility restrictions that physically prevented personnel from approaching the leak source to execute manual isolation. Evidence [EVD-1460] documents that the Chief Engineer (Production) and Junior Engineer (Mechanical) 'saw the heavy cloud in phase I area' but 'could not approach due to dense cloud.' Evidence [EVD-9220] corroborates: 'Visibility was very poor.' Evidence [EVD-9612] documents that the DGM (Production) - GSP Shift In-charge and the GM (Production) - Overall Shift In-charge 'tried to enter the Gas Control room area but could not enter due to heavy gas smell.' This visibility restriction created a 41-minute delay between initial detection at 02:15 and the first isolation action (MOV 54 closure) at 03:01, because no personnel could physically reach the metering skid to locate the exact leak source. Evidence [EVD-9941] shows the cloud spread across the GT CR, road, and piping area within 40 seconds (02:15:35 to 02:16:13), establishing a flammable envelope that personnel could not penetrate. This is NOT a weather-related visibility restriction (PPE 1.1/1.2) — it is a process-release-induced obstruction

CAUSAL EXPLANATION

This visibility restriction is a PRIMARY CAUSE in the incident chain. The vapour cloud created a technological-environment barrier that directly prevented human execution of manual isolation procedures. The causal sequence: (1) Leak commenced at 02:12:50; (2) Personnel detected at 02:15 and attempted to approach; (3) Cloud density forced personnel back, preventing local isolation; (4) Personnel had to retreat, escalate, and coordinate remote valve closure; (5) MOV 54 not closed until 03:01, 46 minutes after leak onset. This 41-minute delay between detection and isolation is the direct consequence of the visibility restriction. The cloud transformed a straightforward 'walk to valve, close valve' response into a complex 'escalate to senior authority, coordinate remote closure' procedure requiring multiple phone calls and hand-offs. This visibility restriction interacted catastrophically with the F&G detection system inadequacy (PTE 2.2 instance 1): automated detection and shutdown would have bypassed the need for human approach entirely. The vapour cloud created a technological-environment precondition where the existing

Linked Evidence: EVD-1460, EVD-9220, EVD-9612, EVD-9941

PSE 3.1 Situation/Tasking - Unusual/New or Not Trained for

SUBSTANTIATED

Maintenance team faced unfamiliar situation of validating non-standard O-ring gasket on 1986 meter without OEM specs or defined testing procedure.

JUSTIFICATION

The maintenance team that performed the 18-Sep-2020 gasket replacement on meter 310FT01 was confronted with a situation not fully covered by their standard procedures: the original OEM rectangular-slot gasket specification was lost [EVD-8645], requiring use of a locally-sourced O-ring substitute without quality control. Evidence shows 'No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure' [EVD-2894]. The committee explicitly notes 'No checklist is followed while carrying such critical jobs of high pressure/risk' [EVD-2506]. This indicates the team was working without a defined procedure for validating non-standard gasket installations. The team appears to have defaulted to operating-pressure testing (a known/trained procedure) rather than the design-pressure hydrostatic test required for pressure-boundary modifications. The situation—validating a substitute component on aging 1986-vintage equipment without OEM specifications—was outside the scope of routine trained procedures, yet no special work plan or engineering review was triggered.

CAUSAL EXPLANATION

This situational unfamiliarity directly enabled ASBE 1.2 (hydrostatic test omission). The maintenance team was not trained for—and had no written procedure governing—the validation of non-standard gasket substitutes on pressure-boundary equipment. Faced with this gap, they applied their known procedure (operating-pressure test) rather than the mandatory design-pressure hydrostatic test. The unfamiliarity created a knowledge vacuum that allowed the checklist omission to occur without triggering supervisory review or engineering involvement.

Linked Evidence: EVD-8645, EVD-2894, EVD-2506

PPE 1.2 Natural Environment Conditions Affecting Vision

SUBSTANTIATED

Night-time darkness combined with dense white vapour cloud restricted visual assessment and source location during the 56-minute emergency response window, compounding response delays.

JUSTIFICATION

Natural darkness at night (02:12:50 leak onset through 03:08:15 ignition) restricted vision during the critical 56-minute emergency response window. Multiple witness statements document severe visibility constraints: Chief Engineer (Production) observed 'heavy cloud' and could not approach the metering area due to dense vapour [EVD-1460]; Junior Engineer (Mechanical) stated 'Visibility was very poor' [EVD-9220]; both engineers were 'physically unable to approach and locate the source on foot due to gas density' (Initial Report §1). The CCTV evidence sequence [EVD-9941] shows rapid vapour cloud expansion across the GT CR, road, and piping area within 40 seconds (02:15:35 to 02:16:13), creating a dense white cloud visible on camera but opaque to personnel on foot. The combination of natural night-time darkness and the white condensate vapour cloud created a compound visibility barrier. The DGM (Production) GSP Shift In-charge and GM (Production) Overall Shift In-charge both reached the Gas Terminal Gate at ~02:35 but 'could not enter the plant area because of gas density' [EVD-9612, EVD-7340]. The Initial Report explicitly states personnel

CAUSAL EXPLANATION

The natural night-time darkness created a baseline environmental precondition that, when combined with the dense white condensate vapour cloud, prevented visual source location and assessment from a safe distance. This environmental constraint contributed to the 46-minute delay between leak onset (02:12:50) and first isolation valve closure (MOV 54 at 03:01). While the primary barrier was the hazardous gas atmosphere itself, the darkness prevented personnel from using visual cues to identify the leak source remotely (e.g., from upwind positions or elevated vantage points) and necessitated close physical approach, which was impossible due to gas density. This environmental precondition did not directly cause the lower-level checklist failures (ASBE 1.2), but it created the emergency-response context in which the consequences of those maintenance failures (unvalidated gasket seal) manifested as a prolonged, uncontrolled vapour release. The darkness was a natural environmental factor beyond operational control, but its impact on emergency response capability is a documented secondary contributing factor to the extended pre-ignition

Linked Evidence: EVD-1460, EVD-9220, EVD-9941, EVD-9612, EVD-7340

PTF 1.4 Over Reliance on Individual/ System or Equipment

SUBSTANTIATED

F&G detectors were annunciator-only with no auto-shutdown interlock; machine detection at 02:12:50 resulted in a manual, delayed response taking 46 minutes.

JUSTIFICATION

The F&G detection system was configured solely as an annunciator alarm, with no automatic interlock to shut down the plant or activate fire suppression. [EVD-8657] records that 'Hydrocarbon/H2S detectors are configured to Annunciator Panel for display but not in DCS for history' and states explicitly, 'F&G detection system does not have provision for auto shut down of plant or activation of fire suppression system.' This design meant that the system's electronic detection of the gas cloud—which occurred at or before the CCTV leak onset at 02:12:50—triggered only a visual/audible alarm, rather than automatically closing isolation valves or initiating deluge. The entire response thereafter depended on human operators to notice the alarm, recognize the severity, and manually execute or authorize isolation—a chain that took 46 minutes from first human detection. The committee recommendation A.1 (p.34) to interlock F&G to automatic shutdown confirms that the existing configuration was a critical gap. This precondition of over-reliance on human intervention in a high-hazard release scenario directly enabled the prolonged vapour inventory accumulation that fuelled the fire.

CAUSAL EXPLANATION

If the F&G system had been configured to automatically close MOV 54 or the 36" PCV upon confirmed gas detection, the release window would have been reduced from 46 minutes to seconds. Machine detection likely occurred before 02:12:50, but without auto-shutdown, the release continued unchecked while operators escalated. The accumulated cloud at 03:00:44 (EVD-8213 'vapour all around even at height') and subsequent ignition at 03:08:15 were the direct result of the 46-minute flow. This over-reliance on the human response chain, itself hampered by other preconditions, converted a potential immediate shutdown into a prolonged emergency.

Linked Evidence: EVD-8657, EVD-3452

PTF 1.4 Over Reliance on Individual/ System or Equipment

SUBSTANTIATED

Gasket replacement had no independent quality checks: no material spec, no QC, no hydrotest, relying entirely on the fitter's execution.

JUSTIFICATION

The maintenance process for meter 310FT01's gasket replacement lacked any independent verification or quality control; its success rested entirely on the individual fitter's actions. [EVD-8645] reports: 'Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings.' Compounding this, [EVD-2894] states, 'No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure... No standard operating procedure is in place for calibration and taking back in service.' Thus, the gasket selection, installation, and pressure-boundary validation all hinged on one person or small team, with no second-check, no supervisor review, and no post-maintenance testing. This over-reliance on unaudited human performance created a single-point failure in the pressure-containment barrier: any error—such as using a round O-ring on a rectangular-slot flange—would pass into service uncaught, exactly as happened.

CAUSAL EXPLANATION

The fitter installing the local O-ring on 18 September 2020 did not possess the correct OEM gasket, specifications, or a checklist. Without independent verification (a second fitter, an inspection, or a hydrostatic test), the defective seal was accepted and the meter returned to service on 19 September. Operating pressure held for 6 days, but the inadequate gasket failed at 02:12:50 on 24 September, initiating the release. Had a mandatory hydrotest or a material check by a designated quality role been in place, the leak path would have been exposed during a controlled test, preventing the catastrophe. This precondition directly enabled the primary unsafe act (wrong gasket) to become the loss of containment.

Linked Evidence: EVD-8645, EVD-2894, EVD-2506

PTF 1.5 Team Roles/ Responsibilities not Allocated or Understood

SUBSTANTIATED

DMP lacked isolation criteria; on-shift operators had no authority to isolate, causing 46-minute delay while they escalated to senior managers.

JUSTIFICATION

The plant's emergency response plan failed to allocate authority or define criteria for immediate isolation of pipelines in the event of a major hydrocarbon leak. [EVD-4052] notes: 'As per DMP On site Hazira Plant, guidelines to be followed in case of hydrocarbon leakage are included but does not include isolation/stoppage of facilities/pipelines' and further, 'decision on isolation/closure of plant/pipelines is not mentioned.' Consequently, the on-shift operating staff had no clear mandate to close the 36" PCV or to direct MOV 54 closure. The Chief Engineer and Junior Engineer detected the alarm at 02:20 but, instead of isolating, attempted a physical approach and then phoned senior managers ([EVD-1460], [EVD-9220]). First isolation occurred only at 03:01 after the GSP Shift In-charge arrived and took the decision ([EVD-9612]). This ambiguity in roles—no one on duty could autonomously isolate—created the operational vacuum that allowed the release to continue for 46 minutes, and the committee's root cause #6 'Non isolation/stoppage of 36 pipeline operation' flows directly from this finding.

CAUSAL EXPLANATION

The absence of a predefined isolation authority meant that the control room operators did not act on the one remote-controlled valve (the 36" PCV) until instructed at 03:04. If they had been explicitly empowered to close that PCV immediately upon confirmed leak, the release rate could have been reduced by 44 minutes. The isolation of MOV 54 also required senior involvement because no remote operation existed, but the PCV could have been shut from the CR. The role allocation failure delayed any operator-initiated isolation, directly contributing to the massive vapour cloud that ignited at 03:08:15. This precondition underlies the lower-level unsafe acts of delayed decision-making (AJDE 2.7, AJDE 2.9) by ensuring no one on shift believed they had the authority to act unilaterally.

Linked Evidence: EVD-4052, EVD-9612, EVD-7340

PTF 1.6 **Rank/ Position Authority Gradient** **SUBSTANTIATED**

A too-flat hierarchy permitted low-level staff to substitute non-conforming O-rings without engineering review or management approval, normalizing the unsafe practice.

JUSTIFICATION

The maintenance material control process suffered from an overly flat authority gradient, where there was no effective supervisory oversight to prevent unauthorized substitutions. [EVD-8645] documents that the switch from OEM gaskets to locally purchased O-rings became a 'regular practice' without any recorded date or authority—indicating that the decision was made informally at a low level and never challenged by managers. The absence of any quality specification or purchase control further demonstrates a flat hierarchy where shop-floor decisions bypassed engineering and procurement standards. [EVD-2986] reinforces this pattern, noting that the Management of Change (MOC) policy was not implemented since 2018, meaning no senior manager enforced the MOC requirement for the gasket substitution. This flat gradient allowed a chronic violation to become normalized, removing the safety barrier that a properly stratified approval process would have provided.

CAUSAL EXPLANATION

If the authority gradient had been steeper and properly enforced, any change in pressure-boundary materials would have required engineering review and management approval. Instead, the maintenance fitters unilaterally adopted a non-conforming O-ring, and this practice was not stopped because no one in the management chain actively supervised material selection. The flat gradient thus enabled the substitution to occur without technical evaluation, directly setting up the defective gasket that failed. This precondition underlies the lower-level unsafe acts (ASBE 1.3, AJDE 2.8) by creating an environment where material standards were not enforced and any worker could deviate from specification with impunity.

Linked Evidence: EVD-8645, EVD-2986

PTF 1.4 **Over Reliance on Individual/ System or Equipment** **SUBSTANTIATED**

The area relied on a single PTZ camera with 90s pan cycle; its cable burned 7s after ignition, terminating all visual monitoring.

JUSTIFICATION

The condensate metering skid area relied on a single PELCO PTZ D5230D CCTV camera for remote visual monitoring. [EVD-0210] records: 'PELCO make PTZ cameras (Model D5230D) are provided at few places and has turn around (360 degree) at the time of about 90 seconds. Visibility from this camera is limited.' The incident narrative confirms this was the sole coverage, and when the cable burned through 7 seconds after ignition at 03:08:22, [EVD-2752] notes 'major fire continued, CCTV capturing discontinued due to burnt of CCTV cable.' The committee's recommendation for per-plant CCTV with master-control displays (p.36) acknowledges this single-camera weakness. The control room thus had no redundant visual source, and once the cable was lost, situational awareness was entirely dependent on personnel reports. While the camera did capture early leak evidence, its non-redundant design meant that any failure would eliminate remote visual assessment—a single point of failure that degraded the emergency response environment.

CAUSAL EXPLANATION

The single-camera configuration meant that when the cable burned at 03:08:22, the control room lost all video feed. This did not cause the release, but it hindered post-ignition assessment and removed the only remote visual tool for the incident commander. The lack of redundancy in monitoring is a systemic precondition that left the plant without fallback surveillance in a critical area during the emergency, a gap the committee's recommendation directly addressed.

Linked Evidence: EVD-0210, EVD-2752

PTF 1.5 **Team Roles/ Responsibilities not Allocated or Understood** **SUBSTANTIATED**

No one was accountable for closing SAP maintenance notifications; the meter replacement went unreviewed and the recurrence pattern undetected.

JUSTIFICATION

The administrative closure of maintenance notifications lacked a clearly accountable individual, resulting in open notifications and missing post-job reviews. [EVD-0591] states: 'Notification no. 3800419840 was raised on 17/09/2020 for malfunction of meter no.1… Notification was not closed after completion of the job' (p.27). The committee also noted via [EVD-5105] that 'few of the work permits were missing. No data is kept on such missing permits,' indicating that no one was tracking or enforcing closure. The recommendation D.2.ii (p.35) calling for 'Sample checks/approval of closure of notifications by Senior Officers' directly addresses this gap. Without assigned responsibility, the record of the gasket replacement—the third in 22 months—was never formally reviewed, allowing the recurrence pattern to go unnoticed and the deficient work to proceed unexamined.

CAUSAL EXPLANATION

If a designated person had been responsible for reviewing and closing the SAP notification, they would have been compelled to examine the job details: the substitute gasket, the missing hydrotest, and the repeated failures. This would likely have triggered an engineering review or halted return-to-service. Instead, because no one owned the closure, the meter went back online on 19 September with no administrative check, and the latent defect persisted. This role allocation shortfall directly enabled the unsafe acts (returning defective meter to service) and contributed to the incident by failing to catch the recurring problem.

Linked Evidence: EVD-0591, EVD-5105

PTE 2.4 Controls and Switches are Inadequate

SUBSTANTIATED

MOV 54 and the gas-terminal MOVs were local manual valves with no emergency remote operation, forcing a ~46-minute physical-access and escalation delay before first isolation.

JUSTIFICATION

The emergency isolation controls at the Gas Terminal were inadequate because the MOVs are local manual valves with no remote operation from the control room during an emergency. [EVD-9612] records that the DGM-GSP "closed the MOV 54 inlet condensate line from slug catcher to 36 metering area at about 0301 hrs" and that "the MOVs in the gas terminal station are not remote-operable from the control room during emergency (p.26)". [EVD-3932] independently confirms the same condition: "MOVs are provided in the gas terminal station for local operations but they are not having remote operation from control room during emergency." Because MOV 54 could only be operated locally, the shift in-charge had to physically reach the Gas Terminal Gate at approximately 02:35 and could only close the valve at 03:01 — 46 minutes after first physical detection at 02:15 and 48 minutes after CCTV-confirmed leak onset at 02:12:50. The correct design for an emergency-isolation control is remote actuation from a manned control room with clear operator authority; the actual design required physical approach into or beside a dense vapour cloud and serial telephonic

CAUSAL EXPLANATION

MOV 54 was the first isolation valve available to stop slug-catcher inventory reaching the failed meter, but with no remote control it could not be closed from the Gas Terminal Control Room. The dense vapour cloud prevented on-foot access, so isolation waited for the GSP Shift In-charge to reach the gate and execute a local close at 03:01. During that 46-minute window the release continued and the cloud expanded from the skid to a vertical column by 03:00:44 (EVD-8213), and the accumulated vapour ignited at 03:08:15 — 4 minutes 15 seconds after the 36" PCV was finally closed from the control room at 03:04. The first isolation was therefore executed at the wrong time and through the wrong access path; this technological-environment control failure translated directly into the lower-level delayed-necessary-action findings that governed the release volume.

Linked Evidence: EVD-9612, EVD-3932

PTE 2.5 Defective Equipment in Use

SUBSTANTIATED

The meter was returned to service with a non-OEM round O-ring in a rectangular-slot flange and no hydrostatic test to design pressure.

JUSTIFICATION

Meter 310FT01 was placed back in hydrocarbon service with a defective pressure-boundary assembly. [EVD-8645] states that "The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket." [EVD-2894] then records the missing verification: "After calibration of the meter 310FE01, the gasket was replaced on 18.09.2020. No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated. Next day meter 310FE01 was taken on online and was in use at the time of Incident." The correct procedure is to fit the OEM rectangular-slot gasket against the 1986 design and hydrostatically test the reassembled joint to design pressure before return to service; the actual procedure substituted a

CAUSAL EXPLANATION

The round-cross-section O-ring could not properly seal the 1986 rectangular gasket slot, leaving a non-conforming leak path at the flange. Because no hydrotest was performed, this installation defect was never exposed at design pressure before hydrocarbon service. The joint then ran for five days under operating conditions and failed at 02:12:50 on 24 September, releasing condensate vapour for 56 minutes until ignition at 03:08:15. This technological-environment condition directly enabled the lower-level wrong-choice (AJDE 2.8), procedure-not-followed (ASBE 1.3), and routine-widespread-violation (AV 1.1) findings.

Linked Evidence: EVD-8645, EVD-2894

PTE 2.5 **Defective Equipment in Use** **SUBSTANTIATED**

Three pressure transmitters for the Gas Terminal metering lines were shown on the drawing but had no data available, eliminating pressure-trend monitoring.

JUSTIFICATION

Gas Terminal pressure instrumentation shown on the P&ID was not actually supplying or retaining data. [EVD-6470] records that "Three Pressure transmitters 310PT4, 310PT5 & 310PT6 are shown in the drawing. It was informed that no pressure transmitter data are available in the system. No pressure data in Gas Terminal area is available/provided to check rise in pressure in the metering lines" (p.26). This is defective or disconnected instrumentation in the very area where the gasket failed: three transmitters existed on paper but produced zero pressure trend for the pre-incident window. The defect did not dictate the immediate outcome because the leak was first detected by perimeter security at 02:15 and by the annunciator at 02:20, but it removed a pressure-based early-warning and forensic-reconstruction layer.

CAUSAL EXPLANATION

With no pressure data from 310PT4/PT5/PT6, the Gas Terminal area had no real-time pressure-trend indication that could have detected the flange failure earlier or confirmed abnormal conditions before the 02:15 human detection. The actual detection chain still worked through CISF and the annunciator, so this deficiency rated present without direct impact on the final outcome, but it compounded the DCS-history and F&G configuration gaps and impaired post-incident sequence reconstruction.

Linked Evidence: EVD-6470

PTE 2.6 **System Automation or System Redundancy** **SUBSTANTIATED**

Gas detectors were not historised in the DCS, eliminating trend/predictive alarm capability and leaving only a display-only annunciator path.

JUSTIFICATION

The same F&G system had a separate automation-data deficiency: detector readings were sent only to the annunciator panel and were not historised in the DCS. [EVD-8657] states that "Hydrocarbon/H2S detectors are configured to Annunciator Panel for display but not in DCS for history," and [EVD-1284] states that at leak onset 02:12:50 "neither the F&G system nor the DCS produced a real-time alarm at this point because hydrocarbon detectors are configured to the annunciator panel only, and not in DCS for history (p.26)." As a result there was no trend data, no predictive or rate-of-rise alarming, and no DCS time-tagged detector history for post-incident reconstruction. The actual first annunciator alarm came at 02:20 — five minutes after human detection at 02:15 and approximately seven minutes after leak onset. This is distinct from the missing auto-shutdown interlock: it is the loss of data continuity and trend capability, which degraded both early-warning potential and forensic analysis.

CAUSAL EXPLANATION

Without DCS historisation, there was no machine trend to detect the developing release at 02:12:50 and no fast, data-driven alarm before the annunciator at 02:20. The first human detection still came from perimeter security at 02:15, so this gap was secondary rather than the primary failure, but it removed an automation layer that could have shortened the detection-to-isolation interval and it compounded the 46-minute response delay by preventing rapid diagnosis across control rooms.

Linked Evidence: EVD-8657, EVD-1284

PTE 2.6 **System Automation or System Redundancy** **SUBSTANTIATED**

The metering area had only one PELCO PTZ camera with a 90-second pan and no redundant coverage or separate control-room displays; it was lost at ignition.

JUSTIFICATION

The metering-skid critical area had no redundant fixed video monitoring; a single panning camera was the sole coverage. [EVD-0210] documents that "PELCO make PTZ cameras (Model D5230D) are provided at few places and has turn around (360 degree) at the time of about 90 seconds. Visibility from this camera is limited" (p.30). [EVD-2752] shows the consequence when the sole camera cable burned through at 03:08:22, seven seconds after ignition, ending the video evidence chain, and notes the committee recommendation at p.36 that "CCTV arrangement shall be provided for each plant including separately with display at plant control rooms as well as master control/CISF station." A single PTZ camera with a 90-second pan turn-around cannot continuously watch any one hazardous area, and the absence of separate displays at the plant control rooms and master/CISF station reduced the value of the video it did capture. This monitoring-redundancy gap is a secondary technological-environment precondition that contributed to the delayed verification architecture and the loss of post-ignition evidence.

CAUSAL EXPLANATION

The single-camera design meant the skid was only intermittently observed, with up to 90 seconds between returned views, and no parallel fixed-camera coverage could continue after the cable burned at 03:08:22. In the incident the camera still recorded leak onset at 02:12:50 and ignition at 03:08:15, so it did not prevent detection, but its lack of redundancy and the absence of plant-control-room displays left no post-ignition visual record and contributed to the CISF-only verification hop between 02:15 and 02:18.

Linked Evidence: EVD-0210, EVD-2752

PTE 2.6 System Automation or System Redundancy

SUBSTANTIATED

DCS clocks across control rooms drifted by 2-12 minutes, complicating event correlation and reconstruction.

JUSTIFICATION

The DCS clocking across control rooms was unreliable and unsynchronised. [EVD-5532] records that "Time stamping in DCS systems provided in various control rooms is varying from 2 minutes to 12 minutes and data matching is difficult" (p.26). In practice the incident reconstruction had to adopt CCTV timestamps as canonical where DCS and CCTV disagreed, because DCS clocks drifted unpredictably. This is an automation-reliability deficiency, not a control-surface deficiency: the DCS was functioning but its time-base was not synchronised. It did not change the physical outcome of the incident because human detection and CCTV independently anchored the timeline, but it could mask correlated alarms across control rooms in real time and materially complicated sequence-of-events reconstruction.

CAUSAL EXPLANATION

The clock drift did not directly cause the release or ignition, but it degraded the automation system's trustworthiness for cross-room alarm correlation and forced the investigation to rely on CCTV timestamps as canonical. Had simultaneous alarms in different control rooms been separated by up to 12 minutes in the logs, operators could have misjudged the sequence and delayed coordinated isolation. In this incident the effect was limited to forensic difficulty, so it is rated present without direct causal impact.

Linked Evidence: EVD-5532

PTE 2.5 Defective Equipment in Use

SUBSTANTIATED

A Zone 1 HPMV flame-proof lamp with three missing bolts and glass case opened 5-6 cm was in service and became the probable ignition source.

JUSTIFICATION

The probable ignition source was a defective HPMV flame-proof lamp kept in service with its flame-proof integrity destroyed. [EVD-6916] records that "three bolts on one of the flame proof lamps were missing resulting in it being open with its integrity compromised, glass case got detached from the main body and had opened out about 5 to 6 cms. The nearby fittings were found intact" (p.32). The qualifier that nearby fittings were intact isolates the anomaly to that single luminaire. [EVD-9587] shows the last clean pre-incident CCTV baseline frame at 02:11:43, which the committee used to bound the last documented state of the fitting's glass-case integrity. In a Zone 1 hydrocarbon area, flame-proof enclosures are the primary barrier between electrical contacts and the flammable atmosphere; a fitting with three missing bolts and a 5-6 cm open glass case is defective equipment in use, and it removed that barrier. This defective luminaire provided the exposed electrical path that ignited the accumulated vapour at 03:08:15, making it a primary causal item in the technological environment.

CAUSAL EXPLANATION

The vapour cloud that accumulated over 56 minutes from leak onset at 02:12:50 did not have to find an uncommon ignition source; it found an electrical fitting whose flame-proof integrity was already absent. Ignition occurred at 03:08:15, after both isolation actions (MOV 54 at 03:01 and the 36" PCV at 03:04), meaning the already-released cloud was sufficient and the defective lamp supplied the ignition energy. This technological precondition directly enabled the transition from a vapour release to a major fire.

Linked Evidence: EVD-6916, EVD-9587

PTE 2.6 **System Automation or System Redundancy** **SUBSTANTIATED**

The F&G system was configured display-only with no auto shutdown or fire suppression interlock, requiring a fully manual response.

JUSTIFICATION

The gas and flame detection automation was designed without any interlock to protective action. [EVD-8657] records that "Hydrocarbon/H2S detectors are configured to Annunciator Panel for display but not in DCS for history" and that "F&G detection system does not have provision for auto shut down of plant or activation of fire suppression system." [EVD-3452] confirms the same gap and that the committee recommendation A.1 at p.34 explicitly calls for "F&G detection system needs to be interlocked to automatic plant shut down and activation of fire suppression system." In this incident the leak was machine-visible on CCTV at 02:12:50, but the automated system produced no automatic isolation, no automatic deluge, and no automatic shutdown; the first alarm reached the control room only at 02:20, and the first isolation occurred at 03:01. A no-action display-only automation design converted a sub-second potential machine response into a fully manual 46-minute response chain, directly enabling the massive vapour accumulation.

CAUSAL EXPLANATION

The leak began at 02:12:50. With no auto-shutdown logic, the system did not close MOV 54, did not initiate deluge, and did not trip the plant; instead the full human detection-verification-escalation-authorisation chain had to execute, delaying first isolation to 03:01 and the 36" PCV to 03:04. The 56-minute accumulation that followed produced the cloud that ignited at 03:08:15. This automation-design deficiency is the Level-2 precondition for the lower-level finding that the F&G alarm-only configuration was a routine/widespread violation (AV 1.1) and for the 46-minute response-delay findings.

Linked Evidence: EVD-8657, EVD-3452

PTE 2.5 **Defective Equipment in Use** **SUBSTANTIATED**

Meter 310FT01, with three gasket replacements in 22 months and a fresh malfunction notification, remained in service and failed a fourth time.

JUSTIFICATION

Meter 310FT01 was degraded equipment that remained in service despite a documented recurrence of sealing failures. [EVD-1494] records that "Cleaning/servicing/calibration of meters 1 & 2 was carried out on 12th & 13th Nov 2018. Afterwards, meter no.1 was cleaned on 25/02/2019 & gasket was changed. Again on 18/09/2019, gasket was changed which leaked within one year" — three gasket replacements in approximately 22 months on a 1986-vintage meter with roughly 34 years in service. [EVD-0591] shows the same meter generated SAP Notification 3800419840 on 17-Sep-2020 for malfunction and that the notification was not closed after the job. Instead of removing the meter from service for redesign, OEM gasket restoration, or replacement, the plant repeated the same locally sourced substitute gasket on each occasion. A known recurrently failing pressure-boundary component was therefore in live condensate service when the fourth failure occurred at 02:12:50 on 24 September. This is distinct from the specific 18-Sep-2020 gasket assembly because it captures the decision to keep the degraded meter itself in use across the full 22-month

CAUSAL EXPLANATION

The recurrence pattern was an equipment-reliability warning that went unheeded. Had the meter been quarantined, redesigned, or replaced after the second or third gasket failure, the fourth failure on 24 September would not have occurred. Instead, the degraded meter remained in service and the inadequate substitute gasket installed on 18-Sep-2020 failed within five days, releasing vapour for 56 minutes before ignition at 03:08:15. This precondition is the technological-environment counterpart to the lower-level AJDE 2.5 finding that the three-failure warning pattern was ignored.

Linked Evidence: EVD-1494, EVD-0591

PTE 2.5 **Defective Equipment in Use** **SUBSTANTIATED**

Three of four 410 m3/hr diesel fire-water pumps were unavailable or unreliable during the incident, though electric pumps carried the load.

JUSTIFICATION

The standby diesel fire-water pump train was defective at the time of the incident, although the electric pumps carried the actual fire-load. [EVD-2432] records that "Both the Tank level and the Fire water header pressure were healthy throughout the incident. Three electrical Fire water pumps namely 47PE003 A, C, D were operated during the cooling activity" but also that "Out of 4 diesel pumps 410 m3/hr, 2 pumps are under maintenance, one operated for 30 minutes & fourth pumps is tripping after 2 minutes. The reason for tripping is not identified or actions have not been recorded." This is defective standby equipment in use: three of four diesel backup pumps were unavailable or unreliable, and the tripping root cause was unknown. The defect did not change the outcome of this fire because the electrical pump train and electrical supply remained available, but it left the firewater system with degraded redundancy.

CAUSAL EXPLANATION

This precondition did not alter the incident outcome because three electrical fire-water pumps maintained tank level and header pressure throughout, and active fire-cooling began at approximately 03:50 with the fire extinguished by 06:00. It was nonetheless a latent redundancy gap: had electrical supply failed or cooling demand been higher, the degraded diesel train would have constrained the response, and the unidentified 2-minute tripping fault was an unrepaired known defect.

Linked Evidence: EVD-2432

PTE 2.7 **Technical Guidance is Unclear/Unavailable or Incorrect** **SUBSTANTIATED**

The OEM rectangular-slot gasket specification was lost and no substitute specification or QC existed, so the 18-Sep-2020 repair used a locally bought O-ring without engineering guidance.

JUSTIFICATION

EVD-8645 records that "The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket" (p.31). The correct technical guidance baseline for this 34-year-old meter is the OEM rectangular-slot gasket drawing and procurement specification; the actual condition was the loss of that specification, no OEM sourcing, and unapproved local O-ring substitution, so a maintainer following the available guidance had no correct part standard to work to. EVD-1494 demonstrates the same defect recurred: "meter no.1 was cleaned on 25/02/2019 & gasket was changed. Again on 18/09/2019, gasket was changed which leaked within one year" (p.27). The PNGRB committee's root cause #2 ("Use of nonstandard O ring") and #3 ("Lack of control over material sourcing"), as carried in EVD-8645,

CAUSAL EXPLANATION

Without the OEM specification, the maintainer had no dimensional or material acceptance basis; the round O-ring could not fill the rectangular slot, leaving a leak path. The same inadequate substitute was reinstalled after the 12-Nov-2018 and 25-Feb-2019 failures, so the defect was recurrent rather than a one-off. After the 18-Sep-2020 installation, the meter operated from 19-Sep to 24-Sep and failed at 02:12:50, releasing vapour for 56 minutes until ignition at 03:08:15. This higher-level precondition enabled the lower-level wrong gasket choice (AJDE 2.8 rating 4), routine substitution violation (AV 1.1 rating 4), and incorrect gasket procedure (ASBE 1.3 rating 4) that caused the loss of containment.

Linked Evidence: EVD-8645, EVD-1494

PTE 2.7 **Technical Guidance is Unclear/Unavailable or Incorrect** **SUBSTANTIATED**

DMP/ERDMP hydrocarbon-leakage sections had no isolation/stoppage decision criteria, leaving shift personnel without response guidance.

JUSTIFICATION

EVD-4052 quotes the On-site DMP: "As per DMP On site Hazira Plant, guidelines to be followed in case of hydrocarbon leakage are included but does not include isolation/stoppage of facilities/pipelines" (p.32). It further records: "In ERDMP, scenario for Level 3 emergencies are taken from the plant failures. Heavy product/gas leakages from pipelines & tanks are considered but decision on isolation/closure of plant/pipelines is not mentioned" (p.27). The guidance therefore describes the hazard but omits the critical decision rule: when a leak requires isolation, who may order it, and which valves must be closed. The committee directly links this gap to root cause #6, "Non isolation/stoppage of 36 pipeline operation" (EVD-4052). This instance is distinct from the metering SOP gap because it governs emergency response rather than maintenance.

CAUSAL EXPLANATION

Absent documented isolation criteria, responders defaulted to physical verification and serial escalation: leak onset at 02:12:50, first detection at 02:15, MOV 54 closure at 03:01 (46 minutes later), 36-inch PCV closure at 03:04, and ignition at 03:08:15. This precondition caused the lower-level findings AJDE 2.9 (decision-making during operation), AJDE 2.10 (inadequate knowledge of isolation criteria), and AJDE 2.11 (unawareness of the policy gap), and is the root that allowed the large vapour cloud to form.

Linked Evidence: EVD-4052

PTE 2.7 **Technical Guidance is Unclear/Unavailable or Incorrect** **SUBSTANTIATED**

No checklist governed the Dec-2019 calibration; the setpoint was changed 16000 to 2540 mmwc and back with no signed documentation.

JUSTIFICATION

EVD-2506 records the calibration event: "During calibration from 17/12/2019 to 18/12/2019, calibration was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of this change in the calibration documents signed by the team. During next calibration for meter 310FT01 was again changed to 16000 MMWC. No checklist is followed while carrying such critical jobs of high pressure/risk" (p.27). The absence of a checklist for high-pressure calibration work meant there was no authoritative sequence requiring the setpoint change to be documented and approved before the job was closed. Committee recommendation D.1.i, carried in EVD-2506, states that "details of Master calibrator/action taken/replacement/deviation from procedure if any, is to be documented in the calibration certificate" (p.35). This is a distinct manifestation of PTE 2.7 in the calibration process, separate from the gasket-specification loss and the return-to-service SOP gap.

CAUSAL EXPLANATION

Without the checklist, an undocumented setpoint change of approximately 6.3x passed without engineering approval, degrading meter measurement and maintenance governance. This produced the lower-level findings ASBE 1.2 (checklist not followed, rating 3) and ASBE 1.5 (under-controlled calibration, rating 3) on the same meter that later became the leak source, contributing to the weakened integrity-management environment.

Linked Evidence: EVD-2506

PTE 2.7 **Technical Guidance is Unclear/Unavailable or Incorrect** **SUBSTANTIATED**

The metering-system P&ID was lost, the vendor replacement was unapproved, and the field configuration did not match the drawing.

JUSTIFICATION

EVD-0933 records: "Reference drawings no. 02-310-AO-101 Rev.1, 2x6 lines & 1x2 lines for metering system are shown but valves & meters configuration are not matching. As old drawings were lost, new drawings are made by a vendor but the same has not been checked & approved by ONGC" (p.26). The P&ID is the fundamental technical guidance for the metering system; here the original was lost, the vendor redraw was never checked or approved by the operator, and the field configuration did not match the drawing, so any maintainer using it would be misled about the actual valve and meter line-up. This is distinct from the OEM gasket specification and the DMP gap because it is the configuration-control guidance for the skid.

CAUSAL EXPLANATION

The unapproved and mismatched drawing denied maintenance a reliable physical configuration baseline, enabling the undocumented O-ring substitution and the no-hydrotest return to service to go unchallenged. It underpins the lower-level finding AV 1.1 on configuration-management breakdown (rating 2), representing a systemic erosion of the technical documentation layer.

Linked Evidence: EVD-0933

PTE 2.7 **Technical Guidance is Unclear/Unavailable or Incorrect** **SUBSTANTIATED**

Baseline pipe-thickness data and original drawings/schedules were unavailable for the 1986 gas-terminal pipelines.

JUSTIFICATION

EVD-8287 records: "It is observed that for a large number of pipes, especially in gas terminal area baseline data of pipe thickness in not available. The original drawings and the schedules of the pipes are also not available. In most cases only two thickness details of 2011 and 2017 is available for a pipe which is commissioned in 1986" (p.29). For a 1986-commissioned pipeline, only two thickness measurement points (2011 and 2017) exist, and the baseline drawings and schedules are gone; this is missing technical guidance for integrity trend analysis. It is distinct from the metering P&ID gap because it concerns the pipeline wall-thickness records and inspection baseline.

CAUSAL EXPLANATION

Without the 1986 baseline and schedules, the 2017 findings of significant wall loss could not be trended or prioritised rigorously, which reinforced the three-year non-action on corrective measures. This precondition supports the lower-level AJDE 2.5 ignored-warning finding (rating 3) and the broader integrity-governance pattern, although the 36-inch pipeline itself remained intact during the incident.

Linked Evidence: EVD-8287

PTE 2.7 **Technical Guidance is Unclear/Unavailable or Incorrect** **SUBSTANTIATED**

The permit system lacked Job Safety Plans in SAP and had untracked missing permits, so task-specific hazard guidance was incomplete.

JUSTIFICATION

EVD-5105 records: "Work Permits are issued in SAP and hard copy of work permits are at sites during execution of work. After completion of work, the originals are filed. On checking the same, it was found that few of the work permits were missing. No data is kept on such missing permits. Job safety Plan is not attached/given in SAP document. Closing of permits & actions indicated in work needs review" (p.26). The Job Safety Plan is the technical guidance identifying hazards and controls for non-routine maintenance; its absence from the SAP permit package means the permit system did not provide complete task-specific guidance. The companion clarification at recommendation B.5, "Maintenance notifications are not be assumed as permit to work" (p.34), implies notification-based execution could substitute for the actual permit, further weakening the guidance available to maintainers. No direct evidence links the absence of a JSP to the hydrotest omission, so this instance is rated 2 rather than 3 or 4.

CAUSAL EXPLANATION

Incomplete permit guidance meant the 18-Sep-2020 high-risk gasket replacement could be executed without a JSP identifying the required hydrostatic test and material-conformance checks, contributing to the lower-level procedural omissions. It is a secondary systemic condition rather than a direct cause, because the available evidence does not tie the specific gasket job to a missing permit.

Linked Evidence: EVD-5105

PTE 2.7 **Technical Guidance is Unclear/Unavailable or Incorrect** **SUBSTANTIATED**

The 2018-19 internal safety audits produced observations without filled checklists and with no checklist format available.

JUSTIFICATION

EVD-7681 records: "Internal Safety Audit was conducted in 2018 & 2019 for 4 & 3 days respectively by a 3-member committee and the observations are given without the filled checklist. The checklist format is not available to check completeness of audit areas" (p.26). The missing filled checklist and absent checklist format mean the technical guidance for these audits cannot be audited after the fact; there is no record showing which areas were inspected versus skipped. This is distinct from the self-safety-audit checklist gap because it concerns the traceability of completed internal-audit work product, not the checklist's plant specificity.

CAUSAL EXPLANATION

The absence of an auditable checklist trail diminished assurance coverage and meant critical equipment-integrity defects could be missed without detection. It contributed to the lower-level ASBE 1.2 audit checklist-not-followed finding (rating 3), allowing the compromised HPMV lamp to remain as the probable ignition source; however, its direct impact on this incident is limited, so it is rated 2.

Linked Evidence: EVD-7681

PTE 2.8 **Workspace Incompatible with Operations** **SUBSTANTIATED**

The sole CCTV camera had a ~90-second pan turnaround and limited visibility, and its cable burned through seven seconds after ignition.

JUSTIFICATION

EVD-0210 documents the monitoring layout: "PELCO make PTZ cameras (Model D5230D) are provided at few places and has turn around (360 degree) at the time of about 90 seconds. Visibility from this camera is limited" (p.30). A single PTZ camera taking about 90 seconds per full pan was the sole video coverage for the metering skid and control-room area, so a continuous view of the leak source was not available to the CCTV operator. EVD-2752 records that at 03:08:22 "major fire continued, CCTV capturing discontinued due to burnt of CCTV cable" (p.25), so the monitoring asset was also not fire-survivable. This design mismatch is distinct from the F&G gap because it denied continuous visual verification and later destroyed the evidence chain.

CAUSAL EXPLANATION

The CCTV operator could only intermittently view the affected area; detection at 02:15 came from perimeter security, and visual verification at 02:18 required the camera to be on-scene. The limited coverage delayed or narrowed the operator's situational confirmation, and the cable loss at 03:08:22 ended the record. This precondition was secondary to the response delay but reduced the monitoring and forensic capability, matching the committee's recommendation for per-plant CCTV with master-control display.

Linked Evidence: EVD-0210, EVD-2752

PTE 2.8 **Workspace Incompatible with Operations** **SUBSTANTIATED**

F&G detection was annunciator-display only with no auto-shutdown or suppression interlocks.

JUSTIFICATION

EVD-3452 records that the F&G detection layout existed, but the operational capability was display-only: "F&G detection system does not have provision for auto shut down of plant or activation of fire suppression system" (p.26). EVD-8657 adds: "Hydrocarbon/H2S detectors are configured to Annunciator Panel for display but not in DCS for history" (p.26). For a high-consequence continuous release, the technological environment should auto-initiate isolation or suppression at gas detection, but the installed design routed detection to an annunciator panel only, leaving detection-to-action entirely to the slow human chain. This is distinct from the CCTV limitation because it concerns automatic protection rather than video monitoring coverage.

CAUSAL EXPLANATION

With no auto-isolation, the 02:12:50 release had to wait for human detection at 02:15, annunciator alarm at 02:20, escalation, and manual MOV 54 closure at 03:01 — a 46-minute window. This interface design converted a potentially immediate machine action into the delayed manual response that produced the large flammable cloud igniting at 03:08:15, and it is a precondition for the lower-level AV 1.1 F&G configuration violation (rating 3).

Linked Evidence: EVD-3452, EVD-8657

PTE 2.8 **Workspace Incompatible with Operations** **SUBSTANTIATED**

The metering skid superstructure was not designed to contain a flange-failure release, allowing fire damage propagation.

JUSTIFICATION

EVD-1394 photographs the post-fire state: "the condensate metering skid was totally damaged & various pipelines have got damaged/shifted" (p.31), and the same item records that "no traces of fire beyond the road inside the plant" (p.32), bounding the damage envelope. The design/layout of the skid did not incorporate containment or separation sufficient to arrest the consequences of a flange-failure release; the fire propagated through the immediately surrounding piping. This is a workspace/installation design condition rather than a maintenance-guidance issue, and it affected consequence severity rather than leak initiation.

CAUSAL EXPLANATION

The skid layout and lack of physical barriers allowed the escaped vapour from 310FE1 to accumulate over the area and, after ignition, caused secondary damage to multiple lines and instrumentation. This raised severity but did not initiate the leak or the ignition, so it is rated 2 rather than 3 or 4.

Linked Evidence: EVD-1394

PTE 2.7 **Technical Guidance is Unclear/Unavailable or Incorrect** **SUBSTANTIATED**

No SOP existed for calibration and return-to-service of the high-pressure meters, so the 18-Sep-2020 repair was returned to line after operating-pressure only, without a hydrotest.

JUSTIFICATION

EVD-2894 records the post-maintenance sequence: "No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated. Next day meter 310FE01 was taken on online and was in use at the time of Incident" (p.31). The same evidence carries the committee's probable-cause statement: "Probable cause of leak is due to use of nonstandard gasket and not testing the system to designed pressure. No standard operating procedure is in place for calibration and taking back in service of these meters which are operating at high pressures" (p.32). Correct pressure-boundary practice after gasket replacement is a hydrostatic test to design pressure with the joint isolated; the actual practice, with no SOP, was pressurisation to operating pressure only, both meters in line, then immediate return to service. This instance is distinct from the lost material specification because it is the missing procedural pathway for re-qualifying a repaired high-pressure boundary, not the part-selection standard itself.

CAUSAL EXPLANATION

The absence of an SOP removed the step that would have subjected the new O-ring joint to design pressure; because only operating pressure was applied and the joint was isolated, the inadequate seal was not detected. The meter returned to service on 19-Sep and failed at 02:12:50 on 24-Sep, about five days later. This precondition directly enabled the lower-level omissions AJDE 2.7 (delayed/omitted hydrotest, rating 4) and ASBE 1.2 (checklist not followed, rating 4), which allowed the defective pressure boundary into live hydrocarbon service.

Linked Evidence: EVD-2894

PTE 2.8 **Workspace Incompatible with Operations** SUBSTANTIATED

Emergency MOVs were local-manual only with no remote control, requiring physical approach that gas density denied until 03:01.

JUSTIFICATION

EVD-3932 records the authority and access design: "MOVs are provided in the gas terminal station for local operations but they are not having remote operation from control room during emergency" (p.26). The MOVs are the emergency isolation mechanism for the 36-inch metering area, but their local-only design required personnel to physically reach the valve in a hydrocarbon atmosphere. EVD-9612 shows what this meant in practice: the GSP Shift In-charge "tried to enter the Gas Control room area but could not enter due to heavy gas smell... [he] closed the MOV 54 inlet condensate line from slug catcher to 36 metering area at about 0301 hrs. Thereafter fire occurred" (p.17). The workspace/equipment design placed the critical isolation actuator beyond safe reach during the release, directly incompatible with the emergency mission.

CAUSAL EXPLANATION

Because MOV 54 could not be closed from the control room, closure waited until the shift in-charge reached the gate and acted at 03:01 — 46 minutes after the 02:15 detection and 48 minutes after the 02:12:50 onset. The unisolated line continued feeding the release; the large cloud visible at 03:00:44 ignited at 03:08:15, seven minutes after the first closure. This is the primary precondition enabling the lower-level AJDE 2.7 isolation delay (rating 4), AJDE 2.9 decision-making failure, and ASBE 1.3 delayed isolation.

Linked Evidence: EVD-3932, EVD-9612

PTE 2.7 **Technical Guidance is Unclear/Unavailable or Incorrect** SUBSTANTIATED

The self-safety audit used a generic non-plant-specific checklist, and approved manuals/SOPs were unavailable.

JUSTIFICATION

EVD-9692 reports the self-safety audit instrument: "The checklist is comprehensive but not related to specific plants and in the process, lot of observations are missing. The duration of audit is not indicated and in most of the items, compliances are indicated, but during our inspection, the data is found to be incomplete & not in line with OISD standards/manuals. Approved manuals/SOPs were not available" (p.26). The audit checklist is a technical guidance document; because it was generic rather than plant-specific and the underlying approved manuals/SOPs were unavailable, the audit could record compliance without verifying plant-specific integrity. This instance is separate from the DMP gap because it concerns the assurance/verification guidance, not the operational isolation guidance.

CAUSAL EXPLANATION

The generic guidance failed to direct auditors to the critical Zone-1 HPMV lamp integrity or the metering gasket-substitution practice, so the latent ignition-source defect persisted undetected. This precondition enabled the lower-level ASBE 1.2 audit checklist-not-followed finding (rating 3) and contributed to the compromised FLP lamp that became the probable ignition source.

Linked Evidence: EVD-9692

PTE 2.10 **Communications Equipment Inadequate/Unavailable** SUBSTANTIATED

The sole PELCO PTZ D5230D CCTV camera panned 360 degrees every ~90 seconds with limited visibility, so the 02:12:50 leak onset was machine-recorded but not continuously monitored or alarmed.

JUSTIFICATION

The Gas Terminal metering area had only one PELCO PTZ D5230D camera with a 360-degree pan turnaround of about 90 seconds and limited visibility. [EVD-0210] records that "PELCO make PTZ cameras (Model D5230D) are provided at few places and has turn around (360 degree) at the time of about 90 seconds. Visibility from this camera is limited" (p.30). Although the camera captured the leak onset at 02:12:50 [EVD-1284], the pan cycle and lack of continuous display meant there was no real-time warning to the control room; first human detection came from CISF perimeter security at 02:15. [EVD-2752] records the committee's subsequent recommendation that "CCTV arrangement shall be provided for each plant including separately with display at plant control rooms as well as master control/CISF station" (p.36), confirming the single-camera arrangement was recognised as inadequate. This is a communication/monitoring equipment inadequacy for early leak detection rather than a direct mechanical cause.

CAUSAL EXPLANATION

Because the camera panned on a ~90-second cycle and was not continuously displayed or linked to an alarm, the leak onset at 02:12:50 produced no immediate operator alert. Detection instead depended on CISF Constable A hearing and seeing the vapour at ~02:15, roughly 145 seconds after onset. This precondition contributed to the initial detection lag and reinforced the lower-level ASBE 1.6 breakdown in visual scan, but it was secondary to the F&G system misconfiguration and the manual escalation chain; human detection still occurred, and the larger 46-minute delay was driven by isolation-authority and configuration factors rather than the camera alone.

Linked Evidence: EVD-0210, EVD-1284, EVD-2752

PTE 2.10 Communications Equipment Inadequate/Unavailable

SUBSTANTIATED

CISF perimeter posts had no direct notification channel to the Gas Terminal Control Room; calls had to relay through the CCTV Control Room, adding roughly 3-5 minutes to initial notification.

JUSTIFICATION

The first detections by CISF Constable A and Head Constable B could not reach the Gas Terminal Control Room directly. [EVD-6234] records that Head Constable B, after hearing the noise and seeing the vapour cloud behind the Gas Terminal building at 02:15-02:16, "informed CCTV Control room and CISF main gate control room" (p.17), not the operating control room. [EVD-8871] records that CISF Sub-Inspector C then verified the release on CCTV and only at 02:18 notified the Gas Terminal Control Room, with the evidence describing "a communication-architecture artefact contributing to the ~5 minute lag between physical detection and Gas Terminal CR notification". The correct emergency-communication configuration would provide a direct path from perimeter security to the affected plant control room; the actual path routed through the CCTV control room as an obligatory verification node, losing about three minutes before the operators were informed.

CAUSAL EXPLANATION

Because the communication architecture did not allow perimeter detection to reach the operating control room directly, the 02:15 physical detections were not received by Gas Terminal Control Room engineers until 02:18-02:20. This consumed the earliest minutes of the response and forced the operators to begin their assessment only after the cloud was already dense. The condition contributed to the initial 5-minute notification lag and is part of the same architecture that later delayed isolation decision-making, although the far larger 46-minute delay was primarily driven by the absence of remote isolation and the lack of DMP isolation criteria. It connects to the lower-level AJDE 2.9 decision-making failure during operation because operators received late, relayed information rather than direct perimeter warning.

Linked Evidence: EVD-6234, EVD-8871

PTE 2.11 Equipment Not configured Correctly

SUBSTANTIATED

Hydrocarbon/H2S detectors were configured as annunciator-only indications with no DCS history and no automatic plant shutdown or fire suppression interlock.

JUSTIFICATION

The flame and gas detection system was configured for display only, not for automatic action or historical data retention. [EVD-8657] quotes the committee finding that "Hydrocarbon/H2S detectors are configured to Annunciator Panel for display but not in DCS for history" and that the "F&G detection system does not have provision for auto shut down of plant or activation of fire suppression system" (p.26). The consequences appear in the timeline: leak onset was machine-recorded at 02:12:50 [EVD-1284], but the annunciator alarm did not activate until about 02:20, roughly 7 minutes 10 seconds later and 5 minutes after CISF human detection at 02:15. [EVD-3452] confirms the regulator's corrective expectation, with recommendation A.1 at p.34 stating that the "F&G detection system needs to be interlocked to automatic plant shut down and activation of fire suppression system". The correct configuration for a gas-terminal hazardous area would write detector data to the DCS historian and automatically initiate isolation or suppression; the actual configuration did neither.

CAUSAL EXPLANATION

Because the F&G system was configured as annunciator-only, the machine witness of the leak at 02:12:50 did not initiate any automatic isolation or suppression. Instead, the entire response depended on a slow human chain: CISF detection at 02:15, CCTV verification and control-room notification at 02:18-02:20, senior engineer call-out at ~02:35, and first isolation only at 03:01 when MOV 54 was closed locally. This misconfiguration is a primary precondition enabling the lower-level AJDE 2.7 delay of necessary action and the AJDE 2.9 operational decision failure, because the operators had no automatic protective action to rely on. The 46-minute release accumulated the vapour cloud seen at 03:00:44 as "vapour all around even at height" [EVD-8213], which ignited at 03:08:15.

Linked Evidence: EVD-8657, EVD-3452, EVD-1284

PTE 2.11 **Equipment Not configured Correctly** **SUBSTANTIATED**

Three metering-line pressure transmitters shown on the P&ID had no data available in the DCS, leaving operators without pressure trend information for leak detection.

JUSTIFICATION

Three pressure transmitters for the Gas Terminal metering lines appear on the P&ID but no pressure data were available in the system. [EVD-6470] records that "Three Pressure transmitters 310PT4, 310PT5 & 310PT6 are shown in the drawing. It was informed that no pressure transmitter data are available in the system. No pressure data in Gas Terminal area is available/provided to check rise in pressure in the metering lines" (p.26). The correct configuration would trend these transmitters in the DCS so operators and investigators could identify abnormal pressure conditions before and during the release; the actual configuration left the Gas Terminal metering area without pressure telemetry. The evidence notes that pre-incident pressure trends could have anchored leak detection earlier than the 02:15 perimeter detection or the 02:20 annunciator alarm, and that the absence also impaired forensic reconstruction.

CAUSAL EXPLANATION

The missing pressure data removed a potential early indicator of the release and forced reliance on perimeter human detection, CCTV verification, and the delayed annunciator. This configuration gap contributed secondarily to the detection lag supporting the lower-level ASBE 1.6 breakdown in visual scan and the overall manual detection chain. It did not directly cause the 46-minute isolation delay, which was driven by valve-actuation and authority factors, but it narrowed the information available to operators during the initial response and to investigators afterward.

Linked Evidence: EVD-6470

PTE 2.11 **Equipment Not configured Correctly** **SUBSTANTIATED**

DCS clocks across control rooms drifted between 2 and 12 minutes, complicating alarm correlation and event reconstruction.

JUSTIFICATION

The DCS time-stamping across multiple control rooms was not synchronised. [EVD-5532] records that "Time stamping in DCS systems provided in various control rooms is varying from 2 minutes to 12 minutes and data matching is difficult" (p.26). The incident narrative itself notes that DCS clocks drifted between approximately 2 and 12 minutes against each other and that CCTV timestamps were taken as canonical. This is a configuration error because the system clocks were not maintained against a common time reference. It did not affect the physical release or the immediate fire response, but it materially complicated sequence-of-events reconstruction and could mask correlated alarms across control rooms in real time.

CAUSAL EXPLANATION

This misconfiguration had no direct impact on the incident outcome because the primary failure sequence was independently established through CCTV, operator logs, and witness statements. Its significance is forensic and systemic: without time synchronisation, correlated alarms or process changes across control rooms could appear separated by minutes, potentially misleading emergency decision-making in future events. It is therefore rated present with no impact on this outcome, but it remains a configuration governance deficiency.

Linked Evidence: EVD-5532

PTE 2.12 **Maintenance Not Conducted/Not Conducted Correctly** **SUBSTANTIATED**

Maintenance installed a locally sourced, unspecified O-ring in place of the OEM rectangular-slot gasket on meter 310FT01, and the seal failed within six days.

JUSTIFICATION

The 18-Sep-2020 maintenance on meter 310FT01 substituted a locally purchased O-ring for a gasket whose 1986 design required a rectangular slot geometry. [EVD-8645] states that "The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings" (p.31). The official committee direct finding follows in [EVD-2894]: "Probable cause of leak is due to use of nonstandard gasket and not testing the system to designed pressure" (p.32). [EVD-0064] further records that meter 310FE1 was "extensively damaged with all bolts out of the flange arrangement" and that "The condensate might have leaked from the meter 310FE1" (p.33). The correct maintenance procedure was to procure and fit the OEM rectangular-slot gasket to specification with quality control; the actual maintenance fitted an unspecified round O-ring into a rectangular groove, leaving a geometry mismatch that could not reliably seal the

CAUSAL EXPLANATION

The wrong-geometry O-ring could not fill the rectangular slot and provide a reliable pressure-boundary seal, so the joint was deficient from installation. Meter 310FT01 was returned to service on 19-Sep-2020 and operated under normal condensate pressure for about six days before the seal failed at 02:12:50 on 24-Sep-2020. This directly produced the lower-level AJDE 2.8 wrong choice of action, AV 1.1 routine substitution violation, and ASBE 1.3 procedure not followed correctly. The release accumulated for 56 minutes and ignited at 03:08:15, making this maintenance error a primary cause of the loss of containment.

Linked Evidence: EVD-8645, EVD-2894, EVD-0064

PTE 2.12 **Maintenance Not Conducted/Not Conducted Correctly** **SUBSTANTIATED**

Meter 310FT01 had three gasket replacements in 22 months (Nov-2018, Feb-2019, Sep-2020) with no root-cause investigation, allowing the same inadequate repair to recur.

JUSTIFICATION

Meter 310FT01 experienced repeated gasket failures that were treated as routine replacement events rather than reliability warnings. [EVD-1494] records that "Cleaning/servicing/calibration of meters 1 & 2 was carried out on 12th & 13th Nov 2018. Afterwards, meter no.1 was cleaned on 25/02/2019 & gasket was changed. Again on 18/09/2019, gasket was changed which leaked within one year" (p.27), with the evidence noting the latest change date should be read as 18-Sep-2020. The three replacements over roughly 22 months on a 1986-vintage meter, combined with [EVD-8645]'s finding that the locally sourced O-ring substitution continued as a regular practice, show that maintenance performed the same inadequate repair repeatedly without investigating why the joints kept failing. The correct maintenance response to a recurring pressure-boundary failure is a root-cause analysis, material review, and design verification; the actual response was repeated routine replacement using the same uncontrolled substitute.

CAUSAL EXPLANATION

The recurring failure pattern was an operational warning that the substitute gasket design was inadequate, but it was not investigated. This precondition enabled the lower-level AJDE 2.5 ignored caution/warning and AV 1.1 routine violation: each replacement repeated the wrong material choice without engineering challenge. The fourth failure, on 24-Sep-2020, initiated the release. This instance is secondary to the immediate 18-Sep-2020 maintenance error because the final wrong gasket and missing hydrotest were the direct mechanical causes, but the failure to learn from the recurrence allowed those direct causes to persist.

Linked Evidence: EVD-1494, EVD-8645

PTE 2.12 **Maintenance Not Conducted/Not Conducted Correctly** **SUBSTANTIATED**

SAP notification 3800419840 for meter 310FT01 was not closed or reviewed after the gasket replacement, removing a supervisory quality gate before return to service.

JUSTIFICATION

The maintenance notification raised for meter 310FT01 remained open after the work was completed. [EVD-0591] records that "Notification no. 3800419840 was raised on 17/09/2020 for malfunction of meter no.1" and that it "was not closed after completion of the job" (p.27). [EVD-0185] records the meter's return to service the next day and the committee recommendation D.2.ii that "Sample checks/approval of closure of notifications by Senior Officers to be introduced" (p.35). The correct maintenance workflow requires closure review that verifies the work scope, materials, test results, and documentation before return to service; the actual workflow left the notification open with no supervisory closure check, allowing the defective gasket work to pass into service without a quality gate.

CAUSAL EXPLANATION

The unclosed notification removed a supervisory checkpoint that could have identified the missing hydrotest, the uncontrolled O-ring substitution, and the recurrence pattern before the meter returned to service. This precondition supports the lower-level AJDE 2.11 misapplication of the SAP closure procedure and AV 1.1 routine non-closure violation. It is secondary because the physical failure was caused by the deficient gasket installation itself, but the absent closure review allowed that installation to pass unchecked.

Linked Evidence: EVD-0591, EVD-0185

PTE 2.12 **Maintenance Not Conducted/Not Conducted Correctly** **SUBSTANTIATED**

A flame-proof HPMV lamp at the metering skid had three missing bolts and a glass case open 5-6 cm, leaving its flame-proof integrity compromised as a probable ignition source.

JUSTIFICATION

The probable ignition source was a flame-proof HPMV lamp whose flame-proof integrity had been lost. [EVD-6916] records that "three bolts on one of the flame proof lamps were missing resulting in it being open with its integrity compromised, glass case got detached from the main body and had opened out about 5 to 6 cms. The nearby fittings were found intact" (p.32). The same condition is described in the narrative as being under forensic assessment because whether it pre-existed the fire or resulted from the fire had not yet been established, but the evidence locker identifies the lamp as the committee's probable ignition source. [EVD-0496] and [EVD-9999] document the electrical preventive maintenance culture: "the electrical maintenance system is a lot more skewed towards breakdown maintenance and lays little emphasis on systematic preventive maintenance and inspection" (p.28). The correct maintenance for Zone 1 flame-proof fittings requires periodic inspection and tightening of all enclosure bolts to preserve explosion protection; the actual condition left three bolts missing and the glass case open by 5-6 cm, exposing internal contacts.

CAUSAL EXPLANATION

The missing bolts and open glass case removed the flame-proof containment barrier on a lamp in a Zone 1 hydrocarbon area. When the vapour cloud reached the lamp after 56 minutes of release, the exposed contacts provided the ignition source at 03:08:15, only 4 minutes after the last isolation action. This maintenance failure is a primary precondition for the ignition and connects to the lower-level ASBE 1.2 audit checklist failure that allowed the compromised fitting to persist unnoticed. The fire severity therefore depended not only on the vapor release but also on this maintenance shortfall.

Linked Evidence: EVD-6916, EVD-0496, EVD-9999

PTE 2.12 **Maintenance Not Conducted/Not Conducted Correctly** **SUBSTANTIATED**

The four-year pipeline thickness measurement schedule was breached, 2017 thickness-loss findings had no corrective plan after three years, and a 2019 pitting finding received only painting on a zero-corrosion-allowance line.

JUSTIFICATION

Integrity maintenance on the gas pipeline was not conducted on schedule or acted upon. [EVD-8307] records that "The laid down schedule for thickness measurement is every four years... As per the records, the last two thickness measurement exercise was carried out in 2011 and 2017 which suggests that the schedule of four years is not being adhered to strictly" (p.29). [EVD-5527] further records that "As per the report of 2017, several pipes have been identified with significant thickness loss and need for corrective action. However, even after three years no plan has been developed for the corrective actions to be taken for those lines" (p.30). [EVD-2397] documents a July 2019 spot measurement finding of 1.03 mm thickness reduction with heavy pitting on the 36" pipeline, a designed thickness of 23.83 mm with zero corrosion allowance, and a corrective action consisting only of painting completed in December 2019. The correct integrity-maintenance process would adhere to the four-year measurement cycle, evaluate fitness-for-service, and implement corrective action plans for identified thickness loss; the actual process breached the

CAUSAL EXPLANATION

These integrity-maintenance failures did not cause the 24-Sep-2020 fire, because the 36" pipeline itself remained intact and undamaged [EVD-5780]. They are significant systemic indicators of the same maintenance-governance culture that allowed the meter gasket substitution and hydrotest omission to persist. The pipeline findings substantiate lower-level AJDE 2.5 ignored caution/warning and AJDE 2.7 delayed necessary action, but they are rated present with no direct impact on this particular loss of containment.

Linked Evidence: EVD-8307, EVD-5527, EVD-2397

PTE 2.10 **Communications Equipment Inadequate/Unavailable** **SUBSTANTIATED**

The sole CCTV camera cable burned through at 03:08:22, seven seconds after ignition, ending all video evidence of the post-ignition phase.

JUSTIFICATION

At 03:08:22, seven seconds after the first flame appeared at 03:08:15, the CCTV recording ended because the camera cable burned through. [EVD-2752] records that "major fire continued, CCTV capturing discontinued due to burnt of CCTV cable" (p.25) and adds that the loss of the single camera and its cable "destroyed the evidence chain at the transition from pre-ignition to post-ignition". The camera cable was routed or protected in a way that could not survive the fire, so the available communication equipment became unavailable at the critical post-ignition phase. This instance is distinct from the pan-cycle inadequacy because it is a fire-induced equipment loss and evidence-chain failure rather than a continuous-coverage limitation.

CAUSAL EXPLANATION

This instance did not alter the incident outcome because the ignition had already occurred at 03:08:15 and the camera was recording evidence rather than controlling the fire response. Its impact was limited to forensic reconstruction: the fire progression from ignition through containment could not be captured visually, complicating confirmation of the ignition source and fire-spread timeline. The loss demonstrates that the single monitoring channel had no redundancy or fire-resilient routing, but this is rated present with no outcome impact rather than a causative precondition.

Linked Evidence: EVD-2752

PTE 2.12 Maintenance Not Conducted/Not Conducted Correctly

SUBSTANTIATED

Calibration maintenance on meter 310FT01 in Dec-2019 changed the setting from 16000 to 2540 mmwc and back without documentation or checklist use.

JUSTIFICATION

Calibration maintenance on the same meter in December 2019 was performed without checklist discipline and left a major setting change undocumented. [EVD-2506] records that "During calibration from 17/12/2019 to 18/12/2019, calibration was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of this change in the calibration documents signed by the team. During next calibration for meter 310FT01 was again changed to 16000 MMWC. No checklist is followed while carrying such critical jobs of high pressure/risk" (p.27). The setting was reduced by roughly 6.3 times and later reversed without either change being recorded in the signed calibration document. The correct calibration procedure would require a checklist, documented justification, and supervisory approval for any setpoint change; the actual work made an undocumented major change to a high-pressure meter in service.

CAUSAL EXPLANATION

This improperly conducted calibration maintenance on the same meter nine months before the incident normalised undocumented changes and checklist bypass on high-pressure equipment. It created measurement uncertainty and demonstrated the same weak procedural discipline that later allowed the substitute O-ring and omitted hydrotest. It connects secondarily to the lower-level ASBE 1.2 checklist not followed, ASBE 1.5 under-controlled system, AV 1.1 routine violation, and AJDE 2.10 inadequate calibration knowledge, though it did not directly cause the September 2020 gasket failure.

Linked Evidence: EVD-2506

PTF 1.1 Crew/ Team Leadership

SUBSTANTIATED

Night-shift leaders at the gate did not establish a shared urgent plan; MOV 54 was closed 26 minutes after their arrival and 46 minutes after detection.

JUSTIFICATION

At approximately 02:35 hrs the two most senior shift officers — GM (Production) as Overall Shift In-charge/SIC and DGM (Production) as GSP Shift In-charge — were both at the Gas Terminal Gate, yet no shared, immediately actionable isolation plan was created. [EVD-7340] records that the SIC "instructed to close the 36 line pressure control valve and get out of the control room. He informed LPG C/R and asked immediately informed to Resident Engineer, Head operations, Plant Manager and all other concerned" (p.17); this was a serial notification and a single PCV instruction, not an order to close the source-isolating MOV 54 immediately. [EVD-9612] shows the GSP Shift In-charge finally "closed the MOV 54 inlet condensate line from slug catcher to 36 metering area at about 0301 hrs" (p.17), 26 minutes after both officers reached the gate. The committee's own characterisation in [EVD-6414] is direct: "Decision of closure of valve was taken after about 40 minutes resulting the spread of vapour cloud in increased area" (p.32). The leadership gap is compounded by [EVD-4052], which records that the on-site DMP "does not include isolation/stoppage of

CAUSAL EXPLANATION

At 02:35 the two senior officers reached the Gas Terminal Gate but did not establish a shared understanding that the metering inlet had to be isolated without waiting for further management authorisation; their instructions covered the 36-inch PCV and serial escalation rather than the immediate closure of MOV 54. As a result the source-isolating valve was closed at 03:01 — 46 minutes after first physical detection at 02:15 and 48 minutes 10 seconds after CCTV-confirmed leak onset at 02:12:50 — and the 36-inch PCV only at 03:04. The vapour released during this leadership-caused delay was already "all around even at height" at 03:00:44 ([EVD-8213]) and ignited at 03:08:15, after isolation, because the accumulated cloud was sufficient fuel. This higher-level leadership failure thus directly enabled the lower-level AJDE 2.7 delayed-action finding and the ASBE 1.3 emergency-isolation procedural failure, turning a containable release into a major fire.

Linked Evidence: EVD-7340, EVD-9612, EVD-6414, EVD-4052, EVD-3932

PTF 1.1 **Crew/ Team Leadership** **SUBSTANTIATED**

Maintenance leadership let an undocumented O-ring substitution and skipped hydro test become routine practice across three gasket replacements.

JUSTIFICATION

Maintenance leadership failed to create and maintain the technical standard for pressure-boundary work on the 1986-vintage meter. [EVD-8645] records six separate governance omissions in one passage: "The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket" (p.31). [EVD-2894] then records that after the 18.09.2020 replacement "No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated" (p.31), and [EVD-1494] documents the recurrence: cleaning/calibration on 12th & 13th Nov 2018, gasket changed 25/02/2019, then 18/09/2020 — three sealing interventions in roughly 22 months with no leadership-driven root-cause review. This failure to establish a proper team

CAUSAL EXPLANATION

The maintenance team leadership left no controlled specification and no quality gate for gasket replacement, so the locally sourced O-ring was fitted to a flange designed for a rectangular-slot gasket and returned to service on 19 Sep 2020 after only an operating-pressure soak. When the inappropriate seal was subjected to six days of live condensate service from 19 to 24 Sep 2020, it failed at 02:12:50, initiating the 56-minute vapour release that ignited at 03:08:15. The recurrence across 12-Nov-2018, 25-Feb-2019 and 18-Sep-2020 shows leadership did not learn from repeat failures; this precondition directly produced the initiating equipment failure and enabled the lower-level AJDE 2.8 and ASBE 1.2 unsafe acts.

Linked Evidence: EVD-8645, EVD-2894, EVD-1494

PTF 1.2 **Cross-Monitoring Performance** **SUBSTANTIATED**

No peer or supervisor caught the non-standard O-ring or the skipped hydro test before the meter returned to service.

JUSTIFICATION

When the retaining-plate gasket of meter 310FT01 was replaced on 18-Sep-2020, no team member cross-checked either the substitute material or the missing post-maintenance pressure test, and the associated SAP notification was never closed by supervisory review. [EVD-8645] documents that the O-rings were purchased "without any specification or quality control during the purchase" and with "no record of date or authority of the decision to switch over" (p.31), while [EVD-2894] records "No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated" (p.31). [EVD-0591] closes the loop: SAP Notification 3800419840 raised on 17.09.2020 "was not closed after completion of the job" (p.27), meaning no second set of eyes reviewed the completed work. This is a distinct manifestation from the Dec-2019 calibration cross-monitoring failure because it involves the flange-joint maintenance execution and return-to-service quality gate on 18–19 Sep 2020 rather than the calibration certificate. The absence of peer and supervisory back-up is the precondition that allowed the

CAUSAL EXPLANATION

Without cross-monitoring, the non-standard O-ring and the skipped hydraulic verification passed into service on 19 Sep 2020. The defective joint then operated for six days before failing at 02:12:50 on 24 Sep, releasing the vapour cloud that ignited at 03:08:15. This instance is the enabling precondition for the initiating lower-level failures: had a peer or supervisor independently verified the part against the rectangular-slot design and the required design-pressure test, the meter would not have been returned to service in that condition.

Linked Evidence: EVD-8645, EVD-2894, EVD-0591

PTE 2.11 **Equipment Not configured Correctly** **SUBSTANTIATED**

MOV 54 and other Gas Terminal emergency isolation valves were available only for local operation and were not remote-operable from the control room.

JUSTIFICATION

The emergency isolation valves in the Gas Terminal were available locally but were not configured for remote operation from the control room during the emergency. [EVD-3932] quotes the committee finding that "MOVs are provided in the gas terminal station for local operations but they are not having remote operation from control room during emergency" (p.26). [EVD-9612] records the consequence: the GSP Shift In-charge had to reach the Gas Terminal Gate before he could act, and "closed the MOV 54 inlet condensate line from slug catcher to 36 metering area at about 0301 hrs". The correct emergency configuration would allow a control-room operator to close MOV 54 immediately upon confirmed release; the actual configuration required a senior officer to travel to the gate and operate the valve locally. This is a distinct equipment-configuration failure from the F&G interlock gap because it concerns the actuation capability of the isolation valves themselves, not the detection-system configuration.

CAUSAL EXPLANATION

Because MOV 54 could not be operated remotely, the control-room engineers who received the alarm at 02:20 could not execute immediate isolation. The DGM was called from quarters at ~02:35, had to reach the Gas Terminal Gate, and could only close MOV 54 at 03:01, 46 minutes after first detection and just 7 minutes before ignition. This configuration gap directly produced the lower-level AJDE 2.7 delayed necessary action and AJDE 2.9 failure to execute immediate remote isolation. The 46-minute release window created by this and the F&G configuration failure allowed the flammable cloud to reach the compromised lamp and ignite at 03:08:15.

Linked Evidence: EVD-3932, EVD-9612

PTE 2.12 Maintenance Not Conducted/Not Conducted Correctly

SUBSTANTIATED

Two of four fire-water diesel pumps were under maintenance and one was tripping after 2 minutes, leaving the diesel backup unreliable during the incident.

JUSTIFICATION

The fire-water system's diesel pump capacity had existing maintenance and reliability problems at the time of the incident. [EVD-2432] records that "Out of 4 diesel pumps 410 m3/hr, 2 pumps are under maintenance, one operated for 30 minutes & fourth pumps is tripping after 2 minutes. The reason for tripping is not identified or actions have not been recorded" (p.27). The same evidence notes that the electrical fire-water pumps 47PE003 A, C, and D operated during cooling and that tank levels and header pressure remained healthy throughout the incident. The correct preventive maintenance posture would have all four diesel pumps available with tripping causes investigated and recorded; the actual condition left the diesel backup compromised with unresolved reliability issues.

CAUSAL EXPLANATION

This maintenance deficiency did not affect the incident outcome because the electrical fire-water pumps maintained tank levels and header pressure throughout the fire. It was a latent condition that could have constrained the response if cooling demand had been higher or if electrical power had failed. It therefore represents degraded equipment availability with no direct impact on the 24-Sep-2020 outcome, but it indicates the same broader preventive-maintenance weakness documented elsewhere.

Linked Evidence: EVD-2432

PTF 1.2 Cross-Monitoring Performance

SUBSTANTIATED

Self-safety audit teams recorded compliance without verifying site data, leaving equipment defects such as the compromised FLP lamp undetected.

JUSTIFICATION

The multi-disciplinary self-safety audit team failed to cross-monitor the accuracy of its own compliance determinations, recording compliances that later inspection showed to be incomplete. [EVD-9692] states: "Safety self-audit is conducted by a multi-disciplinary team based on the checklist prepared by the location. The checklist is comprehensive but not related to specific plants and in the process, lot of observations are missing. The duration of audit is not indicated and in most of the items, compliances are indicated, but during our inspection, the data is found to be incomplete & not in line with OISD standards/manuals. Approved manuals/SOPs were not available" (p.26). [EVD-7681] reinforces the traceability failure for the 2018 and 2019 internal safety audits, which were conducted by three-member committees over 3–4 days but produced observations "without the filled checklist" (p.26), so the completeness of audited areas cannot be verified. This is a distinct manifestation from the calibration and gasket cross-monitoring failures because it involves the audit assurance team rather than maintenance execution, and it removed a detection barrier for latent equipment issues

CAUSAL EXPLANATION

Because the audit teams did not cross-monitor their own compliance findings against actual site conditions, latent preconditions persisted undetected, including the HPMV FLP lamp with three missing bolts and a glass case opened 5–6 cm that became the probable ignition source ([EVD-6916]). This instance is an enabling precondition for the lower-level ASBE 1.2 audit checklist finding and explains why the Zone-1 electrical-integrity defect survived to participate in the ignition at 03:08:15.

Linked Evidence: EVD-9692, EVD-7681

PTE 2.11 **Equipment Not configured Correctly** **SUBSTANTIATED**

The metering-system vendor drawing did not match the field valves/meters configuration and was not checked or approved by ONGC.

JUSTIFICATION

The metering system drawing did not reflect the as-installed configuration and had not undergone operator approval. [EVD-0933] records that "Reference drawings no. 02-310-AO-101 Rev.1, 2x6 lines & 1x2 lines for metering system are shown but valves & meters configuration are not matching. As old drawings were lost, new drawings are made by a vendor but the same has not been checked & approved by ONGC" (p.26). This means the available engineering record of the 1986 measurement skid was a vendor-prepared, unapproved document that did not match the field valves and meters. The correct configuration-management process would maintain controlled as-built drawings verified against the installation; the actual process left a mismatched, uncontrolled drawing as the sole reference.

CAUSAL EXPLANATION

This configuration/documentation deficiency did not directly cause the flange-joint failure, but it contributed to the environment in which the OEM rectangular-slot gasket specification was lost and a locally sourced O-ring was substituted without engineering review [EVD-8645]. Without a controlled, accurate as-built drawing, maintenance personnel lacked the configuration reference needed to verify the correct gasket design. It therefore supported the wrong-choice-of-action precondition chain, though it is rated as present with no direct outcome impact because the physical failure was driven by the incorrect seal and missing hydrotest rather than by the drawing mismatch alone.

Linked Evidence: EVD-0933

PTE 2.12 **Maintenance Not Conducted/Not Conducted Correctly** **SUBSTANTIATED**

The maintenance team omitted the hydrostatic test to design pressure after the gasket replacement and returned meter 310FT01 to service with an unvalidated pressure boundary.

JUSTIFICATION

After the gasket was replaced on 18-Sep-2020, the pressure boundary was not hydrostatically tested to design pressure. [EVD-2894] records that "After calibration of the meter 310FE01, the gasket was replaced on 18.09.2020. No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated. Next day meter 310FE01 was taken on online and was in use at the time of Incident" (p.31). The committee reiterated the omission in its probable-cause statement: "Probable cause of leak is due to use of nonstandard gasket and not testing the system to designed pressure. No standard operating procedure is in place for calibration and taking back in service of these meters which are operating at high pressures" (p.32). The correct post-maintenance procedure for a pressure-boundary repair is a hydrostatic test to design pressure, which would have stressed the joint and likely revealed the defective seal in a controlled environment; the actual procedure pressurised only to operating pressure with both meters in line and isolated, and then returned the meter to service without pressure validation.

CAUSAL EXPLANATION

Because no hydrostatic test was performed, the inadequate O-ring installation was never verified against design pressure. The meter entered service on 19-Sep-2020 and failed on 24-Sep-2020 at 02:12:50. This omission is a primary precondition enabling the lower-level AJDE 2.5 ignored caution/warning, AJDE 2.7 delayed or omitted necessary action, ASBE 1.2 checklist not followed, and AJDE 2.11 misapplied or unaware of existing testing policy. Had the design-pressure test been performed, the defective seal would likely have failed under controlled test conditions rather than in live hydrocarbon service.

Linked Evidence: EVD-2894

PTF 1.2 **Cross-Monitoring Performance** **SUBSTANTIATED**

Calibration team members signed off a 16000!'2540 mmwc setpoint change without recording

JUSTIFICATION

The calibration event of 17–18 Dec 2019 on meter 310FT01 shows a team-level monitoring failure: the setpoint was changed from 16000 mmwc to 2540 mmwc and then back to 16000 mmwc on the next cycle, yet no team member caught or recorded either change. [EVD-2506] states: "During calibration from 17/12/2019 to 18/12/2019, calibration was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of this change in the calibration documents signed by the team. During next calibration for meter 310FT01 was again changed to 16000 MMWC. No checklist is followed while carrying such critical jobs of high pressure/risk" (p.27). The phrase "signed by the team" means multiple individuals endorsed a certificate that omitted an approximately 6.3x setpoint reduction, which is the essence of failed cross-monitoring — no one independently verified what another had done before signing. This instance is distinct from the September 2020 gasket work because it occurred nine months earlier, involved the calibration function rather than the flange-joint maintenance, and is grounded in its own evidentiary record. It established the procedural-discipline

CAUSAL EXPLANATION

The undocumented change meant meter 310FT01 operated with an unapproved calibration for an unknown period, degrading confidence in its readings and, more importantly, demonstrating to the same maintenance organisation that high-pressure work could be performed without cross-check or documentation. This precondition is a direct enabler of the lower-level ASBE 1.2 calibration checklist finding and contributed to the normalisation of procedural shortcuts that culminated in the 18-Sep-2020 gasket replacement without hydrotest. It did not itself cause the 24-Sep release, so it is rated 3 rather than 4.

Linked Evidence: EVD-2506

PTE 2.14 Unfamiliar Equipment/Non-custodial Operations

SUBSTANTIATED

The maintenance crew did not possess the OEM rectangular-slot gasket specification and repeatedly substituted locally procured O-rings without understanding the required seal geometry.

JUSTIFICATION

[EVD-8645] records that "The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket." This is a direct equipment-familiarity failure: the maintenance crew assigned to the 1986-vintage 6-inch condensate meter no longer had the OEM rectangular-slot gasket specification, no controlled drawing, and no procurement path to OEM parts, so they operated without authoritative knowledge of the correct seal geometry. The recurrence record [EVD-1494] - "Cleaning/servicing/calibration of meters 1 & 2 was carried out on 12th & 13th Nov 2018. Afterwards, meter no.1 was cleaned on 25/02/2019 & gasket was changed. Again on 18/09/2019, gasket was changed which leaked within one year" - shows the same unfamiliarity-driven substitution was repeated across three

CAUSAL EXPLANATION

The lost OEM rectangular-slot specification meant the maintenance crew had no authoritative reference for the seal geometry; they substituted a locally bought O-ring whose round cross-section cannot fill the rectangular groove, establishing a leak path at the flange. Because the crew did not understand the equipment's true design requirement, they also omitted the hydrostatic test [EVD-2894] that would have revealed the defective joint, and returned the meter to service on 19 September 2020 after only an operating-pressure pressurisation. The joint then failed at 02:12:50 on 24 September [EVD-1284], releasing vapour that accumulated for 56 minutes and ignited at 03:08:15 [EVD-4507]. This precondition directly enabled lower-level findings AJDE 2.8 Wrong Choice of Action (substitute O-ring) and ASBE 1.3 Procedure Not Followed (wrong gasket), and the 22-month recurrence pattern [EVD-1494] proves it was systemic rather than a one-off.

Linked Evidence: EVD-8645, EVD-1494, EVD-0591, EVD-2894

PTE 2.15 Inadequate Personal Protection Clothing/Survival Equipment

SUBSTANTIATED

A flame-proof HPMV lamp had three missing bolts and a glass case opened 5-6 cm, compromising its explosion-protection integrity and providing the probable ignition source.

JUSTIFICATION

[EVD-6916] records that "three bolts on one of the flame proof lamps were missing resulting in it being open with its integrity compromised, glass case got detached from the main body and had opened out about 5 to 6 cms. The nearby fittings were found intact." This is a confirmed damage state of safety-critical electrical equipment in a Zone 1 hydrocarbon area; the flame-proof enclosure was not intact, so it could expose ignition-capable contacts to the vapour cloud. [EVD-0496] further shows that the post-incident electrical audit clubbed all nine HPMV lamps together, so the unique compromised condition was not separately characterised. [EVD-4507] anchors the flash at 03:08:15 and identifies the compromised lamp as the probable ignition source. The damaged flame-proof lamp is therefore a substantiated, primary safety-equipment failure.

CAUSAL EXPLANATION

The compromised flame-proof enclosure defeated the explosion-protection barrier precisely when a flammable cloud occupied the area; ignition occurred at 03:08:15, 7 minutes after the first isolation action and after the cloud had accumulated for 56 minutes [EVD-8213]. This Level 2 condition enabled the Level 1 ASBE 1.2 audit-checklist failure that missed the missing bolts, and it converted a hydrocarbon release into a major fire by providing the ignition source.

Linked Evidence: EVD-6916, EVD-0496, EVD-4507

PTE 2.15 **Inadequate Personal Protection Clothing/Survival Equipment** **SUBSTANTIATED**

Three of four diesel firewater pumps were under maintenance or unreliable, leaving emergency firewater dependent on electrical pumps during the incident.

JUSTIFICATION

[EVD-2432] records that "Both the Tank level and the Fire water header pressure were healthy throughout the incident. Three electrical Fire water pumps namely 47PE003 A, C, D were operated during the cooling activity" but the committee separately flagged that "Out of 4 diesel pumps 410 m3/hr, 2 pumps are under maintenance, one operated for 30 minutes & fourth pumps is tripping after 2 minutes. The reason for tripping is not identified or actions have not been recorded." Three of four diesel firewater pumps had availability or reliability defects, leaving the emergency firewater system dependent on electrical pumps and grid power. This is a substantiated inadequacy in fire protection/survival equipment; because electrical pumps and header pressure remained healthy during this incident, it did not affect the outcome and is rated 2.

CAUSAL EXPLANATION

Although firewater tank levels and header pressure remained healthy and three electrical pumps carried the cooling load, the compromised diesel pump availability created a latent resilience gap. Had electrical supply failed or cooling demand exceeded electrical capacity, the fire response could have been constrained. This instance did not directly cause the 24 September fire but reflects the same maintenance-governance weakness documented elsewhere in the evidence.

Linked Evidence: EVD-2432

PTE 2.14 **Unfamiliar Equipment/Non-custodial Operations** **SUBSTANTIATED**

The metering system P&ID was a lost-original, unapproved vendor drawing that did not match the field configuration, leaving personnel without an accurate as-built reference.

JUSTIFICATION

[EVD-0933] records that "Reference drawings no. 02-310-AO-101 Rev.1, 2x6 lines & 1x2 lines for metering system are shown but valves & meters configuration are not matching. As old drawings were lost, new drawings are made by a vendor but the same has not been checked & approved by ONGC." With the original drawings lost and the replacement vendor drawing unapproved and mismatched to the field configuration, no maintenance or operating person could establish the as-built configuration of the metering system from controlled documentation. This is a substantiated documentation-level manifestation of equipment unfamiliarity distinct from the lost gasket specification, involving the P&ID/systems view rather than component material specification. Its impact is rated 2 because the direct loss of containment was the flange joint rather than a drawing error, but it removed the configuration baseline that could have challenged the substitute gasket selection.

CAUSAL EXPLANATION

Without an accurate approved as-built drawing, the maintenance group could not independently verify the correct gasket type, valve configuration, or meter design details for the skid. This supports the lower-level AV 1.1 routine configuration-management breakdown rated 2 at Level 1 and is an enabler of the ASBE 1.3 wrong-gasket procedure failure, although it did not directly initiate the loss of containment.

Linked Evidence: EVD-0933

PTE 2.15 **Inadequate Personal Protection Clothing/Survival Equipment** **SUBSTANTIATED**

The F&G detection system was display-only with no auto-shutdown or suppression activation, leaving the plant dependent on slow human detection and manual isolation.

JUSTIFICATION

[EVD-8657] records the committee's finding that "Hydrocarbon/H2S detectors are configured to Annunciator Panel for display but not in DCS for history" and "F&G detection system does not have provision for auto shut down of plant or activation of fire suppression system." [EVD-3452] shows the F&G layout exists but is not interlocked to action, and the committee recommendation A.1 (p.34) calls for interlocking F&G to automatic plant shutdown and fire suppression. Thus the safety monitoring equipment was inadequate in function: at 02:12:50 CCTV recorded leakage onset [EVD-1284], but no automatic alarm or isolation occurred, and the annunciator alarm sounded only at about 02:20, five minutes after human detection. The monitoring system was display-only and lacked the automatic protective action needed for a major release.

CAUSAL EXPLANATION

This inadequacy converted what could have been an automatic sub-second isolation response into a 46-minute human detection-decision-action chain: leak onset at 02:12:50, annunciator alarm at 02:20, and first isolation MOV 54 at 03:01. The continuous release during that interval formed the large flammable cloud documented at 03:00:44 [EVD-8213], which ignited at 03:08:15. This Level 2 condition directly enabled the lower-level AJDE 2.7 delayed necessary action and AJDE 2.9 decision-making failures. Because the gasket failure was the initiating event and the F&G gap mainly magnified severity, it is rated 3.

Linked Evidence: EVD-8657, EVD-3452, EVD-1284

PTF 1.3 Task Delegation

SUBSTANTIATED

Control room emergency tasks were assigned to staff with only four days of general training and no defined qualification.

JUSTIFICATION

Safety-critical control-room tasks were delegated to personnel whose training and qualification were not defined and were materially below the level the committee later required. [EVD-5634] states: "control room operations are carried out by persons inadequately trained in safety, operations, control room handling, logbook records and actions during emergency. It is informed that four days training programme is conducted for new recruitment/transfer staff which is general in nature and later the job training is carried out. Qualification of this staff is also not defined in HR/Operations Policy. Most of the staff have attended only One Basic Course since 2018" (p.30). The committee recommended that control room staff be trained for at least one month, preferably on simulators, before assignment. This is a precondition embodying PTF 1.3 — safety-critical emergency tasks were delegated to under-trained operators — and it directly explains the lower-level AJDE 2.10 knowledge gap that delayed isolation decision-making. It is distinct from the other PTF 1.3 instance because it concerns the standing staffing and training system rather than the minute-to-minute division

CAUSAL EXPLANATION

Because the night-shift operators had not been trained to execute emergency isolation, when the annunciator alarmed at 02:20 they attempted physical verification and then escalated by telephone instead of initiating immediate source isolation. This precondition created and sustained the 46-minute delay from first detection at 02:15 to MOV 54 closure at 03:01, enabling the vapour cloud to grow until it ignited at 03:08:15. It is a direct enabler of the lower-level AJDE 2.10 and AJDE 2.9 decision-making failures.

Linked Evidence: EVD-5634

PTF 1.10 Critical Information not Communicated

SUBSTANTIATED

CISF Constable A's 02:15 detection notification omitted the Gas Terminal Control Room, directing critical leak information only to gate, fire station, and LPG channels.

JUSTIFICATION

The first human detection of the leak was not communicated to the operating control room with authority over the affected area. [EVD-4286] records that CISF Constable A, on duty at Watch Tower 3, at 0215 hrs observed noise followed by vapour cloud from Gas Terminal Area (p.16), and the evidence item states he informed CISF Main Gate, Fire Station, and Main LPG Control Room. The Gas Terminal Control Room, which controlled the leaking metering area, was not among the notified recipients. This omission occurred approximately 145 seconds after CCTV-confirmed leak onset at 02:12:50 and about 5 minutes before the annunciator alarm at 02:20. It preconditioned the lower-level ASBE 1.6 visual-scan/detection failure and contributed to the delayed response.

CAUSAL EXPLANATION

By excluding the Gas Terminal CR from his notification list, Constable A forced the warning through a CISF-internal relay: Sub-Inspector C in the CCTV Control Room had to verify the leak and call the GT CR at 02:18, so the operating crew lost approximately 3-5 minutes. That loss was one input to the 46-minute detection-to-isolation gap, which allowed vapour to accumulate until MOV 54 was closed at 03:01 and the cloud ignited at 03:08:15. More direct notification could have allowed the operating crew to begin assessment and isolation earlier, connecting this communication omission to the lower-level AJDE 2.7 delayed isolation action.

Linked Evidence: EVD-4286

PTF 1.10 Critical Information not Communicated

SUBSTANTIATED

Head Constable B's independent 02:15-02:16 detection at the Gas Terminal Gate also omitted the Gas Terminal CR, duplicating the first notification gap.

JUSTIFICATION

A second independent detection of the same leak also bypassed the Gas Terminal Control Room. [EVD-6234] records that CISF Head Constable B, on night-shift duty at the Gas Terminal Gate east of the Gas Terminal, at 0215 hrs heard the noise and observed the vapour cloud behind the Gas Terminal building, and at 0216 informed the CCTV Control room and CISF main gate control room (p.17). The operating control room for the affected Phase I area was again omitted from the notification list. This is a distinct manifestation from the Watch Tower notification because it involves a different individual, a different post, and a separate notification path, and it shows that no redundancy corrected the first omission.

CAUSAL EXPLANATION

Because Head Constable B also routed the critical leak information through CISF CCTV and main gate rather than directly to the GT CR, the second channel did not compensate for the omission by Constable A. The Gas Terminal CR ultimately learned of the event only after Sub-Inspector C independently verified the leak at 02:18 and phoned the control room, producing the 5-minute lag behind human detection. This duplicated communication gap is a precondition for the lower-level ASBE 1.6 failure to detect by the operating crew and the extended pre-isolation period that ended with ignition at 03:08:15.

Linked Evidence: EVD-6234

PTF 1.11 Standard/ Proper Terminology

SUBSTANTIATED

The CCTV operator notified the Gas Terminal CR of a vapour cloud or steam leakage rather than using standard hydrocarbon-release terminology.

JUSTIFICATION

The first operational notification of the event used non-standard, ambiguous terminology. [EVD-1460] records the exact phrase used by CISF Sub-Inspector C when calling the Gas Terminal Control Room at approximately 02:20: he informed the Chief Engineer about some vapour cloud or steam leakage in Gas Terminal PH I area (p.17). Vapour cloud or steam leakage is not standard emergency terminology for a flammable hydrocarbon release and conflates a relatively benign utility leak with a major process hazard. This ambiguity could understate urgency to a receiver who had not yet visually confirmed the event. It connects to the lower-level AJDE 2.4 misinterpretation and ASBE 1.6 detection and assessment behavior, as the first on-shift operational cue came with a severity-diluting descriptor.

CAUSAL EXPLANATION

The ambiguous vapour cloud or steam leakage phrase arrived at 02:20 at the same moment as the annunciator alarm, but it reinforced uncertainty about what was occurring. The Chief Engineer and Junior Engineer then attempted on-foot approach instead of immediately pursuing remote isolation, which consumed additional minutes within the 46-minute interval between detection and first valve closure. This non-standard terminology is consistent with the training gap identified in [EVD-5634], which documents only a four-day general control-room training programme with no defined qualification.

Linked Evidence: EVD-1460

PTF 1.12 Failed to Effectively Communicate (Feedback/ Acknowledgement)

SUBSTANTIATED

SAP notification 3800419840 was never closed after the 18-Sep-2020 gasket work, leaving no acknowledgement that the corrective job was complete.

JUSTIFICATION

The maintenance communication loop was never closed. [EVD-0591] records that Notification no. 3800419840 was raised on 17/09/2020 for malfunction of meter no.1 (p.27), and the evidence record states the notification was not closed after completion of the job (p.27). [EVD-0185] shows the meter was returned online the next day, yet there is no acknowledgement in SAP that the job was complete or that the original malfunction had been resolved. This is a clear failed feedback and acknowledgement mechanism: the requester and supervisor received no formal confirmation, and the committee's recommendation D.2.ii calls for sample checks/approval of closure of notifications by Senior Officers to be introduced. This preconditioned the lower-level AV 1.1 non-closure and AJDE 2.11 misapplied SAP closure policy.

CAUSAL EXPLANATION

Because no closure was recorded, there was no supervisory gate to inspect the work package, so the missing hydrotest and the non-standard O-ring substitution went unreviewed. The unvalidated meter entered service on 19-Sep-2020 and failed on 24-Sep-2020 at 02:12:50, leading to the 56-minute vapour accumulation and ignition at 03:08:15. This is a secondary but enabling communication failure in the maintenance-to-operations loop, directly tied to the lower-level AV 1.1 non-closure finding and the AJDE 2.11 policy misapplication.

Linked Evidence: EVD-0591, EVD-0185

PTF 1.10 Critical Information not Communicated

SUBSTANTIATED

The 16000 to 2540 mmwc setpoint change on 310FT01 in Dec 2019 and its reversal were never recorded or communicated in the signed calibration documents.

JUSTIFICATION

Critical calibration information was withheld from the signing team and from the formal record. [EVD-2506] records verbatim: During calibration from 17/12/2019 to 18/12/2019, calibration was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of this change in the calibration documents signed by the team. During next calibration for meter 310FT01 was again changed to 16000 MMWC. No checklist is followed while carrying such critical jobs of high pressure/risk (p.27). The 6.3x reduction and its later silent reversal were never communicated to whoever reviewed or signed the certificate, so the approved baseline was false. This omission is a direct manifestation of PTF 1.10 and preconditioned the lower-level ASBE 1.2 calibration checklist failure and AJDE 2.10 calibration knowledge gap on the same meter.

CAUSAL EXPLANATION

Because the signed calibration documents omitted the 16000 to 2540 mmwc change, the formal record gave no warning that the meter's calibration baseline had been altered, and the next calibration silently reverted it to 16000 mmwc. That undisclosed instability created uncertainty about the meter's condition and normalized undocumented changes to critical pressure-boundary equipment. The same meter later experienced the 18-Sep-2020 gasket substitution with no hydrotest, showing that this communication failure was part of a persistent pattern on the same asset. It is a secondary contributor: it did not directly open the leak, but it removed the information gate that should have exposed the meter's unstable maintenance state before the 24-Sep-2020 release.

Linked Evidence: EVD-2506

PTF 1.13 Mission Planning

SUBSTANTIATED

The 18-Sep-2020 gasket job was planned without the OEM specification, part quality control, hydrostatic test, or notification closure — all critical preparatory elements failed.

JUSTIFICATION

The preparatory planning for the 18-Sep-2020 gasket replacement was incomplete on every element that mattered: information collection, specification control, supplier control, test planning, and risk assessment. [EVD-8645] records 'Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket' (p.31) — six planning omissions in one paragraph, including absence of an authorized decision record. [EVD-2894] adds 'No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure... No standard operating procedure is in place for calibration and taking back in service of these meters which are operating at high pressures' (pp.31-32), showing the pressure-boundary verification step was not planned. [EVD-0591] shows SAP Notification 3800419840 'was not closed after completion of the job'

CAUSAL EXPLANATION

Because planning never recovered the OEM rectangular-slot specification, the crew sourced a locally purchased O-ring without QC [EVD-8645], which is the wrong material decision cited by the committee as root cause #2 and the origin of the pressure-boundary weakness. Because the plan omitted hydrostatic testing to design pressure [EVD-2894], the defective seal was never exposed before the meter returned to service on 19-Sep; the joint then operated for 6 days and failed at 02:12:50 on 24-Sep [EVD-1284], releasing vapour for 56 minutes until ignition at 03:08:15 [EVD-4507]. Because the SAP notification was not closed [EVD-0591], the job escaped the supervisory review that might have challenged the missing test and the recurring gasket pattern. This Level-2 planning omission therefore created the environment where the lower-level wrong choices and procedural shortcuts could occur.

Linked Evidence: EVD-8645, EVD-2894, EVD-0591

PTF 1.13 **Mission Planning** SUBSTANTIATED

The plant's disaster management plan named the hydrocarbon-leak scenario but omitted isolation/stoppage decision criteria and authority, leaving no pre-planned emergency action.

JUSTIFICATION

The emergency-response mission was planned with the leak scenario named but its decisive controls unspecified. [EVD-4052] records 'As per DMP On site Hazira Plant, guidelines to be followed in case of hydrocarbon leakage are included but does not include isolation/stoppage of facilities/pipelines' (p.32), and reinforces that in the ERDMP, 'Heavy product/gas leakages from pipelines & tanks are considered but decision on isolation/closure of plant/pipelines is not mentioned' (p.27). This is an explicit planning incompleteness: the plan defines the hazard but not the trigger, authority, or action for isolation. [EVD-6414] quantifies the operational consequence with the committee's finding 'Decision of closure of valve was taken after about 40 minutes resulting the spread of vapour cloud in increased area' (p.32), and the same evidence item ties the delay to the committee's root cause #6, 'Non isolation/stoppage of 36 pipeline operation'. The gap directly explains the lower-level findings AJDE 2.10 (operators lacked knowledge of isolation criteria), AJDE 2.11 (crew believed escalation were the required procedure), and ASBE 1.2 (emergency response checklist omitted

CAUSAL EXPLANATION

Because the plan contained no criteria for when and by whom the 36-inch line or metering area should be isolated, the night crew had no pre-planned authorised action to execute at first detection (02:15) or first alarm (02:20). Personnel defaulted to serial physical verification and management escalation rather than immediate remote/local isolation, which extended the release for 46 minutes until MOV 54 closure at 03:01 and PCV closure at 03:04. The released inventory was visible as a mature cloud at 03:00:44 ('vapour all around even at height' [EVD-8213]) and ignited at 03:08:15 [EVD-4507]. This planning failure is the Level-2 precursor that enabled the lower-level operator indecision and the 46-minute isolation delay.

Linked Evidence: EVD-4052, EVD-6414

PTF 1.13 **Mission Planning** SUBSTANTIATED

Planning failed to provide remote-operated MOVs or delegated standing authority, forcing physical arrival and management escalation before the first isolation at 03:01.

JUSTIFICATION

The planning of emergency isolation resources did not give the on-shift team the ability to act quickly, independent of the missing DMP decision criteria in instance 2. [EVD-3932] records the structural conclusion that 'MOVs are provided in the gas terminal station for local operations but they are not having remote operation from control room during emergency' (p.26), and that no role between Shift In-charge and Plant Manager had standing authority to close MOV 54 without telephoning up the chain. [EVD-9612] confirms the physical consequence: the GSP Shift-in-charge 'tried to enter the Gas Control room area but could not enter due to heavy gas smell... closed the MOV 54 inlet condensate line from slug catcher to 36 metering area at about 0301 hrs' (p.17) — he had to physically reach the gate before execution, 46 minutes after first detection. This is distinct from the DMP instance because even where a closure decision was eventually made, the task could not be executed remotely or by delegated standing authority. It is a secondary but direct contributor to the lower-level AJDE 2.7 delayed closure and AJDE 2.9 decision-making failure.

CAUSAL EXPLANATION

The non-remote MOV design and centralised authority converted what should have been an immediate machine-to-human response into a 46-minute field-transport and approval sequence: leak onset 02:12:50 [EVD-1284], first human detection 02:15, alarm 02:20, supervisor arrival 02:35, and only then closure at 03:01. Each minute of that delay added inventory to the flammable envelope that ignited at 03:08:15. This planning failure created the conditions for the lower-level delayed-action and insufficient-isolation findings at Level 1.

Linked Evidence: EVD-3932, EVD-9612

PTF 1.13 **Mission Planning** SUBSTANTIATED

The Self Safety Audit was planned with a generic, non-plant-specific checklist, causing plant-specific integrity observations to be missed and compliance to be asserted without verification.

JUSTIFICATION

The preparatory planning of the Self Safety Audit mission was incomplete because the instrument used could not sample the plant's specific risk points. [EVD-9692] records 'The checklist is comprehensive but not related to specific plants and in the process, lot of observations are missing. The duration of audit is not indicated and in most of the items, compliances are indicated, but during our inspection, the data is found to be incomplete & not in line with OISD standards/manuals. Approved manuals/SOPs were not available' (p.26). The committee recommendation B.1, that 'Checklist for Self-Safety Audit for individual plants are to be developed' (p.34), confirms the planning instrument was unfit. This planning failure produced the lower-level ASBE 1.2 checklist failure in which audit teams marked compliances without verifying plant-specific integrity details — the same class of missed observation that left the compromised Zone 1 HPMV FLP lamp available as the eventual ignition source. This is distinct from the maintenance and DMP instances because it concerns the assurance/audit mission rather than maintenance or emergency response.

CAUSAL EXPLANATION

Because the audit checklist was not tailored to the Gas Terminal metering area, the audit did not force inspection of the plant-specific details that later became causal — including flameproof electrical fitting integrity and the chronically deficient gasket practice. The generic instrument therefore allowed latent preconditions (such as the HPMV FLP lamp with three missing bolts and case opened 5-6 cm per EVD-6916) to survive unobserved, contributing to the ready ignition source when the vapour cloud formed. This Level-2 planning deficiency directly fed the lower-level audit checklist failures identified at Level 1.

Linked Evidence: EVD-9692

PTF 1.14 Mission Briefing

SUBSTANTIATED

The 02:18 notification to the Gas Terminal CR described a CCTV-confirmed heavy leak as 'some vapour cloud or steam leakage', understating severity and omitting isolation guidance.

JUSTIFICATION

The information transmitted to the Gas Terminal Control Room at the start of the emergency response was materially insufficient relative to what the sender actually observed. [EVD-8871] records that CISF Sub-Inspector C visually verified at 02:18 'heavy leakage and formation of gas cloud through CCTV' (timeline p.12) before notifying the Gas Terminal Control Room and CISF Main Gate. Yet the message received by the operating crew, per [EVD-1460], was only that he was 'informing him about some vapour cloud or steam leakage in Gas Terminal PH I area' (p.17). The word 'some' and the alternative 'steam leakage' understated a confirmed heavy hydrocarbon leak, and the message contained no instruction to isolate or treat the event as a major release. This deficient briefing of the response team contributed to the crew's initial plan to walk out and locate the source rather than immediately execute isolation, feeding the lower-level AJDE 2.4 misinterpretation and AJDE 2.9 decision-making findings.

CAUSAL EXPLANATION

The understated briefing degraded the shared team picture precisely at the moment the emergency plan needed to switch from diagnosis to isolation. The GT CR operators heard the alarm and a vague call simultaneously at 02:20 and, consistent with being told of possible steam, went outside to inspect; when the dense cloud forced them back, they escalated by telephone rather than closing MOV 54, which was not remotely operable. The ensuing 46-minute delay from detection to closure [EVD-6414] allowed the vapour envelope to grow into the cloud that ignited at 03:08:15. This briefing deficiency is therefore a secondary causal contributor to the lower-level response delay findings.

Linked Evidence: EVD-8871, EVD-1460

PTF 1.15 Task/ Mission Re-Planning during Operations

SUBSTANTIATED

The GT CR crew's on-foot source-location plan was invalidated by dense vapour, but they did not re-plan to immediate isolation and instead only escalated by telephone.

JUSTIFICATION

The Gas Terminal Control Room operators' initial plan was to walk to the affected area and locate the source. That plan became unworkable within minutes when the dynamic environment changed, yet no amended plan for immediate isolation was generated by the crew. [EVD-1460] records that after the 02:20 alarm and call, the Chief Engineer 'saw the heavy cloud in phase I area' but could not approach due to the dense cloud, and [EVD-9220] corroborates that visibility was very poor and the smell intense. [EVD-9941] quantifies how fast the situation was changing: the cloud progressed from 'vapour cloud near GT control room' at 02:15:35 to 'vapour cloud intense' at 02:16:13 (pp.19-21). Despite this invalidation of the field-inspection plan, the crew reverted to telephone escalation rather than re-planning toward remote or local isolation; MOV 54 was not closed until 03:01 by the GSP Shift In-charge [EVD-9612]. This failure to re-plan is a secondary contributor because the crew's standing authority and remote control options were constrained by the planning gaps in EVD-3932, but it directly fed the lower-level AJDE 2.9 decision-making finding and the delay before

CAUSAL EXPLANATION

The failed approach at approximately 02:20 was a clear trigger to amend the mission from 'locate and inspect' to 'isolate and contain'. No such amendment was made by the control room crew; their escalation telephoned the chain but did not request or execute isolation, so the leak continued unisolated from 02:20 until 03:01. The resulting 41-minute segment of the release contributed to the mature flammable cloud documented at 03:00:44 [EVD-8213], which ignited at 03:08:15 despite the later valve closures at 03:01 and 03:04.

Linked Evidence: EVD-1460, EVD-9941

PTF 1.15 **Task/ Mission Re-Planning during Operations** **SUBSTANTIATED**

After being unable to enter due to gas density at ~02:35, the shift-in-charge team still took until 03:01 to re-plan and close MOV 54.

JUSTIFICATION

The supervisory team that reached the Gas Terminal Gate at approximately 02:35 also encountered a dynamically changed environment, but did not amend their plan toward immediate isolation for another 26 minutes. [EVD-9612] records the GSP Shift-in-charge 'tried to enter the Gas Control room area but could not enter due to heavy gas smell. He informed Fire station to abstain from any vehicle movement in the plant... closed the MOV 54 inlet condensate line from slug catcher to 36 metering area at about 0301 hrs. Thereafter fire occurred' (p.17). [EVD-7340] shows the Overall Shift-in-charge's immediate response at the gate was to instruct closure of the 36-inch PCV and begin serial management notification, not an immediate comprehensive isolation. The committee's timeline reconstruction in [EVD-6414] states 'Decision of closure of valve was taken after about 40 minutes resulting the spread of vapour cloud in increased area' (p.32). The gas density that prevented entry was the environmental change requiring re-planning; the closure decision made only at 03:01, 26 minutes after the team reached the gate and 46 minutes after first detection, is

CAUSAL EXPLANATION

The inability to enter the plant at 02:35 should have triggered an immediate re-plan to close every available upstream isolation without approaching the release. Instead, the team spent the interval notifying additional management layers and only closed the primary upstream MOV at 03:01. This 26-minute lost segment was the largest single contributor to the 46-minute unisolated release window, directly enlarging the vapour inventory that remained in the air after PCV closure at 03:04 and ignited at 03:08:15. Compared with the lower-level unsafe acts, this supervisory failure to re-plan is the direct amplifier of the incident's severity.

Linked Evidence: EVD-9612, EVD-7340, EVD-6414

PTF 1.13 **Mission Planning** **SUBSTANTIATED**

Maintenance planning never collected and analysed the three gasket replacements in 22 months as a chronic sealing defect, so the same inadequate fix was repeated.

JUSTIFICATION

The information-collection and risk-assessment elements of maintenance planning failed over a much longer window than the 18-Sep-2020 job, because the same meter's repeated sealing failures were never analysed as a systemic reliability warning. [EVD-1494] records 'Cleaning/servicing/calibration of meters 1 & 2 was carried out on 12th & 13th Nov 2018. Afterwards, meter no.1 was cleaned on 25/02/2019 & gasket was changed. Again on 18/09/2019, gasket was changed which leaked within one year' (p.27), and notes the recurrence did not trigger material or procedure review. The initial incident narrative separately directs 'Investigation should determine whether the recurrence pattern was treated as a reliability concern'. This is a distinct manifestation from instance 1 because it is the chronic, multi-year planning analysis gap, not the single job plan. It enabled the lower-level AJDE 2.5 finding that three gasket failures in 22 months constituted a warning that was ignored, and explains why the same locally sourced O-ring continued in use across four failures.

CAUSAL EXPLANATION

Had planning treated the three replacements (12-Nov-2018, 25-Feb-2019, 18-Sep-2020) as a signal to recover the OEM specification, perform root-cause analysis, or replace the meter, the 18-Sep-2020 job could not have repeated the same inadequate O-ring substitution. Instead, the recurrence data sat in SAP without triggering a reliability assessment, so the defective pressure-boundary practice survived until the fourth failure on 24-Sep-2020 initiated the release. This Level-2 analysis gap explains why the lower-level unsafe act of substituting the O-ring persisted as a routine violation rather than being corrected after the second or third failure.

Linked Evidence: EVD-1494

PTF 1.16 **Miscommunication** **SUBSTANTIATED**

CCTV CR verified 'heavy leakage and gas cloud' but GT CR was telephoned about 'some vapour cloud or steam leakage', diluting urgency at 02:18-02:20.

JUSTIFICATION

At 02:18 the CCTV operator possessed verified critical information: [EVD-8871] records he visually verified "heavy leakage and formation of gas cloud through CCTV" (timeline p.12) and then notified the Gas Terminal Control Room and CISF Main Gate. The message actually received by the operating team was materially weaker: [EVD-1460] records that the Chief Engineer was told about "some vapour cloud or steam leakage in Gas Terminal PH I area" (p.17). The CCTV operator's verified heavy-leak report was therefore reduced at the team-to-team interface to an ambiguous 'some vapour cloud or steam leakage' description at the same moment the annunciator alarm first activated at 02:20 — roughly 7 minutes 10 seconds after leak onset at 02:12:50 and 5 minutes after first human detection at 02:15. The Chief Engineer then acted on the diluted message by attempting to approach the source on foot rather than treating the report as requiring immediate isolation, which is exactly the misinterpretation captured in the lower-level finding AJDE 2.4. This is a substantiated PTF 1.16 occurrence because correctly obtained and correctly verified information was either

CAUSAL EXPLANATION

The diluted severity message reached the Chief Engineer at 02:20, just as the annunciator alarmed. Because the phone call framed the vapour cloud as possibly 'steam leakage', the engineer's subsequent choice was source-location on foot rather than immediate remote isolation; the dense cloud forced both engineers back, and the next escalation occurred only at ~02:35 with the GSP/LPG shift in-charges. This miscommunication is a Level-2 precondition that enabled lower-level AJDE 2.4 (decision-making based on misinterpretation) and contributed to AJDE 2.7 and ASBE 1.3: the first isolation valve MOV 54 was not closed until 03:01, 46 minutes after first detection, and the accumulated cloud ignited at 03:08:15, 56 minutes after leak onset. Had the CCTV operator's verified 'heavy leakage and formation of gas cloud' severity been correctly conveyed and interpreted at 02:18-02:20, isolation actions could have been initiated substantially earlier.

Linked Evidence: EVD-8871, EVD-1460

PTF 1.16 Miscommunication

SUBSTANTIATED

Two simultaneous CISF perimeter reports at 02:15 were routed through internal verification nodes; the Gas Terminal CR was not called until 02:18-02:20, losing ~5 minutes.

JUSTIFICATION

Two independent CISF personnel detected the release at 02:15 and correctly communicated it into CISF channels: [EVD-4286] records CISF Constable A "At 0215 hrs, he observed noise followed by vapour cloud from Gas Terminal Area" (p.16), then "he informed CISF Main Gate, Fire Station, and Main LPG Control Room." [EVD-6234] records Head Constable B from the Gas Terminal Gate "At 0215 hrs, he heard the noise and observed vapour cloud behind Gas terminal building. At 0216. He informed CCTV Control room and CISF main gate control room" (p.17). Neither of these two direct perimeter reports was forwarded immediately and directly to the Gas Terminal Control Room; [EVD-8871] shows the operating CR was only notified at 02:18 after the CCTV operator independently verified the leak, and [EVD-1460] shows the call actually received by the operating team at around 02:20. The evidence itself characterises this as a 'communication-architecture artefact contributing to the ~5 minute lag between physical detection and Gas Terminal CR notification.' For PTF 1.16 purposes, the correctly communicated critical update from the two perimeter

CAUSAL EXPLANATION

At 02:15 two independent CISF perimeter detections existed; by 02:15:35-02:16:13 the vapour cloud was already expanding across the Gas Terminal control room, road and piping area (EVD-9941), so every minute before notification mattered. The delayed team-to-team forwarding meant the GT CR alarm and phone action began at 02:20 rather than at 02:15, and the full escalation chain — GSP Shift In-charge called at ~02:35, physically reaching the gate, and closing MOV 54 at 03:01 — started five minutes later than it could have. This PTF 1.16 instance is a precondition for the lower-level delayed-action finding AJDE 2.7 and the procedural isolation failure ASBE 1.3, contributing to the 46-minute interval between first detection and first valve closure and the 56-minute vapour accumulation before ignition at 03:08:15.

Linked Evidence: EVD-4286, EVD-6234, EVD-8871

PTF 1.17 Team Members Unfamiliar with Workspace

SUBSTANTIATED

Control-room staff had only 4 days general training and no simulator qualification, leaving them unfamiliar with plant-specific emergency isolation SOPs.

JUSTIFICATION

[EVD-5634] is the direct documentary basis: "control room operations are carried out by persons inadequately trained in safety, operations, control room handling, logbook records and actions during emergency. It is informed that four days training programme is conducted for new recruitment/transfer staff which is general in nature and later the job training is carried out. Qualification of this staff is also not defined in HR/Operations Policy. Most of the staff have attended only One Basic Course since 2018" (p.30). A four-day general programme, with no defined qualification and no simulator, does not establish plant-specific familiarity with the Gas Terminal workspace, its emergency isolation points, or the DMP response protocols. The committee's recommended expansion to at least one month preferably on simulators supports that the existing orientation left staff unfamiliar with their workspace and emergency SOPs. The night-shift team's actual behaviour — attempting to approach the dense vapour cloud on foot instead of initiating immediate remote isolation, then serially telephoning shift managers — is consistent with operators who had not rehearsed the plant-specific emergency

CAUSAL EXPLANATION

The unfamiliarity precondition showed up at 02:20 when the two control-room engineers, with no simulator-practised emergency response, chose physical source verification and then a telephone escalation chain rather than immediate closure of available isolation valves. The first isolation (MOV 54) occurred at 03:01 — 46 minutes after first detection — and the 36" PCV at 03:04, only 4 minutes before ignition. The accumulated cloud ignited at 03:08:15. This directly provides the precondition for lower-level AJDE 2.10 (inadequate knowledge to make decisions) and AJDE 2.11 (unaware/misapplied policy), and contributes to the ASBE 1.3 procedural isolation failure.

Linked Evidence: EVD-5634, EVD-6414

PTF 1.17 **Team Members Unfamiliar with Workspace** **SUBSTANTIATED**

Maintenance staff did not have the OEM rectangular-slot gasket spec and used locally procured O-rings as a regular practice, repeated three times in 22 months.

JUSTIFICATION

[EVD-8645] demonstrates a durable unfamiliarity with the correct sealing configuration of the 1986 condensate meter: "The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket" (p.31). Maintenance personnel were therefore working without the OEM drawing, specification, or procurement procedure for the rectangular-slot gasket, and the substitute O-ring had become the normal practice. The recurrence record [EVD-1494] — cleaning/servicing/calibration on 12-13 Nov 2018, gasket changed 25-Feb-2019, and gasket changed again 18-Sep-2020 — shows the same unfamiliar or non-conforming practice was repeated three times in ~22 months without any corrective or familiarisation step. This substantiates PTF 1.17 at the maintenance-team level: the

CAUSAL EXPLANATION

Because the team was unfamiliar with the correct gasket specification, they fitted a round O-ring in a rectangular slot on 18-Sep-2020 and, also lacking procedural familiarity, did not hydro-test the joint to design pressure; the system was only pressurised to operating pressure. Meter 310FT01 returned to service on 19-Sep-2020 and operated for six days until the joint failed at 02:12:50 on 24-Sep-2020, initiating the 56-minute vapour accumulation and the ignition at 03:08:15. This Level-2 unfamiliarity is therefore the precondition for lower-level AJDE 2.8 wrong choice, AJDE 2.10 inadequate knowledge, ASBE 1.3 procedure not followed, and the omitted hydro-test findings AJDE 2.7 and ASBE 1.2.

Linked Evidence: EVD-8645, EVD-1494

PPI 5.4 **Misperception of Changing Operational Environment** **SUBSTANTIATED**

The GT Control Room engineers misjudged that they could approach the leaking metering skid on foot through a dense vapour cloud, losing time before escalation.

JUSTIFICATION

The two Gas Terminal Control Room operators formed an inaccurate perception of the operating environment when they left the control room and attempted to walk to the metering area despite a hydrocarbon vapour cloud already expanding across the access route. [EVD-1460] records that at 02:20 the Chief Engineer heard the annunciator alarm, received the CISF call, and “saw the heavy cloud in phase I area” but could not approach due to the dense cloud, while [EVD-9220] records the Junior Engineer went out, “observed leakage and intense smell from Phase-I area. Visibility was very poor”. The CCTV sequence in [EVD-9941] demonstrates why this approach was misjudged: the cloud had already progressed from “vapour cloud near GT control room” (02:15:35) to “vapour cloud intense” (02:16:13) within 38 seconds, meaning the hazardous envelope had fully enveloped the route before the operators left the building. This is the level-2 manifestation of the lower-level APE 3.2 finding that the dense vapour prevented Gas Terminal CR personnel from accurately perceiving distances and safe approach paths, and it is substantiated because the failed approach and forced

CAUSAL EXPLANATION

The operators received the first annunciator alarm at approximately 02:20 but translated it into a perception that a ground-level source could be located and confirmed by walking to the skid. Their misjudgment of the release's physical extent consumed the critical 02:20–02:35 window: only after being forced back by the dense cloud and overpowering smell did the Chief Engineer telephone the GSP Shift In-charge, LPG Shift In-charge, and CFU Panel Engineer, with the GSP Shift In-charge not reaching the Gas Terminal Gate until approximately 02:35. This direct loss of approximately 15 minutes folded into the broader 46-minute gap between first physical detection at 02:15 and first isolation of MOV 54 at 03:01, which the committee found allowed the flammable cloud to accumulate to the point that ignition at 03:08:15 produced a major fire. The finding therefore sits causally beneath the lower-level AJDE 2.7 delayed-isolation finding and AJDE 2.9 decision-making-under-operation failure.

Linked Evidence: EVD-1460, EVD-9220, EVD-9941

PPI 5.5 **Misperceived/ Misread Instrument** **SUBSTANTIATED**

The Chief Engineer received a correct annunciator alarm but treated it as a fresh event to verify on foot rather than a major release requiring immediate isolation.

JUSTIFICATION

The Chief Engineer was presented with a correct hydrocarbon-detection alarm on the Gas Terminal annunciator at approximately 02:20, but the significance of that instrument presentation was not recognized as a major ongoing release requiring immediate isolation. [EVD-8657] establishes that the alarm was based on the F&G system but that "Hydrocarbon/H2S detectors are configured to Annunciator Panel for display but not in DCS for history" and the "F&G detection system does not have provision for auto shut down of plant or activation of fire suppression system"; the alarm was therefore a genuine detection signal stripped of trend, location specificity, or interlock consequence. [EVD-1460] records the Chief Engineer's response: he heard the alarm, received the CISF phone call, saw the heavy cloud, and then went outside to try to reach the metering area rather than closing MOV 54 or the 363 PCV misinterpretation already substantiated at Level 1 as AJDE 2.4, which found that the Chief Engineer "misinterpreted DCS alarm severity timing, treating delayed alarm as recent event requiring field verification rather than 7-minute-old major

CAUSAL EXPLANATION

The hydocarbon detection at the metering skid was valid at 02:12:50, but because the F&G system only generated a generic annunciator alarm at 02:20—approximately 5 minutes after CISF human detection and 7 minutes 10 seconds after leak onset—the Chief Engineer received a delayed, underspecified instrument signal. He interpreted that signal as a newly developing situation requiring source confirmation rather than as evidence of a major ongoing inventory release, so he and the Junior Engineer attempted an on-foot approach, lost approximately 15 minutes, and only then escalated by telephone. This misread is a direct precondition for the lower-level AJDE 2.7 delayed isolation and AJDE 2.10 inadequate knowledge of isolation criteria: had the annunciator presentation been recognized for what it was, MOV 54 closure could have been initiated from the gate or by pre-authorized remote order well before 03:01, materially reducing the 56-minute vapour accumulation that led to ignition at 03:08:15.

Linked Evidence: EVD-1460, EVD-8657

PPI 5.6 **Inaccurate Expectancy** **SUBSTANTIATED**

Safety auditors accepted a generic checklist and indicated compliance where actual data were incomplete, creating a false perception of plant safety that left the compromised FLP lamp undetected.

JUSTIFICATION

The safety audit function exhibited an inaccurate expectancy that the generic audit checklist was adequate and that recorded compliance equated to real compliance, producing a false perception of plant safety. [EVD-9692] states: "The checklist is comprehensive but not related to specific plants and in the process, lot of observations are missing. The duration of audit is not indicated and in most of the items, compliances are indicated, but during our inspection, the data is found to be incomplete & not in line with OISD standards/manuals." This is direct documentary evidence that the audit instrument generated the appearance of compliance while the underlying data were incomplete, an appearance which the audit team accepted rather than challenging. [EVD-7681] reinforces the false perception: the 2018 and 2019 Internal Safety Audits were conducted by 3-member committees but "the observations are given without the filled checklist", leaving no auditable trail of what was actually inspected. The consequential failure is made concrete by [EVD-6916]: the compromised HPMV flame-proof lamp later identified as the probable ignition source had "three bolts on one of the flame

CAUSAL EXPLANATION

The audit teams' expectation that the generic checklist covered plant-specific risks, and that a recorded 'complied' status reflected reality, prevented them from independently verifying high-risk electrical integrity in the Zone 1 hydrocarbon area. This sustained a false perception of safety into 2020, allowing the HPMV FLP lamp with three missing bolts and a 5–6 cm open glass case to remain in place. When the vapour cloud from the failed meter flange reached that lamp at 03:08:15 on 24-Sep-2020, the exposed contacts provided a credible ignition source, directly linking the audit expectancy to the lower-level ASBE 1.2 audit checklist failure and ultimately to the ignition transition from major release to major fire. The rating is 3 because the direct initiating failure was the gasket joint, while the audit failure was a latent condition that converted the release into a fire.

Linked Evidence: EVD-9692, EVD-7681, EVD-6916

PPI 5.7 **Auditory/ Sound Cues** SUBSTANTIATED

The 02:20 annunciator alarm sounded ~7 minutes after leak onset and ~5 minutes after CISF human detection, so the auditory cue masked how mature and severe the release already was.

JUSTIFICATION

At 02:20 the Chief Engineer heard the Gas Terminal annunciator alarm while simultaneously taking the CCTV Control Room telephone report. [EVD-1460] records: 'At around 0220 hrs Mr. [CE] heard alarm on the annunciator panel and the same time he received a phone call from CISF SI [CCTV CR] from CCTV Control room informing him about some vapour cloud or steam leakage in Gas Terminal PH I area.' The auditory cue was misleading because it disguised the age and maturity of the event. Leak onset is fixed at 02:12:50 by [EVD-1284], 'Commencement of Vapour leakage,' so the alarm sounded approximately 7 minutes 10 seconds after loss of containment and about 5 minutes after first human detection by CISF at 02:15. That latency was invisible in the discrete alarm tone because, as [EVD-8657] records, 'Hydrocarbon/H2S detectors are configured to Annunciator Panel for display but not in DCS for history,' meaning the alarm carried no trend, rate, or duration information. The Chief Engineer treated the alarm as a new/recent event requiring field verification rather than an already-56-minute-accumulating major release, which is the same interpretation

CAUSAL EXPLANATION

The alarm tone was correctly heard as an alarm, but its ~7-minute latency relative to the 02:12:50 leak onset and its ~5-minute lag behind CISF detection were not conveyed by the sound itself. This temporal mismatch shaped the Chief Engineer's interpretation: instead of ordering immediate remote isolation, he went outside to locate the source, and the dense plume then forced him back into telephone escalation. The perceptual precondition therefore fed the AJDE 2.4 decision-making failure and contributed to the 46-minute gap between first detection at 02:15 and MOV 54 closure at 03:01. With the release continuing at operating pressure during that interval, the flammable cloud accumulated until ignition at 03:08:15, 56 minutes after leak onset.

Linked Evidence: EVD-1460, EVD-1284, EVD-8657

PPI 5.9 **Spatial Disorientation (Type-2) Recognised** SUBSTANTIATED

The dense vapour plume stripped visual distance and path cues while the intense smell added sensory confusion, but the engineers recognised the disorientation and safely retreated.

JUSTIFICATION

The two Gas Terminal Control Room engineers experienced recognised perceptual confusion when they walked into the dense vapour plume. [EVD-1460] states the Chief Engineer 'saw the heavy cloud in phase I area' but 'could not approach due to dense cloud,' and [EVD-9220] records the Junior Engineer 'went out of the control room and observed leakage and intense smell from Phase-I area. Visibility was very poor.' This is Type-2 rather than Type-1 disorientation: the sensory conflict was visual and olfactory (opaque plume plus overpowering hydrocarbon smell), the engineers explicitly recognised their inability to judge a safe path to the source, and they retained the ability to make appropriate control inputs by retreating to the control room and initiating telephone escalation. The rapid cloud expansion is independently visible in [EVD-9941], whose frames progress from 'vapour cloud near GT control room' at 02:15:35 to 'vapour cloud intense' at 02:16:13, establishing the environmental condition that produced the sensory conflict. This is a substantiated secondary precondition that underlies the lower-level APE 3.2 finding of tactile/sensory misperception

CAUSAL EXPLANATION

The recognised disorientation prevented the only on-shift operators near the scene from physically locating and assessing the leak. Because they correctly retreated rather than pressing deeper into the plume, the source remained unidentified from the ground, and the response shifted to telephone escalation from inside the control room. This sensory condition directly created the environment for the lower-level APE 3.2 misperception and contributed to the serial escalation that delayed first isolation until MOV 54 closed at 03:01, 46 minutes after detection at 02:15. The inventory released during that delay remained in the area and ignited at 03:08:15.

Linked Evidence: EVD-1460, EVD-9220, EVD-9941

PPE 1.12 **Brownout/Whiteout caused by Human Interference** SUBSTANTIATED

Dense condensate vapour cloud from the leak itself prevented personnel from approaching to locate source for 46 minutes, delaying isolation.

JUSTIFICATION

At 02:12:50 the condensate leak commenced [EVD-1284], and by 02:15:35-02:16:13 the CCTV captured progressive vapour cloud expansion across the Gas Terminal CR, road, and piping area [EVD-9941]. Multiple witnesses confirm that the dense hydrocarbon vapour cloud created by the leak itself prevented personnel from approaching and locating the source. Chief Engineer (Production) and Junior Engineer (Mechanical) both state they 'could not get close' due to cloud density and smell [EVD-1460, EVD-9220]. The GSP Shift In-charge and Overall Shift In-charge 'couldn't enter — the smell was overpowering' when they reached the Gas Terminal Gate at ~02:35 [EVD-9612, EVD-7340]. This human-interference-caused obscuration (the leak's own vapour discharge) restricted vision and physical access to the point where isolation actions were delayed by 46 minutes (02:12:50 leak onset to 03:01 MOV 54 closure [EVD-6414]). The vapour cloud was not a natural environmental condition but a particulate obscuration created by the condensate metering system's loss of containment — fitting the definition of brownout/whiteout caused by human interference. The

CAUSAL EXPLANATION

This human-interference brownout directly caused the 46-minute delay in isolating the leak source. The vapour cloud created by the equipment failure itself became a physical barrier preventing personnel from executing the isolation response. This environmental obscuration created by the leak amplified the consequences of the lower-level unsafe acts: the maintenance team's failure to perform hydrostatic testing [ASBE 1.2 instance 2] created a weak pressure boundary that failed at 02:12:50, but the resulting vapour cloud then prevented timely isolation, allowing 46 additional minutes of inventory release. The cloud accumulated sufficient volume that when ignition occurred at 03:08:15 [EVD-4507] — after isolation actions were complete — the already-released inventory sustained a major fire. The physical environment precondition (vapour obscuration) transformed what could have been a controlled leak into an uncontrolled major release by preventing personnel from approaching the source during the critical first 46 minutes.

Linked Evidence: EVD-1284, EVD-9941, EVD-1460, EVD-9220, EVD-9612, EVD-7340, EVD-6414

PPE 1.13 Atmospheric Conditions

SUBSTANTIATED

Dense hydrocarbon vapour cloud prevented Gas Terminal CR operators from approaching metering skid for 41 minutes, delaying leak-source identification and isolation response.

JUSTIFICATION

Between 02:20 and 03:01 hrs, the Chief Engineer (Production) and Junior Engineer (Mechanical) attempted to exit the Gas Terminal Control Room and physically approach the condensate metering skid to locate the leak source. Both personnel were forced to retreat due to the density and overpowering smell of the hydrocarbon vapour cloud. EVD-1460 records the Chief Engineer's statement: 'The Junior Engineer and I went out together to try and reach the metering area. The cloud was so dense and the smell so strong that we could not get close.' EVD-9220 corroborates: the Junior Engineer 'observed leakage and intense smell from Phase-I area. Visibility was very poor.' This atmospheric condition physically prevented the on-shift operating personnel from performing their emergency response duties—specifically, from conducting visual inspection to locate the leak source and assess the extent of the release. The inability to approach the metering area contributed directly to the 46-minute delay between leak onset (02:12:50) and first isolation action (MOV 54 closure at 03:01). The delay allowed continuous inventory release that built the flammable cloud to

CAUSAL EXPLANATION

The adverse atmospheric conditions (dense hydrocarbon vapour, overpowering smell, poor visibility) created a physical barrier that prevented the Gas Terminal CR operators from performing critical emergency response duties. This atmospheric contamination was a DIRECT ENABLING FACTOR for the 46-minute response delay documented at the lower level (EVD-6414). The operators could not visually locate the leak source, assess its magnitude, or determine which isolation valve(s) to close without telephonic escalation to senior personnel who also could not approach (EVD-9612, EVD-7340). The atmospheric degradation forced reliance on remote decision-making chains, adding latency to the isolation response. By the time MOV 54 was closed at 03:01, the accumulated vapour inventory was sufficient to ignite 7 minutes later when it encountered the compromised HPMV lamp (EVD-6916). The atmospheric condition was itself a consequence of the ongoing release from the failed gasket (primary mechanical failure), but it became a critical precondition that degraded human performance during the emergency response phase.

Linked Evidence: EVD-1460, EVD-9220, EVD-1284, EVD-9941

PPE 1.13 **Atmospheric Conditions** **SUBSTANTIATED**

Overpowering hydrocarbon smell prevented senior shift personnel (DGM/GM level) from entering Gas Terminal area at 02:35 hrs, forcing remote emergency management and delaying isolation.

JUSTIFICATION

At approximately 02:35 hrs, the GSP Shift In-charge (DGM-Production) and the Overall Shift In-charge (GM-Production) arrived at the Gas Terminal Gate in response to the escalation from the Gas Terminal CR operators. EVD-9612 records the GSP Shift In-charge's statement: 'Reached the Gas Terminal Gate with the Overall Shift In-charge. We couldn't enter — the smell was overpowering.' EVD-7340 corroborates that the Overall Shift In-charge was present at the Gas Terminal Gate and directed actions remotely, instructing the Gas Terminal CR to close the 36" pressure control valve rather than approaching the area himself. Both senior personnel, who held the operational authority to direct isolation actions, were physically prevented from entering the plant area to conduct direct assessment. The atmospheric contamination forced them to manage the emergency response remotely, relying on telephonic communication and indirect assessment. The GSP Shift In-charge did not close MOV 54 until 03:01 hrs—26 minutes after arriving at the gate and 46 minutes after leak onset. The atmospheric barrier extended the decision-making chain and prevented direct field verification of

CAUSAL EXPLANATION

The atmospheric contamination at the Gas Terminal Gate prevented the most senior on-shift operational authority (Overall Shift In-charge / GM level) and the designated isolation-execution authority (GSP Shift In-charge / DGM level) from conducting direct field assessment and hands-on emergency response. This atmospheric barrier ENABLED the 46-minute response delay by forcing remote command-and-control from a position where direct verification of leak source, magnitude, and isolation-valve effectiveness was impossible. The senior personnel could not independently verify which isolation valves were the correct primary barriers, nor could they confirm closure effectiveness in real-time. The atmospheric precondition degraded the emergency response's speed and effectiveness, directly contributing to the extended inventory release that created the ignition-capable vapour cloud documented at 03:08:15 (EVD-4507). This instance is distinct from Instance 1 in that it affected DIFFERENT personnel (senior decision-makers vs. on-shift operators) at a DIFFERENT location (Gas Terminal Gate vs. Gas Terminal CR) during a DIFFERENT phase of the

Linked Evidence: EVD-9612, EVD-7340, EVD-6414, EVD-8213

PTE 2.2 **Instrumentation/Sensory Feedback Systems and Safety Features** **SUBSTANTIATED**

F&G detection system lagged human detection by 5 minutes, lacked automated shutdown capability, requiring manual response during conditions where personnel could not approach the leak source.

JUSTIFICATION

The Flame & Gas detection system exhibited critical design inadequacies that directly enabled the 56-minute delay between leak onset and isolation. Evidence [EVD-1284] documents that the leak commenced at 02:12:50 per CCTV, but the system was first detected by human perimeter security at 02:15 [EVD-4286] and the automated annunciator alarm did not activate until 02:20 [EVD-8657] — a 5-minute lag behind human detection. Evidence [EVD-8657] explicitly states: 'Hydrocarbon/H2S detectors are configured to Annunciator system only and they do not directly trigger emergency shutdown or fire protection system.' The detection system's configuration required human interpretation and manual action rather than providing automated safety response. Evidence [EVD-3452] documents that the F&G system layout exists but 'does not have provision for auto shut down of plant or activation of fire suppression system.' This design created a critical dependency on human response speed during a rapidly evolving emergency where personnel could not physically approach the source [EVD-1460]. The instrumentation/sensory feedback design failed to provide timely

CAUSAL EXPLANATION

This instrumentation design inadequacy is a PRIMARY CAUSE in the incident chain. The F&G detection system's failure to provide timely automated awareness and protective action created the environmental precondition that enabled the lower-level checklist failures (ASBE 1.2) to cascade into catastrophe. Specifically: (1) The 18-Sep-2020 gasket installation without hydrostatic testing created a latent pressure-boundary weakness; (2) When that weakness manifested as a leak at 02:12:50, the detection system's 5-minute lag and lack of automated isolation meant personnel had to manually locate and isolate the source; (3) Personnel could not approach due to vapour density, creating a 46-minute delay before MOV 54 closure at 03:01; (4) The accumulated inventory ignited 7 minutes later. Had the F&G system provided automated detection and shutdown capability, the leak would have been isolated within seconds to minutes rather than 46 minutes, preventing the accumulation of the flammable cloud that ignited. The detection-system design transformed a maintenance procedural failure into a major fire incident.

Linked Evidence: EVD-1284, EVD-4286, EVD-8657, EVD-3452, EVD-1460, EVD-4507

PTE 2.2 **Instrumentation/Sensory Feedback Systems and Safety Features** **SUBSTANTIATED**

Single PTZ camera with 90-second pan provided intermittent coverage of the metering area, introducing human-verification delays and eliminating visual feedback when the cable burned 7 seconds after ignition.

JUSTIFICATION

The CCTV monitoring system design created critical situational-awareness gaps during the incident. Evidence [EVD-0210] documents that a single 'PELCO make PTZ cameras (Model D5230D)... has turn around (360 degree) at the time of about 90 seconds. Visibility from this camera is limited.' This design constraint meant the metering skid was only visible during a fraction of the 90-second pan cycle. Evidence [EVD-1284] shows the leak was machine-witnessed at 02:12:50, but the PTZ camera's limited-coverage design meant CISF Sub-Inspector C in the CCTV Control Room could not continuously monitor the leak evolution — he required a phone call from the perimeter guard at 02:15 [EVD-8871] before he could visually verify 'heavy leakage' at 02:18. The single-camera design introduced a 5-minute human-in-the-loop verification delay. Evidence [EVD-2752] documents that the CCTV cable burned through at 03:08:22, terminating recording 7 seconds after ignition and eliminating real-time visual feedback for the fire response. The committee's recommendation (p.36) explicitly addresses this deficiency: 'CCTV arrangement shall be provided for each

CAUSAL EXPLANATION

This monitoring-system design inadequacy is a SECONDARY contributing factor. The single-PTZ-camera design created a situational-awareness gap that delayed human detection verification and response coordination. While human perimeter detection occurred at 02:15, the CCTV operator's inability to continuously monitor the leak meant the CISF-internal verification step introduced additional latency before notifying the Gas Terminal Control Room at 02:18. Had multiple fixed cameras with continuous coverage and dedicated plant-CR displays existed (per the committee's recommendation), the operating team would have had real-time visual awareness of the leak evolution independent of CISF notification chains. The early cable burnout at 03:08:22 eliminated visual feedback precisely when responders needed it most — during active fire suppression. The monitoring-system design created preconditions for slower response and degraded situational awareness during the critical first 8 minutes.

Linked Evidence: EVD-0210, EVD-1284, EVD-8871, EVD-2752

PTE 2.2 **Instrumentation/Sensory Feedback Systems and Safety Features** **SUBSTANTIATED**

DCS failed to retain pressure data from three transmitters (310PT4/5/6) at the leak source, preventing forensic pressure-profile reconstruction and likely masking pre-incident anomalies.

JUSTIFICATION

The DCS instrumentation system exhibited a critical data-retention failure that prevented post-incident forensic analysis and likely masked pre-incident pressure anomalies. Evidence [EVD-6470] documents that three pressure transmitters (310PT4, 310PT5, 310PT6) are shown in the P&ID but 'no pressure transmitter data are available in the system. No pressure data in Gas Terminal area is available/provided to check rise in pressure in the metering lines.' This instrumentation design failure has two causal implications: (1) Investigators could not reconstruct the pressure profile leading to the gasket failure, preventing determination of whether the leak was due to overpressure, cyclic fatigue, or installation defect; (2) If pressure data had been retained and alarmed, gradual seal degradation might have been detected before catastrophic failure. The DCS configuration treats critical pressure-boundary instrumentation as unmonitored telemetry rather than safety-critical data requiring retention and alarming. This is compounded by evidence [EVD-5532] showing 2-12 minute time-stamp drift across control rooms, further degrading the instrumentation system's

CAUSAL EXPLANATION

This instrumentation deficiency is a SECONDARY contributing factor. The absence of pressure-data retention prevented detection of seal degradation that may have preceded the 24-Sep-2020 catastrophic failure. If the DCS had been configured to retain and alarm on pressure trends from 310PT4/5/6, the maintenance team might have detected anomalous pressure behavior after the 18-Sep-2020 gasket replacement, triggering investigation before the seal failed. The data gap also prevented post-incident forensic analysis to determine whether the gasket failure was caused by overpressure (indicating a process design problem), installation error (indicating a supervision problem per the existing ASBE 1.2 findings), or material defect (indicating a procurement problem per EVD-8645). The instrumentation design created an observability gap that both masked pre-incident warning signs and prevented definitive root-cause determination post-incident.

Linked Evidence: EVD-6470, EVD-5532

PSE 3.1 **Situation/Tasking - Unusual/New or Not Trained for** **SUBSTANTIATED**

Operating personnel faced untrained scenario of isolating major leak while physically blocked by dense vapour cloud, with no procedure for remote/upstream isolation decision-making.

JUSTIFICATION

The night-shift operating personnel faced an unfamiliar emergency scenario for which standard response procedures proved inadequate. Evidence shows that when the leak was detected at 02:15, both the Chief Engineer (Production) and Junior Engineer (Mechanical) 'went out together to try and reach the metering area. The cloud was so dense and the smell so strong that we could not get close' [EVD-1460]. The GSP Shift In-charge and Overall Shift In-charge similarly 'couldn't enter—the smell was overpowering' when they arrived at 02:35 [EVD-9612, EVD-7340]. The on-site Disaster Management Plan 'does not include isolation/stoppage of facilities/pipelines' as decision criteria for hydrocarbon leakage scenarios [EVD-4052]. The 46-minute delay from leak onset (02:12:50) to first isolation (03:01, MOV 54 closure) represents a response to a situation—massive vapour cloud physically blocking access to isolation points—that was not trained for and not proceduralized. The committee notes 'decision on isolation/closure of upstream facilities/pipelines during such emergencies should be evaluated and communicated to all concerned as a part of emergency

CAUSAL EXPLANATION

This situational unfamiliarity created the preconditions for the 46-minute response delay documented in the timeline [EVD-6414]. The operating personnel had no trained procedure for making upstream isolation decisions when field access to the local isolation point (metering skid) was blocked. They escalated through the authority chain (CR -> Shift In-charge -> SIC) seeking authorization for a non-proceduralized action (closing MOV 54 remotely), consuming critical time. This delay allowed the vapour cloud documented in [EVD-9941, EVD-8213] to accumulate sufficiently to reach the compromised HPMV lamp and ignite at 03:08:15, 4 minutes after isolation was finally achieved.

Linked Evidence: EVD-4052, EVD-1460, EVD-9612, EVD-6414

PSE 3.3 **Time Constraint** **SUBSTANTIATED**

Operating personnel faced time-critical emergency requiring rapid isolation decision within minutes, but 46-minute response delay allowed flammable cloud to accumulate and ignite.

JUSTIFICATION

The operating personnel faced severe time constraint once the leak was detected at 02:15. The CCTV sequence shows the vapour cloud expanding rapidly: from first visible vapour at 02:12:50 [EVD-1284] to 'vapour cloud intense' at 02:16:13 [EVD-9941] (expansion across metering area, road, and GT CR within ~3.5 minutes), to 'vapour all around even at height' at 03:00:44 [EVD-8213] (vertical column formation). The flammable envelope was growing continuously while personnel attempted to locate and isolate the source. Evidence shows multiple personnel attempting field response but being forced back by density [EVD-1460, EVD-9612]. The time window to prevent ignition was closing rapidly—every minute of delay increased the vapour inventory and probability of finding an ignition source. The GSP Shift In-charge closed MOV 54 at 03:01 [EVD-9612], but ignition occurred 7 minutes later at 03:08:15 despite isolation being complete [EVD-4507]. The 46-minute leak-to-isolation interval [EVD-6414] was far too long given the rate of vapour accumulation. The emergency nature of the situation mandated rapid isolation decision-making, but the authority chain and lack of

CAUSAL EXPLANATION

This time constraint is a PRIMARY causal factor in the incident outcome. The emergency required isolation within minutes to prevent flammable-cloud accumulation, but the actual response consumed 46 minutes. This delay was causally driven by the situational unfamiliarity (PSE 3.1 instance 2) and the lack of pre-defined isolation criteria in the DMP [EVD-4052]. By the time MOV 54 was closed at 03:01, the vapour cloud documented in [EVD-8213] had already accumulated sufficient inventory to find the integrity-compromised HPMV lamp [EVD-6916] and ignite at 03:08:15. The time constraint transformed what could have been a contained release into a major fire. It directly created the preconditions for both Level-1 ASBE 1.2 findings: the calibration checklist failure (which created the equipment vulnerability) and the hydrostatic-test omission (which installed an inadequate gasket) combined with this time-delay failure to produce the catastrophic outcome.

Linked Evidence: EVD-6414, EVD-9941, EVD-8213, EVD-4507, EVD-4052

PTF 1.5 Team Roles/ Responsibilities not Allocated or Understood

SUBSTANTIATED

No one was assigned to verify gasket specification, quality, or post-installation integrity, leaving quality control to individual discretion.

JUSTIFICATION

Within the maintenance team, no individual was assigned specific responsibility for ensuring that replacement gaskets met OEM specifications or that quality control checks were performed. [EVD-8645] states: 'Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs… There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over.' Additionally, [EVD-2894] records that no hydro test was done and no SOP existed for return to service, indicating that the role of pressure-boundary validator was also not designated. The calibration record ([EVD-2506]) shows major setpoint changes without checklist or sign-off, further evidence that no reviewer role existed. This vacuum of defined responsibilities meant that every critical check in the maintenance chain was an optional step, and collectively they were omitted, allowing a substandard gasket to be installed without detection.

CAUSAL EXPLANATION

Because no one was tasked with verifying material specifications, the locally-procured O-ring was accepted without challenge. Because no one was assigned to conduct or witness a hydrostatic test, it was skipped. Because no one was responsible for a technical review at job completion, the defective work was passed. This role allocation failure turned the maintenance process into a sequence of individual decisions with no safety net, directly enabling the defective gasket that became the primary leak source. This is the organizational root of the lower-level unsafe acts (ASBE 1.3, AJDE 2.5) where procedures were not followed because no one was assigned to enforce them.

Linked Evidence: EVD-8645, EVD-2894, EVD-2506

PTF 1.4 Over Reliance on Individual/ System or Equipment

SUBSTANTIATED

MOV 54 and other isolation valves required manual on-site closure, with no remote operation from the control room, causing a 46-minute isolation delay.

JUSTIFICATION

The isolation valves in the Gas Terminal area, including MOV 54, were designed for local manual operation only, with no remote actuation capability from the control room. [EVD-9612] narrates that DGM (GSP Shift In-charge) had to physically reach the Gas Terminal Gate and then 'closed the MOV 54 inlet condensate line from slug catcher to 36 metering area at about 0301 hrs.' The committee finding (p.26) further clarifies: 'MOVs are provided in the gas terminal station for local operations but they are not having remote operation from control room during emergency.' This design meant that the fastest possible isolation required a person to travel through the hazardous vapour zone, adding critical minutes. The over-reliance on local manual operation created a physical and temporal barrier to prompt isolation, directly contributing to the 46-minute delay from leak detection to first valve closure.

CAUSAL EXPLANATION

At 02:18, when the CCTV operator confirmed heavy leakage, a remote-operated valve could have been closed from the control room within minutes, containing the leak before 02:20. Instead, because MOV 54 was not remote-capable, the Shift In-charge had to travel to the site and operate it manually, taking until 03:01. This 43-minute gap (from first possible closure to actual closure) allowed massive inventory release, fueling the flammable cloud that ignited at 03:08:15. The manual-only configuration was a primary precondition for the prolonged release and subsequent fire.

Linked Evidence: EVD-9612, EVD-3932

PTF 1.6 Rank/ Position Authority Gradient

SUBSTANTIATED

Operators did not isolate remotely due to a steep authority gradient that required senior management approval before critical actions, adding 46 minutes.

JUSTIFICATION

A steep authority gradient between the control room operators and senior management inhibited swift isolation. [EVD-4052] shows that the DMP omitted isolation decision criteria, so authority to order pipeline isolation effectively resided with the General Manager (Overall Shift In-charge) or above. When the alarm sounded at 02:20, the Chief Engineer and Junior Engineer did not take any remote isolation action; instead they phoned the GSP Shift In-charge, LPG Shift In-charge, and CFU Panel Engineer ([EVD-1460]). Even the GSP Shift In-charge (DGM) only closed MOV 54 after the Overall Shift In-charge (GM) had arrived and assessed the situation; the GM then instructed the closure of the 36" PCV ([EVD-7340]). The 46-minute response delay reflects a culture where junior personnel did not feel empowered to execute time-critical containment actions without senior approval, a direct consequence of the steep authority gradient.

CAUSAL EXPLANATION

In an acceptable response, the Chief Engineer would have closed the 36" PCV immediately upon seeing the vapour cloud at 02:20 and would have directed the MOV 54 closure if remote operation existed. The fact that he instead escalated by telephone suggests the authority to isolate resided exclusively at the Shift In-charge level or higher. This gradient delayed the PCV closure by 44 minutes (02:20 to 03:04) and the MOV closure by 41 minutes. The massive vapour cloud that formed during this period directly fed the ignition, converting a manageable leak into a major fire. The steep authority gradient thus suppressed time-critical action at the lowest competent level.

Linked Evidence: EVD-4052, EVD-1460, EVD-7340

Level 3: Supervision

SVI 1.1 **Leadership/Supervisory Oversight Inadequate** SUBSTANTIATED

SAP notification 3800419840 for meter 310FT01 was never closed or reviewed by a senior officer, removing a supervisory quality gate before return to service.

JUSTIFICATION

No supervisor reviewed and closed the very notification that triggered the fatal repair. [EVD-0591] records that 'Notification no. 3800419840 was raised on 17/09/2020 for malfunction of meter no.1' (p.27) and that it 'was not closed after completion of the job' (p.27). The committee's recommendation D.2.ii - 'Sample checks/approval of closure of notifications by Senior Officers to be introduced' (p.35, per EVD-0185) - is an explicit finding that senior-officer oversight of notification closure was absent. A closure review would have obliged a supervisor to examine the job package, where the substitute O-ring, the missing hydrotest and the third gasket failure in 22 months were all visible in one place. The unclosed notification removed the last administrative gate between defective maintenance work and live hydrocarbon service.

CAUSAL EXPLANATION

Because the notification stayed open with no senior closure check, the 18-Sep-2020 non-OEM gasket and the omitted design-pressure test passed into service unexamined on 19-Sep, and the meter failed on 24-Sep at 02:12:50, leading to ignition at 03:08:15. This oversight directly enabled the lower-level AJDE 2.11 misapplied-SAP-closure-policy finding (rating 3), the AV 1.1 routine non-closure violation (rating 2) and the PTF 1.12 feedback/acknowledgement communication failure.

Linked Evidence: EVD-0591, EVD-0185

SVI 1.1 **Leadership/Supervisory Oversight Inadequate** SUBSTANTIATED

A 16000-to-2540 mmwc calibration setpoint change on meter 310FT01 in Dec-2019 was signed off unrecorded, with no checklist or supervisory challenge.

JUSTIFICATION

Supervision over critical calibration documentation was inadequate nearly ten months before the fire. [EVD-2506] records: 'During calibration from 17/12/2019 to 18/12/2019, calibration was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of this change in the calibration documents signed by the team. During next calibration for meter 310FT01 was again changed to 16000 MMWC. No checklist is followed while carrying such critical jobs of high pressure/risk' (p.27). A setpoint cut of approximately 6.3x (16000 to 2540 mmwc) passed through documents signed by the team with no recording, challenge or approval, and the next cycle silently restored 16000 mmwc. The committee recommendation D.1.i (p.35) - that 'details of Master calibrator/action taken/replacement/deviation from procedure if any, is to be documented in the calibration certificate' - confirms the supervisory gap. This normalised undocumented changes to high-pressure instrumentation on the same meter that later became the leak source.

CAUSAL EXPLANATION

The supervisory tolerance of an unsigned, unexplained 6.3x setpoint change established the precedent that critical meter work needed no documented verification - the same governance hole through which the 18-Sep-2020 substitute gasket and skipped hydrotest later passed. It enabled the lower-level ASBE 1.2 calibration checklist-not-followed finding (rating 3), the ASBE 1.5 under-controlled-system finding (rating 3), and the AJDE 2.10 inadequate calibration knowledge finding (rating 3), contributing to the weakened integrity environment on meter 310FT01 that culminated in the 24-Sep-2020 failure.

Linked Evidence: EVD-2506

SVI 1.1 **Leadership/Supervisory Oversight Inadequate** **SUBSTANTIATED**

The shift in-charges at the gate from 02:35 did not order comprehensive isolation for 26 minutes; MOV 54 closed at 03:01, 46 minutes after first detection.

JUSTIFICATION

The most senior on-shift leaders failed to drive immediate isolation once they reached the scene. [EVD-9612] shows the GSP Shift In-charge arrived at ~02:35, could not enter due to gas smell, yet closed MOV 54 only 'at about 0301 hrs. Thereafter fire occurred' (p.17) - 26 minutes spent at the gate before the single most important isolation action. [EVD-7340] shows the Overall Shift In-charge's only control-room instruction was to 'close the 36 line pressure control valve and get out of the control room', with no pre-authorised comprehensive isolation sequence. The committee's assessment - 'Decision of closure of valve was taken after about 40 minutes resulting the spread of vapour cloud in increased area' (p.32, [EVD-6414]) - is a direct supervisory-oversight conclusion. The result: MOV 54 closed at 03:01, 46 minutes after first detection at 02:15 and 48 minutes after leak onset at 02:12:50, and the 36-inch PCV only at 03:04 - seven and four minutes before ignition respectively. This is the Level-3 cause of the lower-level PTF 1.1 leadership and PTF 1.15 re-planning failures.

CAUSAL EXPLANATION

Between 02:35 and 03:01 the two senior officers serialised notifications instead of executing every available isolation immediately, so the slug-catcher inventory kept feeding the failed meter for 26 additional minutes. The cloud documented at 03:00:44 as 'vapour all around even at height' [EVD-8213] was the direct product of that delay, and it ignited at 03:08:15 - after both valves were finally shut - proving the already-released inventory was sufficient fuel. This Level-3 oversight directly enabled the lower-level AJDE 2.7 delayed-action finding (rating 4), the PTF 1.1 crew-leadership failure (rating 4) and the PTF 1.15 failure-to-re-plan finding (rating 4).

Linked Evidence: EVD-9612, EVD-7340, EVD-6414

SVI 1.1 **Leadership/Supervisory Oversight Inadequate** **SUBSTANTIATED**

The 2018 (4-day) and 2019 (3-day) internal safety audits produced observations without filled checklists, leaving audit completeness unverifiable.

JUSTIFICATION

A separate programme - the 2018 and 2019 Internal Safety Audits - was also run without auditable oversight of what was actually inspected. [EVD-7681] records: 'Internal Safety Audit was conducted in 2018 & 2019 for 4 & 3 days respectively by a 3-member committee and the observations are given without the filled checklist. The checklist format is not available to check completeness of audit areas' (p.26). This is distinct from the self-audit instance: it is a different, three-member committee audit in a different time window, and its deficiency is the absence of any filled checklist trail, so the completeness of the audited areas cannot be verified after the fact. Supervision accepted observations with no evidence of what was actually examined versus skipped, which is how the Zone-1 electrical integrity and gasket-practice defects could be missed without any paper record of the miss.

CAUSAL EXPLANATION

With no checklist trail, supervisors and later managers could not verify whether the metering skid's pressure boundary work or the HPMV lamp enclosures had ever been within audit scope, so latent defects persisted unchallenged into 2020. This gap fed the same lower-level ASBE 1.2 audit-checklist failure (rating 3) and contributed to the compromised flame-proof lamp remaining in place to ignite the accumulated cloud at 03:08:15 on 24-Sep-2020.

Linked Evidence: EVD-7681

SVI 1.1 **Leadership/Supervisory Oversight Inadequate** **SUBSTANTIATED**

A Zone-1 HPMV flame-proof lamp with three missing bolts and a 5-6 cm open glass case remained in service under a maintenance regime skewed to breakdown work.

JUSTIFICATION

Electrical maintenance supervision failed to keep a Zone-1 flame-proof fitting intact, and the committee documented a systemic skew away from preventive maintenance. [EVD-6916] records the probable ignition source: 'three bolts on one of the flame proof lamps were missing resulting in it being open with its integrity compromised, glass case got detached from the main body and had opened out about 5 to 6 cms. The nearby fittings were found intact' (p.32). [EVD-0496] adds: 'the electrical maintenance system is a lot more skewed towards breakdown maintenance and lays little emphasis on systematic preventive maintenance and inspection' (p.28), and the post-incident audit was itself 'too generic and short (One page)' with all nine 125 W HPMV FLP lamps clubbed under the single status 'cable burnt and fitting damaged due to heat' ([EVD-9999], p.28) - the supervising organisation did not even characterise per-equipment failure modes after the event. A flame-proof enclosure with three missing bolts in a hydrocarbon Zone 1 area is a supervisory inspection failure of the first order.

CAUSAL EXPLANATION

Because no supervisor enforced flame-proof integrity inspections in the metering blast radius, the lamp retained exposed contacts through the entire release. When the vapour cloud accumulated over 56 minutes from 02:12:50, it found this ready ignition source at 03:08:15 - four minutes after the last isolation action - and the release became a major fire. This oversight enabled the lower-level PTE 2.12 defective-equipment findings (ratings 2-4) and the ASBE 1.2 audit-checklist failure (rating 3) that left the ignition source in place.

Linked Evidence: EVD-6916, EVD-0496

SVI 1.1 **Leadership/Supervisory Oversight Inadequate** **SUBSTANTIATED**

2017 thickness-loss findings on gas-terminal pipelines had no corrective action plan 3+ years later, and the 4-year inspection schedule itself was breached.

JUSTIFICATION

Supervision over integrity-action tracking allowed a 2017 engineering warning to sit for over three years with no corrective plan. [EVD-5527] records: 'As per the report of 2017, several pipes have been identified with significant thickness loss and need for corrective action. However, even after three years no plan has been developed for the corrective actions to be taken for those lines. Mere measurement of thickness without corrective action does not complete the process of maintaining the integrity of pipes' (p.30). The same oversight environment breached the four-year inspection cycle itself: [EVD-8307] shows the mandated four-year schedule was performed in 2011 and 2017 - a six-year actual gap - and was due again in 2021, not yet done by the 24-Sep-2020 incident. No supervisor maintained an action register with owners and due dates, so measurement data never converted into repair work.

CAUSAL EXPLANATION

While the 36-inch trunk pipeline itself remained intact on 24-Sep-2020 [EVD-5780], this oversight is the clearest organisational evidence of the pattern that also governed the metering skid: findings acknowledged but not actioned. It substantiates the lower-level AJDE 2.5 ignored-warning finding (rating 3) and the AJDE 2.7 three-year delay finding (rating 3), and explains why the maintenance organisation accepted chronic defects without closure pressure.

Linked Evidence: EVD-5527, EVD-8307

SVI 1.1 **Leadership/Supervisory Oversight Inadequate** **SUBSTANTIATED**

The MOC SOP published Aug-2016 was not implemented at all since 2018, so the OEM-to-O-ring gasket change was never engineering-reviewed.

JUSTIFICATION

Supervisory oversight of engineering change control was effectively suspended plant-wide for two years. [EVD-2986] records: 'Management of Change (MOC) philosophy/SOP was published in August 2016. However, the procedure indicated in the document is not implemented while carrying out change of Low-Pressure Gas to Kribhico from Medium Pressure Gas headers in LPG Plant in 2013 & 2018. It is informed no more MOCs are implemented in the plant since 2018' (p.30). The gasket substitution is the clearest demonstration: [EVD-8645] notes 'there is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket' (p.31) - a material change to a pressure boundary with no MOC review whatsoever. Supervisors who should have demanded an MOC for any change in gasket specification instead allowed the substitution to become 'a regular practice', removing the engineering review gate that would have challenged the O-ring geometry.

CAUSAL EXPLANATION

The dormant MOC process removed the only formal gate that could have blocked the O-ring substitution on geometric and material grounds, allowing it to be repeated across three replacements from 12-Nov-2018 to 18-Sep-2020 and to fail catastrophically on 24-Sep-2020 at 02:12:50. This Level-3 oversight enabled the lower-level AJDE 2.11 MOC-policy-misapplication finding (rating 3) and the AV 1.1 routine substitution violation (rating 4), which together created the initiating pressure-boundary defect.

Linked Evidence: EVD-2986, EVD-8645

SVI 1.1 **Leadership/Supervisory Oversight Inadequate** **SUBSTANTIATED**

The metering-system P&ID 02-310-AO-101 Rev.1 was a lost-original, unapproved vendor drawing that mismatched the field configuration.

JUSTIFICATION

Supervision over engineering document control failed to keep a controlled as-built record of the metering system. [EVD-0933] records: 'Reference drawings no. 02-310-AO-101 Rev.1, 2x6 lines & 1x2 lines for metering system are shown but valves & meters configuration are not matching. As old drawings were lost, new drawings are made by a vendor but the same has not been checked & approved by ONGC' (p.26). Three governance failures in one finding: the 1986 originals were lost, the vendor-produced replacement was never checked or approved by the operator, and the field installation did not match even the unapproved drawing. This meant maintenance staff had no authoritative configuration baseline to verify the correct gasket design for the rectangular-slot flange, supporting the substitution error, though the physical failure was driven by the incorrect seal and missing hydrotest rather than the drawing mismatch alone, hence rating 2.

CAUSAL EXPLANATION

The missing controlled drawing compounded the lost OEM gasket specification [EVD-8645], depriving both maintainers and supervisors of the reference needed to recognise the O-ring mismatch before it was installed on 18-Sep-2020. It supports the lower-level AV 1.1 configuration-management breakdown finding (rating 2) but did not directly initiate the 24-Sep-2020 loss of containment.

Linked Evidence: EVD-0933

SVI 1.2 **Improper Role-Modelling** **SUBSTANTIATED**

Supervisors tolerated the locally sourced O-ring for the rectangular-slot flange across three repairs, modelling it as accepted practice to the maintenance crew.

JUSTIFICATION

The unsafe gasket practice was not a single improvised act but a behaviour learned and reinforced through supervisor tolerance. [EVD-8645] records verbatim that 'use of locally purchased O ring was done in its place as a regular practice' (p.31) - the substitute was the established, repeated method, and [EVD-1494] documents its repetition across three gasket replacements on the same meter (12-Nov-2018, 25-Feb-2019, 18-Sep-2020). No supervisor corrected, documented, or escalated the practice at any of those three opportunities, which is itself the modelling act: juniors in the maintenance crew observed the locally sourced O-ring being fitted, accepted and signed off repeatedly, and learned that procuring and installing a non-conforming seal for a rectangular-slot flange was the sanctioned way to do the job. When the same behaviour materialised a fourth time on 18-Sep-2020, the seal was geometrically unable to fill the slot and failed within six days.

CAUSAL EXPLANATION

The modelled substitute produced an inadequate seal that was installed on 18-Sep-2020, ran from 19-Sep, and failed at 02:12:50 on 24-Sep-2020, starting the 56-minute release that ignited at 03:08:15. This Level-3 modelling failure is the mechanism behind the lower-level AV 1.1 routine/widespread violation (rating 4) and the AJDE 2.8 wrong choice of action (rating 4) - an individual error would likely have been corrected after one or two failures, whereas the modelled norm persisted through four.

Linked Evidence: EVD-8645, EVD-1494

SVI 1.2 **Improper Role-Modelling** **SUBSTANTIATED**

The calibration team collectively signed off an unrecorded 16000-to-2540 mmwc setpoint change with no checklist - a shared, modelled bypass.

JUSTIFICATION

The calibration team's collective behaviour demonstrates checklist bypass that was shared and accepted across the group. [EVD-2506] records that the 16000 mmwc-to-2540 mmwc change in Dec-2019 was made 'without any mention of this change in the calibration documents signed by the team' and that 'No checklist is followed while carrying such critical jobs of high pressure/risk' (p.27). Multiple individuals signed documents whose content omitted a ~6.3x setpoint deviation, and no one in the team recorded or challenged it - team-wide participation of this kind is exactly the signature of a behaviour learned from peers and accepted by the supervising instrument engineer. The practice likely propagated from senior technicians who had always worked this way, and the crew absorbed it as the standard method for high-pressure calibration work on meter 310FT01.

CAUSAL EXPLANATION

The modelled checklist bypass on calibration work normalised undocumented modifications to critical metering equipment on the exact meter that later failed, eroding the organisational expectation that deviations must be recorded and approved. It directly enabled the lower-level ASBE 1.2 calibration checklist finding (rating 3), the ASBE 1.5 under-controlled system finding (rating 3), and the AJDE 2.10 calibration knowledge finding (rating 3), contributing to the weak integrity management around meter 310FT01 that culminated in the 24-Sep-2020 failure at 02:12:50.

Linked Evidence: EVD-2506

SVI 1.3 Local Training/ Training Programme Issues

SUBSTANTIATED

No qualification for control-room operators was defined in HR/Operations Policy, and most staff had attended only one basic course since 2018.

JUSTIFICATION

Separate from the course content, the training-programme governance itself had no defined competency threshold. [EVD-5634] records: 'Qualification of this staff is also not defined in HR/Operations Policy. Most of the staff have attended only One Basic Course since 2018' (p.30). There was no formal qualification that an operator had to attain before being assigned control-room duties, and most staff had completed only a single basic course since 2018 - so the plant had no mechanism to certify that the people facing the 02:20 alarm on 24-Sep-2020 were competent to handle it. This is a distinct manifestation from the four-day course content: even if the course were adequate, without a defined qualification the organisation could not verify who had absorbed it, and competency gaps went institutionally undetected.

CAUSAL EXPLANATION

With no defined qualification gate, night-shift staffing was not competency-verified, so the crew that faced the release at 02:20 had no certified capability for emergency isolation, feeding the 46-minute delay and the 03:08:15 ignition. This governance gap underlies the lower-level AJDE 2.10 and AJDE 2.11 findings where operators lacked the knowledge and policy awareness to act.

Linked Evidence: EVD-5634

SVI 1.3 Local Training/ Training Programme Issues

SUBSTANTIATED

Monthly fire drill records carried no actions or observations, making drill effectiveness unmeasurable, though the fire team performed well on the night.

JUSTIFICATION

The fire drill training programme was administered in a way that made its effectiveness unmeasurable. [EVD-2130] records: 'Fire drill reports are available in SAP but without any data on actions taken/recommendations. Separate data is maintained which is not in line with the approved guidelines' (p.26). Monthly drills were recorded only as dates in SAP, while the substantive actions and observations lived in a parallel, non-guideline-compliant dataset - so no one could audit what the drills actually exercised or what corrective actions they produced. This instance had no direct impact on the 24-Sep-2020 outcome because the site fire team performed competently: cooling began at ~03:50, the fire was extinguished by ~06:00, and firewater tank level and header pressure stayed healthy throughout [EVD-2432, EVD-6414]. It is therefore rated present with no outcome impact, but it is a real training-programme administration deficiency that also prevented the drill system from correcting any modelled emergency-response behaviour (see SVI 1.2 instance 4).

CAUSAL EXPLANATION

Because drill records contained no actions or observations, any deficiencies rehearsed in drills - potentially including the escalate-first response pattern observed on 24-Sep-2020 - could not be identified or corrected through the training system. The gap did not change the firefighting outcome (fire extinguished by ~06:00, all-clear 07:30), so its causal role is limited to lost organisational learning capacity rather than a direct contribution to the event.

Linked Evidence: EVD-2130

SVI 1.1 **Leadership/Supervisory Oversight Inadequate** **SUBSTANTIATED**

No supervisor enforced the OEM rectangular-slot gasket specification, purchase QC, or design-pressure hydrotest before meter 310FT01 returned to service on 19-Sep-2020.

JUSTIFICATION

Supervisory oversight was inadequate for the single highest-risk maintenance task in the incident chain. [EVD-8645] records that 'Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs' and that 'there was no specification or quality control during the purchase of the O rings', with 'no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket' (p.31) - a supervisor should have halted a pressure-boundary job with no controlled part specification. [EVD-2894] then records that after the 18.09.2020 replacement 'No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated' (p.31) - the supervisor accepted the repair validated only at operating pressure, not design pressure. The official committee conclusion - 'Probable cause of leak is due to use of nonstandard gasket and not testing the system to designed pressure' (p.32) - and committee root causes #2 'Use of nonstandard O ring' and #3 'Lack of control over material sourcing' (p.33) identify this exact oversight gap. This instance

CAUSAL EXPLANATION

Because no supervisor challenged the unspecified O-ring or enforced the design-pressure test, meter 310FT01 returned to service on 19-Sep-2020 with an unvalidated seal in a flange machined for a rectangular-slot gasket. The joint operated for 6 days (19-24 Sep) and failed at 02:12:50 on 24-Sep-2020 [EVD-1284], releasing vapour that accumulated for 56 minutes and ignited at 03:08:15 [EVD-4507]. This Level-3 oversight is the enabling cause of the lower-level wrong choice of substitute gasket (AJDE 2.8, rating 4), the routine substitution violation (AV 1.1, rating 4), and the omitted hydrostatic-test checklist findings (ASBE 1.2, rating 4) that initiated the loss of containment.

Linked Evidence: EVD-8645, EVD-2894

SVI 1.1 **Leadership/Supervisory Oversight Inadequate** **SUBSTANTIATED**

Three gasket replacements in 22 months (12-Nov-2018, 25-Feb-2019, 18-Sep-2020) were never escalated by maintenance leadership for root-cause investigation.

JUSTIFICATION

Maintenance leadership failed to treat a recurring failure signal on the same asset as a reliability concern. [EVD-1494] records three gasket replacements in approximately 22 months on the 34-year-old meter: 'Cleaning/servicing/calibration of meters 1 & 2 was carried out on 12th & 13th Nov 2018. Afterwards, meter no.1 was cleaned on 25/02/2019 & gasket was changed. Again on 18/09/2019, gasket was changed which leaked within one year' (p.27) - a repetition that should have triggered root-cause analysis, OEM spec recovery, or meter redesign. The initial incident report itself flags this oversight question: 'Investigation should determine whether the recurrence pattern was treated as a reliability concern.' No documented engineering intervention followed the first, second, or third failure, and the same locally sourced O-ring continued in use each time, proving the supervisor's response to a chronic defect was repetition rather than escalation. This is the Level-3 counterpart of the Level-1 AJDE 2.5 ignored-caution/warning finding (rating 4).

CAUSAL EXPLANATION

Had the third failure (18-Sep-2020) triggered the engineering review that the first two (12-Nov-2018, 25-Feb-2019) should have, the marginal O-ring substitute would have been rejected and the meter redesigned or replaced before it could fail a fourth time. Instead the same inadequate seal was installed, the meter ran from 19-Sep to 24-Sep, and the joint failed at 02:12:50, producing the 56-minute vapour accumulation that ignited at 03:08:15. This oversight enabled the lower-level AJDE 2.5 ignored-warning finding and the PTE 2.12 recurring-maintenance-defect precondition.

Linked Evidence: EVD-1494

SVI 1.1 **Leadership/Supervisory Oversight Inadequate** **SUBSTANTIATED**

The self-safety audit used a generic, non-plant-specific checklist and accepted compliance claims that site data contradicted.

JUSTIFICATION

Audit oversight accepted a non-plant-specific instrument and unverified compliance claims. [EVD-9692] records: 'The checklist is comprehensive but not related to specific plants and in the process, lot of observations are missing... in most of the items, compliances are indicated, but during our inspection, the data is found to be incomplete & not in line with OISD standards/manuals. Approved manuals/SOPs were not available' (p.26). The committee's recommendation B.1 - 'Checklist for Self-Safety Audit for individual plants are to be developed' (p.34) - confirms the audit instrument itself was not fit for purpose. Because the generic checklist never forced inspection of the Gas Terminal metering area's specifics, the compromised Zone-1 HPMV flame-proof lamp with three missing bolts and a 5-6 cm open glass case ([EVD-6916]) and the chronic O-ring gasket practice both survived audit undetected, and supervisors accepted 'complied' statuses without verifying the underlying data.

CAUSAL EXPLANATION

The generic audit instrument meant the plant-specific latent defects - the non-OEM gasket practice and the flame-proof lamp integrity loss - were never sampled or challenged, so both survived to participate in the 24-Sep-2020 event. The missing-bolts lamp became the probable ignition source at 03:08:15 after 56 minutes of vapour accumulation. This oversight enabled the lower-level ASBE 1.2 audit-checklist failure (rating 3) and the PPI 5.6 inaccurate-expectancy finding (rating 3), converting a release into a major fire.

Linked Evidence: EVD-9692

SVI 1.1 **Leadership/Supervisory Oversight Inadequate** **SUBSTANTIATED**

Supervision accepted painting alone for 1.03 mm pitted thickness loss on a 23.83 mm zero-corrosion-allowance 36-inch line in the gas terminal.

JUSTIFICATION

A second, distinct integrity-oversight instance is the July-2019 36-inch segment decision. [EVD-2397] records: 'a reduction of thickness of 1.03mm along with heavy pitting and painting peeling off was observed in one of the segments of 36 pipeline in gas terminal in July 2019. The corrective action of painting was completed in Dec. 2019. However, as per the available data, the design thickness of the pipeline is 23.83mm with zero corrosion allowance. It needs to be examined whether painting as a corrective action would suffice in such cases and the mechanism of deciding on a corrective action needs to be reviewed' (p.30). Supervision accepted a coating-restoration action for a structural metal-loss condition on a zero-corrosion-allowance pipe without fitness-for-service evaluation - a decision that any competent supervisor should have scrutinised. This same acceptance-of-inadequate-remediation logic is what permitted the O-ring substitute to stand as a 'repair' for the metering flange.

CAUSAL EXPLANATION

Although this specific pipeline segment did not fail on 24-Sep-2020, the supervisory acceptance of a cosmetic remedy for a structural defect is direct evidence of the decision-making culture that enabled the lower-level AJDE 2.8 wrong-choice-of-action finding on painting (rating 3) and, by extension, the wrong-choice-of-gasket finding that caused the release at 02:12:50. It demonstrates that inadequate corrective actions were routinely approved rather than challenged.

Linked Evidence: EVD-2397

SVI 1.1 **Leadership/Supervisory Oversight Inadequate** **SUBSTANTIATED**

Three of four diesel firewater pumps were under maintenance or tripping, with the tripping cause unidentified and no recorded actions; electrical pumps carried the incident.

JUSTIFICATION

Supervision over the firewater backup train had allowed a known reliability gap to persist unactioned. [EVD-2432] records: 'Out of 4 diesel pumps 410 m3/hr, 2 pumps are under maintenance, one operated for 30 minutes & fourth pumps is tripping after 2 minutes. The reason for tripping is not identified or actions have not been recorded' (p.27). Three of the four 410 m3/hr diesel pumps were unavailable or unreliable, and the unidentified 2-minute tripping fault had been neither diagnosed nor recorded - a textbook supervisory follow-up failure. This instance had no impact on the 24-Sep-2020 outcome because 'Both the Tank level and the Fire water header pressure were healthy throughout the incident' and three electrical pumps (47PE003 A, C, D) carried the cooling load (p.11), so it is rated present with no direct effect on the outcome.

CAUSAL EXPLANATION

This latent gap did not change the outcome because fire water stayed healthy throughout the 5-hour-15-minute emergency (02:15 detection to 07:30 all-clear) on electrical pump capacity, but it exposed the same supervisory pattern - known equipment defects accepted without documented diagnosis or action - that elsewhere allowed the defective gasket and lamp defects to persist into the incident.

Linked Evidence: EVD-2432

SVI 1.1 **Leadership/Supervisory Oversight Inadequate** **SUBSTANTIATED**

Missing permits went untracked, JSPs were absent from SAP, and permit closure went unreviewed, weakening control over the gasket job.

JUSTIFICATION

Supervision over the permit-to-work lifecycle was loose at every stage. [EVD-5105] records: 'it was found that few of the work permits were missing. No data is kept on such missing permits. Job safety Plan is not attached/given in SAP document. Closing of permits & actions indicated in work needs review' (p.26). Missing originals were not even logged, task-specific Job Safety Plans were absent from SAP for hazardous work, and permit closure was unreviewed. The committee's recommendation B.5 - 'Maintenance notifications are not be assumed as permit to work' (p.34) - implies work was being executed under SAP notifications without separate work-permit authorisation. This meant the 18-Sep-2020 high-risk gasket job could proceed without a JSP forcing identification of the required design-pressure hydrotest and material-conformance checks, making this an enabling condition rather than a direct cause, hence rating 2.

CAUSAL EXPLANATION

Without a JSP task hazard analysis, no document forced the maintenance crew and their supervisor to identify the design-pressure test and gasket-conformance steps before the 18-Sep-2020 job, so the omissions passed into service and the joint failed on 24-Sep-2020 at 02:12:50. It contributes secondarily to the lower-level ASBE 1.2 hydrotest-omission finding (rating 4) and the AJDE 2.11 policy-misapplication findings.

Linked Evidence: EVD-5105

SVI 1.6 **Lack of Supervisory Response to Critical Information** **SUBSTANTIATED**

Supervision left SAP notification 3800419840 open after the gasket job, removing the close-out review that could have caught the missing hydrotest and non-standard part.

JUSTIFICATION

[EVD-0591] records that "Notification no. 3800419840 was raised on 17/09/2020 for malfunction of meter no.1" (p.27) and that the notification "was not closed after completion of the job" (p.27). A SAP notification is critical maintenance-governance information: it defined the defect, the 18-Sep-2020 gasket replacement, and the return-to-service decision, yet the supervisory level did not close the loop by reviewing and formally closing it. The committee's recommendation D.2.ii at p.35 — "Sample checks/approval of closure of notifications by Senior Officers to be introduced" — recognises this exact gap. The missing closure removed the only supervisory review point at which the skipped hydrotest [EVD-2894] and the substitute O-ring [EVD-8645] could have been challenged. It enabled the lower-level finding AJDE 2.11 (misapplied SAP closure policy, rating 3) and PTF 1.12 (failure to communicate feedback/acknowledgement, rating 3).

CAUSAL EXPLANATION

Because the notification remained open and unreviewed, the meter returned to service on 19-Sep-2020 with an unvalidated pressure boundary; it operated for six days and failed at 02:12:50 on 24-Sep-2020, initiating the 56-minute vapour accumulation and the ignition at 03:08:15. This supervisory non-response is a secondary enabling cause supporting lower-level AJDE 2.11 and PTF 1.12.

Linked Evidence: EVD-0591

SVI 1.6 **Lack of Supervisory Response to Critical Information** **SUBSTANTIATED**

The 2017 UT report's warning of significant pipe-wall loss drew no corrective-action plan for over three years, showing supervisory failure to close the loop on integrity information.

JUSTIFICATION

[EVD-5527] records "As per the report of 2017, several pipes have been identified with significant thickness loss and need for corrective action. However, even after three years no plan has been developed for the corrective actions to be taken for those lines. Mere measurement of thickness without corrective action does not complete the process of maintaining the integrity of pipes" (p.30). The 2017 thickness report was critical integrity information delivered to the managerial level; the supervisory response was nil — zero corrective-action plan in more than three years. This is distinct from the meter-gasket failures because it concerns the trunk pipeline integrity system rather than the metering skid maintenance. It enabled the lower-level AJDE 2.5 ignored-warning finding (rating 3) and AJDE 2.7 delayed-necessary-action finding (rating 3).

CAUSAL EXPLANATION

While the 36-inch trunk pipeline remained intact during the 24-Sep-2020 fire [EVD-5780], this failure to close the loop on known wall-loss information created the organisational expectation that integrity findings need not be actioned; that same supervisory expectation is the environment in which the meter gasket recurrence and the skipped hydrotest were allowed to persist unaddressed, culminating in the loss of containment at 02:12:50 on 24-Sep-2020.

Linked Evidence: EVD-5527

SVI 1.6 Lack of Supervisory Response to Critical Information

SUBSTANTIATED

Shift supervisors had confirmed major leak information from 02:15-02:35 but did not isolate MOV 54 until 03:01 and the 36-inch PCV until 03:04, producing the fatal 46-minute vapour accumulation.

JUSTIFICATION

[EVD-9612] records that the GSP Shift In-charge received a phone call at "around 0235 hrs" telling him "that heavy gas leakage near 36 metering area" existed and that he "tried to enter the Gas Control room area but could not enter due to heavy gas smell... closed the MOV 54 inlet condensate line from slug catcher to 36 metering area at about 0301 hrs. Thereafter fire occurred." Combined with [EVD-7340], in which the Overall Shift In-charge / SIC "instructed to close the 36 line pressure control valve and get out of the control room" — executed at 03:04 — the supervisory team had critical leak information from 02:15 (CISF), 02:20 (GT CR), and direct verification at 02:35, yet the source-isolating MOV 54 was not closed for another 26 minutes. [EVD-6414] records the committee's characterisation that "Decision of closure of valve was taken after about 40 minutes resulting the spread of vapour cloud in increased area" (p.32). This is the most direct SVI 1.6 failure in the incident: supervisors were the authority level, knew of the major release, and their delayed act-or-escalate response is a primary determinant of the 46-minute release window. It enabled lower-level AJDE 2.7

CAUSAL EXPLANATION

The 26-minute segment from 02:35 to 03:01, plus the earlier 15-minute loss while operators attempted foot access, allowed the release to grow into the vertical column documented at 03:00:44 ("vapour all around even at height" [EVD-8213]); ignition followed at 03:08:15 after isolation actions, proving the already-released inventory was sufficient fuel. Had supervision responded immediately to the 02:35 information with upstream isolation and pre-authorized PCV closure, the flammable cloud mass could have been substantially smaller, potentially preventing or reducing the major fire.

Linked Evidence: EVD-9612, EVD-7340, EVD-6414

SVI 1.4 Failed to Provide Appropriate Policy/ Guidance

SUBSTANTIATED

With the OEM gasket spec lost, management provided no substitute specification, procurement control, or QC policy, leaving maintainers to install locally-bought O-rings in a rectangular-slot flange.

JUSTIFICATION

[EVD-8645] records "The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket" (p.31). The committee's root cause #2 ("Use of nonstandard O ring") and #3 ("Lack of control over material sourcing") (p.33) name exactly this absence of policy/guidance. The correct supervision response would have been to recover the OEM rectangular-slot specification, issue a substitute-specification policy, and establish quality control; the actual situation left maintainers with no acceptance basis, so the wrong-geometry round O-ring was fitted to a rectangular slot. This policy void is the enabling condition for lower-level AJDE 2.8 (wrong choice of action, rating 4), ASBE 1.3 (wrong gasket procedure, rating 4), and PTE 2.7 (technical guidance unavailable, rating 4).

CAUSAL EXPLANATION

Without a substitute-specification and QC policy, the maintenance crew had no authoritative basis to reject the round O-ring that could not fill the 1986 rectangular-slot flange, so the defective seal was installed on 18-Sep-2020, returned to service on 19-Sep-2020, operated for six days under condensate pressure, and failed at 02:12:50 on 24-Sep-2020. The resulting 56-minute vapour accumulation ignited at 03:08:15. This Level-3 policy failure created the conditions for the lower-level wrong-choice, routine-violation, and technical-guidance findings that directly caused the loss of containment.

Linked Evidence: EVD-8645

SVI 1.4 **Failed to Provide Appropriate Policy/ Guidance** **SUBSTANTIATED**

Supervision provided no SOP for high-pressure meter calibration/return-to-service, so the 18-Sep-2020 gasket job went untested to design pressure and returned to service with an unvalidated seal.

JUSTIFICATION

[EVD-2894] records "After calibration of the meter 310FE01, the gasket was replaced on 18.09.2020. No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated. Next day meter 310FE01 was taken on online and was in use at the time of Incident" (p.31). The same evidence states the committee's probable-cause conclusion: "No standard operating procedure is in place for calibration and taking back in service of these meters which are operating at high pressures" (p.32). [EVD-2506] independently shows the same SOP/checklist absence during December 2019 calibration: "No checklist is followed while carrying such critical jobs of high pressure/risk" (p.27). The correct procedure would require a hydrostatic test to design pressure and documented return-to-service gates; the actual practice was an operating-pressure soak only, which cannot validate a pressure boundary. This supervision-level policy gap enabled the lower-level ASBE 1.2 checksheet omission (rating 4), AJDE 2.7 delayed/omitted necessary action (rating 4), and PTE 2.7 technical guidance unavailability (rating 4).

CAUSAL EXPLANATION

Because no return-to-service SOP mandated a hydrostatic test, the inadequate O-ring installation was never verified against design pressure after 18-Sep-2020; the meter entered service on 19-Sep-2020 and the unvalidated pressure boundary failed catastrophically at 02:12:50 on 24-Sep-2020, releasing vapour for 56 minutes until ignition at 03:08:15. This policy gap is the direct supervisory origin of the lower-level ASBE 1.2 and AJDE 2.7 omissions that caused the incident.

Linked Evidence: EVD-2894, EVD-2506

SVI 1.4 **Failed to Provide Appropriate Policy/ Guidance** **SUBSTANTIATED**

Control-room staff received only four days of general training with no defined qualification or simulator practice, leaving them unprepared for emergency isolation decisions.

JUSTIFICATION

[EVD-5634] records "control room operations are carried out by persons inadequately trained in safety, operations, control room handling, logbook records and actions during emergency. It is informed that four days training programme is conducted for new recruitment/transfer staff which is general in nature and later the job training is carried out. Qualification of this staff is also not defined in HR/Operations Policy. Most of the staff have attended only One Basic Course since 2018" (p.30). The committee recommended instead that "operating staffs of control room needs to be trained for at least 1 month (preferably on simulators) before being assigning Control Room duties enabling them to handle operations & emergencies effectively" (p.36) — an almost eightfold expansion. Supervision failed to provide appropriate training policy, so on-shift engineers had no practised emergency-isolation response when the annunciator alarmed at 02:20. This enabled lower-level AJDE 2.10 (inadequate knowledge to decide, rating 3), AJDE 2.9 (decision-making during operation, rating 4), PTF 1.3 task delegation (rating 3), and PTF 1.17 unfamiliar workspace

CAUSAL EXPLANATION

Because the control-room crew had no simulator-practised emergency response and no defined competency, when the annunciator alarmed at 02:20 they attempted physical source verification and then serial telephone escalation rather than initiating immediate upstream isolation; by the time MOV 54 closed at 03:01 the release had continued for 46 minutes and the accumulated cloud ignited at 03:08:15. The training-policy gap is therefore a supervisory enabling condition for the lower-level decision-making and knowledge failures.

Linked Evidence: EVD-5634

SVI 1.4 **Failed to Provide Appropriate Policy/ Guidance** **SUBSTANTIATED**

Supervision provided a generic, non-plant-specific SSA checklist, allowing critical plant-specific defects such as the compromised FLP lamp to go unobserved.

JUSTIFICATION

[EVD-9692] records "Safety self-audit is conducted by a multi-disciplinary team based on the checklist prepared by the location. The checklist is comprehensive but not related to specific plants and in the process, lot of observations are missing. The duration of audit is not indicated and in most of the items, compliances are indicated, but during our inspection, the data is found to be incomplete & not in line with OISD standards/manuals. Approved manuals/SOPs were not available" (p.26). The committee's recommendation B.1 at p.34 — "Checklist for Self-Safety Audit for individual plants are to be developed" — confirms the audit instrument was not fit for purpose. This policy gap meant the audit did not force inspection of plant-specific details such as the Zone-1 HPMV lamp integrity, which later became the probable ignition source [EVD-6916], or the chronic gasket-substitution practice. It enabled the lower-level ASBE 1.2 audit checklist failure (rating 3) and PTF 1.2 cross-monitoring failure (rating 3).

CAUSAL EXPLANATION

Because the self-safety audit instrument was not tailored to the Gas Terminal metering area, the compromised HPMV FLP lamp and the uncontrolled gasket practice remained latent; when the vapour cloud formed after the 02:12:50 release, it found the lamp's exposed contacts and ignited at 03:08:15. This Level-3 audit-policy gap therefore contributed directly to the transition from release to major fire by allowing the ignition source to persist.

Linked Evidence: EVD-9692, EVD-6916

SVI 1.4 **Failed to Provide Appropriate Policy/ Guidance** **SUBSTANTIATED**

Management failed to require automatic shutdown/suppression interlocks for the F&G system, converting a machine-detected 02:12:50 leak into a 46-minute manual response.

JUSTIFICATION

[EVD-8657] records that "Hydrocarbon/H2S detectors are configured to Annunciator Panel for display but not in DCS for history" and "F&G detection system does not have provision for auto shut down of plant or activation of fire suppression system" (p.26). [EVD-3452] quotes the committee recommendation A.1 at p.34: "F&G detection system needs to be interlocked to automatic plant shut down and activation of fire suppression system." The absence of a management policy requiring automatic protective action meant machine detection of the leak at 02:12:50 produced no automatic isolation or suppression; the entire response depended on a slow human detection-escalation-authorisation chain. This is distinct from the DMP isolation-criteria gap because it concerns automated safety-system design policy rather than manual response criteria. It enabled the lower-level AV 1.1 F&G-configuration violation (rating 3), PTE 2.11 equipment-not-configured-correctly finding (rating 4), and PTF 1.4 over-reliance-on-human-response finding (rating 4).

CAUSAL EXPLANATION

Because no supervisory policy required automatic isolation on confirmed gas detection, the leak witnessed by CCTV at 02:12:50 continued unisolated through the full human chain: CISF detection 02:15, GT CR notification 02:18-02:20, shift-in-charge arrival 02:35, MOV 54 closure 03:01, PCV closure 03:04, and ignition 03:08:15. The 46-minute release produced the large flammable cloud documented at 03:00:44 [EVD-8213] that found the compromised HPMV lamp; an automatic interlock would have reduced the release window from minutes to seconds and likely prevented the major fire.

Linked Evidence: EVD-8657, EVD-3452

SVI 1.4 **Failed to Provide Appropriate Policy/ Guidance** **SUBSTANTIATED**

Supervision maintained a breakdown-centric electrical PM policy and even post-incident grouped all nine HPMV lamps under one generic status, letting the compromised flame-proof fitting go uncharacterised.

JUSTIFICATION

[EVD-9999] records that after the incident the electrical audit "has clubbed all nine 125 W HPMV FLP lamps together and reported 'cable burnt and fitting damaged due to heat' as status for all of them" and that "the electrical maintenance system is a lot more skewed towards breakdown maintenance and lays little emphasis on systematic preventive maintenance and inspection" (p.28). [EVD-0496] adds that "the unique condition of one particular lamp due to the type of damage suffered by it with respect to its area of installation was not mentioned in the report" (p.28). This is a supervision-level failure to provide a preventive-maintenance policy adequate for Zone-1 flame-proof electrical fittings; even the post-incident audit did not differentiate the per-equipment failure modes. It allowed the HPMV FLP lamp with three missing bolts and a glass case opened 5–6 cm [EVD-6916] to remain as a compromised ignition source. It enabled the lower-level PTE 2.12 maintenance-not-conducted-correctly finding (rating 4) and the ASBE 1.2 audit checklist failure (rating 3).

CAUSAL EXPLANATION

Without systematic preventive inspection of flame-proof electrical fittings, the HPMV lamp with three missing bolts and an open glass case remained in service in a Zone-1 hydrocarbon area; when the vapour cloud from the failed meter flange enclosed the area at 03:08:15 on 24-Sep-2020, the compromised fitting supplied the ignition source and converted the release into a major fire. This policy gap is therefore a secondary but direct enabler of the fire's ignition transition.

Linked Evidence: EVD-9999, EVD-6916

SVI 1.2 Improper Role-Modelling

SUBSTANTIATED

Skipping the design-pressure hydrotest and accepting an operating-pressure-only pressurisation had become the modelled return-to-service practice.

JUSTIFICATION

The omission of the design-pressure hydrotest was itself the modelled standard behaviour, not a single lapse. [EVD-2894] records that after the 18.09.2020 gasket replacement 'No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated' (p.31), and the committee adds that 'No standard operating procedure is in place for calibration and taking back in service of these meters' (p.32). The lower-level AV 1.1 finding rated this at 4 and characterises it specifically as 'routine practice across multiple maintenance events' - meaning the shortcut of skipping design-pressure verification had become the accepted, observable method passed down within the maintenance team. The correct procedure is to subject a repaired pressure boundary to a hydrostatic test to design pressure before hydrocarbon service; the ACTUAL practiced procedure substituted an operating-pressure soak only, omitting the verification step. A new maintainer observing this would learn the shortcut as the way the organisation returns meters to service.

CAUSAL EXPLANATION

Because the modelled norm substituted an operating-pressure soak for a design-pressure test, the defective O-ring joint passed post-maintenance verification on 18-Sep-2020, entered service on 19-Sep, and failed at 02:12:50 on 24-Sep, leading to ignition at 03:08:15 after 56 minutes of accumulation. This modelling failure is the behavioural basis of the lower-level ASBE 1.2 checklist findings (ratings 3-4), the AJDE 2.7 omitted-test finding (rating 4) and the AJDE 2.11 policy-misapplication finding (rating 4).

Linked Evidence: EVD-2894

SVI 1.3 Local Training/ Training Programme Issues

SUBSTANTIATED

Control-room operators received only a 4-day general training course, versus the committee's recommended minimum of one month on simulators.

JUSTIFICATION

The control-room training programme was materially inadequate for the task demands of running the Gas Terminal. [EVD-5634] records verbatim: 'control room operations are carried out by persons inadequately trained in safety, operations, control room handling, logbook records and actions during emergency. It is informed that four days training programme is conducted for new recruitment/transfer staff which is general in nature and later the job training is carried out' (p.30). The committee's recommendation at p.36 - that 'operating staffs of control room needs to be trained for at least 1 month (preferably on simulators) before being assigning Control Room duties enabling them to handle operations & emergencies effectively' - is a near-eightfold expansion of the current four-day baseline, quantifying how far the programme fell short. A four-day general course cannot build the procedural automaticity needed to respond to a night-time vapour release, which is why the night shift defaulted to on-foot verification and telephoned escalation rather than immediate isolation at 02:20 on 24-Sep-2020.

CAUSAL EXPLANATION

The four-day programme gave the 02:20 responders no trained emergency-isolation script, so they reverted to source-location by foot and serial escalation while the release continued unisolated until MOV 54 closed at 03:01 - 46 minutes after first detection. This training gap is the Level-3 enabler of the lower-level PTF 1.3 task-delegation finding (rating 3), the PTF 1.17 unfamiliar-workspace finding (rating 3), and the AJDE 2.10 inadequate-knowledge finding (rating 3), all of which contributed to the vapour accumulation that ignited at 03:08:15.

Linked Evidence: EVD-5634

SVI 1.6 Lack of Supervisory Response to Critical Information

SUBSTANTIATED

A 16000-to-2540 mmwc calibration change on meter 310FT01 passed through signed records without any supervisory intervention, normalising undocumented critical changes.

JUSTIFICATION

[EVD-2506] records that during calibration from 17/12/2019 to 18/12/2019, the setting "was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of this change in the calibration documents signed by the team. During next calibration for meter 310FT01 was again changed to 16000 MMWC. No checklist is followed while carrying such critical jobs of high pressure/risk" (p.27). This is a documented instance of critical technical information — a roughly 6.3x setpoint change on a high-pressure meter — passing through signed calibration records without any supervisory challenge or close-out action. It is distinct from the gasket recurrence and the SAP non-closure because it occurred in December 2019 on the same meter and concerns calibration integrity rather than mechanical sealing. It enabled the lower-level ASBE 1.2 calibration-checklist finding (rating 3), ASBE 1.5 under-controlled system finding (rating 3), and PTF 1.2 cross-monitoring failure (rating 3).

CAUSAL EXPLANATION

The unreviewed setpoint change normalised undocumented, check-list-free work on meter 310FT01 nine months before the incident; the same procedural-discipline failure reappeared in the 18-Sep-2020 gasket replacement, which omitted the hydrotest and introduced the defective seal that failed on 24-Sep-2020 at 02:12:50, leading to the vapour release and fire at 03:08:15.

Linked Evidence: EVD-2506

SVI 1.3 Local Training/ Training Programme Issues

SUBSTANTIATED

No simulator-based emergency training meant the crew could not execute isolation actions under a live vapour release on 24-Sep-2020.

JUSTIFICATION

The control-room programme lacked any simulator-based rehearsal of emergency actions, which is the specific training deficiency that showed up in the response. [EVD-5634]'s recommendation states the requirement the plant failed to meet: 'operating staffs of control room needs to be trained for at least 1 month (preferably on simulators) before being assigning Control Room duties enabling them to handle operations & emergencies effectively' (p.36). On the night, the inadequacy is behavioural: at 02:20 the Chief and Junior Engineers heard the annunciator, attempted an on-foot approach, were forced back by gas density, and then only telephoned the chain - the un-rehearsed crew lost the 02:20-02:35 window while the release continued, and first isolation followed only at 03:01. A crew drilled in emergency isolation would have moved directly to remote isolation orders from the control room the moment a vapour cloud was confirmed. This instance is distinct from the four-day course content finding because it isolates the emergency-response phase of training rather than routine operations competency.

CAUSAL EXPLANATION

Untrained in emergency isolation behaviour, the crew treated the 02:20 alarm as a diagnosis task instead of an isolation trigger, attempting physical approach and then serial escalation; the first valve did not close until 03:01 (46 minutes after the 02:15 detection). This emergency-phase training gap directly enabled the lower-level PSE 3.1 untrained-scenario finding (rating 3), the AJDE 2.9 operational decision failure (rating 4), and the ASBE 1.2 emergency-checklist/procedure findings, and determined the size of the vapour cloud that ignited at 03:08:15.

Linked Evidence: EVD-5634

SVI 1.6 Lack of Supervisory Response to Critical Information

SUBSTANTIATED

Maintenance supervisors received three gasket-failure signals on meter 310FT01 in 22 months and took no root-cause or corrective supervisory action, allowing the same inadequate O-ring repair to continue.

JUSTIFICATION

[EVD-1494] records that "Cleaning/servicing/calibration of meters 1 & 2 was carried out on 12th & 13th Nov 2018. Afterwards, meter no.1 was cleaned on 25/02/2019 & gasket was changed. Again on 18/09/2019, gasket was changed which leaked within one year" (p.27), i.e. three gasket replacements in approximately 22 months on the same 1986-vintage meter. This recurrence was safety-critical information available to maintenance supervision, yet no supervisory response closed the loop: no root-cause investigation, no reliability review, and no escalation to recover the OEM specification. [EVD-8645] confirms the same inadequate practice continued — "use of locally purchased O ring was done in its place as a regular practice" — so supervisors repeatedly accepted a non-conforming pressure-boundary repair as normal. This instance directly enabled the lower-level finding AJDE 2.5 (three-failure warning ignored, rating 4) and the wrong-choice unsafe act AJDE 2.8, because the final substitute seal installed 18-Sep-2020 was the fourth recurrence rather than an isolated error.

CAUSAL EXPLANATION

Because supervision never acted on the 12-Nov-2018, 25-Feb-2019 and 18-Sep-2020 replacement pattern, the same non-standard O-ring practice was repeated for the 18-Sep-2020 job; the defective seal entered service on 19-Sep-2020 and failed at 02:12:50 on 24-Sep-2020, releasing vapour for 56 minutes until ignition at 03:08:15. This higher-level supervisory omission is the enabling condition for lower-level AV 1.1 (routine/widespread violation, rating 4) and PTF 1.13 (mission planning failure, rating 3) that caused the loss of containment.

Linked Evidence: EVD-1494, EVD-8645

SVI 1.6 **Lack of Supervisory Response to Critical Information** **SUBSTANTIATED**

Supervisors accepted painting as sufficient for a 1.03 mm wall-loss finding with zero corrosion allowance, repeating the pattern of inadequate supervisory response to critical integrity information.

JUSTIFICATION

[EVD-2397] records that "a reduction of thickness of 1.03mm along with heavy pitting and painting peeling off was observed in one of the segments of 36 pipeline in gas terminal in July 2019. The corrective action of painting was completed in Dec. 2019. However, as per the available data, the design thickness of the pipeline is 23.83mm with zero corrosion allowance. It needs to be examined whether painting as a corrective action would suffice in such cases and the mechanism of deciding on a corrective action needs to be reviewed" (p.30). Supervision received a specific, quantified integrity finding on a zero-corrosion-allowance line and accepted a coating-only corrective action without fitness-for-service analysis. This is distinct from instance 4's 2017 report because it is a later spot-UT finding with an explicit inadequate action. It is the same pattern as the lower-level AJDE 2.8 wrong-choice finding (painting instead of engineering evaluation, rating 3).

CAUSAL EXPLANATION

This instance did not directly cause the 24-Sep-2020 gasket failure, but it is direct evidence that supervision accepted cosmetic mitigations for structural degradation; that same acceptance logic allowed the O-ring substitution, the omitted hydrotest, and the three recurring gasket failures to persist, contributing to the organisational environment that produced the loss of containment.

Linked Evidence: EVD-2397

SVI 1.6 **Lack of Supervisory Response to Critical Information** **SUBSTANTIATED**

Twelve February 2018 external-audit recommendations remained overdue on extensions for which no Director Offshore approval existed, evidencing supervisory failure to close safety actions.

JUSTIFICATION

[EVD-6852] records "External safety audit was conducted in February 2018, and 296 out of 308 recommendations are complied with. Balance are in various stages and dates are extended till 30/04/2021. Document indicating approval from Director Offshore for extending the target date was not available as per the procedure" (p.26). This is an audit-level instance of supervision not closing the loop: twelve outstanding critical recommendations were allowed to continue on extended schedules without the required competent-authority approval. It did not directly cause the meter gasket failure, but it substantiates the system-level supervision pattern in which known safety items did not receive supervisory action. It supports the lower-level finding that overdue actions were tolerated without enforcement.

CAUSAL EXPLANATION

The 12 unapproved overdue recommendations are a systemic manifestation of the same SVI 1.6 pattern shown in the gasket recurrence and pipeline findings; they indicate supervision at multiple levels accepted unresolved safety-critical items as routine, which weakened the overall management-of-change and assurance culture that should have caught the meter gasket defect before 24-Sep-2020.

Linked Evidence: EVD-6852

SVI 1.4 **Failed to Provide Appropriate Policy/ Guidance** **SUBSTANTIATED**

The DMP/ERDMP defined the hydrocarbon-leak scenario but omitted isolation/stoppage decision criteria, leaving shift personnel without authority or guidance to stop the release.

JUSTIFICATION

[EVD-4052] records "As per DMP On site Hazira Plant, guidelines to be followed in case of hydrocarbon leakage are included but does not include isolation/stoppage of facilities/pipelines" (p.32), and reinforces at p.27 that "In ERDMP, scenario for Level 3 emergencies are taken from the plant failures. Heavy product/gas leakages from pipelines & tanks are considered but decision on isolation/closure of plant/pipelines is not mentioned." The committee's root cause #6 ("Non isolation/stoppage of 36 pipeline operation") flows directly from this policy gap. The correct emergency policy would have specified when a hydrocarbon release warrants immediate isolation, who may authorise it, and which valves to close; the actual DMP named the scenario but provided no decision criteria at all. This supervision-level failure to provide policy guidance enabled the lower-level AJDE 2.9 (decision-making during operation, rating 4), AJDE 2.10 (inadequate knowledge of isolation criteria, rating 4), and AJDE 2.11 (unaware of policy gap, rating 4), and is the procedural origin of the 46-minute isolation delay.

CAUSAL EXPLANATION

Without DMP isolation criteria, personnel at first detection (02:15) and first annunciator alarm (02:20) had no pre-authorized action to execute, so they defaulted to on-foot verification and serial management escalation; MOV 54 was not closed until 03:01 and the 36-inch PCV until 03:04. The 46-minute unisolated release filled the area with vapour documented at 03:00:44 ("vapour all around even at height" [EVD-8213]) and that cloud ignited at 03:08:15, converting the release into a major fire.

Linked Evidence: EVD-4052

SVI 1.4 **Failed to Provide Appropriate Policy/ Guidance** **SUBSTANTIATED**

The 2016 MOC policy was a paper control: no MOCs had been implemented since 2018, so the gasket material substitution bypassed engineering review.

JUSTIFICATION

[EVD-2986] records "Management of Change (MOC) philosophy/SOP was published in August 2016. However, the procedure indicated in the document is not implemented while carrying out change of Low-Pressure Gas to Kribhico from Medium Pressure Gas headers in LPG Plant in 2013 & 2018. It is informed no more MOCs are implemented in the plant since 2018" (p.30). The same failure is visible in [EVD-8645]: "There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket" (p.31). Supervision failed to enforce an existing policy, meaning material changes such as the gasket substitution could occur as undocumented regular practice. This is distinct from the lost OEM specification because the guidance existed on paper; the failure was non-implementation. It enabled lower-level AJDE 2.11 (MOC policy misapplied, rating 3) and the routine O-ring substitution violation AV 1.1 (rating 4).

CAUSAL EXPLANATION

Because the MOC policy was inactive after 2018, the switch from the OEM rectangular-slot gasket to a locally procured O-ring was never subjected to engineering review, specification control, or documented approval; the deficient seal was installed on 18-Sep-2020 and failed at 02:12:50 on 24-Sep-2020, initiating the release and subsequent fire at 03:08:15. This Level-3 policy non-implementation is a foundational enabling condition for the lower-level AJDE 2.11 and AV 1.1 findings.

Linked Evidence: EVD-2986, EVD-8645

SVI 1.4 **Failed to Provide Appropriate Policy/ Guidance** **SUBSTANTIATED**

The permit/JSA system lacked tracking of missing permits and did not attach Job Safety Plans in SAP, leaving high-risk maintenance with incomplete task-specific hazard guidance.

JUSTIFICATION

[EVD-5105] records "Work Permits are issued in SAP and hard copy of work permits are at sites during execution of work. After completion of work, the originals are filed. On checking the same, it was found that few of the work permits were missing. No data is kept on such missing permits. Job safety Plan is not attached/given in SAP document. Closing of permits & actions indicated in work needs review" (p.26). The committee recommendation B.5 at p.34 notes "Maintenance notifications are not be assumed as permit to work", implying work was being executed under SAP notifications without separate work-permit authorisation. This policy deficiency meant the 18-Sep-2020 high-risk gasket replacement could proceed without a task-specific Job Safety Plan identifying the required hydrotest and material-conformance checks. It is a secondary systemic condition rather than a direct cause, so it is rated 2.

CAUSAL EXPLANATION

Without a mandated JSP and auditable permit closure, the 18-Sep-2020 gasket job was not forced to specify the hydrostatic test and material-verification steps required for pressure-boundary work; this contributed to lower-level ASBE 1.2 and AJDE 2.7 omissions that allowed the defective seal into service, although the primary causes were the missing SOP and uncontrolled material substitution.

Linked Evidence: EVD-5105

SVI 1.4 **Failed to Provide Appropriate Policy/ Guidance** **SUBSTANTIATED**

Supervision did not maintain a controlled P&ID baseline: original drawing lost, vendor drawing unapproved, and field configuration mismatched.

JUSTIFICATION

[EVD-0933] records "Reference drawings no. 02-310-AO-101 Rev.1, 2x6 lines & 1x2 lines for metering system are shown but valves & meters configuration are not matching. As old drawings were lost, new drawings are made by a vendor but the same has not been checked & approved by ONGC" (p.26). Engineering drawing control is a foundational supervisory/management policy; here the original P&ID was lost, the replacement drawing was never approved by the operator, and the field configuration did not match the unapproved drawing. Without a controlled as-built reference, maintenance personnel could not independently verify the correct gasket design or meter configuration, which compounded the lost OEM gasket specification [EVD-8645]. This enabled lower-level AV 1.1 configuration-management breakdown (rating 2) and PTE 2.14 unfamiliar equipment (rating 2).

CAUSAL EXPLANATION

The absence of an approved, accurate as-built drawing for the metering skid meant the maintenance group could not consult an authoritative configuration document to identify the correct rectangular-slot gasket design or verify the substitute; this policy gap supported the uncontrolled O-ring substitution that failed on 24-Sep-2020 at 02:12:50, initiating the release. It is a secondary enabling condition for the lower-level ASBE 1.3 wrong-gasket procedure findings.

Linked Evidence: EVD-0933

SVI 1.6 **Lack of Supervisory Response to Critical Information** **SUBSTANTIATED**

Three SAP-recorded gasket failures on 310FT01 in 22 months never triggered supervisory root-cause investigation, allowing the same nonconforming O-ring to be reinstalled in Sep-2020.

JUSTIFICATION

EVD-1494 records that 'Cleaning/servicing/calibration of meters 1 & 2 was carried out on 12th & 13th Nov 2018. Afterwards, meter no.1 was cleaned on 25/02/2019 & gasket was changed. Again on 18/09/2019, gasket was changed which leaked within one year' (p.27), with the evidence note resolving the three replacements as 12-Nov-2018, 25-Feb-2019 and 18-Sep-2020. These three gasket replacements on the same 1986 meter over roughly 22 months were recorded in SAP and were therefore available to maintenance supervisors and plant management as a persistent pressure-boundary alarm on the exact meter that later became the leak source. Instead of closing the loop by recovering the OEM rectangular-slot gasket specification, investigating the root cause, or replacing the meter, supervision allowed the same local O-ring substitute to continue; EVD-8645 states the locally purchased O-ring was used 'as a regular practice' with 'no specification or quality control during the purchase of the O rings' and no record of the decision to abandon OEM supply, and the evidence links the committee's root cause #2 ('Use of nonstandard O ring') and #3 ('Lack

CAUSAL EXPLANATION

The three replacements over ~22 months were the critical information that supervision failed to close the loop on; each event was an opportunity to reject the nonconforming O-ring and recover the 1986 OEM rectangular-slot specification, but the same substitute was accepted every time. This supervisory inaction propagated directly into the 18-Sep-2020 repair, which returned meter 310FT01 to service on 19-Sep-2020 without a hydrostatic test; the joint failed at 02:12:50 on 24-Sep-2020 and the release accumulated for 56 minutes until ignition at 03:08:15. This Level 3 non-response directly enabled the lower-level AJDE 2.5 ignored warning (rating 4), ASBE 1.3 wrong gasket procedure (rating 4), PTF 1.1 maintenance leadership (rating 3), and PTF 1.13 mission planning (rating 3).

Linked Evidence: EVD-1494, EVD-8645

SVI 1.6 **Lack of Supervisory Response to Critical Information** **SUBSTANTIATED**

The DGM/GM shift supervisors received verified heavy-leak reports by 02:35 but closed no isolation valve until 03:01, leaving the release unisolated for 46 minutes.

JUSTIFICATION

EVD-6414 reconstructs the response chain: leak onset 02:12:50 (CCTV), first physical detection 02:15 (CISF), Gas Terminal CR notification 02:18-02:20, shift-in-charge phone call ~02:35, shift-in-charge reaches gate, MOV 54 closure 03:01, 36-inch PCV closure 03:04, and ignition 03:08:15, and it quotes the committee directly: 'Decision of closure of valve was taken after about 40 minutes resulting the spread of vapour cloud in increased area' (p.32). EVD-9612 records that the DGM (GSP Shift In-charge) was telephoned at ~02:35, attempted to enter the area but 'could not enter due to heavy gas smell', and only closed MOV 54 'at about 0301 hrs. Thereafter fire occurred' (p.17). Critical information about a heavy gas leak was therefore in the hands of the two senior shift authorities from ~02:35 at the latest, yet the first isolation valve was not closed until 03:01 and the PCV until 03:04. This failure of the supervisory level to close the loop on escalation information is a direct manifestation of this nanocode and is rated 4 because the delay determined the size of the flammable inventory.

CAUSAL EXPLANATION

With the leak confirmed by CCTV at 02:18 and the two senior supervisors on scene at ~02:35, a prompt supervisory decision to isolate upstream inventory should have closed MOV 54 far earlier than 03:01. Instead, the supervisors continued serial management notification, and the release continued unisolated for 46 minutes after first detection; the cloud expanded within ~40 seconds of first detection (02:15:35-02:16:13 per EVD-9941), was already 'vapour all around even at height' at 03:00:44, and the accumulated inventory ignited at 03:08:15, only 4 minutes after the last isolation action. This Level 3 supervisory non-response directly enabled the lower-level AJDE 2.7 delayed necessary action (rating 4), PTF 1.1 crew leadership failure at the gate (rating 4), and PTF 1.15 failure to re-plan immediately after being blocked at 02:35 (rating 4).

Linked Evidence: EVD-6414, EVD-9612

SVI 1.6 Lack of Supervisory Response to Critical Information

SUBSTANTIATED

A 1.03 mm wall-loss and heavy-pitting finding on the zero-corrosion-allowance 36-inch line received only painting as supervisory-approved corrective action.

JUSTIFICATION

EVD-2397 records that 'a reduction of thickness of 1.03mm along with heavy pitting and painting peeling off was observed in one of the segments of 36 pipeline in gas terminal in July 2019. The corrective action of painting was completed in Dec. 2019. However, as per the available data, the design thickness of the pipeline is 23.83mm with zero corrosion allowance. It needs to be examined whether painting as a corrective action would suffice in such cases and the mechanism of deciding on a corrective action needs to be reviewed' (p.30). The July 2019 spot measurement was critical integrity information delivered to the supervisory and engineering levels, and the supervisor-approved response was solely cosmetic repainting completed in Dec 2019, without fitness-for-service evaluation of metal loss on a zero-corrosion-allowance line. This is a distinct manifestation of non-response to critical information because it is a later, separate pipeline finding from the 2017 report, grounded in independent measurement evidence.

CAUSAL EXPLANATION

By accepting painting as the corrective action for 1.03 mm metal loss with zero design corrosion allowance, the supervisory level closed the loop with an inadequate response, signaling that surface treatment was a legitimate resolution for structural integrity findings. This is the same decision-making pattern that accepted locally sourced O-rings as a legitimate substitute for OEM rectangular-slot gaskets, and it directly enabled the lower-level AJDE 2.8 wrong choice of corrective action (rating 3) and PTE 2.12 maintenance-not-conducted-correctly finding for pipeline integrity (rating 2). It did not directly cause the 24-Sep-2020 flange failure because the 36-inch line remained intact, so it is rated 3 as a systemic contributor.

Linked Evidence: EVD-2397

SVI 1.6 Lack of Supervisory Response to Critical Information

SUBSTANTIATED

The 17-Sep-2020 meter malfunction notification was not closed or reviewed after the gasket job, removing the supervisory quality gate before return-to-service.

JUSTIFICATION

EVD-0591 records that 'Notification no. 3800419840 was raised on 17/09/2020 for malfunction of meter no.1' (p.27) and that the notification 'was not closed after completion of the job' (p.27), meaning the work package returned to service without a supervisory closure gate. EVD-0185 confirms the operational consequence: 'Next day meter 310FE01 was taken on online and was in use at the time of Incident' (p.31) while the notification remained open. The notification contained the maintenance trigger that should have forced a supervisor to review the job details, including the locally sourced substitute O-ring and the absence of a hydrostatic test to design pressure, before the meter was handed back to operations. The supervisor's failure to close the loop on this critical maintenance information is a distinct manifestation from the recurrence findings because it concerns the specific administrative control for the 17-18 Sep 2020 repair.

CAUSAL EXPLANATION

Without supervisory closure of SAP Notification 3800419840, the substitute O-ring and the omitted hydrostatic test passed into service without any management review. The meter was back online on 19-Sep-2020 and operated under normal condensate pressure until the flange joint failed at 02:12:50 on 24-Sep-2020, releasing vapour for 56 minutes until ignition at 03:08:15. This instance enabled the lower-level AV 1.1 routine non-closure violation (rating 3) and AJDE 2.11 misapplication of the SAP closure procedure (rating 3), and it removed the last administrative barrier that might have caught the defective work.

Linked Evidence: EVD-0591, EVD-0185

SVI 1.4 **Failed to Provide Appropriate Policy/ Guidance** **SUBSTANTIATED**

No supervisory policy existed for retaining the OEM gasket spec, procuring certified replacement seals, or controlling the O-ring substitution.

JUSTIFICATION

EVD-8645 documents that 'The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket' (p.31). Supervision therefore provided no policy or guidance requiring retention of the OEM rectangular-slot specification, procurement from approved sources, or quality control of pressure-boundary materials, and no Management-of-Change record existed for the substitution. The correct control would have been to maintain the OEM specification and procure certified gaskets against it, with QC verification and an approved MOC for any deviation; the actual practice had none of these safeguards. The committee's root cause #2 ('Use of nonstandard O ring') and #3 ('Lack of control over material sourcing') as cited in this evidence name exactly this policy failure, and it is

CAUSAL EXPLANATION

Because supervision never provided a material specification or procurement policy, the maintenance crew had no authoritative basis to select the correct rectangular-slot gasket; they fitted a locally purchased O-ring whose round cross-section could not fill the machined rectangular groove, leaving a leak path at the pressure boundary. The meter operated from 19-Sep-2020 to 24-Sep-2020 and failed at 02:12:50, releasing vapour for 56 minutes until ignition at 03:08:15. This Level 3 policy gap directly enabled the lower-level AJDE 2.8 wrong choice of substitute O-ring (rating 4), ASBE 1.3 wrong gasket procedure (rating 4), AV 1.1 routine substitution violation (rating 4), and PTE 2.7 technical guidance unavailable (rating 4).

Linked Evidence: EVD-8645

SVI 1.4 **Failed to Provide Appropriate Policy/ Guidance** **SUBSTANTIATED**

No checklist or documented-deviation guidance was provided for the Dec-2019 high-pressure calibration of 310FT01, so a 16000-to-2540 mmwc setpoint change went undocumented and unchallenged.

JUSTIFICATION

EVD-2506 records that during calibration from 17/12/2019 to 18/12/2019 the meter 310FT01 setpoint 'was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of this change in the calibration documents signed by the team', that the next calibration silently returned it to '16000 MMWC', and that 'No checklist is followed while carrying such critical jobs of high pressure/risk' (p.27). The same evidence records committee recommendation D.1.i (p.35) that 'details of Master calibrator/action taken/replacement/deviation from procedure if any, is to be documented in the calibration certificate'. Supervision therefore failed to provide a mandatory checklist or structured guidance for critical high-pressure calibration work, allowing the setpoint change to occur and reverse without documented justification. This is a distinct policy gap from the absent return-to-service SOP because it concerns the December 2019 calibration event and its specific documentation control, and it is rated 3 because it weakened governance on the same meter rather than directly opening the September 2020 leak.

CAUSAL EXPLANATION

The absence of a supervisory-provided calibration checklist removed the structured step that would have forced documentation and approval of the 16000-to-2540 mmwc change and its later reversal, so the nonconformance passed without engineering review. This normalized undocumented changes on meter 310FT01 and contributed to the same procedural erosion that later allowed the 18-Sep-2020 gasket replacement without hydrostatic testing. It enabled the lower-level ASBE 1.2 calibration checklist-not-followed finding (rating 3), ASBE 1.5 under-controlled calibration finding (rating 3), and AJDE 2.10 inadequate calibration knowledge finding (rating 3), but it did not itself open the leak, so it is rated 3.

Linked Evidence: EVD-2506

SVI 1.4 **Failed to Provide Appropriate Policy/ Guidance** **SUBSTANTIATED**

Supervision provided only a four-day general training programme with no defined control-room qualification and no simulator emergency training.

JUSTIFICATION

EVD-5634 records that 'control room operations are carried out by persons inadequately trained in safety, operations, control room handling, logbook records and actions during emergency. It is informed that four days training programme is conducted for new recruitment/transfer staff which is general in nature and later the job training is carried out. Qualification of this staff is also not defined in HR/Operations Policy. Most of the staff have attended only One Basic Course since 2018' (p.30). The committee's recommendation at p.36 that 'operating staffs of control room needs to be trained for at least 1 month (preferably on simulators) before being assigning Control Room duties enabling them to handle operations & emergencies effectively' shows the existing four-day general programme was a policy-level provision failure. Supervision failed to provide a training and qualification standard appropriate to time-critical emergency isolation duties.

CAUSAL EXPLANATION

Because control-room operators had only a four-day general course and no defined qualification or simulator training in emergency isolation, the night-shift engineers at 02:20 could not fall back on a trained immediate-isolation drill; they attempted physical verification and then escalated by telephone, contributing to the 46-minute delay before MOV 54 closure at 03:01 and ignition at 03:08:15. This policy gap enabled the lower-level AJDE 2.10 inadequate knowledge to make isolation decisions (rating 3), AJDE 2.11 unawareness of policy (rating 4), PTF 1.3 task delegation to inadequately trained staff (rating 3), and PTF 1.17 team unfamiliarity with workspace (rating 3).

Linked Evidence: EVD-5634

SVI 1.4 **Failed to Provide Appropriate Policy/ Guidance** **SUBSTANTIATED**

The permit system lacked JSPs in SAP, had untracked missing permits, and had no reviewed closure, and maintenance notifications could be mistaken for work permits.

JUSTIFICATION

EVD-5105 records that 'Work Permits are issued in SAP and hard copy of work permits are at sites during execution of work. After completion of work, the originals are filed. On checking the same, it was found that few of the work permits were missing. No data is kept on such missing permits. Job safety Plan is not attached/given in SAP document. Closing of permits & actions indicated in work needs review' (p.26). The committee recommendation B.5 at p.34 that 'Maintenance notifications are not be assumed as permit to work' further indicates the supervisory failure to provide a clear permit-based authorisation policy. The correct control is a permit-to-work system with attached JSPs, tracked originals, and reviewed closure; the actual system had missing originals with no tracking, no JSP in SAP, and unreviewed closure.

CAUSAL EXPLANATION

The incomplete permit guidance meant the 18-Sep-2020 high-risk gasket replacement could be executed without a JSP identifying the required hydrostatic test and material-conformance checks, and without a reviewed permit closure that would have exposed the missing verification. This enabled the lower-level PTE 2.7 permit guidance incomplete finding (rating 2) and contributed to the procedural shortcuts on the pressure boundary, but the evidence does not tie the specific gasket job to a missing permit, so it is rated 2 as a systemic condition rather than a direct cause.

Linked Evidence: EVD-5105

SVI 1.4 **Failed to Provide Appropriate Policy/ Guidance** **SUBSTANTIATED**

Pipeline integrity policy failed to preserve 1986 baseline thickness data and original pipe schedules, leaving only 2011 and 2017 data for gas-terminal pipes.

JUSTIFICATION

EVD-8287 records that 'It is observed that for a large number of pipes, especially in gas terminal area baseline data of pipe thickness in not available. The original drawings and the schedules of the pipes are also not available. In most cases only two thickness details of 2011 and 2017 is available for a pipe which is commissioned in 1986' (p.29). Supervision failed to provide the policy requiring preservation of baseline thickness data and original pipe schedules across the asset life, leaving only two data points for a 34-year-old pipeline. The correct control is a documented integrity-management policy retaining baseline measurements and schedules; the actual record had no 1986 baseline and no original drawings.

CAUSAL EXPLANATION

Without baseline thickness data and original schedules, the 2011 and 2017 measurements could not be rigorously trended or prioritised, which reinforced the three-year non-action on the 2017 corrective-action findings and weakened the integrity-management environment. This gap supported the lower-level AJDE 2.5 ignored warning on pipeline thickness (rating 3) and PTE 2.7 baseline-data availability finding (rating 3), but the 36-inch pipeline itself remained intact during the incident, so this instance is rated 2 with no direct outcome impact.

Linked Evidence: EVD-8287

SVI 1.4 **Failed to Provide Appropriate Policy/ Guidance** **SUBSTANTIATED**

The Aug-2016 MOC SOP was never applied in practice and no MOC had been implemented since 2018, so the O-ring substitution bypassed change approval.

JUSTIFICATION

EVD-2986 records that 'Management of Change (MOC) philosophy/SOP was published in August 2016. However, the procedure indicated in the document is not implemented while carrying out change of Low-Pressure Gas to Kribhico from Medium Pressure Gas headers in LPG Plant in 2013 & 2018. It is informed no more MOCs are implemented in the plant since 2018' (p.30). Although a paper SOP existed, supervision failed to provide it as an active operational control, so there was no functioning MOC policy in practice for exactly the kind of material substitution made at the metering skid. EVD-8645 confirms the same gap in relation to the gasket substitution: 'There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket' (p.31). This is distinct from the lost-specification finding because it concerns the general change-management policy rather than the gasket specification itself.

CAUSAL EXPLANATION

Because the MOC process was not functioning, the substitution of locally purchased O-rings for OEM rectangular-slot gaskets was treated as routine maintenance rather than as a change requiring engineering review, material specification, and approval. This removed the management gate that could have challenged the substitute before it was installed on 18-Sep-2020; the unvalidated joint then failed at 02:12:50 on 24-Sep-2020. This policy failure enabled the lower-level AJDE 2.11 misapplied MOC policy finding (rating 3), AV 1.1 routine substitution violation (rating 4), and PTF 1.13 mission planning failure (rating 4), and it is rated 3 because the initiating defect was ultimately the physical seal itself.

Linked Evidence: EVD-2986, EVD-8645

SVI 1.4 **Failed to Provide Appropriate Policy/ Guidance** **SUBSTANTIATED**

The 2018 and 2019 internal safety audits produced observations without filled checklists, so audit coverage could not be verified.

JUSTIFICATION

EVD-7681 records that 'Internal Safety Audit was conducted in 2018 & 2019 for 4 & 3 days respectively by a 3-member committee and the observations are given without the filled checklist. The checklist format is not available to check completeness of audit areas' (p.26). Supervision failed to provide a controlled checklist format and required the filled checklist to be retained, so the audit output could not be verified for completeness after the fact. The correct control is a standard checklist format completed and retained for every audited area; the actual audit produced observations with no filled checklist and no format available to establish what was inspected. This is distinct from the self-safety-audit instance because it concerns the internal audit reports for 2018 and 2019.

CAUSAL EXPLANATION

Without filled checklists, the supervision-level audit process could not demonstrate that plant-specific integrity items had been inspected, allowing latent defects such as the compromised HPMV flame-proof lamp to persist undetected until it became the probable ignition source at 03:08:15. This gap contributed to the lower-level ASBE 1.2 audit checklist-not-followed finding (rating 3) and PTE 2.12 maintenance-not-conducted-correctly finding for electrical fittings (rating 4), but it was an assurance-process weakness rather than a direct cause of the release, so it is rated 2.

Linked Evidence: EVD-7681

SVI 1.7 **Lack of Standardisation** **SUBSTANTIATED**

No one controlled standard governed gasket selection/procurement; locally purchased O-rings were substituted for the OEM rectangular-slot gasket as a regular, undocumented practice.

JUSTIFICATION

Supervision permitted materially different gasket practices to coexist on the same pressure boundary without one organisational standard governing which applied. [EVD-8645] records: “The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket” (p.31). The same inconsistent practice repeated on the same meter across the [EVD-1494] replacements of 12-Nov-2018, 25-Feb-2019 and 18-Sep-2020, so each fitter was left to improvise material selection instead of following one controlled standard. Without a standardised rule there was no uniform basis on which any supervisor could accept or reject the substitute seal; this lack of standardisation directly produced the lower-level AV 1.1 routine-violation and ASBE 1.3 wrong-gasket findings that

CAUSAL EXPLANATION

The absence of a single standardized gasket specification and procurement rule meant maintenance was governed by local habit rather than engineering control. The round O-ring installed on 18-Sep-2020 could not fill the rectangular slot, leaving a leak path from installation; with no uniform post-replacement test requirement applied, the meter returned to service on 19-Sep and failed at 02:12:50 on 24-Sep, releasing vapour for 56 minutes until ignition at 03:08:15. This Level-3 lack of standardisation enabled the lower-level wrong-choice-of-action, procedure-not-followed-correctly and routine-violation findings, and corresponds to the committee's root causes #2 and #3 on nonstandard O-ring use and lack of material-sourcing control.

Linked Evidence: EVD-8645, EVD-1494

SVI 1.7 **Lack of Standardisation** **SUBSTANTIATED**

No standardized isolation/stoppage criteria existed in the DMP/ERDMP, so the night-shift response varied by role and improvised serially rather than executing a pre-planned isolation sequence.

JUSTIFICATION

[EVD-4052] states that “As per DMP On site Hazira Plant, guidelines to be followed in case of hydrocarbon leakage are included but does not include isolation/stoppage of facilities/pipelines” (p.32), and reinforces at p.27 that “Heavy product/gas leakages from pipelines & tanks are considered but decision on isolation/closure of plant/pipelines is not mentioned.” Without a standardized decision rule, identical leakage indications were handled differently by each role: CISF officers could only relay, the Gas Terminal CR engineers attempted an on-foot approach, the SIC ordered only a PCV closure, and the GSP Shift In-charge finally closed MOV 54 at 03:01. [EVD-6414] reconstructs the resulting staggered timeline — leak onset 02:12:50, first detection 02:15, CR alarm 02:20, supervisor arrival ~02:35, MOV 54 closure 03:01 and PCV closure 03:04 — proving there was no standard pre-planned isolation sequence applied identically by all responders. The same authority-chain finding in [EVD-3932] confirms that no standing role below senior management had authority to close the emergency MOVs. The committee's root cause #6, “Non isolation/stoppage of 36

CAUSAL EXPLANATION

Had the organisational response been standardised — with a DMP rule such as “on confirmed vapour release, immediately close MOV 54/PCV and stop trunk-line supply,” pre-authorising the control room — the same situation would have elicited the same rapid action from whichever role detected it. Instead each actor applied an improvised response, consuming 46 minutes during which the cloud expanded to a vertical column by 03:00:44 and ignited at 03:08:15. This Level-3 lack of standardisation created the conditions for the lower-level AJDE 2.9, AJDE 2.10 and AJDE 2.11 decision-making failures and the ASBE 1.3 procedural failure in emergency isolation.

Linked Evidence: EVD-4052, EVD-6414, EVD-3932

SVI 1.7 **Lack of Standardisation** **SUBSTANTIATED**

Fire-drill records were split between incomplete SAP entries and a separate informal dataset not aligned with approved guidelines, preventing standardized measurement of emergency readiness.

JUSTIFICATION

[EVD-2130] records: “Fire drill reports are available in SAP but without any data on actions taken/recommendations. Separate data is maintained which is not in line with the approved guidelines” (p.26). This is a direct example of the organisation applying different documentation rules to the same drill activity: SAP records dates only, while a separate informal data set tracks actions and observations outside the approved guidelines. Because no single standardized record existed, drill effectiveness could not be measured and supervisory follow-up on emergency-response actions could not be audited. This lack of standardisation did not initiate the 24-Sep-2020 release, but it left emergency-response readiness unverified and undocumented, reinforcing the same governance weakness that left the DMP isolation criteria absent and the night-shift response improvised. It is therefore present and substantiated but rated 2 for outcome impact.

CAUSAL EXPLANATION

The two parallel, non-conforming drill records meant lessons and action-trackers that should have exposed and corrected the isolation-response deficiency were never consolidated in an auditable system. Emergency drills are the supervisory tool for standardising responses, and their documentation non-conformance explains partly why no corrective action was generated for the deficiencies that produced the 46-minute response gap. This condition connects weakly to the lower-level response-delay findings rather than to the initiating gasket failure, hence rating 2.

Linked Evidence: EVD-2130

SVI 1.8 **Failed to Identify/ Correct Risky or Unsafe Practices** **SUBSTANTIATED**

Three gasket replacements in 22 months using locally sourced O-rings were never escalated into a root-cause investigation or corrective engineering change.

JUSTIFICATION

Supervision had three documented opportunities to identify the meter's sealing problem and institute a durable correction, but the same inadequate local O-ring fix was repeated. [EVD-1494] records maintenance on 12-13 Nov 2018, a gasket change on 25-Feb-2019, and an 18-Sep replacement “which leaked within one year” — three replacements in ~22 months on the 1986-vintage meter. [EVD-8645] further establishes that the risky practice was accepted as normal: “use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket” (p.31). [EVD-0591] shows the 17-Sep-2020 SAP notification for meter malfunction “was not closed after completion of the job” (p.27), so no supervisor reviewed the recurrence. Failing to identify and correct this recurring unsafe practice is exactly the lower-level AJDE 2.5 ignored-caution/warning finding. A supervisor treating the repeat failures as a chronic defect would have required OEM gasket procurement, a

CAUSAL EXPLANATION

Each replacement produced another data point that the round O-ring in the rectangular slot was not sealing: 12-Nov-2018, 25-Feb-2019, 18-Sep-2020. Instead of requiring root-cause analysis after the second failure — which would have stopped the substitution practice and driven OEM procurement — supervision allowed the same repair to recur. The 18-Sep-2020 joint then ran from 19 Sep until it failed at 02:12:50 on 24 Sep, releasing vapour for 56 minutes before ignition at 03:08:15. This Level-3 failure permitted the lower-level wrong-choice-of-action and routine-violation unsafe acts to persist until they caused the loss of containment.

Linked Evidence: EVD-1494, EVD-8645, EVD-0591

SVI 1.8 **Failed to Identify/ Correct Risky or Unsafe Practices** **SUBSTANTIATED**

Supervision failed to require or verify a hydrostatic test after the 18-Sep-2020 gasket replacement, allowing an untested pressure boundary back into service.

JUSTIFICATION

Supervisors did not identify or correct the omission of the mandatory pressure-boundary verification after the 18-Sep-2020 gasket replacement. [EVD-2894] records: “After calibration of the meter 310FE01, the gasket was replaced on 18.09.2020. No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated. Next day meter 310FE01 was taken on online and was in use at the time of Incident” (p.31). The warning signs were visible to supervision: the SAP notification remained open ([EVD-0591]) and there was no post-maintenance pressure validation captured ([EVD-0185]). The committee explicitly links this to the probable cause: “Probable cause of leak is due to use of nonstandard gasket and not testing the system to designed pressure” (p.32). The required remedial action — a hydrostatic test to design pressure — would have revealed the inadequate seal in a controlled environment; instead no supervisor required it before the meter returned to hydrocarbon service. This supervisory failure directly enabled the lower-level AJDE 2.5 ignored-caution/warning, AJDE 2.7

CAUSAL EXPLANATION

The hydrostatic test was the final opportunity to detect the inadequate O-ring joint before hydrocarbon service. Supervisory failure to identify and correct the omission meant the defective pressure boundary entered live service on 19-Sep and operated for six days, failing at 02:12:50 on 24-Sep. The resulting 56-minute release and ignition at 03:08:15 can be traced directly to this uncorrected risky practice. Because the test requirement was neither enforced nor replaced by an SOP for high-pressure meter calibration and return-to-service, the defect passed unseen into operation.

Linked Evidence: EVD-2894, EVD-0185, EVD-0591

SVI 1.8 **Failed to Identify/ Correct Risky or Unsafe Practices** **SUBSTANTIATED**

Supervision never corrected a flame-proof lamp with three missing bolts and a glass case open 5-6 cm in a Zone 1 area; it became the ignition source at 03:08:15.

JUSTIFICATION

Supervisors failed to identify and correct the hazardous condition of the flame-proof lamp that supplied the ignition energy. [EVD-6916] records the probable ignition source: “three bolts on one of the flame proof lamps were missing resulting in it being open with its integrity compromised, glass case got detached from the main body and had opened out about 5 to 6 cms. The nearby fittings were found intact” (p.32). The same supervisory failure is manifest in the post-incident electrical PM audit, which [EVD-0496] describes as clubbing “all nine 125 W HPMV FLP lamps together and reported ‘cable burnt and fitting damaged due to heat’ as status for all of them,” while omitting the unique missing-bolt condition; the committee also finds “the electrical maintenance system is a lot more skewed towards breakdown maintenance and lays little emphasis on systematic preventive maintenance and inspection” (p.28). The missing preventive inspection and correction of a compromised Zone 1 fitting enabled the lower-level ASBE 1.2 audit-checklist failure and PTE 2.15 compromised-equipment condition. Had the lamp been restored to flame-proof

CAUSAL EXPLANATION

The three missing bolts and 5-6 cm open glass case defeated the flame-proof protection in a Zone 1 hydrocarbon area, so when the vapour cloud accumulated over 56 minutes (02:12:50 to 03:08:15) it found an uncontained electrical ignition source. Because supervision never identified or corrected this condition, the release became a fire. This Level-3 failure substantiates the lower-level audit checklist and equipment-integrity findings, and it converted a loss of containment into the major fire that destroyed the metering skid.

Linked Evidence: EVD-6916, EVD-0496

SVI 1.8 **Failed to Identify/ Correct Risky or Unsafe Practices** **SUBSTANTIATED**

A 1.03 mm thickness reduction with heavy pitting on a zero-corrosion-allowance pipe was treated with painting only; supervision never reviewed whether that correction was sufficient.

JUSTIFICATION

[EVD-2397] records: “a reduction of thickness of 1.03mm along with heavy pitting and painting peeling off was observed in one of the segments of 36 pipeline in gas terminal in July 2019. The corrective action of painting was completed in Dec. 2019. However, as per the available data, the design thickness of the pipeline is 23.83mm with zero corrosion allowance. It needs to be examined whether painting as a corrective action would suffice in such cases and the mechanism of deciding on a corrective action needs to be reviewed” (p.30). The supervisor accepted a coating restoration for metal loss on a zero-corrosion-allowance line instead of requiring fitness-for-service analysis or replacement, and no one reviewed whether that decision was adequate. This is the same supervisory tendency that accepted the local O-ring substitution as a corrective fix, and it substantiates the lower-level AJDE 2.8 wrong-choice-of-action finding for painting-only correction. It is a secondary contributor because the 36-inch pipeline survived the incident, but it illustrates the uncorrected-risky-decision culture that enabled the initiating meter failure.

CAUSAL EXPLANATION

Selecting painting in Dec 2019 left a 1.03 mm pitted section with no fitness-for-service evaluation, and supervision failed to identify the inadequacy of this corrective action. This non-engineering remedy mirrors the O-ring substitution: in both cases the visible symptom was addressed while the underlying integrity requirement was ignored. While the line survived the fire, this pattern of supervisory correction failure was a systemic enabler of the uncorrected meter gasket defect that caused the 24-Sep-2020 release.

Linked Evidence: EVD-2397

SVI 1.8 **Failed to Identify/ Correct Risky or Unsafe Practices** **SUBSTANTIATED**

Three of four diesel fire-water pumps were unavailable or unreliable, with the tripping cause unidentified and no recorded corrective actions, left uncorrected by supervision.

JUSTIFICATION

[EVD-2432] records the committee's observation: “Out of 4 diesel pumps 410 m3/hr, 2 pumps are under maintenance, one operated for 30 minutes & fourth pumps is tripping after 2 minutes. The reason for tripping is not identified or actions have not been recorded” (p.27). Supervision allowed a safety-critical backup system to remain degraded without root-cause identification or recorded corrective actions. This risky condition did not alter the 24-Sep outcome because three electrical fire-water pumps maintained tank levels and header pressure throughout, but it left the plant without a reliable diesel backup and reflects the same failure-to-correct pattern documented elsewhere. It is rated 2 for outcome impact.

CAUSAL EXPLANATION

The uncorrected diesel pump condition meant that had electrical power failed or cooling demand escalated, firefighting capacity would have been constrained. It did not influence this incident's outcome because the electric pumps performed adequately, but it reinforces the systemic supervisory failure to identify and correct known unsafe or unavailable equipment, contributing to the overall weakened safety environment at Hazira.

Linked Evidence: EVD-2432

SVI 1.9 **Selected Individual with Lack of Proficiency** **SUBSTANTIATED**

Control-room operators were assigned night-shift emergency duties with only a four-day general program and no defined qualification, leaving them non-proficient for time-critical isolation decisions.

JUSTIFICATION

Supervision selected and maintained control-room operators who were not proficient for the emergency-response role they occupied. [EVD-5634] records: “control room operations are carried out by persons inadequately trained in safety, operations, control room handling, logbook records and actions during emergency. It is informed that four days training programme is conducted for new recruitment/transfer staff which is general in nature and later the job training is carried out. Qualification of this staff is also not defined in HR/Operations Policy. Most of the staff have attended only One Basic Course since 2018” (p.30). The two engineers on duty — the Chief Engineer (Production) and Junior Engineer (Mechanical) — were the selected individuals who, at the 02:20 alarm, attempted an on-foot approach into the dense cloud and then telephoned the chain rather than initiating urgent isolation ([EVD-1460], [EVD-9220]). The committee's recommendation for at least one month of simulator-based training before control-room duties underscores how far the assigned personnel were below the required proficiency. This supervisory selection failure directly produced the

CAUSAL EXPLANATION

Because supervisors selected under-proficient individuals for the critical night-shift control-room position, the 02:20 annunciator was interpreted as a fresh event to verify on foot rather than an existing 7-minute-old major release requiring immediate isolation. The lost time fed the serial escalation: supervisor called ~02:35, MOV 54 closed at 03:01, PCV closed at 03:04, ignition at 03:08:15. This Level-3 selection failure created the conditions for the lower-level inadequate-knowledge and misapplied-policy findings, and it is one of the principal enabling causes of the 46-minute response delay that allowed the flammable cloud to accumulate.

Linked Evidence: EVD-5634, EVD-1460, EVD-9220

SVI 1.6 **Lack of Supervisory Response to Critical Information** **SUBSTANTIATED**

The 2017 UT report flagged pipes with significant thickness loss, but no corrective-action plan was developed in over three years.

JUSTIFICATION

EVD-5527 records the critical integrity information that was provided in the 2017 thickness survey and the supervisory failure to act on it: 'As per the report of 2017, several pipes have been identified with significant thickness loss and need for corrective action. However, even after three years no plan has been developed for the corrective actions to be taken for those lines. Mere measurement of thickness without corrective action does not complete the process of maintaining the integrity of pipes' (p.30). The 2017 report functioned as a formal warning to the engineering and management levels, and the evidence confirms that the supervising organisation did not close the loop for more than three years up to the 24-Sep-2020 incident. This is distinct from the metering gasket findings because it occurred on the pipeline integrity system and is grounded in a separate formal report; it is rated 3 rather than 4 because the 36-inch pipeline itself remained intact during the incident.

CAUSAL EXPLANATION

The 2017 report was a formal supervisory input that should have produced risk-based remediation planning; its three-year silence normalised the deferral of known integrity defects. Although the trunk pipeline survived the fire (EVD-5780 confirms it was intact and undamaged), the same management pattern of perceiving warnings but not acting on them was present in the meter 310FT01 recurrence, where three gasket failures were similarly not investigated. This instance therefore enabled the lower-level AJDE 2.5 ignored caution/warning on pipeline thickness (rating 3) and AJDE 2.7 delayed corrective action (rating 3), and reinforced the governance culture behind the initiating gasket failure.

Linked Evidence: EVD-5527

SVI 1.6 Lack of Supervisory Response to Critical Information

SUBSTANTIATED

Management left 12 of 308 external-audit recommendations open and extended their dates without the required Director Offshore approval.

JUSTIFICATION

EVD-6852 records that 'External safety audit was conducted in February 2018, and 296 out of 308 recommendations are complied with. Balance are in various stages and dates are extended till 30/04/2021. Document indicating approval from Director Offshore for extending the target date was not available as per the procedure' (p.26). The twelve outstanding recommendations were critical audit information delivered to the supervisory and managerial levels, yet the target-date extensions were granted without the required competent-authority approval, meaning the loop was not properly closed and the overdue items remained open without an approved basis. This is a distinct manifestation from the internal and self-audit findings because it is grounded in the February 2018 external audit report and concerns recommendation-extension governance.

CAUSAL EXPLANATION

The external audit findings were a formal supervision-level control, and the failure to close them with proper authority shows the same weak closure discipline seen in the SAP notification and the 2017 thickness findings. This supervisory pattern allowed plant-specific integrity defects to persist without management challenge, including the compromised HPMV flame-proof lamp with three missing bolts and a glass case opened 5-6 cm that became the probable ignition source at 03:08:15. It enabled the lower-level ASBE 1.2 audit checklist-not-followed finding (rating 3) and the broader PTF 1.2 audit cross-monitoring failure, although the gasket failure itself was the initiating cause, so it is rated 3.

Linked Evidence: EVD-6852

SVI 1.4 Failed to Provide Appropriate Policy/ Guidance

SUBSTANTIATED

The plant DMP/ERDMP defined hydrocarbon-leak scenarios but gave no isolation/stoppage decision criteria or authority, leaving responders without policy guidance.

JUSTIFICATION

EVD-4052 records that 'As per DMP On site Hazira Plant, guidelines to be followed in case of hydrocarbon leakage are included but does not include isolation/stoppage of facilities/pipelines' (p.32), and reinforces this at p.27: 'In ERDMP, scenario for Level 3 emergencies are taken from the plant failures. Heavy product/gas leakages from pipelines & tanks are considered but decision on isolation/closure of plant/pipelines is not mentioned.' Supervision failed to provide the emergency-operations policy defining when a hydrocarbon leak must trigger pipeline isolation and who holds the authority to order it. The evidence directly traces the committee's root cause #6 ('Non isolation/stoppage of 36 pipeline operation') to this guidance gap. The correct control is a DMP specifying quantitative or scenario-based isolation criteria and pre-delegated authority; the actual DMP named the scenario but left the decision unmade, so shift personnel had no authorised action to execute.

CAUSAL EXPLANATION

With no isolation criteria or delegated authority in the DMP, the night-shift operators and supervisors had no pre-authorized action to execute when the leak was detected at 02:15 and the alarm sounded at 02:20; they defaulted to physical verification and serial management escalation instead of immediate isolation. This is what stretched the response to 46 minutes between first detection and MOV 54 closure at 03:01, with the 36-inch PCV closed only at 03:04. The cloud was already 'vapour all around even at height' at 03:00:44 (EVD-8213) and ignited at 03:08:15. This policy failure therefore directly enabled the lower-level AJDE 2.9 decision-making under operation (rating 3), AJDE 2.10 inadequate knowledge of isolation criteria (rating 3), AJDE 2.11 unawareness of the policy gap (rating 4), PTF 1.5 roles not allocated (rating 4), and PTE 2.7 technical guidance unavailable (rating 4).

Linked Evidence: EVD-4052

SVI 1.4 **Failed to Provide Appropriate Policy/ Guidance** **SUBSTANTIATED**

Supervision provided no SOP for calibrating and returning the high-pressure condensate meters to service, so the required post-gasket hydrostatic test was never performed.

JUSTIFICATION

EVD-2894 records that 'After calibration of the meter 310FE01, the gasket was replaced on 18.09.2020. No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated. Next day meter 310FE01 was taken on online and was in use at the time of Incident' (p.31), and the same evidence quotes the committee conclusion that 'Probable cause of leak is due to use of nonstandard gasket and not testing the system to designed pressure. No standard operating procedure is in place for calibration and taking back in service of these meters which are operating at high pressures' (p.32). Supervision failed to provide the procedure that would have required a design-pressure hydrostatic test and formal acceptance before returning a pressure boundary to hydrocarbon service. The correct control is an SOP mandating hydro-test to design pressure, documented verification of gasket conformance, and supervisory sign-off; the actual practice was an operating-pressure pressurisation with no SOP at all.

CAUSAL EXPLANATION

Because there was no SOP defining the pressure-boundary validation required after gasket replacement, the maintenance crew applied only operating pressure on 18-Sep-2020 and isolated the meter, so the inadequate O-ring joint was never tested to design pressure. The unvalidated meter returned to service on 19-Sep-2020 and failed at 02:12:50 on 24-Sep-2020, releasing vapour for 56 minutes until ignition at 03:08:15. This policy gap directly enabled the lower-level ASBE 1.2 omission of the hydrostatic test checklist step (rating 4), AJDE 2.7 delayed/omitted hydrotest (rating 4), AJDE 2.11 misapplied or unaware testing policy (rating 4), and PTE 2.7 technical guidance unavailable for meter return-to-service (rating 4).

Linked Evidence: EVD-2894

SVI 1.4 **Failed to Provide Appropriate Policy/ Guidance** **SUBSTANTIATED**

The self-safety audit checklist was generic and not plant-specific, and approved manuals/SOPs were unavailable, so auditors recorded compliance without verifying plant-specific hazards.

JUSTIFICATION

EVD-9692 records that 'Safety self-audit is conducted by a multi-disciplinary team based on the checklist prepared by the location. The checklist is comprehensive but not related to specific plants and in the process, lot of observations are missing. The duration of audit is not indicated and in most of the items, compliances are indicated, but during our inspection, the data is found to be incomplete & not in line with OISD standards/manuals. Approved manuals/SOPs were not available' (p.26). The committee's recommendation B.1 at p.34 that 'Checklist for Self-Safety Audit for individual plants are to be developed' confirms the audit guidance itself was not fit for purpose. Supervision failed to provide plant-specific audit criteria and approved manuals, so the self-audit asserted compliance without verifying the plant-specific integrity conditions that later became causal.

CAUSAL EXPLANATION

Because the supervisory-provided audit checklist was generic, it did not force the audit team to inspect the Zone 1 HPMV flame-proof lamp whose three missing bolts and open glass case became the probable ignition source, nor did it expose the gasket substitution practice. The compromised lamp remained in service until the vapour cloud, accumulated over 56 minutes, reached it and ignited at 03:08:15 on 24-Sep-2020. This policy gap enabled the lower-level ASBE 1.2 audit checklist-not-followed finding (rating 3), PTF 1.2 audit cross-monitoring failure (rating 3), and PTE 2.7 self-audit guidance unavailable finding (rating 3), converting the release into a major fire.

Linked Evidence: EVD-9692

SVI 1.4 **Failed to Provide Appropriate Policy/ Guidance** **SUBSTANTIATED**

The metering-system P&ID was a lost-original, unapproved vendor drawing that did not match the field, leaving maintenance without controlled configuration guidance.

JUSTIFICATION

EVD-0933 records that 'Reference drawings no. 02-310-AO-101 Rev.1, 2x6 lines & 1x2 lines for metering system are shown but valves & meters configuration are not matching. As old drawings were lost, new drawings are made by a vendor but the same has not been checked & approved by ONGC' (p.26). Supervision therefore failed to maintain or provide a controlled, approved as-built drawing for the condensate metering system, leaving maintenance without an authoritative configuration reference. The correct control is a maintained and approved as-built drawing verified against the field; the actual drawing was a lost-original, unapproved vendor product that did not match the installed valves and meters. This is a real policy gap, but the physical failure was driven by the wrong gasket and omitted hydrotest rather than by the drawing mismatch alone, so it is rated present with no direct outcome impact.

CAUSAL EXPLANATION

Without an approved as-built drawing, maintenance personnel had no controlled configuration baseline to verify the correct rectangular-slot gasket design or the meter flange details, which reinforced the environment in which the OEM specification was lost and the O-ring substitution went unchallenged. This gap supported the lower-level AV 1.1 routine configuration-management breakdown (rating 2) and PTE 2.14 unfamiliar equipment/noncustodial operations finding (rating 2), but it did not directly initiate the 24-Sep-2020 release, so it is rated 2.

Linked Evidence: EVD-0933

SVI 1.10 **Failed to Communicate Intent** **SUBSTANTIATED**

No communicated SOP required a design-pressure hydrotest after the 18-Sep-2020 gasket replacement; the meter was returned to service after operating-pressure only.

JUSTIFICATION

The maintenance supervision chain did not formally communicate the intended post-maintenance testing sequence for a high-pressure meter after gasket replacement. [EVD-2894] records that after the 18.09.2020 gasket change 'No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated. Next day meter 310FE01 was taken on online and was in use at the time of Incident' (p.31), and the committee directly found that 'No standard operating procedure is in place for calibration and taking back in service of these meters which are operating at high pressures' (p.32). [EVD-8645] confirms that the original OEM gasket specification was lost and no substitute specification was communicated. The intent that the pressure boundary had to be proved by a design-pressure hydrostatic test before return to service was therefore never transmitted to the fitters; the actual practice stopped at an operating-pressure soak with both meters in line. This instance is distinct from the emergency-response and material-substitution instances because it concerns the return-to-service intent for the specific

CAUSAL EXPLANATION

With no formal statement of the required design-pressure test, the maintenance fitter repeated the established but inadequate practice and returned meter 310FT01 to service on 19-Sep-2020 with an unvalidated pressure boundary. The defective O-ring joint then ran for six days under live condensate pressure and failed at 02:12:50 on 24-Sep-2020, initiating the 56-minute vapour accumulation that ignited at 03:08:15. This Level-3 non-communication directly enabled the lower-level findings ASBE 1.2 (checklist not followed - hydrotest omitted), AJDE 2.7 (delayed/omitted necessary action), AJDE 2.11 (misapplied or unaware of existing policy), and PTF 1.13 (mission planning failure) by removing the formal instruction that should have subjected the substitute O-ring to design pressure before it entered hydrocarbon service.

Linked Evidence: EVD-2894, EVD-8645

SVI 1.10 **Failed to Communicate Intent** **SUBSTANTIATED**

Dec-2019 calibration setpoint changes and reversals were never communicated through the signed calibration documents or a checklist.

JUSTIFICATION

Calibration supervision failed to communicate a material setpoint decision to the team and the formal record. [EVD-2506] records that 'During calibration from 17/12/2019 to 18/12/2019, calibration was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of this change in the calibration documents signed by the team. During next calibration for meter 310FT01 was again changed to 16000 MMWC. No checklist is followed while carrying such critical jobs of high pressure/risk' (p.27). The supervisor's intended rationale, authority, and impact of the approximately 6.3x setpoint reduction and the later reversion were never put into the signed calibration certificate, and the committee's recommendation D.1.i later explicitly required such deviations to be documented in the certificate. This missing communication of intent left the meter's measurement baseline ungoverned and normalized undocumented changes on the same asset that later became the leak source.

CAUSAL EXPLANATION

The uncommunicated setpoint changes on meter 310FT01 in December 2019 did not directly open the leak, but they reinforced the same supervision pattern on the same asset: high-pressure maintenance was executed without documented intent, checklist direction, or supervisory challenge. This enabled the lower-level findings ASBE 1.2 (calibration checklist not followed), ASBE 1.5 (under-controlled calibration), and AJDE 2.10 (inadequate calibration knowledge) and contributed to an integrity-management environment in which the later 18-Sep-2020 gasket substitution and omitted hydrotest could proceed unchallenged.

Linked Evidence: EVD-2506

SVI 1.11 **Support Not Provided by Supervisor** **SUBSTANTIATED**

Control-room operators received only four days of general training with no defined qualification and no simulator mentoring, leaving them unsupported for emergency isolation decisions.

JUSTIFICATION

Supervisors did not provide adequate mentoring, coaching, or structured development for control-room operators, leaving them unprepared for emergency isolation decisions. [EVD-5634] records that 'control room operations are carried out by persons inadequately trained in safety, operations, control room handling, logbook records and actions during emergency. It is informed that four days training programme is conducted for new recruitment/transfer staff which is general in nature and later the job training is carried out. Qualification of this staff is also not defined in HR/Operations Policy. Most of the staff have attended only One Basic Course since 2018' (p.30). The committee subsequently recommended at least one month of training, preferably on simulators, before assigning control-room duties. This instance is distinct from the maintenance-support instances because it concerns the development support provided to the operating crew that had to manage the 24-Sep-2020 emergency.

CAUSAL EXPLANATION

Because their supervisors had not mentored or qualified them beyond a general four-day course with no simulator practice, the two engineers who received the 02:20 alarm and phone call had no drilled emergency-isolation response to fall back on. They attempted on-foot source verification, were forced back by the dense vapour cloud, and then resorted to telephone escalation rather than closing available isolation promptly; MOV 54 was not closed until 03:01 and the accumulated cloud ignited at 03:08:15. This Level-3 support gap directly enabled the lower-level findings PTF 1.17 (team members unfamiliar with workspace), AJDE 2.10 (knowledge/experience inadequate to make decisions), AJDE 2.11 (misapplied/unaware of policy), and PSE 3.1 (situation not trained for).

Linked Evidence: EVD-5634

SVI 1.11 **Support Not Provided by Supervisor** **SUBSTANTIATED**

Electrical supervisors provided no effective preventive-maintenance support for the nine HPMV FLP lamps, and the compromised lamp with three missing bolts and open glass case remained as the probable ignition source.

JUSTIFICATION

Supervisory support for systematic preventive maintenance of flame-proof electrical fittings was absent, leaving the compromised ignition-source lamp in service. [EVD-0496] records that the post-incident PM audit 'clubbed all nine 125 W HPMV FLP lamps together and reported cable burnt and fitting damaged due to heat as status for all of them' without noting the unique condition of one lamp, and states that 'the electrical maintenance system is a lot more skewed towards breakdown maintenance and lays little emphasis on systematic preventive maintenance and inspection' (p.28). [EVD-9999] confirms the audit report was 'too generic and short (One page)' and lacked per-equipment detail. The consequence is documented in [EVD-6916]: 'three bolts on one of the flame proof lamps were missing resulting in it being open with its integrity compromised, glass case got detached from the main body and had opened out about 5 to 6 cms' (p.32). Electrical supervisors did not provide the PM support that would have identified and corrected this Zone 1 integrity loss before it could act as an ignition source.

CAUSAL EXPLANATION

Without supervisory drive for systematic preventive maintenance and inspection, the missing bolts and open glass case on the HPMV flame-proof lamp persisted undetected in a Zone 1 hydrocarbon area. When the vapour cloud released by the failed meter gasket reached the lamp at 03:08:15, the exposed contacts provided the probable ignition energy that converted the release into a major fire. This Level-3 support failure enabled the lower-level findings ASBE 1.2 (audit checklist not followed correctly), PPI 5.6 (inaccurate expectancy), and PTF 1.2 (cross-monitoring performance failure), and determined the ignition transition of the incident.

Linked Evidence: EVD-0496, EVD-9999, EVD-6916

SVI 1.7 **Lack of Standardisation** **SUBSTANTIATED**

Calibration setpoint changes of 16000!'2540!'16000 mmwc on meter 310FT01 were made witho signed record, showing inconsistent documentation rules for the same critical job.

JUSTIFICATION

[EVD-2506] shows the same meter was treated inconsistently from one calibration event to the next: “During calibration from 17/12/2019 to 18/12/2019, calibration was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of this change in the calibration documents signed by the team. During next calibration for meter 310FT01 was again changed to 16000 MMWC. No checklist is followed while carrying such critical jobs of high pressure/risk” (p.27). A setpoint was changed roughly 6.3-fold and then changed back, with neither event recorded, because no standardized checklist or documentation rule was applied to this critical high-pressure job. This is a distinct manifestation from the gasket and DMP instances: it concerns calibration documentation on the same meter in December 2019, nine months before the failure. It shows supervisory staff accepted differing documentation practices for identical work on the same equipment, normalising procedural shortcuts; this enabled the lower-level ASBE 1.2 calibration-checklist and ASBE 1.5 under-controlled-calibration findings, and contributed to the normalisation of undocumented changes on the meter that

CAUSAL EXPLANATION

The 16000!'2540!'16000 mmwc sequence created measurement uncertainty and demonstrate workforce that critical changes could be made without a standardized record or checklist. That normative shift was an enabling condition for the later 18-Sep-2020 gasket replacement, where the hydrostatic test was omitted without being caught by any checklist or supervisory review. The unchecked calibration instability on the same meter also degraded its reliability and contributed to the recurrence pattern that culminated in the 24-Sep-2020 release.

Linked Evidence: EVD-2506

SVI 1.14 **Supervisor Training Inadequate** **SUBSTANTIATED**

On-shift supervisors received only 4 days of general control-room training with no simulator qualification, leaving them unprepared to direct immediate isolation during the release.

JUSTIFICATION

EVD-5634 states that 'control room operations are carried out by persons inadequately trained in safety, operations, control room handling, logbook records and actions during emergency', with only a four-day general training programme and no simulator training; the committee recommended at least one month on simulators before assigning control room duties. The supervisors' performance on 24-Sep-2020 matches this gap: EVD-9612 shows the GSP Shift In-charge had to physically reach the Gas Terminal Gate and closed MOV 54 only at 03:01, 46 minutes after first detection, and EVD-7340 shows the Overall Shift In-charge's instructions covered the 36-inch PCV and notifications rather than an immediate, comprehensive isolation order. These senior on-shift supervisors were not trained to execute or direct rapid emergency isolation, which directly enabled the lower-level AJDE 2.10 inadequate knowledge to make a decision and AJDE 2.11 unawareness of the applicable emergency policy.

CAUSAL EXPLANATION

Because the supervisors lacked trained, rehearsed emergency isolation actions, the night-shift response defaulted to physical verification and management escalation rather than closure of available upstream valves. Leak onset was at 02:12:50, alarm at 02:20, senior arrival at ~02:35, MOV 54 closure at 03:01 and PCV closure at 03:04, with ignition at 03:08:15. This supervision training deficiency is the Level-3 enabler of the lower-level AJDE 2.10/2.11 decision failures and PTF 1.3 delegation to under-trained staff.

Linked Evidence: EVD-5634, EVD-9612, EVD-7340

SVIO 2.1 **Ordered/ Led on Mission Beyond Capability** **SUBSTANTIATED**

A four-day generally trained CR crew was assigned an emergency isolation mission it could not execute, with only local manual MOVs and no practiced emergency response.

JUSTIFICATION

Supervision led the Gas Terminal Control Room night-shift operators into an emergency-response role that exceeded both their training and their equipment capability. [EVD-5634] records on p.30 that 'control room operations are carried out by persons inadequately trained in safety, operations, control room handling, logbook records and actions during emergency. It is informed that four days training programme is conducted for new recruitment/transfer staff which is general in nature and later the job training is carried out. Qualification of this staff is also not defined in HR/Operations Policy. Most of the staff have attended only One Basic Course since 2018', while the committee's recommendation is 'at least 1 month (preferably on simulators)' before assigning control-room duties. The rostered team on 23-24 Sep 2020 was a Chief Engineer (Production) and a Junior Engineer (Mechanical) [EVD-1460, EVD-9220], and at 02:20 they were expected to respond to a confirmed vapour-cloud notification with only this four-day general programme. In addition, [EVD-3932] confirms that 'MOVs are provided in the gas terminal station for local operations but they are not having

CAUSAL EXPLANATION

When the annunciator alarm and CCTV telephone call arrived at 02:20, the Chief Engineer and Junior Engineer treated the event as a source-location task rather than an isolation action, consistent with their lack of practiced emergency training. They went on foot, were forced back by dense vapour, and then telephoned the GSP/LPG shift in-charges at ~02:35. The GSP Shift In-charge closed MOV 54 only at 03:01, 46 minutes after first physical detection at 02:15 and 48 minutes after leak onset at 02:12:50, and the 36-inch PCV followed at 03:04. By 03:00:44 the cloud was 'vapour all around even at height' [EVD-8213], and it ignited at 03:08:15. This supervision-planned mismatch therefore produced the Level-1 decision failures (AJDE 2.10, AJDE 2.9) and the Level-2 competence and manual-valve preconditions (PTF 1.3, PTE 2.8).

Linked Evidence: EVD-5634, EVD-3932, EVD-1460

SVIO 2.1 **Ordered/ Led on Mission Beyond Capability** **SUBSTANTIATED**

The 18-Sep-2020 gasket job was led with no OEM specification, no conforming gasket, no quality control and no hydrotest, exceeding the material and testing capability available to the crew.

JUSTIFICATION

Supervision planned and led the 18-Sep-2020 gasket replacement on meter 310FT01 as a mission that exceeded the available equipment and material capability. [EVD-8645] records on p.31 that 'The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket'; the committee consequently identified root cause #2 'Use of nonstandard O ring' and root cause #3 'Lack of control over material sourcing'. [EVD-2894] adds that after the 18.09.2020 replacement 'No hydro test was done... The system was only pressurised to operating pressure', so the job was not merely under-specified but lacked the test equipment and test plan required to validate the pressure boundary. The correct planning would have recovered the OEM rectangular-slot specification, procured

CAUSAL EXPLANATION

The rectangular-slot flange requires a flat slot-filling gasket; the round O-ring substituted could not fill the slot, leaving a leak path. The joint was returned to service on 19-Sep-2020 after only an operating-pressure pressurisation, and it failed at 02:12:50 on 24-Sep-2020, releasing vapour for 56 minutes until ignition at 03:08:15. The recurrence record [EVD-1494] shows the same incorrect material choice had already been made on 12-Nov-2018, 25-Feb-2019 and 18-Sep-2020, so this was a sustained supervision-led practice rather than an isolated mistake. This Level-3 finding directly underlies the Level-1 wrong-choice (AJDE 2.8) and routine-substitution (AV 1.1) findings and the Level-2 lost-technical-guidance and defective-equipment preconditions (PTE 2.7, PTE 2.5).

Linked Evidence: EVD-8645, EVD-2894, EVD-1494

SVIO 2.2 **Inappropriate Crew / Team/ Flight Composition** **SUBSTANTIATED**

The two-person night-shift CR team contained no member with simulator-practiced emergency-isolation competence, leaving the first response entirely unqualified.

JUSTIFICATION

The night-shift team composition at the Gas Terminal Control Room should have raised safety concerns because both rostered members came from a training population the committee found inadequate for emergency operations. [EVD-5634] states that 'control room operations are carried out by persons inadequately trained in safety, operations, control room handling, logbook records and actions during emergency... four days training programme is conducted for new recruitment/transfer staff which is general in nature... Most of the staff have attended only One Basic Course since 2018', with no defined qualification in HR/Operations Policy and no simulator training. The team assembled a Chief Engineer (Production) and a Junior Engineer (Mechanical) [EVD-1460, EVD-9220], so neither member supplied the missing emergency-isolation competence and no third member was present to compensate. This skill composition mapped directly onto the 02:20 response, where both engineers tried to approach the vapour cloud on foot and then only escalated at ~02:35. The correct composition would have included at least one operator current in emergency isolation

CAUSAL EXPLANATION

Because no member of the two-person night crew possessed practiced emergency isolation competence, the alarm at 02:20 produced foot-based source location rather than immediate closure of the available 36-inch PCV or direction to close MOV 54. Escalation consumed the interval from 02:20 to 02:35, the GSP Shift In-charge reached the gate at ~02:35 and closed MOV 54 only at 03:01, and the accumulated cloud ignited at 03:08:15. This compositional failure connects causally to the Level-1 inadequate-knowledge and operational-decision findings (AJDE 2.10, AJDE 2.9) and to the Level-2 task-delegation precondition (PTF 1.3).

Linked Evidence: EVD-5634, EVD-1460, EVD-9220

SVIO 2.2 **Inappropriate Crew / Team/ Flight Composition** **SUBSTANTIATED**

The repair team had no material inspector, no hydrotest witness and no closing supervisor, leaving the pressure boundary dependent on a single fitter's judgement.

JUSTIFICATION

The maintenance team assigned to the 18-Sep-2020 pressure-boundary repair was composed without any quality-control or engineering-check role. [EVD-8645] records that there was 'no specification or quality control during the purchase of the O rings' and that the OEM rectangular-slot specification was unavailable; [EVD-2894] records that after replacing the gasket 'No hydro test was done after replacement of the gasket'; and [EVD-0591] records that SAP Notification 3800419840 raised on 17.09.2020 for checking meter 1 'was not closed after completion of the job'. With no material inspector, no test witness, and no closing supervisor in the team, the only control left was the executing fitter's individual judgement, which the 22-month recurrence history [EVD-1494] shows had already failed three times (12-Nov-2018, 25-Feb-2019, 18-Sep-2020). A suitable team for a high-pressure condensate pressure boundary would have included a materials/QA checker, a test engineer to hydrotest to design pressure, and a supervisory signatory linked to notification closure; the actual team had none. This inappropriate composition is the Level-3 origin of the

CAUSAL EXPLANATION

Without a QC or engineering member in the team, the locally procured O-ring was accepted without material conformance, the hydrostatic test was omitted, and the notification was left open, removing the last supervisory gate. Meter 310FT01 returned to service on 19-Sep-2020 with an unvalidated pressure boundary, operated through 24-Sep-2020, and failed at 02:12:50; the resulting vapour accumulated for 56 minutes and ignited at 03:08:15. This Level-3 composition failure enabled the Level-1 wrong material choice (AJDE 2.8), the routine substitution (AV 1.1), and the omitted checklist step (ASBE 1.2 rating 4).

Linked Evidence: EVD-8645, EVD-2894, EVD-0591

SVIO 2.3 **Selected Individual with Lack of Current or Limited Experience** **SUBSTANTIATED**

The senior night-shift production operator was selected without current emergency training, and his 02:20 response showed no practiced isolation action.

JUSTIFICATION

The Chief Engineer (Production) selected for the 24-Sep-2020 night shift in the Gas Terminal Control Room lacked current emergency experience. [EVD-5634] establishes the population's limited access to training: 'Most of the staff have attended only One Basic Course since 2018', with only a four-day general programme for new/transfer staff and no defined qualification in HR/Operations Policy. [EVD-1460] places this specific individual in the response: at 02:20 he heard the annunciator and the CCTV telephone call, went outside and saw the heavy cloud, but could not approach due to dense vapour, then telephoned the GSP Shift In-charge, LPG Shift In-charge and CFU Panel Engineer. The behaviour is consistent with an individual whose emergency-response experience was not current: no immediate isolation order and no use of a pre-planned remote-action protocol. The correct selection practice would verify recent simulator or emergency-event exposure before assigning a night-shift control-room position; the actual rostering accepted him on a shift where he was the senior production operator at the emergency centre. This selection is therefore a Level-3

CAUSAL EXPLANATION

Because this selected operator did not immediately direct available isolation or dispatch an isolation action, the critical interval 02:20-02:35 passed with only attempted foot approach and telephone calls. The downstream shift-in-charge arrived at ~02:35, closed MOV 54 at 03:01 and the 36-inch PCV at 03:04; ignition occurred at 03:08:15. This selection failure connects to the Level-1 misinterpretation and delayed-action findings (AJDE 2.4, AJDE 2.7) and the Level-2 task-delegation precondition (PTF 1.3).

Linked Evidence: EVD-5634, EVD-1460

SVIO 2.3 **Selected Individual with Lack of Current or Limited Experience** **SUBSTANTIATED**

The second night-shift CR operator was selected with the same limited emergency training record, leaving the crew with no current emergency operator.

JUSTIFICATION

The Junior Engineer (Mechanical) rostered alongside the Chief Engineer had the same limited experience baseline. [EVD-9220] records that at 02:20 he 'heard alarm and phone call simultaneously. He went out of the control room and observed leakage and intense smell from Phase-I area. Visibility was very poor'; he joined the in-progress response rather than initiating an independent one. With no defined qualification in HR/Operations Policy [EVD-5634] and no simulator exposure, his selection for night-shift emergencies left the two-person crew with no practiced emergency operator. This is a separate individual manifestation from the Chief Engineer instance because it concerns a different role and a different witness statement, although it shares the same training-system evidence.

CAUSAL EXPLANATION

Because this second selected operator also lacked current emergency experience, the team had no redundancy of competence when the alarm sounded at 02:20. He followed the Chief Engineer into an unsuccessful foot approach and did not initiate isolation, so the response remained dependent on telephone escalation until MOV 54 was closed at 03:01, 46 minutes after first detection at 02:15. This individual selection failure contributed to the same Level-1 decision failures (AJDE 2.7, AJDE 2.9) and to the Level-2 team-unfamiliarity precondition (PTF 1.17).

Linked Evidence: EVD-9220, EVD-5634

SVIO 2.3 Selected Individual with Lack of Current or Limited Experience

SUBSTANTIATED

The fitter assigned to the 18-Sep-2020 repair was current only in the incorrect O-ring practice, with no demonstrated currency against the OEM rectangular-slot design.

JUSTIFICATION

The individual selected to execute the 18-Sep-2020 gasket replacement had current experience only with the wrong substitute practice, not with the correct OEM rectangular-slot gasket. [EVD-8645] records the original 1986 gasket specification was not available and local O-rings were used 'as a regular practice' with no specification or QC; [EVD-1494] documents that cleaning/servicing/calibration was carried out on 12-13 Nov 2018, another gasket change on 25-Feb-2019, and the third on 18-Sep-2020, so the selected maintenance personnel had repeatedly performed the same non-conforming repair without being re-qualified against the true design. Selecting an individual with this limited, incorrect experience to return a high-pressure pressure boundary to service was inappropriate; the correct selection would have required demonstrated currency against the rectangular-slot OEM design and the hydrotest procedure. This is the Level-3 selection root for the initial loss of containment.

CAUSAL EXPLANATION

The selected fitter installed a round O-ring into a rectangular-slot flange because his available reference was the repeated local-substitute practice, not the OEM design. No hydrotest was performed, and the meter returned to service on 19-Sep-2020; the inadequate seal failed at 02:12:50 on 24-Sep-2020, releasing vapour for 56 minutes until ignition at 03:08:15. This Level-3 selection failure underlies the Level-1 wrong-choice (AJDE 2.8), the procedure-not-followed (ASBE 1.3) and knowledge-inadequacy (AJDE 2.10) findings.

Linked Evidence: EVD-8645, EVD-1494

SVI 1.8 Failed to Identify/ Correct Risky or Unsafe Practices

SUBSTANTIATED

Pipes with significant 2017-measured thickness loss had no corrective-action plan for over three years, showing supervision accepted identified degradation without remediation.

JUSTIFICATION

[EVD-5527] states: “As per the report of 2017, several pipes have been identified with significant thickness loss and need for corrective action. However, even after three years no plan has been developed for the corrective actions to be taken for those lines. Mere measurement of thickness without corrective action does not complete the process of maintaining the integrity of pipes” (p.30). Supervision had an inspection-based warning identifying degraded piping yet failed to institute any remedial action plan over three years, exactly the SVI 1.8 definition. [EVD-8307] further shows the four-year thickness-measurement schedule was itself breached, with measurements in 2011 and 2017 rather than on the mandated cycle. This instance did not cause the 24-Sep-2020 fire because the 36-inch pipeline remained intact, but it demonstrates a systemic supervisory tolerance for uncorrected integrity findings, which is the same culture that allowed the meter gasket defects to persist. It is a secondary contributing factor that substantiates the lower-level AJDE 2.5 ignored-warning and AJDE 2.7 delayed-action findings.

CAUSAL EXPLANATION

Treating identified pipe degradation as a measurement exercise rather than an intervention demonstrates an organisational failure to convert findings into corrections. This same pattern appeared on meter 310FT01, where recurrence findings were collected but never acted upon. The uncorrected pipeline conditions remained latent, and the behaviour normalised non-action on integrity warnings, contributing to the environment in which the gasket defect went uncorrected until 24-Sep-2020. Its direct causal share is secondary because the 36-inch line did not fail.

Linked Evidence: EVD-5527, EVD-8307

SVI 1.8 **Failed to Identify/ Correct Risky or Unsafe Practices** **SUBSTANTIATED**

Supervision signed off calibration work in which meter 310FT01's setpoint was changed 160 record or checklist, and did not correct the practice.

JUSTIFICATION

Supervisors signed calibration documents that omitted the setpoint changes and did not require correction or engineering justification. [EVD-2506] records: “During calibration from 17/12/2019 to 18/12/2019, calibration was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of this change in the calibration documents signed by the team. During next calibration for meter 310FT01 was again changed to 16000 MMWC. No checklist is followed while carrying such critical jobs of high pressure/risk” (p.27). The supervisory signatory with approval authority did not identify the undocumented 6.3-fold change or require a checklist-based correction. This failure to correct risky calibration practice on the same meter later normalised checklist bypass, contributing to the Sep-2020 hydrotest omission. It substantiates the lower-level AJDE 2.10 inadequate-knowledge, ASBE 1.2 checklist-not-followed and ASBE 1.5 under-controlled-calibration findings, but is rated 3 because it did not directly open the leak.

CAUSAL EXPLANATION

The uncorrected setpoint change degraded meter 310FT01's measurement reliability and demonstrated that critical high-pressure jobs could be executed without checklist or documentation. That norm carried into the 18-Sep-2020 gasket replacement, where the hydrostatic test was skipped with no supervision catching it, ultimately allowing the defective joint to fail at 02:12:50 on 24-Sep and ignite at 03:08:15. Because the calibration error did not itself fail the flange, it is a secondary supervisory failure.

Linked Evidence: EVD-2506

SVI 1.10 **Failed to Communicate Intent** **SUBSTANTIATED**

No DMP criteria or standing orders communicated when and who should isolate on confirmed leak, forcing escalation that delayed MOV 54 closure by 46 minutes.

JUSTIFICATION

Supervisory intent for the emergency isolation task was never formally communicated: the written disaster-management plan named hydrocarbon leakage as a scenario but omitted the isolation/stoppage decision criteria and authority. [EVD-4052] records that 'guidelines to be followed in case of hydrocarbon leakage are included but does not include isolation/stoppage of facilities/pipelines' (p.32) and that ERDMP Level-3 scenarios considered heavy product/gas leakages but 'decision on isolation/closure of plant/pipelines is not mentioned' (p.27). Because no standing order or night order communicated to the Gas Terminal Control Room that a confirmed vapour cloud had to trigger immediate isolation, the Chief Engineer and Junior Engineer defaulted to on-foot source verification after the 02:20 annunciator alarm and then to telephone escalation. [EVD-3932] documents the resulting authority vacuum: no role between Shift In-charge and Plant Manager had standing authority to close an emergency-isolation valve without telephoning up the chain, and the MOVs were local-manual only. The consequence is anchored in [EVD-6414], which records leak onset at 02:12:50, first

CAUSAL EXPLANATION

Because no supervisor or management system had communicated the intent that a major vapour-cloud report or alarm must trigger immediate isolation without waiting for senior approval, the night shift had no pre-authorized action to execute. The sequence shows the cost of that gap: leak onset at 02:12:50, first human detection at 02:15, control-room notification at 02:18-02:20, shift-in-charge arrival at the gate at ~02:35, and first isolation only at 03:01, with ignition at 03:08:15 only seven minutes later. This Level-3 omission directly created the lower-level findings AJDE 2.7 (delayed necessary action), AJDE 2.9 (decision-making during operation), AJDE 2.10 (inadequate knowledge of isolation criteria), ASBE 1.3 (delayed emergency isolation), PTF 1.5 (roles/responsibilities not allocated), and PTF 1.13 (mission planning failure) by leaving the most time-critical emergency decision to an undocumented, authority-gated escalation chain while the vapour cloud grew continuously.

Linked Evidence: EVD-4052, EVD-3932, EVD-6414

SVIO 2.4 **Risk Assessment - Formal** **SUBSTANTIATED**

The high-risk gasket replacement was executed via an unreviewed SAP notification with no JSA, no permit and no SOP, so the substitute O-ring and the omitted hydrotest were never controlled by a formal risk assessment.

JUSTIFICATION

Supervision ordered the 18-Sep-2020 gasket replacement on meter 310FT01 — pressure-boundary work on a 1986 high-pressure hydrocarbon meter — without any adequate formal risk assessment. [EVD-5105] records that "Job safety Plan is not attached/given in SAP document" and that original permits were missing with "No data... kept on such missing permits"; the committee's recommendation B.5 at p.34 is explicit that "Maintenance notifications are not be assumed as permit to work", meaning the job was executed under SAP notification 3800419840 without a separate risk-assessed permit. [EVD-2894] confirms the absence of any governing risk-evaluated procedure: "No standard operating procedure is in place for calibration and taking back in service of these meters which are operating at high pressures" (p.32), and the system was "only pressurised to operating pressure" after the repair. A competent formal risk assessment would have identified the lost OEM rectangular-slot gasket specification, the locally procured O-ring substitute, and the mandatory design-pressure hydrostatic test as controls; none of these were prescribed because the

CAUSAL EXPLANATION

Because no JSA, permit review, or SOP existed for this job, the maintainer installed an uncontrolled locally purchased O-ring into a rectangular-slot flange and no design-pressure hydrotest was required or performed. The meter returned to service on 19-Sep-2020 and operated for six days before the seal failed at 02:12:50 on 24-Sep-2020, releasing the vapour that accumulated for 56 minutes and ignited at 03:08:15. This Level-3 supervision gap directly enabled the Level-1 findings ASBE 1.2 checklist-not-followed, AJDE 2.7 omission of the hydrostatic test, and AJDE 2.11 misapplication of the testing policy, and it supplies the supervisory origin for the Level-2 PTF 1.13 mission-planning failure.

Linked Evidence: EVD-5105, EVD-2894

SVIO 2.4 **Risk Assessment - Formal** **SUBSTANTIATED**

The O-ring-for-OEM-gasket substitution was never processed through the Plant's MOC risk assessment, which had not been applied to any change since 2018, leaving the non-conforming seal unevaluated.

JUSTIFICATION

The change from the OEM rectangular-slot gasket to a locally purchased O-ring was a material and geometry change to a hydrocarbon pressure boundary that should have been processed through the Management of Change formal risk assessment program. [EVD-8645] documents that "The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket." [EVD-2986] establishes that the risk-assessment program was effectively abandoned: the MOC SOP was published in August 2016 but "the procedure indicated in the document is not implemented" and "no more MOCs are implemented in the plant since 2018." The MOC tool is precisely the formal risk evaluation that would have challenged the substitution on pressure/temperature, chemical compatibility, and

CAUSAL EXPLANATION

MOC is the plant's formal risk screen for substitutions that could degrade process safety; because no MOC was raised, the round O-ring was installed in a rectangular sealing groove without any evaluation of sealing mechanics or material suitability. This supervisory omission enabled the lower-level findings AJDE 2.8 wrong choice of gasket, AV 1.1 routine substitution violation, and ASBE 1.3 procedure not followed correctly. The defective joint was installed on 18-Sep-2020, returned to service on 19-Sep, and failed on 24-Sep at 02:12:50, producing the release that ignited at 03:08:15. The same unevaluated-substitution risk assessment gap was repeated across three replacements in 22 months (12-Nov-2018, 25-Feb-2019, 18-Sep-2020).

Linked Evidence: EVD-8645, EVD-2986

SVIO 2.4 **Risk Assessment - Formal** **SUBSTANTIATED**

The formal DMP/ERDMP risk assessment for hydrocarbon leakage omitted isolation/stoppage decision criteria and authority, leaving responders without a pre-assessed control action.

JUSTIFICATION

The plant's formal emergency risk-assessment document named the hydrocarbon-leakage scenario but did not evaluate or prescribe the isolation/stoppage decisions required to control it. [EVD-4052] records that "As per DMP On site Hazira Plant, guidelines to be followed in case of hydrocarbon leakage are included but does not include isolation/stoppage of facilities/pipelines" (p.32), reinforced at p.27: "Heavy product/gas leakages from pipelines & tanks are considered but decision on isolation/closure of plant/pipelines is not mentioned." A formal risk assessment of the major-leak scenario should have predetermined the trigger criteria, the valves to close, the authority to act, and the maximum allowable delay; none of these controls existed. This is not merely a documentation gap — it is a failed pre-mission risk evaluation for a known Level-3 emergency, and the committee's root cause #6 ("Non isolation/stoppage of 36 pipeline operation") derives from it.

CAUSAL EXPLANATION

Because the formal risk assessment never specified isolation decision criteria, the night crew had no pre-authorized protective action at first detection (02:15) or first annunciator alarm (02:20). Personnel defaulted to physical verification and serial management escalation, so MOV 54 was not closed until 03:01 and the 36-inch PCV not until 03:04, 46 and 49 minutes after first detection. The vapour released during that period was visible as a mature cloud at 03:00:44 (EVD-8213) and ignited at 03:08:15. This Level-3 failure caused the lower-level findings AJDE 2.9 decision-making during operation, AJDE 2.10 inadequate knowledge of isolation criteria, AJDE 2.11 unawareness of the policy gap, ASBE 1.3 delayed isolation, and the Level-2 PTF 1.13 planning failure.

Linked Evidence: EVD-4052, EVD-6414

SVIO 2.4 **Risk Assessment - Formal** **SUBSTANTIATED**

The self-safety audit risk-assessment checklist was generic and not plant-specific, missing the hazardous-area electrical-integrity defect that became the probable ignition source.

JUSTIFICATION

The plant's formal self-safety audit instrument was not tailored to the specific hazards of the Gas Terminal metering area, so it could not direct auditors to the plant-specific integrity conditions that later became causal. [EVD-9692] states: "The checklist is comprehensive but not related to specific plants and in the process, lot of observations are missing. The duration of audit is not indicated and in most of the items, compliances are indicated, but during our inspection, the data is found to be incomplete & not in line with OISD standards/manuals." The committee's recommendation B.1 at p.34 calls for a "Checklist for Self-Safety Audit for individual plants are to be developed" — acknowledging the risk-assessment tool itself was unfit for purpose. A plant-specific checklist would have forced inspection of Zone-1 flameproof fitting integrity, which [EVD-6916] shows was compromised on the HPMV lamp later identified as the probable ignition source ("three bolts... missing... glass case got detached... opened out about 5 to 6 cms").

CAUSAL EXPLANATION

Because the formal audit tool was generic, it did not force verification of the HPMV FLP lamp integrity in the metering blast zone, allowing a non-flameproof fitting to remain in service in a Zone 1 hydrocarbon area. When the vapour cloud from the failed 310FT01 seal reached the exposed contacts at 03:08:15, the release transitioned into a major fire. This supervision-level risk-tool failure enabled the Level-1 ASBE 1.2 audit checklist failure and the Level-2 PTF 1.2/PSE 5.6 inaccurate-expectancy preconditions, making the ignition-source condition a direct downstream consequence.

Linked Evidence: EVD-9692, EVD-6916

SVIO 2.4 **Risk Assessment - Formal** **SUBSTANTIATED**

Inspection findings on pipeline thickness loss were not processed through a formal risk assessment for three years, and a zero-corrosion-allowance pitting defect was addressed with painting only.

JUSTIFICATION

Supervision did not apply a formal risk assessment to the integrity findings produced by the pipeline thickness measurement program, leaving known degradation unactioned for three years. [EVD-5527] records: "As per the report of 2017, several pipes have been identified with significant thickness loss and need for corrective action. However, even after three years no plan has been developed for the corrective actions to be taken for those lines. Mere measurement of thickness without corrective action does not complete the process of maintaining the integrity of pipes" (p.30). [EVD-2397] further shows the risk-evaluation mechanism was weak for the July 2019 spot finding of 1.03 mm thickness reduction with "heavy pitting and painting peeling off" on a line with design thickness 23.83 mm and "zero corrosion allowance", noting that "the mechanism of deciding on a corrective action needs to be reviewed." The supervision failure is not the absence of the raw inspection data but the absence of a formal risk-based evaluation and action plan after the data highlighted defects.

CAUSAL EXPLANATION

This pipeline-integrity risk assessment gap did not directly cause the 24-Sep-2020 fire because the 36-inch trunk line remained intact, but it is a substantiated supervisory risk-management failure that normalised acknowledging findings without risk-based action. It enabled the lower-level findings AJDE 2.5 ignored warning, AJDE 2.7 delayed necessary action, and AJDE 2.8 wrong choice of corrective action, and it demonstrates the same supervision culture that allowed the meter 310FT01 sealing defect to remain unanalysed until it failed at 02:12:50 on 24-Sep-2020.

Linked Evidence: EVD-5527, EVD-2397

SVIO 2.5 **Authorised Unnecessary Hazard** **SUBSTANTIATED**

Supervision allowed a geometry-incompatible locally purchased O-ring to become regular practice for the rectangular-slot meter seal, an avoidable hazard repeated three times in 22 months.

JUSTIFICATION

Supervision authorised — through sustained acceptance and repetition — a sealing substitution for meter 310FT01 that was unnecessarily hazardous. The meter flange was designed for a rectangular-slot gasket, yet [EVD-8645] records that "use of locally purchased O ring was done in its place as a regular practice" with "no specification or quality control during the purchase of the O rings" and no record of the authority for the switch. The safer and practicable alternative was to recover the OEM specification or hold the meter out of service until an engineered equivalent was qualified; instead the hazardous substitute was installed repeatedly on 12-Nov-2018, 25-Feb-2019, and 18-Sep-2020 ([EVD-1494]). Authorising a round-cross-section seal in a rectangular groove introduced an avoidable leak path in hydrocarbon service without any demonstrated need for the substitution. This is a primary supervisory authorisation failure that produced the committee's root cause #2 "Use of nonstandard O ring" and root cause #3 "Lack of control over material sourcing".

CAUSAL EXPLANATION

The authorised O-ring substitution created a pressure boundary that could not reliably seal the rectangular slot; after the 18-Sep-2020 installation, the joint survived only six days in service before failing at 02:12:50 on 24-Sep-2020. The release accumulated for 56 minutes and ignited at 03:08:15. This supervision-level authorisation of unnecessary hazard directly produced the lower-level unsafe acts AJDE 2.8 wrong choice of action, AV 1.1 routine/widespread violation, and ASBE 1.3 incorrect gasket procedure, and it was enabled by the SVIO 2.4 failure to apply MOC risk assessment to the substitution.

Linked Evidence: EVD-8645, EVD-1494

SVIO 2.5 **Authorised Unnecessary Hazard** **SUBSTANTIATED**

The meter was returned to service on 19-Sep-2020 after only an operating-pressure pressurisation, with the design-pressure hydrotest waived by supervision.

JUSTIFICATION

Supervision authorised meter 310FT01 to return to live hydrocarbon service after the 18-Sep-2020 gasket replacement without the design-pressure hydrostatic test that would have verified the pressure boundary. [EVD-2894] records: "After calibration of the meter 310FE01, the gasket was replaced on 18.09.2020. No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated. Next day meter 310FE01 was taken on online and was in use at the time of Incident" (p.31). The committee then cites the probable cause: "Probable cause of leak is due to use of nonstandard gasket and not testing the system to designed pressure" (p.32). A safe alternative was readily available — perform a hydrostatic test to design pressure on the isolated meter before placing it in service. By accepting an operating-pressure pressurisation as sufficient validation of a non-standard seal, supervision authorised an unnecessary hazard that placed the plant at risk without justification.

CAUSAL EXPLANATION

The authorised return-to-service decision eliminated the last technical test capable of detecting the defective O-ring seal before live operation. The meter entered service on 19-Sep-2020 and failed at 02:12:50 on 24-Sep-2020, initiating the hydrocarbon release. This supervisory authorisation of unnecessary hazard directly enabled the lower-level findings AJDE 2.5 ignored warning, AJDE 2.7 omitted hydrotest, ASBE 1.2 checklist not followed, and AV 1.1 routine violation, and it was made possible by the SVIO 2.4 absence of an SOP/JSA and by the SVIO 2.6 failure to provide independent verification manning.

Linked Evidence: EVD-2894, EVD-0185

SVIO 2.5 **Authorised Unnecessary Hazard** **SUBSTANTIATED**

A pitted 36-inch line segment with 1.03 mm metal loss and zero corrosion allowance was returned to continued service after painting only.

JUSTIFICATION

Supervision authorised painting as the corrective action for a pipeline segment with 1.03 mm thickness loss, heavy pitting, and peeling coating on a line designed with zero corrosion allowance, without requiring a fitness-for-service or replacement evaluation. [EVD-2397] records: "a reduction of thickness of 1.03mm along with heavy pitting and painting peeling off was observed in one of the segments of 36 pipeline in gas terminal in July 2019. The corrective action of painting was completed in Dec. 2019. However, as per the available data, the design thickness of the pipeline is 23.83mm with zero corrosion allowance. It needs to be examined whether painting as a corrective action would suffice in such cases and the mechanism of deciding on a corrective action needs to be reviewed" (p.30). Painting addresses coating restoration, not the accumulated metal loss on a zero-corrosion-allowance pressure envelope; authorising it as the sole remedy accepted an avoidable structural-integrity hazard without engineering justification.

CAUSAL EXPLANATION

This supervisory authorisation did not itself cause the 24-Sep-2020 fire because the 36-inch trunk line remained intact, but it is a substantiated example of supervision approving a cosmetic remedy for a serious integrity finding without demonstrated need or risk acceptance. It enabled the lower-level AJDE 2.8 wrong choice of action and AJDE 2.5 ignored warning, and it reinforces the same acceptance-of-inadequate-remedy culture that permitted painting-style superficial repairs to substitute for engineered risk control on the meter gasket at the centre of the incident.

Linked Evidence: EVD-2397

SVIO 2.6 Ordered Mission without Sufficient Manning

SUBSTANTIATED

The gasket replacement job was authorised without any independent QC, inspection, or permit-verification manning, leaving the substitute O-ring and omitted hydrotest wholly unchecked.

JUSTIFICATION

Supervision ordered the safety-critical gasket replacement on meter 310FT01 without providing the independent quality-control and inspection manning needed to verify material conformity and test performance. [EVD-8645] records that for the locally purchased O-rings "There was no specification or quality control during the purchase of the O rings", and [EVD-2894] records that no hydrotest was performed after the 18-Sep-2020 replacement, with the system "only pressurised to operating pressure" before the meter was returned to line. [EVD-5105] adds that "Job safety Plan is not attached/given in SAP document" and original permits were missing, meaning no independent permit authority or QC role was allocated to challenge the substitute part. For a 1986-vintage high-pressure hydrocarbon boundary, adequate manning would include a designated QC/inspector to verify part geometry/material against the specification and a supervisor to witness the design-pressure test; the evidence shows these verification functions were not provided at all.

CAUSAL EXPLANATION

With no QC manning assigned to verify the round O-ring against the rectangular-slot design and no test-witness manning to enforce the hydrostatic test, the defective pressure boundary passed directly from fitter to operations. The meter entered service on 19-Sep-2020 and failed on 24-Sep at 02:12:50, releasing vapour for 56 minutes before ignition at 03:08:15. This supervisory manning failure directly enabled the lower-level PTF 1.4 over-reliance on individual execution, PTF 1.5 unallocated quality roles, AJDE 2.8 wrong-choice, ASBE 1.2 checklist omission, and AV 1.1 routine violation, and it is the supervisory counterpart to the SVIO 2.4 JSA/MOC risk-assessment gaps.

Linked Evidence: EVD-8645, EVD-2894, EVD-5105

SVI 1.10 Failed to Communicate Intent

SUBSTANTIATED

The switch from OEM gaskets to locally purchased O-rings was never recorded as a MOC or otherwise communicated, allowing an unapproved material change to become regular practice.

JUSTIFICATION

The decision to replace the OEM rectangular-slot gasket with locally purchased O-rings was never reduced to a formal, communicated management instruction. [EVD-8645] records 'There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket' (p.31) and that the switch was practised as a 'regular practice'. [EVD-2986] shows the formal mechanism that should have carried and communicated that change was dormant: the MOC philosophy/SOP was published in August 2016, but 'no more MOCs are implemented in the plant since 2018' (p.30). Without a recorded and communicated change decision, subordinates had no management statement that the O-ring substitution was acceptable, no engineering constraints communicated for the substitute part, and no authority trail to hold any supervisor accountable. This is distinct from the testing-intent gap because it addresses the material-substitution decision itself, which the committee identified as the origin of the pressure-boundary weakness.

CAUSAL EXPLANATION

Because no supervisor formally communicated the material change and no MOC review was applied, the locally sourced O-ring substitution persisted across the 12-Nov-2018, 25-Feb-2019, and 18-Sep-2020 replacements without engineering challenge. The final improperly specified seal was installed on 18-Sep-2020, returned to service on 19-Sep-2020, and failed at 02:12:50 on 24-Sep-2020, producing the release that ignited at 03:08:15. This Level-3 failure directly enabled the lower-level findings AJDE 2.8 (wrong choice of action), AV 1.1 (routine/widespread violation), ASBE 1.3 (procedure not followed correctly), and PTE 2.12 (maintenance not conducted correctly) by leaving an unapproved material change in place without any communicated authorization or control.

Linked Evidence: EVD-8645, EVD-2986

SVIO 2.7 **Inappropriate Asset Management** **SUBSTANTIATED**

Three gasket replacements in 22 months were not escalated to root-cause analysis, OEM seal restoration, or meter replacement, so the same inadequate O-ring repair was repeated until the fourth failure.

JUSTIFICATION

This is a distinct asset-reliability supervision failure spanning 22 months, separate from the one-off material-procurement gap. [EVD-1494] records that 'Cleaning/servicing/calibration of meters 1 & 2 was carried out on 12th & 13th Nov 2018. Afterwards, meter no.1 was cleaned on 25/02/2019 & gasket was changed. Again on 18/09/2019, gasket was changed which leaked within one year' (p.27), with the evidence pack clarifying that the most recent gasket change is dated 18-Sep-2020. Three gasket replacements in approximately 22 months on a high-pressure meter were not treated as a chronic asset defect: the correct supervision response is root-cause analysis, recovery of the OEM sealing configuration, or replacement/rehabilitation of the meter; the actual response was to repeat the same locally sourced O-ring replacement each time. [EVD-0591] independently confirms the monitoring loop was not closed: SAP Notification 3800419840 raised on 17/09/2020 for the meter malfunction 'was not closed after completion of the job' (p.27). The recurrence was thus visible in the SAP record but never triggered asset-level intervention.

CAUSAL EXPLANATION

Because supervision did not convert the three-replacement pattern (12-Nov-2018, 25-Feb-2019, 18-Sep-2020) into a decision to restore the original sealing design or replace the meter, the defective repair practice survived and was repeated in September 2020. The fourth failure occurred at 02:12:50 on 24-Sep-2020, producing a 56-minute vapour release that ignited at 03:08:15. This finding is the Level-3 enabler of the lower-level AJDE 2.5 ignored warning finding on the three-failure history, PTE 2.12 maintenance-not-conducted-correctly recurrence, and AV 1.1 routine substitution: had the asset-reliability signal been managed at supervisory level, the later unsafe acts could not have recurred.

Linked Evidence: EVD-1494, EVD-0591

SVIO 2.7 **Inappropriate Asset Management** **SUBSTANTIATED**

Two of four diesel fire-water pumps were under maintenance and the remaining two were unreliable, leaving fire-water backup dependent on electrical pumps, though electric pumps maintained adequate supply.

JUSTIFICATION

Supervision failed to assure the availability of the diesel fire-water pump train as a critical emergency asset. [EVD-2432] records that 'Out of 4 diesel pumps 410 m3/hr, 2 pumps are under maintenance, one operated for 30 minutes & fourth pumps is tripping after 2 minutes. The reason for tripping is not identified or actions have not been recorded.' The same evidence also records that during the incident 'Both the Tank level and the Fire water header pressure were healthy throughout' and that 'Three electrical Fire water pumps namely 47PE003 A, C, D were operated during the cooling activity.' The correct supervision expectation is that the full redundant firewater capacity, including diesel-driven backup, is maintained or its unavailability is risk-assessed and recorded; the actual state was two pumps out for maintenance, one capable of only 30 minutes of operation, and one tripping after 2 minutes with no identified cause. This is a substantiated asset-availability deficiency, but the electrical pumps carried the entire cooling load and the incident outcome was not altered.

CAUSAL EXPLANATION

This supervision gap did not change the 24-Sep-2020 outcome because the three operating electrical fire-water pumps (47PE003 A, C, D) kept tank level and header pressure healthy throughout the fire, which was brought under control by approximately 06:00. It is nonetheless a latent resilience failure: had electrical supply failed or cooling demand exceeded electric-pump capacity, the diesel backup was materially degraded, with two pumps under maintenance, one limited to 30 minutes, and one tripping after 2 minutes for an unrecorded reason. The finding illustrates the same supervisory failure pattern underlying the lower-level asset-availability findings on the fire-water system, and its recurrence in the fire-water train reinforces the broader maintenance-governance weakness.

Linked Evidence: EVD-2432

SVIO 2.7 **Inappropriate Asset Management** **SUBSTANTIATED**

The mandated four-year pipeline thickness measurement schedule was not adhered to: measurements occurred in 2011 and 2017 only, with the next schedule already overdue by 2020.

JUSTIFICATION

Supervision did not enforce the mandated four-year thickness-monitoring schedule on the gas-terminal pipeline assets. [EVD-8307] records that 'The laid down schedule for thickness measurement is every four years... As per the records, the last two thickness measurement exercise was carried out in 2011 and 2017 which suggests that the schedule of four years is not being adhered to strictly' (p.29). The actual intervals were six years between 2011 and 2017, and more than three years had elapsed since 2017 by the incident date of 24-Sep-2020 without the next campaign. This is a substantiated asset-management schedule failure: the measurement program that should characterise degrading pipe-wall condition was not maintained. However, the 36-inch trunk pipeline itself was confirmed intact after the fire [EVD-5780], so this deficiency did not contribute to the 24-Sep-2020 loss of containment.

CAUSAL EXPLANATION

The missed schedule did not directly cause the 24-Sep-2020 fire because the main 36-inch pipeline remained intact and the loss of containment occurred at the metering-skid flange, not at a pipe-wall failure. Its significance is systemic and supervisory: because the schedule was not enforced, the 2017 thickness-loss findings (see instance 6) lacked timely follow-up measurement, and degraded-wall assets remained uncharacterised. This condition supported the lower-level AJDE 2.7 delayed necessary action pattern and is an example of the same supervisory tolerance of deferred integrity actions that allowed the chronic meter sealing defect to persist.

Linked Evidence: EVD-8307

SVIO 2.7 **Inappropriate Asset Management** **SUBSTANTIATED**

The 2017 inspection identified pipes with significant thickness loss requiring corrective action, but no plan was developed for over three years.

JUSTIFICATION

Supervision measured but did not manage a known degraded asset condition. [EVD-5527] records that 'As per the report of 2017, several pipes have been identified with significant thickness loss and need for corrective action. However, even after three years no plan has been developed for the corrective actions to be taken for those lines. Mere measurement of thickness without corrective action does not complete the process of maintaining the integrity of pipes' (p.30). The correct asset-management response to significant wall loss on high-pressure pipelines is fitness-for-service evaluation, repair, replacement, or a documented risk-based deferral; the actual response was identification followed by three years of inaction as of 24-Sep-2020. This is a distinct supervisory failure from the schedule breach because it concerns the decision-to-remediate phase after inspection data already existed. Although the 36-inch pipeline remained intact during the incident, this finding demonstrates that identified asset deficiencies were not converted into resourced corrective plans.

CAUSAL EXPLANATION

This finding did not directly cause the metering-skid fire, since the affected trunk pipeline survived with no damage [EVD-5780]. Its causal role is systemic: it shows the organisation had a supervision-level pattern of collecting integrity data without generating corrective actions, mirroring the meter 310FT01 recurrence where three gasket failures were recorded but no engineering intervention followed. It therefore enabled the lower-level AJDE 2.5 ignored warning pattern and is part of the management environment in which the directly causal meter defects were allowed to persist.

Linked Evidence: EVD-5527

SVIO 2.7 **Inappropriate Asset Management** **SUBSTANTIATED**

A 1.03 mm thickness reduction with heavy pitting on a zero-corrosion-allowance 36-inch pipe was treated with repainting rather than fitness-for-service analysis or repair.

JUSTIFICATION

Supervision selected an inadequate corrective action for a specific degraded pipeline asset. [EVD-2397] records that 'a reduction of thickness of 1.03mm along with heavy pitting and painting peeling off was observed in one of the segments of 36 pipeline in gas terminal in July 2019. The corrective action of painting was completed in Dec. 2019. However, as per the available data, the design thickness of the pipeline is 23.83mm with zero corrosion allowance. It needs to be examined whether painting as a corrective action would suffice in such cases and the mechanism of deciding on a corrective action needs to be reviewed' (p.30). The correct response to measured metal loss with heavy pitting on a zero-corrosion-allowance line is fitness-for-service analysis, repair, or replacement; the actual response was repainting within five months, which restores coating but not lost wall thickness. This is a distinct, time-separated asset-management decision from the 2017 findings because it occurred in 2019 and involved a specific 1.03 mm defect on the 36-inch line.

CAUSAL EXPLANATION

The painting-only decision did not contribute directly to the 24-Sep-2020 fire because the pipeline remained intact, but it is a supervisory demonstration of the same wrong-choice pattern seen at the metering skid: treating an integrity deficiency with a non-remedial action without engineering challenge. It underpins the lower-level AJDE 2.8 wrong choice of action finding on this pipe segment and reinforces the management climate in which the substitute O-ring, itself an inadequate material choice for a pressure boundary, was accepted without review.

Linked Evidence: EVD-2397

SVIO 2.7 **Inappropriate Asset Management** **SUBSTANTIATED**

Zone-1 flame-proof lamp maintenance was breakdown-oriented rather than preventive, and one lamp with three missing bolts and a glass case opened 5-6 cm remained in service as the probable ignition source.

JUSTIFICATION

Supervision failed to assure the integrity of explosion-protected electrical assets in a Zone 1 hydrocarbon area. [EVD-6916] records the resulting defect: 'three bolts on one of the flame proof lamps were missing resulting in it being open with its integrity compromised, glass case got detached from the main body and had opened out about 5 to 6 cms. The nearby fittings were found intact.' The condition therefore was unique to one lamp and consistent with a pre-existing maintenance failure rather than uniform fire damage. [EVD-0496] explains the systemic supervision gap: the post-incident electrical maintenance audit 'has clubbed all nine 125 W HPMV FLP lamps together and reported cable burnt and fitting damaged due to heat as status for all of them' and failed to mention the unique condition of the affected lamp (p.28), while 'the electrical maintenance system is a lot more skewed towards breakdown maintenance and lays little emphasis on systematic preventive maintenance and inspection' (p.28). The correct asset-management practice is periodic verification of flame-proof enclosure integrity on every Zone 1 fitting; the actual practice did not detect three

CAUSAL EXPLANATION

Because supervision did not enforce systematic preventive maintenance on the nine 125 W HPMV FLP lamps in the metering-area blast radius, a lamp with three missing flange bolts and a glass case opened 5-6 cm remained energised in a Zone 1 area. When the vapour cloud released by the failed meter accumulated over 56 minutes, the compromised enclosure supplied exposed contacts and the cloud ignited at 03:08:15 [EVD-4507], 4 minutes and 15 seconds after the 36-inch PCV closed at 03:04, proving the already-released cloud was sufficient fuel. This supervision failure converted a major vapour release into a major fire and enabled the lower-level ASBE 1.2 audit checklist failure and PTE 2.12/PTE 2.15 findings on the compromised FLP lamp.

Linked Evidence: EVD-6916, EVD-0496

SVIO 2.7 **Inappropriate Asset Management** **SUBSTANTIATED**

Three pressure transmitters for the Gas Terminal metering lines were shown on the drawing but no pressure data were retained or available, leaving the metering area without pressure trend monitoring.

JUSTIFICATION

Supervision did not ensure that the pressure-monitoring assets shown on the metering-system drawings were actually delivering data. [EVD-6470] records that 'Three Pressure transmitters 310PT4, 310PT5 & 310PT6 are shown in the drawing. It was informed that no pressure transmitter data are available in the system. No pressure data in Gas Terminal area is available/provided to check rise in pressure in the metering lines' (p.26). The correct asset-management state is that every transmitter shown on the P&ID is available in the DCS with trending and history; the actual state was three identified transmitters with zero data retention in the Gas Terminal area. This is a substantiated supervision gap in monitoring-asset availability. It did not determine the 24-Sep-2020 outcome because detection was achieved by CISF at approximately 02:15 and the annunciator at approximately 02:20, but it removed a potentially earlier leak indicator and impaired forensic reconstruction.

CAUSAL EXPLANATION

This asset-availability failure did not directly cause the release or ignition, so it is rated 2. Its causal significance is that the absence of pressure-trend data denied operators and investigators a real-time or reconstructed pressure profile that could have indicated abnormal conditions before the 02:15 perimeter detection. It compounded the lower-level PTE 2.5/PTE 2.11 configuration findings on the missing transmitter data and contributed to the detection and diagnosis limitations documented in the 46-minute response delay, though the primary drivers of that delay were the non-remotely operable MOV 54 and the missing DMP isolation criteria.

Linked Evidence: EVD-6470

SVI 1.11 **Support Not Provided by Supervisor** **SUBSTANTIATED**

Maintenance supervisors did not provide the OEM gasket specification, quality control, or coaching for the 18-Sep-2020 gasket replacement, leaving fitters to install an unsupported local O-ring.

JUSTIFICATION

The maintenance workforce was left without the supervisory technical support needed to select and verify the correct pressure-boundary seal. [EVD-8645] records that 'Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs' and that locally purchased O-rings were used 'as a regular practice' with 'no specification or quality control during the purchase of the O rings' (p.31). [EVD-2894] then documents that no SOP existed for calibration and return to service of the high-pressure meters (p.32). A supervisor providing adequate mentoring and coaching would have supplied the OEM rectangular-slot geometry and material requirements, or obtained engineering support, rather than leaving the fitter to substitute a round O-ring with no specification. This instance is distinct from the control-room and electrical-maintenance instances because it concerns the technical support owed to the maintenance fitters who executed the 18-Sep-2020 gasket replacement.

CAUSAL EXPLANATION

Without the correct gasket specification or any coaching on the rectangular-slot design, the maintenance team fitted a locally sourced O-ring that could not seat properly in the 1986 meter's rectangular slot, and without support they also omitted the hydrostatic test that would have exposed the defective joint. Meter 310FT01 returned to service on 19-Sep-2020, ran for six days, and failed at 02:12:50 on 24-Sep-2020, producing the vapour cloud that ignited at 03:08:15. This supervision-level support failure directly enabled the lower-level findings ASBE 1.3 (procedure not followed correctly), PTE 2.14 (unfamiliar equipment/non-custodial operations), AJDE 2.8 (wrong choice of action), and AV 1.1 (routine violation).

Linked Evidence: EVD-8645, EVD-2894

SVI 1.13 **Policy/ SOP/ Regulation/ Alert Not Communicated by Supervisor** **SUBSTANTIATED**

The MOC SOP existed from Aug 2016 but was not communicated or applied, so the 18-Sep-2020 gasket substitution occurred without MOC review or recorded authority.

JUSTIFICATION

EVD-2986 records that the MOC SOP was published in August 2016, but 'the procedure indicated in the document is not implemented while carrying out change of Low-Pressure Gas to Kribhico from Medium Pressure Gas headers in LPG Plant in 2013 & 2018' and 'no more MOCs are implemented in the plant since 2018'. For the exact gasket substitution that preceded this incident, EVD-8645 records 'There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket' and that locally purchased O-rings were used 'as a regular practice' without specification or quality control. The correct process would have been for the supervisor to communicate the MOC SOP to maintenance staff and initiate an MOC review before substituting the OEM rectangular-slot gasket; the actual process shows no MOC activity and no recorded authority. This failure to communicate and enforce the MOC policy removed the engineering review gate and enabled the wrong-gasket substitution identified in the committee's root cause #2 'Use of nonstandard O ring' and root cause #3 'Lack of control over material sourcing'.

CAUSAL EXPLANATION

Because the MOC policy was not communicated to the maintenance supervisor or crew, the locally purchased O-ring substitution on 18-Sep-2020 was never subjected to engineering review. This directly enabled the lower-level AJDE 2.11 misapplication/unawareness of the MOC policy and the ASBE 1.3 wrong-gasket procedure, producing the non-conforming seal that returned to service on 19-Sep-2020 and failed at 02:12:50 on 24-Sep-2020, releasing vapour for 56 minutes until ignition at 03:08:15.

Linked Evidence: EVD-2986, EVD-8645

SVIO 2.2 **Inappropriate Crew / Team/ Flight Composition** **SUBSTANTIATED**

The Dec-2019 calibration crew executed and reversed a 16000-to-2540-mmwc setpoint change with no checklist and no independent checker, then signed records that omitted it.

JUSTIFICATION

The December 2019 calibration team on meter 310FT01 was also an inappropriate composition for critical high-risk work. [EVD-2506] records that 'During calibration from 17/12/2019 to 18/12/2019, calibration was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of this change in the calibration documents signed by the team. During next calibration for meter 310FT01 was again changed to 16000 MMWC. No checklist is followed while carrying such critical jobs of high pressure/risk.' A 6.3x setpoint change on a high-pressure meter was executed, reversed, and left out of the signed record with no independent verifier in the team and no checklist. The correct calibration team would have paired the executing technician with an independent checker and a controlled checklist requiring documented approval of any setpoint change; the actual team could sign its own work without challenge. This instance is distinct from the 18-Sep-2020 gasket team by date, work type, and responsible roles, even though it affected the same meter and contributed to the same weak governance culture.

CAUSAL EXPLANATION

The same meter that later became the leak source was subjected to undocumented calibration changes because no independent team member challenged the work and no checklist guided the task. This instance normalised bypassing recording and verification on 310FT01 and is the Level-3 compositional counterpart to the Level-1 calibration checklist finding (ASBE 1.2) and the under-controlled calibration finding (ASBE 1.5). It did not directly open the 24-Sep-2020 release but contributed to the accumulation of procedural shortcuts on the same asset.

Linked Evidence: EVD-2506

SVIO 2.4 Risk Assessment - Formal

SUBSTANTIATED

Three gasket replacements on meter 310FT01 in 22 months were treated as routine reseals without any formal risk assessment, root-cause analysis, or reliability review.

JUSTIFICATION

Supervision had three documented sealing failures on the same meter in 22 months but never convened or recorded a formal risk assessment or root-cause analysis. [EVD-1494] records "Cleaning/servicing/calibration of meters 1 & 2 was carried out on 12th & 13th Nov 2018. Afterwards, meter no.1 was cleaned on 25/02/2019 & gasket was changed. Again on 18/09/2019, gasket was changed which leaked within one year" (p.27); the latest application was 18-Sep-2020. [EVD-0591] adds that the 17-Sep-2020 SAP notification 3800419840 "was not closed after completion of the job". A formal risk assessment after the second or third recurrence would have challenged the substitute O-ring design and the missing hydrostatic test; instead, each failure was treated as a routine reseal, so the identical inadequate fix was reinstalled. This is a secondary but demonstrable supervisory risk-assessment failure specific to repetition/reliability management.

CAUSAL EXPLANATION

The failure to formally risk-assess a known recurrence allowed the same unsafe material substitution and the same omitted hydrotest to be repeated. This enabled the lower-level findings AJDE 2.5 ignored warning, AV 1.1 routine violation, and PTF 1.13 planning-without-analysis. The fourth failure on 24-Sep-2020 at 02:12:50 initiated the 56-minute vapour release and the 03:08:15 ignition, demonstrating that the absence of a risk-based response to the recurrence directly preserved the incident-causing condition.

Linked Evidence: EVD-1494, EVD-0591

SVIO 2.4 Risk Assessment - Formal

SUBSTANTIATED

Electrical hazardous-area risk assessment was reduced to a one-page generic report clubbing all nine HPMV lamps into one status line, masking the unique integrity loss of the probable ignition-source lamp.

JUSTIFICATION

The formal electrical inspection and risk assessment program for hazardous-area flameproof fittings was not fit to identify or characterise the integrity loss that became the ignition mechanism. [EVD-9999] records that even the post-incident audit "is too generic and short (One page) and does not cover details of damages... the report has clubbed all nine 125 W HPMV FLP lamps together and reported 'cable burnt and fitting damaged due to heat' as status for all of them" (p.28). [EVD-0496] adds that "the unique condition of one particular lamp due to the type of damage suffered by it with respect to its area of installation was not mentioned in the report" and that "the electrical maintenance system is a lot more skewed towards breakdown maintenance and lays little emphasis on systematic preventive maintenance and inspection" (p.28). The assessment tool therefore collapsed nine independent equipment items into one generic status, erasing the distinction of the lamp with three missing bolts and a 5-6 cm open glass case that [EVD-6916] identifies as the probable ignition source.

CAUSAL EXPLANATION

A competent hazardous-area risk assessment would have classified the meter skid as Zone 1, verified each flameproof enclosure's bolt and glass integrity, and raised a corrective work order for the deficient lamp. Instead, the generic assessment left the compromised fitting in service, so the vapour cloud released from the failed gasket found a ready ignition source at 03:08:15, converting the loss of containment into a major fire. This is a secondary supervisory factor that enabled the Level-1 ASBE 1.2 audit-checklist failure and the Level-2 PSE 5.6 false-safety-expectancy condition.

Linked Evidence: EVD-9999, EVD-0496

SVF 3.1 **Personnel Management** **SUBSTANTIATED**

Maintenance fitters repeatedly substituted locally purchased O-rings for the lost OEM rectangular-slot gasket spec; supervisors never identified or remediated the unsafe tendency across three replacements in 22 months.

JUSTIFICATION

[EVD-8645] records that the 1986 meter 'is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket'. This is a direct description of maintenance personnel repeatedly making an unauthorised materials substitution with no specification or QC, and supervisory failure to identify and remedy that unsafe tendency. [EVD-1494] confirms the same individuals repeated the practice at least three times in 22 months — 12-Nov-2018, 25-Feb-2019 and 18-Sep-2020. No supervisor stopped the practice, recovered the OEM specification, or instituted remedial material-control actions, so the risky behaviour was allowed to become 'regular practice'. This instance is the personnel-management root of the lower-level unsafe acts AJDE 2.8 Wrong Choice of Action, ASBE 1.3

CAUSAL EXPLANATION

Because no supervisor intervened to stop the unauthorised O-ring substitution, the unsafe practice persisted through three gasket changes on 12-Nov-2018, 25-Feb-2019 and 18-Sep-2020. The final O-ring installed on 18-Sep-2020 was returned to service on 19-Sep-2020 with only an operating-pressure check, and the defective seal failed at 02:12:50 on 24-Sep-2020, releasing vapour for 56 minutes until ignition at 03:08:15. This instance enabled the lower-level AJDE 2.8 and ASBE 1.3 findings that directly produced the loss of containment, and it explains why the wrong material choice was repeated rather than escalated.

Linked Evidence: EVD-8645, EVD-1494

SVF 3.1 **Personnel Management** **SUBSTANTIATED**

Audit teams reported compliances not supported by site data using a non-plant-specific checklist; supervisory remediation of this unsafe audit behaviour was absent.

JUSTIFICATION

[EVD-9692] records that the self-safety audit team used a checklist that 'is comprehensive but not related to specific plants and in the process, lot of observations are missing' and that 'in most of the items, compliances are indicated, but during our inspection, the data is found to be incomplete & not in line with OISD standards/manuals'. [EVD-7681] adds that the 2018 and 2019 internal safety audits 'are given without the filled checklist. The checklist format is not available to check completeness of audit areas'. This describes auditors repeatedly asserting compliance without evidence and without plant-specific scrutiny — a recognisable unsafe tendency — while supervisors failed to correct the audit team's behaviour or replace the defective audit instrument. Because the audits did not independently verify Zone 1 electrical integrity, the compromised HPMV FLP lamp survived to provide the probable ignition source. This enabled the lower-level PTF 1.2 audit cross-monitoring failure and ASBE 1.2 audit checklist failure.

CAUSAL EXPLANATION

Because supervisors did not correct the audit team's practice of recording compliance without verification, the audits repeatedly failed to detect plant-specific hazards. The HPMV FLP lamp with three bolts missing and a glass case opened 5-6 cm, described in [EVD-6916], remained in place in Zone 1 until the vapour cloud reached it at 03:08:15 and ignited. This supervision failure is the precursor to the lower-level PTF 1.2 and ASBE 1.2 audit findings that left the ignition source undetected.

Linked Evidence: EVD-9692, EVD-7681

SVF 3.1 **Personnel Management** **SUBSTANTIATED**

Electrical maintenance staff operated a breakdown-skewed PM system and produced a one-page generic audit clubbing all nine FLP lamps; supervisors did not correct this behaviour.

JUSTIFICATION

[EVD-0496] records the committee's direct finding that 'the electrical maintenance system is a lot more skewed towards breakdown maintenance and lays little emphasis on systematic preventive maintenance and inspection'. [EVD-9999] documents the behaviour of the post-incident audit team, which produced a report that 'is too generic and short (One page) and does not cover details of damages... the report has clubbed all nine 125 W HPMV FLP lamps together and reported "cable burnt and fitting damaged due to heat" as status for all of them'. The unique condition of the one lamp with missing bolts and open glass case was not mentioned. This shows electrical maintenance personnel repeatedly substituting generic reporting for per-equipment inspection, and no supervisor corrected that behaviour or imposed systematic PM. The unsafe tendency persisted long enough that a flame-proof lamp with its integrity compromised remained in service in a Zone 1 hydrocarbon area. This is the personnel-management root of the lower-level ASBE 1.2 audit checklist failure and PTE 2.12 maintenance-not-conducted finding that left the ignition source available at 03:08:15.

CAUSAL EXPLANATION

The electrical maintenance team's habitual generic reporting and breakdown-driven culture meant the unique compromised state of the HPMV FLP lamp — three missing bolts and a glass case opened 5-6 cm per [EVD-6916] — was never identified and corrected. When the vapour cloud from the failed meter gasket accumulated over 56 minutes, it found this ready ignition source at 03:08:15, converting the release into a major fire. This instance enabled the lower-level PTE 2.12 maintenance-not-conducted and ASBE 1.2 audit-checklist failures.

Linked Evidence: EVD-9999, EVD-0496

SVF 3.2 **Operations Management** **SUBSTANTIATED**

Operations management allowed the locally-sourced O-ring substitution to continue as regular practice without specification or QC, through three gasket replacements in 22 months.

JUSTIFICATION

[EVD-8645] establishes that the use of locally purchased O-rings in place of OEM rectangular-slot gaskets was a known and continuous practice: 'use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket'. [EVD-1494] shows the same hazardous practice was applied on 12-Nov-2018, 25-Feb-2019 and 18-Sep-2020 across three replacements of the same meter. Despite this, operations management never corrected the practice, recovered the specification, restored OEM procurement, or imposed quality control. The committee classified this failure as root cause #2 'Use of nonstandard O ring' and root cause #3 'Lack of control over material sourcing' (p.33). This operations-management failure is the enabling condition for the lower-level unsafe acts AJDE 2.8 Wrong Choice of Action, AV 1.1 Routine Violation, ASBE 1.3 Procedure Not Followed, and the Level-2 precondition PTE 2.7 technical guidance unavailable.

CAUSAL EXPLANATION

The known hazardous practice persisted because no manager corrected it, so the 18-Sep-2020 repair again installed a non-conforming round O-ring into a flange designed for a rectangular-slot gasket. The meter went back online on 19-Sep-2020 and the defective seal failed at 02:12:50 on 24-Sep-2020, releasing condensate vapour. The release accumulated for 56 minutes and ignited at 03:08:15. This higher-level failure created the environment in which the lower-level AJDE 2.8, ASBE 1.3, AV 1.1 and PTE 2.7 findings could combine to produce the initiating loss of containment.

Linked Evidence: EVD-8645, EVD-1494

SVF 3.2 **Operations Management** **SUBSTANTIATED**

Management left a known gap where high-pressure meter gasket replacement was returned to service without a hydrostatic test and with no SOP governing calibration or start-up.

JUSTIFICATION

[EVD-2894] records the known hazardous practice and procedural void in management's own terms: 'After calibration of the meter 310FE01, the gasket was replaced on 18.09.2020. No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated.' The same evidence item quotes the committee's formal conclusion: 'Probable cause of leak is due to use of nonstandard gasket and not testing the system to designed pressure. No standard operating procedure is in place for calibration and taking back in service of these meters which are operating at high pressures' (p.32). This is a known hazardous condition — high-pressure pressure-boundary work with no design-pressure validation and no governing SOP — that operations management failed to correct before the 18-Sep-2020 job. It enabled the lower-level ASBE 1.2 hydrotest omission, AJDE 2.7 delayed/omitted necessary action, AJDE 2.11 misapplication of testing policy, and PTE 2.7 unavailable technical guidance findings.

CAUSAL EXPLANATION

Because operations management had not corrected the missing hydrotest requirement or written an SOP for high-pressure meter return-to-service, the 18-Sep-2020 repair was only pressurised to operating pressure and the defective O-ring seal was never validated to design pressure. The meter went online on 19-Sep-2020 and the joint failed at 02:12:50 on 24-Sep-2020. The vapour released for 56 minutes and ignited at 03:08:15. This higher-level failure directly enabled the lower-level ASBE 1.2 and AJDE 2.7 testing omissions that allowed the initiating failure.

Linked Evidence: EVD-2894

SVF 3.2 **Operations Management** SUBSTANTIATED

The 4-year pipeline thickness measurement schedule was breached (2011-2017 gap of six years) and not enforced by supervision.

JUSTIFICATION

[EVD-8307] records that 'The laid down schedule for thickness measurement is every four years... As per the records, the last two thickness measurement exercise was carried out in 2011 and 2017 which suggests that the schedule of four years is not being adhered to strictly'. The mandated 4-year cycle was breached (2011 to 2017 was six years), and the post-2017 cycle had not been completed by 24-Sep-2020. Operations management knew the schedule and did not enforce it, leaving pipe-thickness characterisation outside the mandated window. This condition did not directly cause the gasket failure or the fire, but it is part of the same integrity-management deterioration. It supports the lower-level PTE 2.12 maintenance-not-conducted finding and the AJDE 2.7 delayed-action pattern.

CAUSAL EXPLANATION

The unenforced schedule meant degraded pipeline thickness was not characterised on time. In the specific 24-Sep-2020 incident the 36-inch pipeline remained intact, so this instance had no direct impact on the outcome, but it reinforced an operating environment in which known mandatory integrity actions could be deferred without supervisory consequence — the same environment that enabled the omitted hydrotest and ignored gasket recurrence.

Linked Evidence: EVD-8307

SVF 3.2 **Operations Management** SUBSTANTIATED

Gas detectors were configured for annunciator display only, with no auto-shutdown or suppression interlock, and management never corrected this known design deficiency.

JUSTIFICATION

[EVD-8657] records the known configuration deficiency: 'Hydrocarbon/H2S detectors are configured to Annunciator Panel for display but not in DCS for history' and 'F&G detection system does not have provision for auto shut down of plant or activation of fire suppression system'. [EVD-3452] quotes the committee's recommendation A.1 at p.34: 'F&G detection system needs to be interlocked to automatic plant shut down and activation of fire suppression system.' Operations management left this display-only configuration in place despite the area's criticality, so the machine-witnessed leak onset at 02:12:50 produced no automatic protective action; the annunciator alarm did not arrive until 02:20, five minutes after human detection at 02:15, and first valve closure was delayed to 03:01. This known hazardous condition enabled the lower-level PTE 2.11 equipment-not-configured-correctly finding (rating 4), AV 1.1 F&G configuration violation, PTF 1.4 over-reliance on the human response chain, and the 46-minute release window that created the ignitable cloud.

CAUSAL EXPLANATION

Because the F&G system could not act automatically, the leak that began at 02:12:50 was left entirely to a slow manual chain: CISF detection at 02:15, CCTV verification at 02:18, CR notification at 02:20, senior call-out at 02:35, MOV 54 closure at 03:01, and PCV closure at 03:04. The cloud accumulated for 56 minutes and ignited at 03:08:15. This supervision failure directly enabled the lower-level PTE 2.11 and PTF 1.4 findings and is a primary amplifier of the incident's severity.

Linked Evidence: EVD-8657, EVD-3452

SVF 3.2 **Operations Management** SUBSTANTIATED

The DMP named the hydrocarbon leak scenario but omitted isolation/stoppage decision criteria and authority, a known procedural gap management never closed.

JUSTIFICATION

[EVD-4052] records that 'As per DMP On site Hazira Plant, guidelines to be followed in case of hydrocarbon leakage are included but does not include isolation/stoppage of facilities/pipelines' and that in the ERDMP 'the scenario for Level 3 emergencies are taken from the plant failures. Heavy product/gas leakages from pipelines & tanks are considered but decision on isolation/closure of plant/pipelines is not mentioned'. The committee's root cause #6, 'Non isolation/stoppage of 36 pipeline operation', flows from this gap. Operations management knew the plant's major-leak emergency could occur — the scenario was explicitly included — but never specified when and by whom isolation must be executed, leaving shift personnel without pre-planned authority. This enabled the lower-level AJDE 2.11 unawareness of policy, PTF 1.5 unclear roles, PTE 2.7 missing technical guidance, and the 46-minute first-detection-to-isolation delay.

CAUSAL EXPLANATION

Without documented isolation criteria, the night crew at 02:15-02:20 had no pre-planned authorised action and defaulted to serial escalation and physical verification. The GSP Shift In-charge closed MOV 54 only at 03:01, 46 minutes after first detection, and the 36-inch PCV at 03:04. The vapour inventory released during that interval was already 'all around even at height' at 03:00:44 [EVD-8213] and ignited at 03:08:15. This operations-management failure is the Level-3 root of the lower-level delayed-isolation findings.

Linked Evidence: EVD-4052

SVF 3.2 **Operations Management** **SUBSTANTIATED**

The meter malfunction notification raised 17-Sep-2020 was not closed or reviewed after the gasket replacement, and management had not enforced notification closure.

JUSTIFICATION

[EVD-0591] records that 'Notification no. 3800419840 was raised on 17/09/2020 for malfunction of meter no.1' and that the notification 'was not closed after completion of the job' (p.27). [EVD-0185] confirms the meter was 'taken on online and was in use at the time of Incident' without any post-maintenance validation captured, and notes committee recommendation D.2.ii at p.35: 'Sample checks/approval of closure of notifications by Senior Officers to be introduced'. Operations management therefore knew that maintenance notifications were not being closed and reviewed, yet did not correct the practice before the 18-Sep-2020 gasket job. The missing closure removed the supervisory quality gate that could have identified the unapproved O-ring, the absent hydrotest, and the recurrence pattern. This enabled the lower-level AJDE 2.11 misapplication of the SAP closure procedure, AV 1.1 non-closure violation, and PTF 1.12 communication-failure findings.

CAUSAL EXPLANATION

Because no supervisor reviewed and closed the notification, the defective 18-Sep-2020 gasket replacement passed without scrutiny. The meter returned to service on 19-Sep-2020 and failed at 02:12:50 on 24-Sep-2020, releasing vapour that ignited at 03:08:15. This failure directly enabled the lower-level AJDE 2.11 and PTF 1.12 findings that let the maintenance-to-operations loop remain unverified.

Linked Evidence: EVD-0591, EVD-0185

SVF 3.2 **Operations Management** **SUBSTANTIATED**

Missing permits were not tracked, JSPs were absent from SAP, and permit closure was not reviewed — a known permit-to-work control gap left uncorrected.

JUSTIFICATION

[EVD-5105] records that 'Work Permits are issued in SAP and hard copy of work permits are at sites during execution of work. After completion of work, the originals are filed. On checking the same, it was found that few of the work permits were missing. No data is kept on such missing permits. Job safety Plan is not attached/given in SAP document. Closing of permits & actions indicated in work needs review'. This is a known deficiency in the permit-to-work control that operations management had not corrected: missing permits were untracked, JSPs were absent from SAP for hazardous work, and closure was not reviewed. The committee's recommendation B.5 reinforced that 'Maintenance notifications are not be assumed as permit to work'. This failure contributed to the environment in which the 18-Sep-2020 high-risk gasket replacement could be executed without a JSP identifying the required hydrostatic test and material-conformance checks, supporting the lower-level PTE 2.7 technical-guidance and ASBE 1.2 procedural findings. It did not directly cause the leak, so it is rated 2.

CAUSAL EXPLANATION

The weak permit system meant the 18-Sep-2020 gasket replacement did not carry task-specific hazard controls such as a JSP calling for a hydrostatic test. That absence enabled the lower-level ASBE 1.2 omission, although the direct physical cause was the non-conforming gasket itself. The permit gap therefore had no direct impact on the outcome but was part of the supervision environment that allowed the unsafe maintenance work to proceed.

Linked Evidence: EVD-5105

SVF 3.2 **Operations Management** **SUBSTANTIATED**

Three of four diesel fire-water pumps were under maintenance or unreliable, with a tripping cause unidentified and unaddressed by supervision.

JUSTIFICATION

[EVD-2432] records that 'Both the Tank level and the Fire water header pressure were healthy throughout the incident. Three electrical Fire water pumps namely 47PE003 A, C, D were operated during the cooling activity' but that 'Out of 4 diesel pumps 410 m3/hr, 2 pumps are under maintenance, one operated for 30 minutes & fourth pumps is tripping after 2 minutes. The reason for tripping is not identified or actions have not been recorded'. This is a known hazardous condition — three of four diesel fire-water pumps unavailable or unreliable with an unidentified tripping fault — that operations management had not corrected before 24-Sep-2020. Because the electrical pumps carried the incident load and fire-water pressure remained healthy, this condition did not affect the outcome, but it represented an uncorrected redundancy gap in the emergency suppression system. It supports the lower-level PTE 2.12 and PTE 2.15 diesel-pump unreliability findings.

CAUSAL EXPLANATION

This uncorrected reliability gap did not change the 24-Sep-2020 outcome because the three electrical fire-water pumps maintained tank level and header pressure throughout cooling. It is nevertheless a known hazardous condition that could have constrained the response had electrical supply failed or cooling demand been higher, and it exemplifies the same supervision failure to close out known maintenance deficiencies as the meter gasket recurrence.

Linked Evidence: EVD-2432

SVF 3.2 **Operations Management** **SUBSTANTIATED**

Baseline thickness data and original pipe drawings/schedules for 1986 gas-terminal lines were not preserved or reconstructed by supervision.

JUSTIFICATION

[EVD-8287] records that 'for a large number of pipes, especially in gas terminal area baseline data of pipe thickness in not available. The original drawings and the schedules of the pipes are also not available. In most cases only two thickness details of 2011 and 2017 is available for a pipe which is commissioned in 1986'. This is a known records-management condition — no 1986 baseline and no pipe schedules for most gas-terminal lines — that supervision had not corrected across 34 years of asset life. The missing baseline degraded the ability to trend the 2017 thickness losses and support fitness-for-service decisions. In this incident the 36-inch pipeline remained intact, so the finding had no direct impact on the outcome, but it underpinned the lower-level PTE 2.7 technical-guidance and AJDE 2.7 delayed-action patterns in pipeline integrity management.

CAUSAL EXPLANATION

The absent baseline data prevented rigorous trending of the 2017 wall-loss findings and allowed corrective action to be postponed for three years. It did not contribute directly to the 24-Sep-2020 gasket failure or fire, but it formed part of the integrity-record environment that permitted known hazards to remain unquantified and unactioned, supporting the lower-level PTE 2.7 finding.

Linked Evidence: EVD-8287

SVF 3.3 **Workspace Management** **SUBSTANTIATED**

The meter maintenance workspace lacked OEM specs, SOPs, approved drawings, and enforced test requirements — a negative work environment supervisors never corrected.

JUSTIFICATION

[EVD-8645] records that the meter's 'Original Specifications of gasket are not available' and that locally purchased O-rings were used 'without any specification or quality control'. [EVD-2894] records that 'No standard operating procedure is in place for calibration and taking back in service of these meters which are operating at high pressures', and [EVD-0933] records that the metering-system P&ID did not match field configuration because 'As old drawings were lost, new drawings are made by a vendor but the same has not been checked & approved by ONGC'. These conditions describe a maintenance workspace stripped of the engineering references and procedures needed for safe pressure-boundary work. Supervision failed to prevent this negative workspace from developing and persisting, so maintenance staff operated with no correct gasket spec, no SOP, no controlled drawing, and no hydrotest requirement enforced. This workspace failure enabled the lower-level PTE 2.7 technical guidance unavailable, PTE 2.14 unfamiliar equipment, PTF 1.13 mission planning failures, and the ASBE 1.3 wrong-gasket procedure that initiated the release.

CAUSAL EXPLANATION

The workspace deficiency removed every technical defence that should have guided the 18-Sep-2020 repair: no OEM spec to identify the correct rectangular-slot gasket, no SOP to require a design-pressure hydrotest, and no approved drawing to verify the configuration. The fitter therefore installed a non-conforming O-ring, and the meter returned to service on 19-Sep-2020 before failing at 02:12:50 on 24-Sep-2020 and igniting at 03:08:15. This Level-3 workspace failure is the environment that enabled the lower-level PTE 2.7 and ASBE 1.3 findings.

Linked Evidence: EVD-8645, EVD-2894, EVD-0933

SVF 3.3 **Workspace Management** **SUBSTANTIATED**

The emergency workspace combined display-only gas alarms, manual-only MOVs, a single 90-second-pan camera, and no isolation criteria — management never corrected these workspace deficiencies.

JUSTIFICATION

[EVD-8657] records that 'Hydrocarbon/H2S detectors are configured to Annunciator Panel for display but not in DCS for history' and that 'F&G detection system does not have provision for auto shut down of plant or activation of fire suppression system'. [EVD-3932] records that the MOVs 'are not having remote operation from control room during emergency'. [EVD-0210] records that the sole 'PELCO make PTZ cameras (Model D5230D) ... has turn around (360 degree) at the time of about 90 seconds. Visibility from this camera is limited'. [EVD-4052] records that the DMP 'does not include isolation/stoppage of facilities/pipelines'. Together these describe a control-room workspace that was structurally unable to support time-critical emergency response: no auto-isolation, no remote valve actuation, intermittent CCTV coverage, and no written isolation decision authority. Supervisors failed to prevent this negative operational workspace from developing and left it in place until the incident. This enabled the lower-level PTE 2.4, PTE 2.8, PTE 2.11, PTF 1.5, and AJDE 2.7 findings that produced the 46-minute detection-to-isolation delay.

CAUSAL EXPLANATION

At 02:12:50 the leak began, but the workspace gave the operators no automatic or remote way to stop it. The annunciator alarmed only at 02:20, operators could not approach through the vapour cloud, and MOV 54 could not be closed remotely, delaying first isolation to 03:01. The cloud accumulated for 56 minutes and ignited at 03:08:15. This workspace-management failure is the Level-3 root that enabled the lower-level PTE 2.4 and PTE 2.11 response-delay findings.

Linked Evidence: EVD-8657, EVD-3932, EVD-0210, EVD-4052

SVF 3.3 **Workspace Management** **SUBSTANTIATED**

The Zone-1 electrical workspace contained a compromised FLP lamp (three bolts missing, glass open 5-6 cm) because PM inspection was never systematically enforced by supervision.

JUSTIFICATION

[EVD-6916] records that in the metering area 'three bolts on one of the flame proof lamps were missing resulting in it being open with its integrity compromised, glass case got detached from the main body and had opened out about 5 to 6 cms. The nearby fittings were found intact'. [EVD-0496] records that the post-incident audit reported nine 125 W HPMV FLP lamps as 'cable burnt and fitting damaged due to heat' but did not mention the unique condition of the one lamp, and that 'the electrical maintenance system is a lot more skewed towards breakdown maintenance and lays little emphasis on systematic preventive maintenance and inspection'. This describes a physical electrical workspace in a Zone 1 hydrocarbon area where flame-proof integrity was not systematically inspected or maintained, and a compromised fitting remained in service. Supervisors failed to prevent this unsafe workspace condition from developing and persisting. It enabled the lower-level PTE 2.15 inadequate explosion-protection, PTE 2.12 maintenance-not-conducted, and ASBE 1.2 audit findings, and it supplied the probable ignition source at 03:08:15.

CAUSAL EXPLANATION

The unsafe physical workspace — a flame-proof lamp with its explosion-protection integrity already lost — was the destination for the vapour cloud that accumulated over 56 minutes from the failed gasket. Ignition occurred at 03:08:15, 7 minutes after the first isolation action and 4 minutes after the PCV closure, so the already-released cloud alone was sufficient. This workspace-management failure converted a major vapour release into a major fire and enabled the lower-level PTE 2.15 and PTE 2.12 findings.

Linked Evidence: EVD-6916, EVD-0496

SVF 3.3 Workspace Management

SUBSTANTIATED

Missing permits, unfilled audit checklists, and incomplete drill records created a documentation workspace that could not surface or verify safety actions.

JUSTIFICATION

[EVD-5105] records that 'few of the work permits were missing. No data is kept on such missing permits', [EVD-7681] records that internal safety audit observations were 'given without the filled checklist', and [EVD-2130] records that fire drill reports 'are available in SAP but without any data on actions taken/recommendations'. These are symptoms of a documentation and assurance workspace where critical control records were absent or incomplete, and supervision did not prevent this negative environment from developing. The records that should have exposed the non-conforming gasket substitution, the missing hydrotest, and the compromised electrical fitting were either not created or not traceable. This workspace failure enabled the lower-level PTF 1.12 communication, ASBE 1.2 audit, and AV 1.1 documentation findings, and it denied the organisation the audit trail that would have flagged the latent hazards before 24-Sep-2020.

CAUSAL EXPLANATION

The deficient documentation workspace removed the evidentiary controls that would have forced the organisation to confront the known hazards: the three gasket replacements, the absent hydrotest, and the compromised lamp. Without those records, management could not detect or correct the accumulating unsafe conditions, and the latent defects survived until the 24-Sep-2020 release and 03:08:15 ignition. This instance is the workspace-level enabler of the lower-level AV 1.1 and PTF 1.12 findings.

Linked Evidence: EVD-5105, EVD-7681, EVD-2130

SVIO 2.5 Authorised Unnecessary Hazard

SUBSTANTIATED

Supervision staffed the Gas Terminal control room with personnel whose qualifications were undefined and who had only a four-day general training course, despite the emergency decision load.

JUSTIFICATION

Supervision intentionally assigned emergency-critical control-room duties to personnel whose qualification had never been defined and whose training was a four-day general programme. [EVD-5634] records: "control room operations are carried out by persons inadequately trained in safety, operations, control room handling, logbook records and actions during emergency. It is informed that four days training programme is conducted for new recruitment/transfer staff which is general in nature... Qualification of this staff is also not defined in HR/Operations Policy. Most of the staff have attended only One Basic Course since 2018" (p.30). The committee recommends control-room staff be trained "for at least 1 month (preferably on simulators)" before duties (p.36) — an eightfold expansion over the current four days. This falls within the code's specific criterion of "intentionally authorising personnel for execution of mission or operation, despite knowing that they are not appropriately qualified": the organisation knew the training was inadequate and qualifications undefined, yet continued to place these individuals in control of a major gas terminal.

CAUSAL EXPLANATION

When the annunciator alarmed at 02:20, the two Gas Terminal control-room engineers had no simulator-practised isolation response and no defined qualification benchmark for emergency handling, so they attempted on-foot source location and then telephonic escalation rather than immediate isolation. The first valve closure did not occur until 03:01, 46 minutes after first detection, and the accumulated cloud ignited at 03:08:15. This supervisory authorisation of under-qualified personnel enabled lower-level AJDE 2.10 inadequate knowledge for decision, PTF 1.3 task delegation to untrained staff, and PTF 1.17 unfamiliarity with the workspace, extending the vapour accumulation period that made ignition possible.

Linked Evidence: EVD-5634

SVIO 2.7 Inappropriate Asset Management

SUBSTANTIATED

Supervision never secured the OEM rectangular-slot gasket specification or correct replacement seal for meter 310FT01, forcing repeat use of a locally purchased round O-ring with no specification or QC.

JUSTIFICATION

Supervision did not ensure the correct pressure-boundary material was available for the 1986 condensate meter. [EVD-8645] records that 'The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket' (p.31). The correct asset-management control is a controlled OEM or equivalent rectangular-slot gasket stock with specification and receipt QC for a high-pressure boundary; the actual state was a lost specification, no OEM procurement, and an unspecified locally bought round O-ring. The official committee root cause #2 ('Use of nonstandard O ring') and root cause #3 ('Lack of control over material sourcing') flow directly from this supervision failure. This is the Level-3 material-asset failure that created the lower-level wrong-choice and routine-substitution

CAUSAL EXPLANATION

Because supervision failed to make the correct gasket asset available, the 18-Sep-2020 repair fitted a round O-ring into a rectangular-slot flange, creating a leak path that could not seal reliably. Meter 310FT01 returned to service on 19-Sep-2020 and operated for about five days under normal condensate pressure before the joint failed at 02:12:50 on 24-Sep-2020 [EVD-1284]. The resulting release accumulated for 56 minutes and ignited at 03:08:15 [EVD-4507]. This finding is the Level-3 origin of the lower-level AJDE 2.8 wrong choice of substitute gasket, AV 1.1 routine widespread substitution, ASBE 1.3 procedure not followed correctly, and PTE 2.7 unavailable technical guidance: the supervision-level absence of the correct material removed the precondition that would have prevented those unsafe acts.

Linked Evidence: EVD-8645

SVF 3.4 **Failed to Mitigate/ Ineffective Action against Identified Risk** **SUBSTANTIATED**

Three gasket replacements in 22 months were each treated as routine; supervisors never investigated why the same non-conforming O-ring kept leaking.

JUSTIFICATION

The meter had a known, documented recurrence of sealing failures that supervision repeatedly treated with the same ineffective routine replacement rather than escalating or engineering out. [EVD-1494] records the chronology directly: "Cleaning/servicing/calibration of meters 1 & 2 was carried out on 12th & 13th Nov 2018. Afterwards, meter no.1 was cleaned on 25/02/2019 & gasket was changed. Again on 18/09/2019, gasket was changed which leaked within one year" (p.27), with the contextual note establishing the latest replacement as 18-Sep-2020. Each repetition used the same locally purchased O-ring described in [EVD-8645] as "a regular practice" even though the meter is a 1986 rectangular-slot design. The identified risk was therefore a leak that had already recurred at least twice, yet the mitigation selected by maintenance supervision was identical each time — replace the gasket with a non-conforming substitute and return the meter to service — with no root-cause investigation, engineering review, OEM spec recovery, or meter replacement decision. That is the textbook SVF 3.4 pattern of an ineffective fix for a known leak compounded by no

CAUSAL EXPLANATION

Because the maintenance supervision accepted the recurring gasket replacement as a routine job, the known leak risk was never escalated to engineering or procurement authority for a correct OEM rectangular-slot gasket or meter replacement. The same inadequate local O-ring was installed on 18-Sep-2020, the meter returned to service on 19-Sep-2020, and the fourth failure occurred at 02:12:50 on 24-Sep-2020, initiating a 56-minute vapour release that ignited at 03:08:15. This supervision-level failure directly enabled the lower-level findings AJDE 2.5 (three gasket failures ignored, rating 4), AJDE 2.8 (wrong choice of action, rating 4), ASBE 1.3 (wrong gasket procedure, rating 4) and PTE 2.12 (recurrent maintenance failure, rating 3), and it is the organizational reason the initiating equipment defect survived to the day of the incident.

Linked Evidence: EVD-1494, EVD-8645

SVF 3.4 **Failed to Mitigate/ Ineffective Action against Identified Risk** **SUBSTANTIATED**

The 1986 OEM gasket spec was lost and supervisors allowed locally bought O-rings without specification or QC to become the regular repair material.

JUSTIFICATION

The loss of the OEM gasket specification and the uncontrolled switch to locally purchased O-rings constituted an identified material-control hazard that supervision never corrected. [EVD-8645] states: "The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket" (p.31). This single passage documents six separate omissions — lost specification, no OEM procurement, market O-ring substitution, no purchasing specification, no quality control, and no authorized change decision — all of which were known conditions of the material supply chain at the time the 18-Sep-2020 gasket was fitted. The committee's root cause #2 "Use of nonstandard O ring" and root cause #3 "Lack of control over material sourcing" flow directly from this finding, confirming that the unresolved

CAUSAL EXPLANATION

Without the OEM rectangular-slot specification and without any procurement quality gate, the site had no basis to ensure the replacement seal matched the 1986 flange geometry or hydrocarbon service requirements. The round O-ring substituted for the flat rectangular-slot gasket could not fill the machined groove, leaving a leak path at the pressure boundary. That substitution then passed into service on 19-Sep-2020 and the joint failed at 02:12:50 on 24-Sep-2020. This higher-level failure directly enabled the lower-level wrong choice of action AJDE 2.8 (rating 4), the routine violation AV 1.1 (rating 4), the wrong gasket procedure ASBE 1.3 (rating 4), and the technical guidance precondition PTE 2.7 (rating 4); it is the supervision-level root that allowed the identified material hazard to persist unchecked.

Linked Evidence: EVD-8645

SVF 3.4 **Failed to Mitigate/ Ineffective Action against Identified Risk** **SUBSTANTIATED**

The corrective notification for meter 310FT01 was never closed or reviewed, so the defective repair escaped a supervisory quality gate.

JUSTIFICATION

The maintenance notification system created an identified follow-up action on 17-Sep-2020, but supervision never closed the loop after the repair. [EVD-0591] records: "Notification no. 3800419840 was raised on 17/09/2020 for malfunction of meter no.1" (p.27) and further states that the notification "was not closed after completion of the job" (p.27). The sequence is critical: the performance check on 14-17 September observed jerks from meter 1, the notification was raised on 17-Sep, the gasket was replaced on 18-Sep, and the meter was back online on 19-Sep with the notification still open. A closed-loop review would have compelled a supervisor or engineer to verify what was actually done — the non-conforming O-ring, the missing hydrostatic test, and the recurrence history — before accepting the work. Its absence removed a known quality gate exactly when the meter's repair quality needed scrutiny.

CAUSAL EXPLANATION

The unclosed notification meant no senior officer reviewed the repair package, so the non-standard O-ring, the missing hydrotest, and the third-in-22-months recurrence all passed without challenge. The meter returned to service on 19-Sep-2020 and failed at 02:12:50 on 24-Sep-2020. This supervision failure enabled the lower-level findings AV 1.1 (routine non-closure, rating 3), AJDE 2.11 (misapplied SAP closure policy, rating 3), PTF 1.12 (failed feedback/acknowledgement, rating 3) and PTE 2.12 (no supervisory quality gate, rating 3), and it was a secondary but important contributor to the opportunity lost to detect the defective gasket before failure.

Linked Evidence: EVD-0591

SVF 3.4 **Failed to Mitigate/ Ineffective Action against Identified Risk** **SUBSTANTIATED**

The 2017 thickness survey flagged pipes needing corrective action, but supervisors developed no remediation plan for more than three years.

JUSTIFICATION

The 2017 thickness analysis was an identified integrity risk that supervision acknowledged but never actioned. [EVD-5527] states: "As per the report of 2017, several pipes have been identified with significant thickness loss and need for corrective action. However, even after three years no plan has been developed for the corrective actions to be taken for those lines. Mere measurement of thickness without corrective action does not complete the process of maintaining the integrity of pipes" (p.30). The period from the 2017 report to the 24-Sep-2020 incident is more than three years, during which no corrective action plan was developed. The committee's own phrasing that measurement without corrective action does not complete the integrity process is an explicit organizational finding that the identified hazard was not mitigated. Although the specific 2017 findings concerned the trunk pipelines rather than the metering skid flange, the same supervisory behavior of receiving an integrity warning and doing nothing about it is part of the environment in which the meter's recurrent seal failures were also left uncorrected.

CAUSAL EXPLANATION

The failure to translate a known thickness-loss finding into a corrective action plan normalized the practice of identifying risks without resolving them. This did not directly cause the metering gasket failure — the 36-inch line itself remained intact per [EVD-5780] — but it created and reinforced the organizational pattern that allowed the recurrent gasket warning on meter 310FT01 to be ignored for 22 months. It is the supervision-level counterpart of lower-level finding AJDE 2.5 (2017 warning ignored, rating 3) and AJDE 2.7 (corrective action delayed, rating 3), and it shows that risk-register follow-up was systematically ineffective, not merely missed once.

Linked Evidence: EVD-5527

SVF 3.4 **Failed to Mitigate/ Ineffective Action against Identified Risk** **SUBSTANTIATED**

The four-year UT thickness-monitoring cycle was breached (2011 to 2017 gap of six years), and no catch-up or enforcement action is recorded.

JUSTIFICATION

The site had a defined four-year thickness-monitoring cycle, but supervision allowed the cycle to slip from 2011 to 2017 without correction. [EVD-8307] states: "The laid down schedule for thickness measurement is every four years... As per the records, the last two thickness measurement exercise was carried out in 2011 and 2017 which suggests that the schedule of four years is not being adhered to strictly" (p.29). The gap between 2011 and 2017 was six years rather than four, and by the time of the 24-Sep-2020 incident more than three years had elapsed since the last measurement. Maintaining the defined monitoring interval is a basic risk-control measure for aging 1986-vintage pipelines, and the schedule breach was an identified deficiency on the integrity record. There is no evidence supervision took action to recover the lost interval, reschedule the overdue survey, or escalate the non-compliance.

CAUSAL EXPLANATION

The lapsed monitoring schedule reduced the site's ability to trend wall loss on the gas-terminal pipelines and to act on degradation before it became urgent. This particular failure did not directly change the 24-Sep-2020 outcome — the trunk pipeline remained intact and the failure was at the meter flange — so it is rated present with no direct outcome impact. It nevertheless shows that pipeline integrity controls were allowed to drift without supervisory enforcement, part of the same broader pattern in which known control requirements, such as the hydrostatic test for the meter gasket, were treated as optional.

Linked Evidence: EVD-8307

SVF 3.4 **Failed to Mitigate/ Ineffective Action against Identified Risk** **SUBSTANTIATED**

Three of four diesel fire-water pumps were unavailable or unreliable, and one unexplained two-minute tripping fault was never diagnosed or actioned.

JUSTIFICATION

The diesel fire-water backup was an identified reliability risk whose defects were known but not corrected. [EVD-2432] records: "Out of 4 diesel pumps 410 m3/hr, 2 pumps are under maintenance, one operated for 30 minutes & fourth pumps is tripping after 2 minutes. The reason for tripping is not identified or actions have not been recorded" (p.27). At the time of the incident, therefore, three of the four diesel pumps were unavailable or unreliable, and the fourth had an unexplained two-minute tripping fault for which no diagnosis or corrective action had been recorded. This is a direct example of identified equipment deficiencies remaining unresolved, with no follow-up to determine whether the emergency backup would function if called upon. In this incident the electrical fire-water pumps carried the load and tank level/header pressure stayed healthy, so the latent defect did not affect the outcome.

CAUSAL EXPLANATION

The diesel backup was degraded by known, unrepaired defects, but this did not change the 24-Sep-2020 outcome because three electrical fire-water pumps maintained header pressure and tank levels throughout, active cooling began around 03:50, and the fire was extinguished by about 06:00. The unresolved diesel-pump faults are therefore rated present with no direct impact on this fire, but they represent the same supervision pattern of not completing corrective actions and not recording follow-up on known equipment risks that also characterized the meter gasket and pipeline integrity failures.

Linked Evidence: EVD-2432

SVF 3.4 **Failed to Mitigate/ Ineffective Action against Identified Risk** **SUBSTANTIATED**

A generic self-safety audit checklist marked items compliant without supporting data, missing plant-specific risks and producing false assurance.

JUSTIFICATION

The self-safety audit was an identified control intended to catch plant-specific hazards, but its instrument and follow-up were ineffective. [EVD-9692] records: "Safety self-audit is conducted by a multi-disciplinary team based on the checklist prepared by the location. The checklist is comprehensive but not related to specific plants and in the process, lot of observations are missing. The duration of audit is not indicated and in most of the items, compliances are indicated, but during our inspection, the data is found to be incomplete & not in line with OISD standards/manuals. Approved manuals/SOPs were not available" (p.26). The audit claimed compliance for most items while the underlying data did not support it, and the generic checklist did not force review of plant-specific risks such as Zone 1 flame-proof fitting integrity or the metering skid's non-standard gasket practice. This false assurance is a direct supervision failure to verify that identified safety measures were actually effective, and the committee's recommendation B.1 to develop plant-specific checklists confirms the instrument was not fit for purpose.

CAUSAL EXPLANATION

Because the self-safety audit recorded compliance without verification and was not tailored to the Gas Terminal metering area, it gave management an assurance of safety that did not reflect reality. This allowed the existing equipment-integrity hazards — including the HPMV FLP lamp later found with three missing bolts and its glass case open 5-6 cm per [EVD-6916] — to survive undetected. When the vapour cloud released from the failed meter flange accumulated for 56 minutes, the uncompromised-assurance assumption was falsified by ignition at 03:08:15. This supervision failure directly enabled the lower-level ASBE 1.2 audit checklist finding (rating 3) and PTE 2.7 generic-guidance precondition (rating 3).

Linked Evidence: EVD-9692

SVF 3.4 **Failed to Mitigate/ Ineffective Action against Identified Risk** **SUBSTANTIATED**

The MOC SOP published in 2016 was not used after 2018, so the non-OEM gasket substitution escaped formal change review.

JUSTIFICATION

A formal Management of Change control existed but was not being applied, so the material substitution on the meter was never subjected to engineering review. [EVD-2986] records: "Management of Change (MOC) philosophy/SOP was published in August 2016. However, the procedure indicated in the document is not implemented while carrying out change of Low-Pressure Gas to Kribhico from Medium Pressure Gas headers in LPG Plant in 2013 & 2018. It is informed no more MOCs are implemented in the plant since 2018" (p.30). The switch from the OEM rectangular-slot gasket to locally purchased O-rings was a material change with pressure-boundary significance, but with no MOC practice in place after 2018 it was recorded as no more than a routine replacement. The same evidence item explicitly connects the gap to [EVD-8645]'s finding that there is no record of the decision to switch gasket sourcing, demonstrating that the identified MOC procedure was a paper control whose non-implementation supervision failed to correct.

CAUSAL EXPLANATION

Without an active MOC process, the substitution of locally sourced O-rings for OEM rectangular-slot gaskets was never formally assessed for geometry, material compatibility, or pressure rating. That governance absence allowed the wrong-material decision documented in [EVD-8645] to become regular practice and led to the seal installed on 18-Sep-2020 failing at 02:12:50 on 24-Sep-2020. This supervision-level gap enabled the lower-level AJDE 2.11 misapplication of MOC policy (rating 3) and ASBE 1.3 wrong-procedure finding (rating 4), and it is a foundational reason the known material change was never corrected.

Linked Evidence: EVD-2986

SVF 3.4 **Failed to Mitigate/ Ineffective Action against Identified Risk** **SUBSTANTIATED**

Missing work permits were not tracked and Job Safety Plans were absent from SAP, leaving hazardous tasks without documented hazard analysis.

JUSTIFICATION

The permit system had known traceability and job-safety-plan deficiencies that supervision did not remedy. [EVD-5105] records: "Work Permits are issued in SAP and hard copy of work permits are at sites during execution of work. After completion of work, the originals are filed. On checking the same, it was found that few of the work permits were missing. No data is kept on such missing permits. Job safety Plan is not attached/given in SAP document. Closing of permits & actions indicated in work needs review" (p.26). Missing original permits with no log of which permits were missing, JSPs absent from SAP for hazardous/non-routine work, and permit closure not reviewed are all known control gaps. The committee's recommendation B.5 that maintenance notifications are not to be assumed as permits implies that maintenance could be executed under SAP notifications without a separate work-permit authorization, which would have applied directly to the 18-Sep-2020 gasket replacement.

CAUSAL EXPLANATION

The permit/JSP system gaps meant the high-risk 18-Sep-2020 gasket replacement could proceed without a documented Job Safety Plan identifying the required design-pressure hydrotest and material-conformance checks. This is rated present with no direct outcome impact because the evidence does not tie the specific gasket job to a missing permit, but it contributed to the procedural environment in which the lower-level ASBE 1.2 hydrotest omission and AJDE 2.11 policy misapplication could occur without permit-level challenge.

Linked Evidence: EVD-5105

SVF 3.4 **Failed to Mitigate/ Ineffective Action against Identified Risk** **SUBSTANTIATED**

Fire drill records in SAP had no action or observation data, and an informal parallel dataset did not follow approved guidelines.

JUSTIFICATION

The site's fire drill program produced records that could not demonstrate whether identified drill findings had been corrected. [EVD-2130] states: "Fire drill reports are available in SAP but without any data on actions taken/recommendations. Separate data is maintained which is not in line with the approved guidelines" (p.26). Two parallel record sets existed — SAP records with dates only and no actions or observations, and an informal separate dataset not compliant with approved guidelines — so drill effectiveness and follow-up on drill-identified deficiencies could not be measured from either source. This is an identified documentation deficiency in an emergency-preparedness control, left uncorrected by supervision, and it means the site could not verify that drill recommendations were being closed out.

CAUSAL EXPLANATION

The unverifiable drill records did not directly cause the 24-Sep-2020 fire, but they removed the ability of supervision to confirm that emergency-response lessons were followed up. This is rated present with no direct outcome impact because the evidence does not link any specific drill finding to the incident response. It does, however, reinforce the broader SVF 3.4 pattern across the site: identified issues were allowed to persist with no auditable follow-up.

Linked Evidence: EVD-2130

SVF 3.1 Personnel Management

SUBSTANTIATED

Calibration staff changed meter 310FT01 from 16000 to 2540 mmwc and back without documentation or checklist; supervisors took no remedial action.

JUSTIFICATION

[EVD-2506] records that during calibration on 17-18 Dec 2019 the setpoint of meter 310FT01 was changed 'from 16000 mmwc to 2540 mmwc... without any mention of this change in the calibration documents signed by the team. During next calibration for meter 310FT01 was again changed to 16000 MMWC. No checklist is followed while carrying such critical jobs of high pressure/risk'. This is a recognised risky behaviour by the calibration personnel — making a 6.3x setpoint change without documentation or checklist — and there is no evidence that any supervisor instituted remedial action such as retraining, disciplinary action, or a mandatory checklist. The same meter later became the leak source, and the same procedural-discipline weakness reappeared in the Sep-2020 gasket work. This supervision failure enabled the lower-level ASBE 1.2 calibration checklist finding, ASBE 1.5 under-controlled calibration finding, and AJDE 2.10 inadequate calibration knowledge finding.

CAUSAL EXPLANATION

The December 2019 undocumented setpoint instability normalised checklist-free work on meter 310FT01 and demonstrated that high-pressure work was not independently verified. Nine months later the same maintenance organisation installed a non-conforming O-ring on the same meter and omitted the hydrostatic test, returning it to service on 19-Sep-2020. The defective joint failed on 24-Sep-2020 at 02:12:50, releasing vapour that ignited at 03:08:15. This instance is therefore a secondary but direct enabler of the lower-level ASBE 1.2, AJDE 2.10 and AV 1.1 findings that permitted the initiating failure.

Linked Evidence: EVD-2506

SVV 4.1 Failure to Enforce Existing Rules (Supervisory act of Omission)

SUBSTANTIATED

Supervision did not enforce closure of SAP Notification 3800419840 after the gasket replacement, removing the post-job supervisory review gate.

JUSTIFICATION

Supervisors failed to enforce the SAP notification closure rule for maintenance job completion on meter 310FT01. [EVD-0591] records: "Notification no. 3800419840 was raised on 17/09/2020 for malfunction of meter no.1" (p.27), and the same evidence states the notification "was not closed after completion of the job" (p.27). Because closure was not enforced, there was no supervisory review gate that would have inspected the work package, challenged the missing hydrotest, or noticed the recurrence of gasket replacement. This is a distinct supervisory omission from the test omission itself: the failure to close the notification removed the administrative checkpoint that should have independently caught the defective job.

CAUSAL EXPLANATION

Because the SAP notification was never closed, the 18-Sep-2020 gasket replacement escaped the supervisory review that would have checked whether the non-standard O-ring, the absent design-pressure test, and the three-replacement recurrence were acceptable. This enabled the lower-level findings AV 1.1 (routine non-closure violation) and AJDE 2.11 (misapplication of the notification closure policy). The unclosed work record allowed the defective meter to return to service on 19-Sep-2020 and fail at 02:12:50 on 24-Sep-2020; it is rated secondary rather than primary because the physical failure was initiated by the deficient seal itself, while the missing closure review was a supervisory control that could have caught it.

Linked Evidence: EVD-0591, EVD-0185

SVV 4.1 **Failure to Enforce Existing Rules (Supervisory act of Omission)** **SUBSTANTIATED**

Supervision accepted a signed calibration record that omitted a 16000-to-2540 mmwc setpoint change and used no checklist for critical high-pressure work.

JUSTIFICATION

The supervisor who signed the December 2019 calibration record of meter 310FT01 failed to enforce the rules requiring checklist use and documentation of setting changes on high-pressure equipment. [EVD-2506] records: "During calibration from 17/12/2019 to 18/12/2019, calibration was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of this change in the calibration documents signed by the team. During next calibration for meter 310FT01 was again changed to 16000 MMWC. No checklist is followed while carrying such critical jobs of high pressure/risk" (p.27). The setting change of roughly 6.3 times occurred without any entry in the signed calibration document and was silently reversed later; the team's signed document was accepted by supervision as complete when it was not. This omission is distinct by phase and evidence item from the September 2020 gasket-job failure, but it occurred on the same meter that later became the leak source.

CAUSAL EXPLANATION

By accepting an incomplete signed calibration record, supervision normalised checklist bypass and undocumented changes on the same meter that later failed. This created the environment for the lower-level findings ASBE 1.2 (calibration checklist not followed) and ASBE 1.5 (under-controlled calibration setting), and demonstrated to the maintenance organisation that critical high-pressure work could be executed without documentation or cross-check. That same weak supervision culture allowed the 18-Sep-2020 gasket replacement to proceed without a hydrotest, connecting this phase to the eventual release at 02:12:50 on 24-Sep-2020.

Linked Evidence: EVD-2506

SVV 4.1 **Failure to Enforce Existing Rules (Supervisory act of Omission)** **SUBSTANTIATED**

Supervisors did not enforce flame-proof integrity rules or systematic electrical PM, leaving a Zone 1 HPMV lamp with three bolts missing and its glass case open 5-6 cm in service as the probable ignition source.

JUSTIFICATION

Supervision did not enforce the existing flame-proof integrity requirements for electrical fittings in the Zone 1 hydrocarbon area, and did not enforce a meaningful preventive-maintenance inspection regime that would have detected the compromised fitting. [EVD-6916] records: "three bolts on one of the flame proof lamps were missing resulting in it being open with its integrity compromised, glass case got detached from the main body and had opened out about 5 to 6 cms. The nearby fittings were found intact" (p.32). The post-incident audit itself demonstrates the supervisory failure: [EVD-9999] states "the report is too generic and short (One page) and does not cover details of damages... the report has clubbed all nine 125 W HPMV FLP lamps together and reported 'cable burnt and fitting damaged due to heat' as status for all of them" (p.28), and that "the electrical maintenance system is a lot more skewed towards breakdown maintenance and lays little emphasis on systematic preventive maintenance and inspection" (p.28). Existing Zone 1 explosion-protection rules were therefore unenforced, leaving a lamp with its flame-proof integrity already absent in

CAUSAL EXPLANATION

Because supervision did not enforce flame-proof integrity rules in the Zone 1 area, the lamp's missing bolts and 5-6 cm open glass case removed the explosion-protection barrier before the incident. When the vapour cloud from the failed meter flange reached the lamp at 03:08:15 on 24-Sep-2020 — 56 minutes after leak onset at 02:12:50 — the exposed fitting supplied the ignition energy that converted the hydrocarbon release into a major fire. This supervisory omission is the higher-level enabler of the lower-level findings PTE 2.15 (inadequate protection equipment) and ASBE 1.2 (audit checklist failure that missed the defect), and it directly produced the ignition transition of the incident.

Linked Evidence: EVD-6916, EVD-9999

SVV 4.1 **Failure to Enforce Existing Rules (Supervisory act of Omission)** **SUBSTANTIATED**

Supervision allowed the mandated four-year UT thickness cycle to be breached and left 2017 significant-thickness-loss findings without a corrective action plan for over three years.

JUSTIFICATION

Integrity-management supervision did not enforce the mandated four-year pipeline thickness measurement schedule nor the follow-up corrective actions demanded by the 2017 survey. [EVD-8307] records: "The laid down schedule for thickness measurement is every four years... As per the records, the last two thickness measurement exercise was carried out in 2011 and 2017 which suggests that the schedule of four years is not being adhered to strictly" (p.29). [EVD-5527] further states: "As per the report of 2017, several pipes have been identified with significant thickness loss and need for corrective action. However, even after three years no plan has been developed for the corrective actions to be taken for those lines. Mere measurement of thickness without corrective action does not complete the process of maintaining the integrity of pipes" (p.30). This is distinct from the metering-skid maintenance failures because it involves the 36-inch and 42-inch pipeline systems and a separate integrity-management process; the 2011-to-2017 gap of six years and the 2017-to-2020 gap of over three years show repeated non-enforcement.

CAUSAL EXPLANATION

This supervisory omission did not directly cause the 24-Sep-2020 release because the 36-inch trunk pipeline itself was confirmed intact [EVD-5780]. However, it created the systemic integrity-governance environment in which inspection findings were acknowledged but not actioned, directly enabling the lower-level findings AJDE 2.5 (ignored caution/warning) and AJDE 2.7 (delayed necessary action) on the pipeline system. The same management pattern of allowing known defects and missed schedules to persist explains why the recurring meter 310FT01 gasket failures were also not escalated, so this instance is rated as a secondary systemic contributor.

Linked Evidence: EVD-8307, EVD-5527

SVV 4.1 Failure to Enforce Existing Rules (Supervisory act of Omission)

SUBSTANTIATED

Supervision allowed control room operators to staff emergencies with only a 4-day general training course and no defined qualification.

JUSTIFICATION

Operations supervision and HR failed to enforce defined training and qualification requirements for control room operators before assigning them emergency-response duties. [EVD-5634] records: "control room operations are carried out by persons inadequately trained in safety, operations, control room handling, logbook records and actions during emergency. It is informed that four days training programme is conducted for new recruitment/transfer staff which is general in nature and later the job training is carried out. Qualification of this staff is also not defined in HR/Operations Policy. Most of the staff have attended only One Basic Course since 2018" (p.30). The committee recommended at p.36 that control room staff be trained for at least one month, preferably on simulators. The night-shift Chief Engineer and Junior Engineer thus performed emergency response with only this minimal general training and no defined qualification, and supervision did not enforce a higher standard or simulator practice.

CAUSAL EXPLANATION

Because supervision did not enforce a defined training and qualification standard, the night-shift crew at 02:20 on 24-Sep-2020 attempted on-foot source verification and then telephone escalation instead of initiating immediate isolation. This training/qualification omission is the supervisory-level enabler of the lower-level findings AJDE 2.9 (decision-making during operation) and AJDE 2.10 (inadequate knowledge of isolation criteria), and it contributed to the 46-minute gap between detection at 02:15 and MOV 54 closure at 03:01, after which the accumulated cloud ignited at 03:08:15.

Linked Evidence: EVD-5634

SVV 4.2 Allowing Unwritten Policies to Become Standard

SUBSTANTIATED

Use of locally sourced O-rings instead of OEM rectangular-slot gaskets became an undocumented 'regular practice' on meter 310FT01.

JUSTIFICATION

The substitution of locally procured round O-rings for the lost OEM rectangular-slot gasket specification was an unwritten practice that became the accepted standard for meter 310FT01 maintenance. [EVD-8645] records: "Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket" (p.31). The phrase "as a regular practice" and the absence of any formal decision record are the defining signature of this code: the unofficial practice was perceived and followed by the maintenance workforce without any properly constituted authority having established it. [EVD-1494] confirms it was applied repeatedly on 12-Nov-2018, 25-Feb-2019 and 18-Sep-2020, embedding it as the standard repair method.

CAUSAL EXPLANATION

This unwritten policy of O-ring substitution became the standard repair method, so the 18-Sep-2020 gasket replacement was executed without any expectation of OEM specification recovery, material conformance, or engineering review. It is the higher-level enabler of the Level-1 findings AJDE 2.8 (wrong choice of action), ASBE 1.3 (procedure not followed correctly) and AV 1.1 (routine widespread violation). The round O-ring in the rectangular slot failed under operating pressure at 02:12:50 on 24-Sep-2020, initiating the 56-minute vapour release that ignited at 03:08:15. The lack of any recorded authority or date for the change also meant the practice could not be traced or revoked, making it a primary causal factor.

Linked Evidence: EVD-8645, EVD-1494

SVF 3.2 **Operations Management** **SUBSTANTIATED**

Three gasket replacements in 22 months on meter 310FT01 were not escalated as a chronic reliability problem; the same inadequate repair was repeated until the fourth failure caused the leak.

JUSTIFICATION

[EVD-1494] documents the recurrence pattern in the SAP record: 'Cleaning/servicing/calibration of meters 1 & 2 was carried out on 12th & 13th Nov 2018. Afterwards, meter no.1 was cleaned on 25/02/2019 & gasket was changed. Again on 18/09/2019, gasket was changed which leaked within one year'. Together with the 18-Sep-2020 replacement mentioned in [EVD-2894], this is three gasket replacements in approximately 22 months on a 1986-vintage meter. Operations management failed to treat this recurring sealing failure as a reliability concern, failed to perform root-cause analysis, and failed to change the repair strategy or replace the meter. The pattern was visible in SAP yet never triggered corrective engineering. This directly enabled the lower-level AJDE 2.5 Ignored a Caution/Warning (rating 4), PTE 2.5 Defective Equipment in Use, and PTF 1.13 planning failures, and it allowed the fourth failure on 24-Sep-2020 to initiate the release.

CAUSAL EXPLANATION

The known recurrence was the organisation's clearest warning that the substitute gasket design was inadequate. By failing to correct it after the second and third failures, managers guaranteed the same defect would be reintroduced on 18-Sep-2020. The meter operated from 19-Sep to 24-Sep-2020 and the joint failed at 02:12:50, initiating a 56-minute release that ignited at 03:08:15. This supervision-level failure directly caused the lower-level AJDE 2.5 ignored-warning and PTF 1.13 planning failures.

Linked Evidence: EVD-1494, EVD-0591

SVF 3.2 **Operations Management** **SUBSTANTIATED**

A flame-proof HPMV lamp with three missing bolts and a 5-6 cm open glass case remained in service because electrical PM was breakdown-skewed and never systematically corrected.

JUSTIFICATION

[EVD-6916] describes the probable ignition source: 'three bolts on one of the flame proof lamps were missing resulting in it being open with its integrity compromised, glass case got detached from the main body and had opened out about 5 to 6 cms. The nearby fittings were found intact'. [EVD-0496] records that the electrical maintenance system 'is a lot more skewed towards breakdown maintenance and lays little emphasis on systematic preventive maintenance and inspection', and that the post-incident audit 'clubbed all nine 125 W HPMV FLP lamps together and reported "cable burnt and fitting damaged due to heat" as status for all of them' without noting the unique condition of the one lamp. This is a known hazardous condition in a Zone 1 hydrocarbon area that operations management failed to correct through systematic PM inspection, and the supervisory failure allowed the compromised flame-proof enclosure to remain in service. It enabled the lower-level PTE 2.12 maintenance-not-conducted finding, PTE 2.15 inadequate explosion-protection finding, and ASBE 1.2 audit-checklist failure, and it is the reason a vapour release became a fire.

CAUSAL EXPLANATION

Operations management's failure to correct the breakdown-skewed electrical PM system left the flame-proof lamp with its integrity compromised in Zone 1. After the meter gasket failed at 02:12:50 on 24-Sep-2020, the accumulating vapour cloud reached this lamp and ignited at 03:08:15, 7 minutes after the first isolation action. This supervising-level condition therefore enabled the lower-level PTE 2.12 and PTE 2.15 findings that converted a major release into a major fire.

Linked Evidence: EVD-6916, EVD-0496, EVD-9999

SVV 4.4 **Authorised Unqualified Individuals for Task** **SUBSTANTIATED**

The night-shift GT Control Room crew held emergency duty with only a general 4-day training course, no defined qualification and no simulator practice.

JUSTIFICATION

Supervision authorised the Chief Engineer (Production) and Junior Engineer (Mechanical) to hold the Gas Terminal Control Room on the night of 23–24 Sep 2020 even though the plant's own evidence shows those operators were not qualified for emergency control-room duty. [EVD-5634] records that "control room operations are carried out by persons inadequately trained in safety, operations, control room handling, logbook records and actions during emergency. It is informed that four days training programme is conducted for new recruitment/transfer staff which is general in nature and later the job training is carried out. Qualification of this staff is also not defined in HR/Operations Policy. Most of the staff have attended only One Basic Course since 2018" (p.30), and the committee's own recommendation (p.36) states "operating staffs of control room needs to be trained for at least 1 month (preferably on simulators) before being assigning Control Room duties enabling them to handle operations & emergencies effectively" — a near-eightfold expansion of the four-day program actually given. The authorised crew's behaviour at 02:20 on 24 Sep, when the

CAUSAL EXPLANATION

Because supervision rostered a crew whose only preparation was a four-day general course with no simulator and no defined qualification, the two engineers who received the 02:20 alarm treated the event as a locate-and-inspect problem rather than an isolate-immediately problem: they walked toward the source, were repelled by the dense vapour, and then spent the 02:20–02:35 window telephoning senior managers instead of closing anything. Their trained-for-emergency absence of any practised isolation drill fed the serial escalation chain — GSP/LPG shift in-charge reached the gate only at ~02:35, and MOV 54 was not closed until 03:01, 46 minutes after first physical detection at 02:15 and 48 minutes 10 seconds after CCTV-confirmed leak onset at 02:12:50 — during which the vapour cloud grew to the "vapour all around even at height" state captured at 03:00:44 [EVD-8213] and then ignited at 03:08:15. This Level-3 authorisation failure therefore created the precondition for the Level-2 PTF 1.3/PTF 1.17 and Level-1 AJDE 2.10 findings, acting as a secondary but material contributor to the 46-minute unisolated release window that made the fire possible.

Linked Evidence: EVD-5634, EVD-1460, EVD-9220

SVV 4.4 **Authorised Unqualified Individuals for Task** **SUBSTANTIATED**

A locally-sourced O-ring was fitted three times in 22 months by a crew with no OEM spec, no QC and no design-pressure testing — repeatedly authorised by maintenance supervision.

JUSTIFICATION

Supervision repeatedly authorised a maintenance crew to perform pressure-boundary work on meter 310FT01 for which the crew demonstrably lacked the technical qualification. [EVD-8645] records that "The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket" (p.31) — six qualifying omissions in two sentences, and the committee's root cause #2 ("Use of nonstandard O ring") and #3 ("Lack of control over material sourcing") flow directly from this. The same non-qualified crew then skipped the one verification step that could have exposed their error: [EVD-2894] records "After calibration of the meter 310FE01, the gasket was replaced on 18.09.2020. No hydro test was done after replacement of the gasket. The system was only pressurised to operating

CAUSAL EXPLANATION

The authorisation of an under-qualified maintainer for the 18-Sep-2020 gasket replacement produced the initiating equipment failure. The round cross-section O-ring could not fill the 1986 rectangular slot, and because the crew also lacked the competence to recognise the design-pressure test obligation, the joint was only soaked at operating pressure and never hydro-tested. Meter 310FT01 returned to service on 19-Sep-2020 and ran for six days until the joint failed at 02:12:50 on 24-Sep-2020 [EVD-1284], releasing condensate vapour that accumulated for 56 minutes and ignited at 03:08:15 [EVD-4507]. The supervisory choice was made three times (12-Nov-2018, 25-Feb-2019, 18-Sep-2020), converting what should have been a one-off error into a normalised practice — the Level-3 failure that directly produced the Level-1 AJDE 2.8 wrong-choice, ASBE 1.3 incorrect gasket procedure and AV 1.1 routine-violation findings and therefore the loss of containment itself.

Linked Evidence: EVD-8645, EVD-2894, EVD-1494

SVV 4.4 **Authorised Unqualified Individuals for Task** **SUBSTANTIATED**

A calibration crew changed a 16000-to-2540 mmwc setpoint and back with no checklist and no signed record — authorised nevertheless for critical high-pressure work.

JUSTIFICATION

A second, distinct manifestation of SVV 4.4 occurred in December 2019 when supervision authorised a calibration crew for critical high-pressure work on the same meter that later failed. [EVD-2506] records: "During calibration from 17/12/2019 to 18/12/2019, calibration was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of this change in the calibration documents signed by the team. During next calibration for meter 310FT01 was again changed to 16000 MMWC. No checklist is followed while carrying such critical jobs of high pressure/risk" (p.27). The team signed calibration certificates that omitted an approximately 6.3x setpoint change and its silent reversal — evidence the individuals were functionally non-current for a task that required documented, checklist-controlled execution — yet they were authorised to perform and sign off the job. The committee's recommendation D.1.i (p.35) — "details of Master calibrator/action taken/replacement/deviation from procedure if any, is to be documented in the calibration certificate" — is the corrective requirement for exactly this authorisation failure. This instance is distinct from instance 2

CAUSAL EXPLANATION

Authorising a non-qualified, checklist-less calibration team normalised undocumented changes to critical pressure-boundary instrumentation on the very meter that later became the leak source. excursions went unrecorded and unchallenged, degrading confidence in 310FT01's measurement and — more importantly — teaching the maintenance organisation that high-pressure work could pass without cross-check or documentation. That lesson carried forward to the 18-Sep-2020 gasket replacement on the same meter, where the same organisation omitted the design-pressure hydrotest and accepted an unspecified O-ring; the joint then failed at 02:12:50 on 24-Sep-2020, 56 minutes before ignition at 03:08:15. This Dec-2019 authorisation is therefore a secondary but real contributor: it did not open the leak itself, but it entrenched the checklist-bypass pattern (ASBE 1.2, AJDE 2.10) that made the fatal September job possible.

Linked Evidence: EVD-2506

SVV 4.4 Authorised Unqualified Individuals for Task

SUBSTANTIATED

The post-fire electrical audit was assigned to the general maintenance team, which returned a one-page report clubbing all nine HPMV lamps — no impact on the incident outcome, but the same authorisation pattern.

JUSTIFICATION

A final, distinct manifestation occurred after the fire when the plant maintenance team — whose competence for per-equipment failure-mode characterisation the very report calls into question — was authorised to conduct the post-incident electrical preventive-maintenance audit of the affected area. [EVD-9999] records: "Post the incident, an electrical audit of the affected area was carried out by the plant maintenance team. Considering that the audit was carried out to identify the damage to the equipment, the report is too generic and short (One page) and does not cover details of damages... the report has clubbed all nine 125 W HPMV FLP lamps together and reported 'cable burnt and fitting damaged due to heat' as status for all of them" (p.28), and the same evidence adds that "the electrical maintenance system is a lot more skewed towards breakdown maintenance and lays little emphasis on systematic preventive maintenance and inspection" (p.28). The one-page, nine-lamps-clubbed output demonstrates that the team authorised for this specialised damage characterisation lacked the skill to discriminate per-equipment failure modes —

CAUSAL EXPLANATION

This instance sits outside the causal chain to the 24-Sep-2020 fire — the incident had already occurred and the audit was damage characterisation — so it has no bearing on the outcome and is rated present without impact. Its significance is diagnostic: the same supervision that authorised an unqualified electrical-maintenance function for a specialised audit task is the supervision under whose authority the Zone-1 HPMV FLP lamp with three missing bolts and a 5–6 cm open glass case [EVD-6916] went uninspected pre-incident and later supplied the ignition at 03:08:15.

Linked Evidence: EVD-9999, EVD-0496

SVV 4.5 No Action/ Mitigation Measures for Identified Risk

SUBSTANTIATED

Repeat gasket failures on 310FT01 (Nov-2018, Feb-2019, Sep-2020) were recorded but never investigated or mitigated — supervision allowed the same inadequate fix three times.

JUSTIFICATION

The most direct supervisory failure to act on an identified risk is the repeated gasket-failure pattern on the meter that ultimately caused the release. [EVD-1494] records: "Cleaning/servicing/calibration of meters 1 & 2 was carried out on 12th & 13th Nov 2018. Afterwards, meter no.1 was cleaned on 25/02/2019 & gasket was changed. Again on 18/09/2019, gasket was changed which leaked within one year" (p.27) — three gasket replacements on the 1986 meter within roughly 22 months (12-Nov-2018, 25-Feb-2019, 18-Sep-2020 with the text/narrative date caveat). These were formally recorded maintenance events: the risk of a chronic sealing defect was not hidden, it sat in SAP as repeat work on the same asset, yet no supervisor initiated a root-cause analysis, no material review, no return to OEM procurement, and no meter replacement or quarantine. The initial incident report itself flags the exact supervisory question: "This gasket replacement is the third on this meter in the past 22 months (previous replacements: November 2018, February 2019). Investigation should determine whether the recurrence pattern was treated as a reliability concern." The same theme

CAUSAL EXPLANATION

Each recorded gasket replacement was an identified warning that the substitute O-ring did not seal the rectangular slot; by taking no mitigation after the 12-Nov-2018 and 25-Feb-2019 failures, supervision guaranteed the 18-Sep-2020 job would repeat the same defect. The third (Sep-2020) job was executed identically — locally purchased O-ring [EVD-8645], no hydrostatic test [EVD-2894] — and the joint failed within six days of returning to service on 19-Sep-2020: leak onset at 02:12:50 on 24-Sep [EVD-1284], 56 minutes of vapour accumulation, and ignition at 03:08:15 [EVD-4507]. This no-action finding is thus the primary supervisory cause of the initiating loss of containment, directly enabling the Level-1 AJDE 2.5, AJDE 2.8 and AV 1.1 findings that produced the release.

Linked Evidence: EVD-1494, EVD-0591

SVV 4.5 **No Action/ Mitigation Measures for Identified Risk** **SUBSTANTIATED**

The DMP and ERDMP named the major-leak scenario but management never wrote the isolation criteria or authority into the plan — the gap behind the 46-minute response delay.

JUSTIFICATION

Management formally identified the major hydrocarbon-leakage risk in the site Disaster Management Plan and the ERDMP scenarios, yet never took the step of defining the isolation/stoppage decision criteria and authority that the scenario required. [EVD-4052] records: "As per DMP On site Hazira Plant, guidelines to be followed in case of hydrocarbon leakage are included but does not include isolation/stoppage of facilities/pipelines" (p.32), reinforced at p.27: "In ERDMP, scenario for Level 3 emergencies are taken from the plant failures. Heavy product/gas leakages from pipelines & tanks are considered but decision on isolation/closure of plant/pipelines is not mentioned." The committee made this a formal root cause — root cause #6 "Non isolation/stoppage of 36 pipeline operation" — and [EVD-6414] characterises the consequence: "Decision of closure of valve was taken after about 40 minutes resulting the spread of vapour cloud in increased area" (p.32). Because the scenario was written into the DMP but the operational response criteria were left blank, the night crew had no pre-planned authorised action at first detection (02:15) or first alarm (02:20)

CAUSAL EXPLANATION

With no documented criterion for when a hydrocarbon leak warranted pipeline isolation and no named authority below the senior chain, the night-shift response became a discovery process: leak onset 02:12:50 [EVD-1284], human detection 02:15, annunciator 02:20, shift-in-charge called at ~02:35 and arriving at the gate, and only then MOV 54 closure at 03:01 — 46 minutes after first detection — followed by the 363 PCV at 03:04. released live inventory, producing the mature cloud visible at 03:00:44 [EVD-8213] which ignited at 03:08:15, after isolation, proving the already-released inventory was the fuel. Had management completed the identified scenario with an isolation decision rule and pre-delegated authority, the first closure could have occurred within roughly 5–10 minutes of 02:15, hugely reducing the flammable envelope. This is therefore a primary supervisory cause of the incident's severity, directly enabling the response-delay findings at Levels 1 and 2.

Linked Evidence: EVD-4052, EVD-6414

SVV 4.5 **No Action/ Mitigation Measures for Identified Risk** **SUBSTANTIATED**

A Zone 1 flame-proof lamp with three missing bolts and the glass case open 5–6 cm went uninspected under a breakdown-skewed electrical-maintenance system and became the probable ignition source.

JUSTIFICATION

Management maintained no preventive-maintenance or inspection action for the flame-proof integrity of electrical fittings in a Zone 1 hydrocarbon area, leaving a demonstrably compromised lamp in service to act as the ignition source. [EVD-6916] records the physical condition: "three bolts on one of the flame proof lamps were missing resulting in it being open with its integrity compromised, glass case got detached from the main body and had opened out about 5 to 6 cms. The nearby fittings were found intact" (p.32) — the committee identified this single fitting as the probable ignition source, and the "nearby fittings were intact" qualifier is the committee's basis for treating the missing-bolt/open-glass condition as pre-existing integrity loss rather than fire-created damage. The supervisory failure to act on this reasonably apparent Zone 1 hazard is documented in [EVD-9999] and [EVD-0496], which record that "the electrical maintenance system is a lot more skewed towards breakdown maintenance and lays little emphasis on systematic preventive maintenance and inspection" (p.28), and that the post-incident PM audit "clubbed all nine 125 W HPMV FLP lamps together and reported

CAUSAL EXPLANATION

Because supervision directed no systematic preventive maintenance toward the Zone 1 flame-proof fittings, the lamp with three missing bolts and a glass case open 5–6 cm remained energised and exposed in the metering blast radius. When the vapour cloud accumulated over 56 minutes from leak onset at 02:12:50 and persisted after both isolation actions (MOV 54 at 03:01, 363 PCV at 03:04), the already-released cloud reached that co at 03:08:15 [EVD-4507] — 4 minutes 15 seconds after the last isolation and 7 minutes after the first. Correct protection would have been a Zone 1 flame-proof enclosure with all bolts fitted and the glass case secured; the actual state supplied exposed electrical contacts inside a flammable envelope. This supervisory no-mitigation therefore converted a vapour release into a major fire and is a primary cause of the incident's escalation to fire.

Linked Evidence: EVD-6916, EVD-9999, EVD-0496

SVV 4.5 **No Action/ Mitigation Measures for Identified Risk** **SUBSTANTIATED**

1.03 mm wall loss with heavy pitting on a zero-corrosion-allowance pipeline was 'mitigated' with painting only — an inadequate response to an identified integrity risk.

JUSTIFICATION

When the July 2019 spot UT measurement identified real metal loss on a critical zero-corrosion-allowance segment, supervision selected a coating-restoration measure instead of any fitness-for-service analysis or replacement — an inadequate mitigation of an identified risk. [EVD-2397] records: "a reduction of thickness of 1.03mm along with heavy pitting and painting peeling off was observed in one of the segments of 36 pipeline in gas terminal in July 2019. The corrective action of painting was completed in Dec. 2019. However, as per the available data, the design thickness of the pipeline is 23.83mm with zero corrosion allowance. It needs to be examined whether painting as a corrective action would suffice in such cases and the mechanism of deciding on a corrective action needs to be reviewed" (p.30). A pipe with zero corrosion allowance has no margin for wall loss by design, so 1.03 mm of measured loss with heavy pitting required engineering evaluation of remaining strength; painting alone restores the coating, not the wall. This is the Level-3 origin of the lower-level AJDE 2.8 wrong-choice-of-action finding (rating 3) and contributes to the systemic AJDE

CAUSAL EXPLANATION

Supervision accepted painting as closure for a 1.03 mm metal-loss finding on a design that carries zero corrosion allowance, leaving the degraded segment in service without any remaining-strength evaluation from Dec-2019 through the incident. This action-adequacy failure did not cause the 02:12:50 metering release, but it is the clearest documented case of the decision pattern — treating a structural-integrity risk with a protective coating — that the same organisation applied when it repeatedly accepted unspecified O-rings instead of the OEM rectangular-slot gasket. It thereby sits beneath the AJDE 2.8 wrong-choice finding and forms part of the supervisory culture that enabled the initiating gasket failure and the 56-minute release culminating in ignition at 03:08:15.

Linked Evidence: EVD-2397

SVV 4.5 **No Action/ Mitigation Measures for Identified Risk** **SUBSTANTIATED**

The mandatory four-year UT thickness cycle was breached — 2011 to 2017 was six years — with no supervisory enforcement of the schedule.

JUSTIFICATION

Supervision failed to enforce the laid-down four-year thickness-measurement schedule for the gas-terminal pipelines, allowing a six-year gap between the 2011 and 2017 surveys to occur and pass without correction. [EVD-8307] records: "The laid down schedule for thickness measurement is every four years... As per the records, the last two thickness measurement exercise was carried out in 2011 and 2017 which suggests that the schedule of four years is not being adhered to strictly" (p.29). The 2011–2017 gap of six years is a 50% overrun of the mand was still outstanding when the incident occurred on 24-Sep-2020 (3+ years after 2017). No evidence shows any supervisor holding the schedule owner to the four-year cadence. This instance is distinct from instance 5 (no action on the 2017 findings) because it concerns the inspection cadence itself rather than the follow-up on results; it is rated 2 because the pipeline remained intact and no causal path from the missed survey to the 02:12:50 metering release is demonstrated.

CAUSAL EXPLANATION

The breached schedule meant wall-loss on the 1986-vintage pipelines was being characterised at best every six years rather than every four, so degradation could run undetected between surveys; this fed the same integrity-governance weakness as the missing baseline data (instance 12) and the ignored 2017 findings (insta line was confirmed intact and undamaged after the fire [EVD-5780] and the release originated at the metering flange, this instance is rated present without direct impact on the 24-Sep-2020 outcome, while remaining a substantiated supervisory no-action on a formally mandated risk-control cadence.

Linked Evidence: EVD-8307

SVF 3.2 **Operations Management** **SUBSTANTIATED**

Significant thickness losses identified in 2017 received no corrective action plan for three years despite a formal recommendation.

JUSTIFICATION

[EVD-5527] records that 'As per the report of 2017, several pipes have been identified with significant thickness loss and need for corrective action. However, even after three years no plan has been developed for the corrective actions to be taken for those lines. Mere measurement of thickness without corrective action does not complete the process of maintaining the integrity of pipes'. This is a formally reported, known hazardous condition on gas-terminal piping that operations management failed to correct for over three years, with no remediation plan developed by 24-Sep-2020. Although the 36-inch pipeline itself remained intact during this incident per [EVD-5780], the failure demonstrates the same supervision pattern that allowed the meter gasket defect to persist. It enabled the lower-level AJDE 2.5 Ignored a Caution/Warning (rating 3) and AJDE 2.7 Delayed a Necessary Action (rating 3) findings.

CAUSAL EXPLANATION

The three-year non-action on documented thickness loss showed operators that inspection findings did not have to be resolved. That same management tolerance explains why the recurring meter gasket failure and the no-hydrotest practice were also left uncorrected. While the 2017 pipeline findings did not initiate the 24-Sep-2020 release, this instance directly enabled the lower-level AJDE 2.5 and AJDE 2.7 patterns of ignoring and delaying corrective action that were replicated on meter 310FT01.

Linked Evidence: EVD-5527

SVF 3.2 **Operations Management** **SUBSTANTIATED**

Heavy pitting plus 1.03 mm thickness loss on a zero-corrosion-allowance segment was answered only with painting; management did not require engineering evaluation.

JUSTIFICATION

[EVD-2397] records that 'a reduction of thickness of 1.03mm along with heavy pitting and painting peeling off was observed in one of the segments of 36 pipeline in gas terminal in July 2019. The corrective action of painting was completed in Dec. 2019. However, as per the available data, the design thickness of the pipeline is 23.83mm with zero corrosion allowance. It needs to be examined whether painting as a corrective action would suffice in such cases and the mechanism of deciding on a corrective action needs to be reviewed'. This is a known loss of metal on a zero-corrosion-allowance line that management treated with a cosmetic coating-restoration action instead of fitness-for-service evaluation or replacement. The supervision failure to correct this inadequate response enabled the lower-level AJDE 2.8 Wrong Choice of Action (rating 3) and demonstrated the same decision-making pattern that accepted substitute O-rings for pressure-boundary gaskets. The pipeline survived the fire per [EVD-5780], so this instance is secondary to the incident outcome but a direct operational-management failure.

CAUSAL EXPLANATION

Management accepted painting as sufficient for a pitted, zero-corrosion-allowance pipeline, institutionalising the norm that structural integrity deficiencies could be addressed symptomatically. That same management judgement pattern allowed a locally-sourced O-ring to be used for a pressure-boundary gasket on meter 310FT01, which failed at 02:12:50 on 24-Sep-2020 and led to the 03:08:15 ignition. The pipeline remained intact in this incident, so the finding is rated secondary, but it is causally connected to the lower-level AJDE 2.8 wrong-choice pattern.

Linked Evidence: EVD-2397

SVF 3.2 **Operations Management** **SUBSTANTIATED**

Critical isolation valves such as MOV 54 had no remote emergency operation, requiring physical approach that the vapour cloud denied until 03:01; management did not correct this.

JUSTIFICATION

[EVD-3932] records that 'MOVs are provided in the gas terminal station for local operations but they are not having remote operation from control room during emergency'. [EVD-9612] documents the consequence: the GSP Shift In-charge 'tried to enter the Gas Control room area but could not enter due to heavy gas smell' and 'closed the MOV 54 inlet condensate line from slug catcher to 36 metering area at about 0301 hrs'. The first isolation therefore occurred 46 minutes after first detection at 02:15 and 48 minutes after CCTV-confirmed leak onset at 02:12:50. This was a known design/operations condition — critical emergency isolation valves requiring physical approach — that management had not corrected, and it directly forced the delayed manual response. It enabled the lower-level PTE 2.4 controls-and-switches-inadequate finding (rating 4), PTE 2.8 workspace incompatible with operations, PTF 1.4 over-reliance on manual operation, and the AJDE 2.7 delayed-action finding.

CAUSAL EXPLANATION

The manual-only MOV configuration meant the control-room engineers who received the alarm at 02:20 could not remotely close the source-isolating valve. The DGM had to be called at 02:35 and physically reach the gate, closing MOV 54 only at 03:01. During the 46-minute delay the release continued, producing the cloud seen at 03:00:44 as 'vapour all around even at height' [EVD-8213], which ignited at 03:08:15. This higher-level failure is a primary enabler of the lower-level AJDE 2.7 and PTF 1.4 response-delay findings.

Linked Evidence: EVD-3932, EVD-9612

SVF 3.2 **Operations Management** **SUBSTANTIATED**

Control-room staff received only four days of general training with no defined qualification or simulator practice; management did not remediate this known competency gap.

JUSTIFICATION

[EVD-5634] records that 'control room operations are carried out by persons inadequately trained in safety, operations, control room handling, logbook records and actions during emergency. It is informed that four days training programme is conducted for new recruitment/transfer staff which is general in nature and later the job training is carried out. Qualification of this staff is also not defined in HR/Operations Policy. Most of the staff have attended only One Basic Course since 2018'. The committee recommended 'at least 1 month (preferably on simulators)' (p.36), nearly eight times the current initial training. Operations management knew the training was general and four days long, with no defined qualification, yet did not correct it. This supervision failure enabled the lower-level AJDE 2.10 inadequate knowledge to make decisions, PTF 1.3 task delegation to untrained staff, and PTF 1.17 unfamiliarity with workspaces, which together produced the initial on-foot verification and telephone escalation instead of immediate isolation.

CAUSAL EXPLANATION

The 4-day general training left the night-shift engineers without practised emergency isolation procedures, so at 02:20 they attempted physical source-location and then telephoned the management chain instead of initiating isolation. The first valve closure occurred at 03:01, 46 minutes after first detection, and the accumulated cloud ignited at 03:08:15. This Level-3 training-management failure directly enabled the lower-level AJDE 2.10 and PTF 1.3 response-decision failures.

Linked Evidence: EVD-5634

SVF 3.2 **Operations Management** **SUBSTANTIATED**

The 2016 MOC SOP was not used after 2018, and management took no action to enforce it, allowing un-reviewed material changes such as the O-ring substitution.

JUSTIFICATION

[EVD-2986] records that 'Management of Change (MOC) philosophy/SOP was published in August 2016. However, the procedure indicated in the document is not implemented while carrying out change of Low-Pressure Gas to Kribhico from Medium Pressure Gas headers in LPG Plant in 2013 & 2018. It is informed no more MOCs are implemented in the plant since 2018'. This is a formally published management control that operations management allowed to lapse into non-use, with zero MOCs implemented since 2018. The same governance void is visible in [EVD-8645]: the switch from OEM gaskets to locally purchased O-rings had 'no record of date or authority of the decision'. This failure enabled the lower-level AJDE 2.11 MOC misapplication finding and ASBE 1.3 wrong-gasket procedure, removing the engineering review that would have challenged the substitute O-ring.

CAUSAL EXPLANATION

With the MOC process dormant since 2018, the substitution of locally-sourced O-rings for OEM gaskets was never subjected to engineering change review. That allowed the non-conforming material to be installed on 18-Sep-2020 and returned to service on 19-Sep-2020 without challenge, leading to failure at 02:12:50 on 24-Sep-2020 and ignition at 03:08:15. This instance is a foundational governance failure enabling the lower-level AJDE 2.11 and ASBE 1.3 findings.

Linked Evidence: EVD-2986, EVD-8645

SVF 3.2 **Operations Management** **SUBSTANTIATED**

Three metering-line pressure transmitters had no data available in the DCS, a known observability gap left uncorrected.

JUSTIFICATION

[EVD-6470] records that 'Three Pressure transmitters 310PT4, 310PT5 & 310PT6 are shown in the drawing. It was informed that no pressure transmitter data are available in the system. No pressure data in Gas Terminal area is available/provided to check rise in pressure in the metering lines'. This is a known instrumentation/data-retention condition — three transmitters on the P&ID with zero pressure data available — that operations management had not corrected. The missing pressure trend removed a potential early indicator of the 24-Sep-2020 release and impaired both real-time diagnosis and forensic reconstruction. It enabled the lower-level PTE 2.11 equipment-not-configured-correctly (rating 3) and PTE 2.5 defective-equipment (rating 3) findings and compounded the F&G system's annunciator-only limitation.

CAUSAL EXPLANATION

Without retained pressure data, operators had no pressure-trend confirmation of the developing flange failure before the 02:15 perimeter detection or the 02:20 annunciator alarm. While the primary response delay arose from isolation-authority and manual valve constraints, the missing data removed an independent diagnostic layer, supporting the lower-level PTE 2.11 finding and narrowing the information available during the 46-minute pre-isolation window.

Linked Evidence: EVD-6470

SVF 3.2 **Operations Management** **SUBSTANTIATED**

Fire drill reports in SAP had no action or observation data and separate records were not in line with approved guidelines; management did not correct this.

JUSTIFICATION

[EVD-2130] records that 'Fire drill reports are available in SAP but without any data on actions taken/recommendations. Separate data is maintained which is not in line with the approved guidelines' (p.26). This created two parallel records: an SAP record containing only dates with no actions or observations, and an informal dataset not compliant with approved guidelines. Emergency-response readiness could not be measured from either record. Operations management left this known documentation-control deficiency uncorrected, meaning drill findings could not be used to test or improve the emergency response that later failed during the 46-minute isolation delay. This supports the lower-level organisational documentation weakness but had no direct mechanical impact on the incident, so it is rated 2.

CAUSAL EXPLANATION

The incomplete drill records meant the organisation could not evidence that emergency response drills had tested the very actions — rapid isolation and escalation of a major hydrocarbon release — that failed on 24-Sep-2020 between 02:15 and 03:01. While not a direct cause of the release or ignition, this known supervision gap left emergency readiness unverified before the incident.

Linked Evidence: EVD-2130

SVF 3.4 **Failed to Mitigate/ Ineffective Action against Identified Risk** **SUBSTANTIATED**

No hydrostatic test to design pressure was required or performed after the 18-Sep-2020 gasket replacement, and no SOP existed for returning the meter to high-pressure service.

JUSTIFICATION

The pressure-boundary verification step required after a flange gasket replacement was a known integrity control, yet supervision allowed the 18-Sep-2020 job to close without it and had not established any SOP for the task. [EVD-2894] records: "After calibration of the meter 310FE01, the gasket was replaced on 18.09.2020. No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated. Next day meter 310FE01 was taken on online and was in use at the time of Incident" (p.31). The same evidence item carries the committee's causal formulation: "Probable cause of leak is due to use of nonstandard gasket and not testing the system to designed pressure. No standard operating procedure is in place for calibration and taking back in service of these meters which are operating at high pressures" (p.32). The supervisory failure here is twofold: the correct post-maintenance test was not enforced for this specific job, and the broader absence of a written return-to-service SOP for high-pressure meters was never corrected, so there was no procedural backstop to catch the

CAUSAL EXPLANATION

Because no design-pressure hydrostatic test was performed, the inadequate O-ring installation was never verified at its design load, and the meter entered hydrocarbon service on 19-Sep-2020 with an unvalidated pressure boundary. It then operated for about five days before the joint failed at 02:12:50 on 24-Sep-2020, releasing vapour for 56 minutes until ignition at 03:08:15. This supervision failure is the direct enabler of the lower-level findings ASBE 1.2 (hydrostatic test omitted, rating 4), AJDE 2.7 (delayed/omitted necessary action, rating 4), AJDE 2.11 (misapplied testing policy, rating 4), and PTE 2.7 (no SOP for return-to-service, rating 4), and it is the reason the defective seal was not caught in a controlled test before live operation.

Linked Evidence: EVD-2894

SVF 3.4 **Failed to Mitigate/ Ineffective Action against Identified Risk** **SUBSTANTIATED**

Heavy pitting and 1.03 mm wall loss on a zero-corrosion-allowance 36-inch line were treated with painting only, not fitness-for-service evaluation.

JUSTIFICATION

A specific known defect on the 36-inch pipeline was mitigated with an action that did not address the underlying integrity risk. [EVD-2397] records: "a reduction of thickness of 1.03mm along with heavy pitting and painting peeling off was observed in one of the segments of 36 pipeline in gas terminal in July 2019. The corrective action of painting was completed in Dec. 2019. However, as per the available data, the design thickness of the pipeline is 23.83mm with zero corrosion allowance. It needs to be examined whether painting as a corrective action would suffice in such cases and the mechanism of deciding on a corrective action needs to be reviewed" (p.30). The pipeline has zero corrosion allowance, so a measured 1.03 mm metal loss with heavy pitting is a fitness-for-service concern, but supervision selected repainting — a coating-restoration measure that cannot restore lost wall thickness or address active pitting growth. This is precisely an ineffective action taken against an identified risk, and the evidence explicitly flags the corrective-action decision mechanism as deficient.

CAUSAL EXPLANATION

The painting-only response left known metal loss and active pitting on a zero-corrosion-allowance pipeline unaddressed, while the site recorded the action as complete. Although this segment is not the source of the 24-Sep-2020 release, the decision pattern — selecting a cosmetic or temporary measure instead of an engineering fix for a measured integrity defect — is the same pattern that allowed the non-conforming O-ring to be fitted repeatedly to meter 310FT01 without engineering evaluation. It directly supports lower-level finding AJDE 2.8 (wrong choice of painting as corrective action, rating 3) and illustrates the supervision-level tolerance for inadequate mitigation of identified risks.

Linked Evidence: EVD-2397

SVF 3.4 **Failed to Mitigate/ Ineffective Action against Identified Risk** **SUBSTANTIATED**

Twelve external-audit recommendations remained open past their dates, with extensions to 30-Apr-2021 lacking the required Director Offshore approval.

JUSTIFICATION

The February 2018 external safety audit produced 308 identified corrective actions, and supervision left 12 of them outstanding with target dates pushed to 30-Apr-2021 without the procedural approval. [EVD-6852] records: "External safety audit was conducted in February 2018, and 296 out of 308 recommendations are complied with. Balance are in various stages and dates are extended till 30/04/2021. Document indicating approval from Director Offshore for extending the target date was not available as per the procedure" (p.26). The control failure is not only that 12 recommendations remained open for more than two years, but that the extensions lacked the competent-authority approval required by the procedure. This is a clear SVF 3.4 manifestation of inadequate follow-up on identified risks and failure to escalate overdue actions through the proper authority channel.

CAUSAL EXPLANATION

The overdue recommendations did not have an identified direct link to the 24-Sep-2020 failure, so this instance is rated present with no direct outcome impact. It nevertheless demonstrates that the site's recommendation-tracking system allowed identified risks to age for years without procedurally valid extension authority, which is the same follow-up weakness that left the recurrent meter gasket problem and the missing hydrotest unflagged in SAP. It supports the broader supervision finding that corrective actions were routinely not closed out under control.

Linked Evidence: EVD-6852

SVF 3.4 **Failed to Mitigate/ Ineffective Action against Identified Risk** **SUBSTANTIATED**

Internal safety audits in 2018 and 2019 issued observations without completed checklists, so follow-up and completeness could not be verified.

JUSTIFICATION

The internal safety audit process identified observations but left no auditable record of what was inspected or whether follow-up was complete. [EVD-7681] states: "Internal Safety Audit was conducted in 2018 & 2019 for 4 & 3 days respectively by a 3-member committee and the observations are given without the filled checklist. The checklist format is not available to check completeness of audit areas" (p.26). Without completed checklists, supervision could not verify which areas were actually covered, whether critical gas-terminal equipment such as the meter gasketing or the flame-proof lighting was examined, or whether prior audit observations had been remedied. The audit therefore generated recommendations without the documentary basis needed to follow them up effectively, a direct failure of supervision to maintain a verifiable mitigation loop for identified findings.

CAUSAL EXPLANATION

The missing checklist trail weakened the site's ability to confirm that identified hazards had been corrected, and it allowed compliance to be asserted without evidence. This did not directly cause the 24-Sep-2020 release, but it is part of the assurance-layer failure that let the compromised HPMV lamp and the gasket substitution practice persist unnoticed until the vapour cloud found the ignition source at 03:08:15. It is rated present with no direct outcome impact because its contribution is systemic and indirect rather than tied to a specific causal step.

Linked Evidence: EVD-7681

SVV 4.1 **Failure to Enforce Existing Rules (Supervisory act of Omission)** **SUBSTANTIATED**

Supervisors allowed locally procured, unspecified O-rings to replace lost OEM rectangular-slot gaskets with no specification or quality control, across three replacements in 22 months.

JUSTIFICATION

Maintenance supervisors and the constituted authority responsible for material control failed to enforce the existing rules requiring OEM-specification replacement parts and quality control on pressure-boundary components. [EVD-8645] records: "The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket" (p.31). The evidential package itself identifies this supervisory omission as the source of the committee's root cause #2 "Use of nonstandard O ring" and root cause #3 "Lack of control over material sourcing". A round O-ring cannot fill a rectangular-slot groove, so the substituted seal left a continuous leak path; the recurrence pattern in [EVD-1494] shows the same locally sourced substitute was installed on 12-Nov-2018, 25-Feb-2019 and 18-Sep-2020 across a

CAUSAL EXPLANATION

By not enforcing the material specification and purchase-control rules, supervision permitted a round O-ring to be substituted for the rectangular-slot gasket on the 1986 meter flange; the geometric mismatch meant the joint could not seal reliably under operating pressure. The joint installed on 18-Sep-2020 was therefore defective from installation, returned to service on 19-Sep-2020 without a design-pressure test, and failed at 02:12:50 on 24-Sep-2020 [EVD-1284]. This supervisory omission is the higher-level enabler of the Level-1 unsafe acts AJDE 2.8 (wrong choice of action), ASBE 1.3 (procedure not followed correctly) and AV 1.1 (routine widespread violation), because the maintenance team's wrong material choice could have been blocked had supervision enforced the procurement and specification rules. The released vapour accumulated for 56 minutes and ignited at 03:08:15 [EVD-4507], making this a primary causal factor in the loss of containment.

Linked Evidence: EVD-8645, EVD-1494

SVV 4.1 **Failure to Enforce Existing Rules (Supervisory act of Omission)** **SUBSTANTIATED**

Maintenance supervision allowed the 18-Sep-2020 gasket job to close out with only an operating-pressure soak and no hydrostatic test to design pressure.

JUSTIFICATION

The supervisor responsible for the 18-Sep-2020 meter 310FT01 gasket job failed to enforce the existing pressure-boundary integrity rule that a hydrostatic test to design pressure must be performed after gasket replacement. [EVD-2894] records: "After calibration of the meter 310FE01, the gasket was replaced on 18.09.2020. No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated. Next day meter 310FE01 was taken on online and was in use at the time of Incident" (p.31). The committee itself concludes: "Probable cause of leak is due to use of nonstandard gasket and not testing the system to designed pressure. No standard operating procedure is in place for calibration and taking back in service of these meters which are operating at high pressures" (p.32). The supervisor's omission of enforcement is distinct from the material-sourcing omission because it occurred at the return-to-service phase, and it extinguished the last verification barrier that could have detected the wrong-geometry O-ring before hydrocarbon exposure.

CAUSAL EXPLANATION

Had supervision enforced the hydrostatic test to design pressure after the 18-Sep-2020 gasket replacement, the non-conforming round O-ring in the rectangular slot would have been exposed under controlled water-pressure conditions before hydrocarbon service. Instead, the meter returned to online service on 19-Sep-2020 with an unvalidated pressure boundary, operated for approximately six days under condensate pressure, and the joint failed at 02:12:50 on 24-Sep-2020. This omission is the supervisory-level cause of the lower-level findings AJDE 2.7 (delayed/omitted necessary action) and ASBE 1.2 (checklist not followed), and it directly produced the initiating release that accumulated for 56 minutes and ignited at 03:08:15.

Linked Evidence: EVD-2894, EVD-0185

SVV 4.1 **Failure to Enforce Existing Rules (Supervisory act of Omission)** **SUBSTANTIATED**

Supervisors allowed three gasket replacements in 22 months without triggering root-cause review, treating a chronic sealing failure as routine maintenance.

JUSTIFICATION

Maintenance supervision did not enforce the reliability-management rule that repeated failures must trigger root-cause investigation and corrective engineering. [EVD-1494] records: "Cleaning/servicing/calibration of meters 1 & 2 was carried out on 12th & 13th Nov 2018. Afterwards, meter no.1 was cleaned on 25/02/2019 & gasket was changed. Again on 18/09/2019, gasket was changed which leaked within one year" (p.27). With three gasket replacements in approximately 22 months on the same 1986-vintage meter, the pattern was a clear warning that the substitute seal design was failing repeatedly; supervisors nevertheless treated each event as a routine replacement and did not require OEM specification recovery, meter replacement, or design review. This is distinct from the single-job omission because it spans the 2018-2020 recurrence history and represents a failure to enforce the process for managing chronic reliability problems.

CAUSAL EXPLANATION

The three-failure recurrence (12-Nov-2018, 25-Feb-2019, 18-Sep-2020) was a reliability warning that the locally sourced O-ring substitution was inadequate, but no supervisor escalated it for investigation. This failure to enforce the rule created the environment for the lower-level AJDE 2.5 (ignored caution/warning) finding: the same inadequate repair was repeated a third time and failed within six days on 24-Sep-2020 at 02:12:50, initiating the release that ignited at 03:08:15. It is rated secondary because the immediate physical cause was the 18-Sep-2020 installation itself, but the supervisory omission allowed that immediate cause to recur.

Linked Evidence: EVD-1494

SVV 4.1 **Failure to Enforce Existing Rules (Supervisory act of Omission)** **SUBSTANTIATED**

Management extended 12 external audit recommendation dates to 30-Apr-2021 without the required Director Offshore approval.

JUSTIFICATION

Management failed to enforce the procedure requiring documented approval from the competent authority before extending the target dates of outstanding external safety audit recommendations. [EVD-6852] records: "External safety audit was conducted in February 2018, and 296 out of 308 recommendations are complied with. Balance are in various stages and dates are extended till 30/04/2021. Document indicating approval from Director Offshore for extending the target date was not available as per the procedure" (p.26). Although 96% of the items were reported complied, the process-control failure lay in the unapproved extensions for the remaining 12 items, which meant deadlines were moved without the required competent-authority authorisation. This is a distinct supervisory-omission instance in the audit-closeout system, separate from the maintenance and electrical findings.

CAUSAL EXPLANATION

This supervisory omission did not directly initiate the 24-Sep-2020 fire, but it demonstrates that the organisation's management did not enforce its own procedure for controlling overdue safety actions. By allowing unapproved date extensions, supervision normalised the deferral of known findings. This same non-enforcement pattern reappeared in the 2017 pipeline thickness findings that had no corrective plan after three years [EVD-5527] and in the unclosed SAP notification on meter 310FT01; therefore it is rated present but with no direct impact on this incident's physical outcome.

Linked Evidence: EVD-6852

SVV 4.1 **Failure to Enforce Existing Rules (Supervisory act of Omission)** **SUBSTANTIATED**

Supervisors allowed missing, untracked work permits and absent Job Safety Plans, with maintenance notifications treated as permits.

JUSTIFICATION

Supervision failed to enforce the work permit system requirements that would have required a hazard-specific Job Safety Plan for high-risk maintenance. [EVD-5105] records: "Work Permits are issued in SAP and hard copy of work permits are at sites during execution of work. After completion of work, the originals are filed. On checking the same, it was found that few of the work permits were missing. No data is kept on such missing permits. Job safety Plan is not attached/given in SAP document. Closing of permits & actions indicated in work needs review" (p.26). The committee's recommendation B.5 at p.34 was quoted as: "Maintenance notifications are not be assumed as permit to work", indicating that maintenance was being executed under SAP notifications without the separate work-permit authorisation the rules required. This distinct supervisory omission in the permit-control system removed the task-level hazard review that should have identified the required hydrotest and material-conformance checks for the gasket job.

CAUSAL EXPLANATION

Because the permit and JSP rules were not enforced, the 18-Sep-2020 gasket replacement could be treated as a routine notification-driven job without a hazard-specific plan identifying the design-pressure test and material-conformance checks. This created the environment for the lower-level procedural omissions ASBE 1.2 (hydrotest checklist omitted) and AJDE 2.11 (misapplication of policies). It is rated secondary rather than primary because the available evidence does not tie a specific missing permit directly to the failed joint, but the systemic permit-control failure removed a barrier that would otherwise have challenged the defective repair before it returned to service on 19-Sep-2020.

Linked Evidence: EVD-5105

SVV 4.1 **Failure to Enforce Existing Rules (Supervisory act of Omission)** **SUBSTANTIATED**

The published Aug-2016 MOC SOP was not enforced; no MOCs were implemented since 2018, allowing changes including the gasket substitution to bypass engineering review.

JUSTIFICATION

Management did not enforce the Management of Change SOP that had been formally published in August 2016, allowing significant plant changes to proceed without MOC review. [EVD-2986] records: "Management of Change (MOC) philosophy/SOP was published in August 2016. However, the procedure indicated in the document is not implemented while carrying out change of Low-Pressure Gas to Kribhico from Medium Pressure Gas headers in LPG Plant in 2013 & 2018. It is informed no more MOCs are implemented in the plant since 2018" (p.30). The same evidence cross-links this non-enforcement to the gasket substitution: there was "no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket" (p.31). Supervisors therefore allowed a published, mandatory engineering-control process to become a paper artefact, and this omission is distinct from the gasket-procurement finding because it concerns the management-of-change governance layer across the plant.

CAUSAL EXPLANATION

By failing to enforce the MOC SOP, supervision removed the engineering-review gate that should have challenged the substitution of a non-OEM O-ring for the OEM rectangular-slot gasket as a significant material change. This created the environment for the lower-level finding AJDE 2.11 (misapplication of MOC policy) and enabled the undocumented 'regular practice' of locally sourced gaskets to persist. Because the MOC bypass was a foundational governance failure rather than the immediate mechanical cause, it is rated secondary, but it directly enabled the wrong-material decision that failed at 02:12:50 on 24-Sep-2020.

Linked Evidence: EVD-2986, EVD-8645

SVV 4.1 **Failure to Enforce Existing Rules (Supervisory act of Omission)** **SUBSTANTIATED**

Internal and self-safety audits were accepted as complete without filled checklists, allowing plant-specific integrity deficiencies to be missed.

JUSTIFICATION

Audit supervision did not enforce the requirement for completed, auditable checklists in internal and self-safety audits. [EVD-7681] records: "Internal Safety Audit was conducted in 2018 & 2019 for 4 & 3 days respectively by a 3-member committee and the observations are given without the filled checklist. The checklist format is not available to check completeness of audit areas" (p.26). [EVD-9692] adds: "Safety self-audit is conducted by a multi-disciplinary team based on the checklist prepared by the location. The checklist is comprehensive but not related to specific plants and in the process, lot of observations are missing. The duration of audit is not indicated and in most of the items, compliances are indicated, but during our inspection, the data is found to be incomplete & not in line with OISD standards/manuals" (p.26). These omissions are distinct from the electrical-PM audit finding because they concern the general audit assurance process across the plant, not the post-incident electrical damage report.

CAUSAL EXPLANATION

Because audit supervision did not enforce checklist completion and verifiable compliance data, the audit assurance layer never systematically inspected plant-specific risks such as Zone 1 flame-proof lamp integrity or the metering gasket substitution practice. This is the supervisory-level cause of the lower-level ASBE 1.2 audit-checklist finding, and it allowed the compromised HPMV lamp with three missing bolts and a 5-6 cm open glass case [EVD-6916] to remain in service. Without that supervisory omission, the ignition source present when the cloud reached the lamp at 03:08:15 might have been identified and corrected before the fire.

Linked Evidence: EVD-7681, EVD-9692

SVV 4.5 No Action/ Mitigation Measures for Identified Risk

SUBSTANTIATED

The 2017 thickness report flagged pipes needing corrective action; three years later no plan existed — measurement without remediation.

JUSTIFICATION

A formal 2017 integrity report identified pipes requiring corrective action, and supervision took no mitigation step for more than three years until the 24-Sep-2020 incident. [EVD-5527] records the committee's direct phrasing: "As per the report of 2017, several pipes have been identified with significant thickness loss and need for corrective action. However, even after three years no plan has been developed for the corrective actions to be taken for those lines. Mere measurement of thickness without corrective action does not complete the process of maintaining the integrity of pipes" (p.30). The risk was formally identified in writing, yet from 2017 to 24-Sep-2020 no remediation plan, risk-based priority, or fitness-for-service evaluation was produced. This is a textbook SVV 4.5 instance at the pipeline-integrity system level and is the Level-3 origin of the lower-level AJDE 2.5 ignored-warning finding (rating 3), AJDE 2.7 delayed-necessary-action finding (rating 3) and PTE 2.12 integrity-maintenance finding (rating 2). It is rated 3 because the 363 pipeline itself remained intact [EVD-5780] and this inaction did not directly caus

CAUSAL EXPLANATION

The 2017 report functioned as a formal integrity warning; by leaving its findings without a corrective plan from 2017 to 24-Sep-2020, supervision demonstrated the organisational default of acknowledging findings without acting on them — the same default that let the metering gasket recurrence (three replacements in 22 months) go unexamined. This instance did not itself open the leak — the 363 pipeline survived the fire intact — but it supervisory culture in which identified risks evaporated without mitigation, directly enabling the AJDE 2.5 and AJDE 2.7 lower-level findings on pipeline integrity and reinforcing the environment in which the gasket hazard persisted until 02:12:50 on 24-Sep-2020.

Linked Evidence: EVD-5527

Level 4: Organizational Influences

OIR 1.1 Command and Control Resources are Deficient

SUBSTANTIATED

The F&G system was display-only with no auto-shutdown or suppression interlock, forcing a 46-minute manual isolation response after the 02:12:50 leak.

JUSTIFICATION

The plant's flame and gas detection system commanded no protective action: it could detect but could not act. [EVD-3452] records the design implication at p.26 that the "F&G detection system does not have provision for auto shut down of plant or activation of fire suppression system", and the committee's recommendation A.1 at p.34 explicitly calls for "F&G detection system needs to be interlocked to automatic plant shut down and activation of fire suppression system." [EVD-8657] repeats the same configuration observation alongside the alarm record: "Hydrocarbon/H2S detectors are configured to Annunciator Panel for display but not in DCS for history" and "F&G detection system does not have provision for auto shut down of plant or activation of fire suppression system." The consequence is visible in the timeline: the leak commenced at 02:12:50 and was machine-visible on CCTV, but with no interlock to close MOV 54 or trip the plant, the entire 46-minute manual response (CISF detection 02:15, annunciator alarm 02:20, shift-in-charge arrival ~02:35, MOV 54 closure 03:01, 36-inch PCV closure 03:04) had to execute before isolation, and the cloud ignited

CAUSAL EXPLANATION

With no auto-shutdown interlock, the machine witness of the release at 02:12:50 produced zero protective response; the system could annunciate but could not close a valve, trip a unit, or start deluge. The incident therefore depended entirely on the slow human chain that followed: CISF perimeter detection at 02:15, CCTV verification and control-room call at 02:18, annunciator alarm at 02:20, senior-shift call-out at ~02:35, and first isolation only at 03:01 when MOV 54 was closed locally, with the 36-inch PCV closing at 03:04. The 46-minute unisolated release from 02:15 to 03:01 (48 minutes from leak onset) is what filled the area with the mature cloud documented at 03:00:44 as "vapour all around even at height" [EVD-8213], and that already-released inventory ignited at 03:08:15 after both valves were shut, proving that an automatic interlock could have collapsed the release window from minutes to seconds. This Level-4 deficiency created the environment for the lower-level findings PTE 2.6/PTE 2.11 (equipment not configured correctly — no interlock), PTE 2.2 (F&G system no automated protection), SVI 1.4 (management never required automatic shutdown/suppression), and

OIR 1.1 **Command and Control Resources are Deficient** **SUBSTANTIATED**

Detectors displayed on the annunciator only with no DCS history, so the first control-room alarm came at ~02:20, about 7 minutes after the 02:12:50 leak onset.

JUSTIFICATION

The gas and H2S detectors were wired for annunciator display only, not for DCS history, so the control room had no trend or predictive alarm on the developing release. [EVD-8657] records the configuration verbatim: "Hydrocarbon/H2S detectors are configured to Annunciator Panel for display but not in DCS for history." The quantitative consequence is in the same evidence item: leak onset at 02:12:50 (confirmed by [EVD-1284]) versus the first annunciator alarm at ~02:20 — the automated alarm lagged machine-visible leak onset by approximately 7 minutes 10 seconds and was 5 minutes behind the first human detection by CISF at 02:15. The command-and-control resource therefore provided the least responsive possible alarm path (display-only) and no data retention for diagnosis, although independent human detection and the eventual annunciator still reached the operators. This instance is distinct from the missing auto-shutdown interlock: here the deficiency is the detection/alarm latency and the absence of historised trend data, not the absence of a protective action.

CAUSAL EXPLANATION

Because the detectors had no DCS historisation and no trending alarm, the 02:12:50 onset produced no control-room alarm for about 7 minutes. The operators first learned of the event through the CISF relay phone call and the annunciator together at ~02:18-02:20, more than five minutes after the perimeter guard had detected it at 02:15. While this 5-7 minute detection lag was not the dominant contributor to the 46-minute isolation delay — the absence of remote movable valves and DMP isolation criteria dominated — it consumed the earliest minutes of the response and forced the operating crew to begin assessment only after the cloud was already dense enough to block foot access. This Level-4 gap underlies the lower-level findings PTE 2.2 (detection system lagged human detection by 5 minutes), PTE 2.6 (gas detectors not historised in DCS), PPI 5.7 (auditory cue masked the maturity of the release), and the early ASBE 1.6 visual-scan / detection-latency findings, adding secondary latency to an already manual response chain that ended at 03:01 isolation and 03:08:15 ignition.

Linked Evidence: EVD-8657, EVD-1284

OIR 1.1 **Command and Control Resources are Deficient** **SUBSTANTIATED**

MOV 54 and the Gas Terminal MOVs could only be closed locally on-site, so first isolation took 46 minutes instead of being remotely executable at the 02:20 alarm.

JUSTIFICATION

The first isolation-valve resource available to stop slug-catcher inventory reaching the failed meter could not be operated from the control room. [EVD-9612] records that the GSP Shift In-charge "closed the MOV 54 inlet condensate line from slug catcher to 36 metering area at about 0301 hrs" and that "the MOVs in the gas terminal station are not remote-operable from the control room during emergency (p.26)." [EVD-3932] reinforces the same architecture: "MOVs are provided in the gas terminal station for local operations but they are not having remote operation from control room during emergency." The consequence is measurable in the timeline: the control-room alarm sounded at 02:20, but because MOV 54 was local-manual only, the DGM had to be called from quarters at ~02:35, physically reached the Gas Terminal Gate, and could only execute the local close at 03:01 — 46 minutes after first human detection at 02:15 and 48 minutes 10 seconds after CCTV-confirmed leak onset at 02:12:50, with ignition following at 03:08:15. This is the single most direct command-and-control resource deficiency driving the prolonged pre-isolation release window.

CAUSAL EXPLANATION

Because the source-isolating valve MOV 54 was local-manual only, the control-room engineers who received the 02:20 alarm could not close it; the same was true for the other Gas Terminal MOVs. Closure instead required the physical arrival of the GSP Shift In-charge at the gate and a local operation at 03:01, which cost 46 minutes from first physical detection at 02:15. Throughout that interval the failed gasket at 310FT01 continued releasing under operating pressure, and the cumulative effect is visible in the CCTV sequence: by 03:00:44 the cloud was "vapour all around even at height" [EVD-8213] and it ignited at 03:08:15, only 7 minutes after the first closure and 4 minutes 15 seconds after the 36-inch PCV closed at 03:04 — proving the already-accumulated inventory was the fire fuel. This Level-4 resource gap directly produced the lower-level findings PTE 2.4/PTE 2.8 (controls inoperable remotely / workspace incompatible with operations), PTE 2.11 (MOV not remotely operable), PTF 1.4 over-reliance on manual local access, SVF 3.2 operations management left the manual-only design uncorrected, and the Level-1 delayed-necessary-action findings that governed

Linked Evidence: EVD-9612, EVD-3932

OIR 1.1 **Command and Control Resources are Deficient** **SUBSTANTIATED**

A single 360-degree panning camera with ~90-second turnaround was the sole CCTV coverage; its cable burned at 03:08:22, ending the only visual monitoring channel.

JUSTIFICATION

The metering-skid and Gas Terminal Control Room area had only one panning camera as its video command-and-control resource, and that camera's view was intermittent by design. [EVD-0210] identifies the equipment: "PELCO make PTZ cameras (Model D5230D) are provided at few places and has turn around (360 degree) at the time of about 90 seconds. Visibility from this camera is limited" (p.30). [EVD-2752] records the terminal consequence: at 03:08:22, "major fire continued, CCTV capturing discontinued due to burnt of CCTV cable" (p.25), ending the sole video chain seven seconds after the first flame at 03:08:15. The committee's p.36 recommendation that "CCTV arrangement shall be provided for each plant including separately with display at plant control rooms as well as master control/CISF station" confirms the coverage was recognised as inadequate. Detection in this incident still occurred through CISF hearing and sight at 02:15 and the annunciator at 02:20, so this deficiency was secondary rather than the primary detection failure, but it removed continuous remote situational awareness and destroyed the post-ignition evidence

CAUSAL EXPLANATION

The single panning camera meant the metering area was only intermittently observed, with up to 90 seconds between returned views, and there was no fixed redundant coverage or plant-control-room display. In the event, human detection at 02:15 and the annunciator at 02:20 compensated, so the camera did not cause the primary detection lag; its causal contribution was the CISF-internal verification hop between 02:15 and 02:18 and the loss of continuous situational awareness for the incident commander. At 03:08:22 the cable burned through seven seconds after the 03:08:15 ignition, terminating the sole visual record exactly at the release-to-fire transition, which handicapped both live post-ignition response and later forensic confirmation of the fire spread and ignition source. This Level-4 condition therefore created the environment for the lower-level PTE 2.6/PTE 2.8/PTE 2.10 camera-coverage findings and the recommendation that plant control rooms receive dedicated CCTV displays; its direct causal share is secondary because detection occurred independent of the camera.

Linked Evidence: EVD-0210, EVD-2752

OIR 1.1 **Command and Control Resources are Deficient** **SUBSTANTIATED**

Perimeter security reports were routed through the CCTV Control Room rather than directly to the Gas Terminal Control Room, adding a 3-5 minute notification lag.

JUSTIFICATION

The communication architecture forced the earliest and most direct leak information through CISF-internal relay nodes instead of the operating control room that could act on it. [EVD-6234] shows Head Constable B at the Gas Terminal Gate, on detecting the 02:15 incident, "informed CCTV Control room and CISF main gate control room" (p.17) — not the Gas Terminal Control Room. [EVD-8871] describes the resulting relay: the CCTV Control Room Sub-Inspector C received the call at 02:15, visually verified "heavy leakage and formation of gas cloud through CCTV" (timeline p.12) at 02:18, and only then notified the Gas Terminal Control Room. The consequence is a ~3-5 minute loss between the 02:15 physical detections and the 02:18-02:20 operating-CR notification, at a time when [EVD-9941] shows the cloud expanding across the control room, road, and piping area within ~40 seconds. The resource deficiency — no direct perimeter-to-control-room link — therefore delayed the start of the operating crew's response rather than preventing detection entirely, making it a secondary contributing factor.

CAUSAL EXPLANATION

Because the CISF perimeter posts and gate had no direct line to the Gas Terminal Control Room, both simultaneous 02:15 detections were sent only to CISF internal nodes. The operating crew became aware only after the CCTV Control Room had independently verified the leak and placed a separate call at 02:18-02:20, so the response started about five minutes after the first human detection and about seven minutes after the 02:12:50 onset, with the vapour cloud already expanding across the area. That lost margin fed the serial escalation that followed — shift-in-charge called at ~02:35, MOV 54 closed at 03:01, PCV at 03:04, ignition at 03:08:15. This Level-4 resource gap underlies the lower-level findings PTF 1.10 (critical leak information omitted from the operating CR by two independent CISF reporters), PTF 1.16 (the two perimeter reports were routed through internal verification nodes, losing ~5 minutes), and PTE 2.10 (communications equipment/infrastructure inadequate for direct notification), and it contributed its share of latency to the 46-minute pre-isolation window.

Linked Evidence: EVD-6234, EVD-8871

OIR 1.1 **Command and Control Resources are Deficient** **SUBSTANTIATED**

DCS clocks across control rooms differed by 2-12 minutes, complicating cross-room alarm correlation and event reconstruction without directly changing the outcome.

JUSTIFICATION

The multiple DCS installations did not share a common time base, degrading the command-and-control system's ability to correlate alarms and audit events. [EVD-5532] records the condition: "Time stamping in DCS systems provided in various control rooms is varying from 2 minutes to 12 minutes and data matching is difficult" (p.26). The incident narrative confirms the operational consequence — "where DCS and CCTV disagree the CCTV timestamp is taken as canonical; we have observed that DCS clocks across the multiple control rooms drift between approximately 2 and 12 minutes against each other and this is being corrected." In this incident the drift did not change the physical outcome because the sequence was independently anchored by CCTV frames and witness statements, and no decision was shown to have been misled by a false inter-room alarm correlation. Its impact was forensic and systemic: it complicated event reconstruction and means simultaneous alarms in different control rooms could appear separated by up to 12 minutes, undermining trust in the automation resource.

CAUSAL EXPLANATION

The 2-12 minute drift did not directly cause the release, the 46-minute isolation delay, or the 03:08:15 ignition; those were driven by the failed gasket, the manual-only valves, the F&G interlock gap, and the missing isolation criteria. The deficiency's causal role was the degradation of the command-and-control information environment: operators could not trust cross-control-room time correlation in real time, and investigators had to fall back on CCTV timestamps as canonical to reconstruct the sequence. It is rated present without direct outcome impact, consistent with the lower-level PTE 2.6/PTE 2.11 clock-drift findings rated 2, but it is a substantiated weakness of the plant's DCS command-and-control resource that should be corrected to prevent future misdiagnosis of correlated alarms.

Linked Evidence: EVD-5532

OIR 1.1 **Command and Control Resources are Deficient** **SUBSTANTIATED**

The three metering-line pressure transmitters shown on the P&ID delivered no data to the DCS, leaving the Gas Terminal area without pressure-trend monitoring.

JUSTIFICATION

The Gas Terminal metering area was nominally instrumented with pressure transmitters, but the control room received no data from them, removing a basic pressure-trend resource for both operation and investigation. [EVD-6470] records: "Three Pressure transmitters 310PT4, 310PT5 & 310PT6 are shown in the drawing. It was informed that no pressure transmitter data are available in the system. No pressure data in Gas Terminal area is available/provided to check rise in pressure in the metering lines" (p.26). The incident therefore proceeded with no pressure-based indication of the developing flange failure before the 02:15 perimeter detection or the 02:20 annunciator alarm, and with no pre-incident pressure profile to establish whether the loss of containment was driven by overpressure, installation error, or material defect. The deficiency did not create the leak and the detection chain still functioned through CISF and the annunciator, but it removed an independent early-warning and diagnostic layer from the command-and-control resource environment, making it a secondary contributing factor.

CAUSAL EXPLANATION

With 310PT4/PT5/PT6 reporting nothing to the DCS, the Gas Terminal operators had no pressure-trend warning that the metering lines were behaving abnormally after the 18-Sep-2020 gasket replacement, and the investigators were denied the pre-incident pressure record that could have separated overpressure, installation error, and material defect as competing failure mechanisms. The primary detection and isolation chain still operated — CISF heard and saw the vapour at 02:15 and the annunciator alarmed at 02:20 — so this resource deficiency did not by itself extend the 46-minute delay before MOV 54 closure at 03:01 or prevent the ignition at 03:08:15. It did, however, compound the F&G display-only configuration and the missing DCS history to narrow the information available to the operating crew during the pre-isolation window, and it underlies the lower-level PTE 2.5/PTE 2.11 findings on the absent transmitter data and the secondary detection-diagnosis limitations documented in the response-delay chain.

Linked Evidence: EVD-6470

OIR 1.5 **Failure in Timely Acquisition/ Procurement/ Upgrade** **SUBSTANTIATED**

Gas detectors were left as annunciator-display only with no auto-shutdown or suppression interlock, converting a machine-detected 02:12:50 leak into a 46-minute manual response.

JUSTIFICATION

A distinct acquisition/upgrade failure is documented for the flame-and-gas detection system, independent of the gasket procurement instance. [EVD-8657] records the design state verbatim: 'Hydrocarbon/H2S detectors are configured to Annunciator Panel for display but not in DCS for history' and 'F&G detection system does not have provision for auto shut down of plant or activation of fire suppression system.' The detector layer thus existed only as a display-and-annunciator function: it could alert but could not act. The regulator's own recommendation in [EVD-3452], committee recommendation A.1 at p.34, confirms the upgrade gap: 'F&G detection system needs to be interlocked to automatic plant shut down and activation of fire suppression system.' The quantitative consequence is in the timeline: the leak was machine-witnessed at 02:12:50 (CCTV), the annunciator only alarmed at 02:20 — about 7 minutes 10 seconds later and 5 minutes after human detection — and no automatic protective action fired at any point; the first isolation was a manual MOV 54 closure at 03:01 and the 36-inch PCV at 03:04, with ignition at 03:08:15. The correct design would

CAUSAL EXPLANATION

With no automatic interlock in place, the 02:12:50 leak was left entirely to the slow human detection-verification-escalation-authorization chain: CISF detection at 02:15, CCTV verification 02:18, control-room notification 02:18-02:20, senior call-out at ~02:35, MOV 54 closure at 03:01 and PCV closure at 03:04 — a 46-minute release window during which the cloud grew to the 'vapour all around even at height' state captured at 03:00:44 [EVD-8213] and ignited at 03:08:15, after isolation, proving the already-released inventory was the fuel. This Level-4 upgrade failure created the conditions for the Level-3 finding SVF 3.2 operations management (rating 4), the Level-2 findings PTE 2.11 misconfiguration (rating 4), PTF 1.4 over-reliance on the human system (rating 4) and PTE 2.2 instrumentation inadequacy (rating 4), and the Level-1 AV 1.1 F&G configuration violation (rating 3).

Linked Evidence: EVD-8657, EVD-3452

OIR 1.5 **Failure in Timely Acquisition/ Procurement/ Upgrade** **SUBSTANTIATED**

MOV 54 and the gas-terminal MOVs remained local-manual with no emergency remote control, forcing physical attendance and delaying first isolation to 03:01 — 46 minutes after detection.

JUSTIFICATION

A separate, independently evidenced acquisition failure applies to the emergency isolation valve infrastructure. [EVD-3932] records the committee's structural finding: 'MOVs are provided in the gas terminal station for local operations but they are not having remote operation from control room during emergency' (p.26). [EVD-9612] corroborates the operational consequence: the DGM-GSP shift in-charge 'closed the MOV 54 inlet condensate line from slug catcher to 36 metering area at about 0301 hrs' only after being called from quarters at ~02:35 and physically reaching the Gas Terminal Gate — closure 46 minutes after first physical detection at 02:15 and 48 minutes after the 02:12:50 leak onset. The correct configuration for a time-critical hydrocarbon isolation layer would have made MOV 54 (and the gas-terminal MOV set) remotely operable from the Gas Terminal Control Room so that the engineers who heard the annunciator at 02:20 could have closed the source-isolating valve within minutes; the ACTUAL configuration required physical attendance, which the dense vapour cloud denied. The failure to fund and execute this upgrade is a primary

CAUSAL EXPLANATION

Because MOV 54 had no remote operation, the control-room engineers who received the 02:20 annunciator alarm could not close it themselves; the DGM had to be called from quarters at ~02:35, travel to the Gas Terminal Gate, and execute a local close at 03:01, while the 36-inch PCV followed at 03:04. During those 46 minutes the release continued unisolated, producing the vertical vapour column visible at 03:00:44 [EVD-8213] that ignited at 03:08:15. This Level-4 upgrade failure is the direct enabler of the Level-2 findings PTE 2.4 controls inadequate (rating 4), PTE 2.11 equipment not configured correctly (rating 4), PTF 1.4 over-reliance on manual field operation (rating 4), and PTE 1.5 roles/authority gap, and it underlies the Level-1 AJDE 2.7 delayed necessary action (rating 4) and AJDE 2.9 operational decision failures that extended the release.

Linked Evidence: EVD-3932, EVD-9612

OIR 1.5 **Failure in Timely Acquisition/ Procurement/ Upgrade** **SUBSTANTIATED**

Sole PELCO PTZ camera with a ~90-second pan and limited visibility provided the only monitoring of the skid; it was lost with its cable at 03:08:22 and had no control-room displays.

JUSTIFICATION

A distinct upgrade failure is evidenced for the visual monitoring system serving the metering skid. [EVD-0210] records the equipment constraint: 'PELCO make PTZ cameras (Model D5230D) are provided at few places and has turn around (360 degree) at the time of about 90 seconds. Visibility from this camera is limited.' The same single camera was the only video coverage of the skid and gas-terminal area, and its cable burned through at 03:08:22, seven seconds after ignition, terminating the evidence chain mid-transition [EVD-2752]. The committee's own recommendation — 'CCTV arrangement shall be provided for each plant including separately with display at plant control rooms as well as master control/CISF station' (p.36 in [EVD-2752]) — confirms that per-plant redundant CCTV with control-room displays had never been acquired. The ACTUAL configuration also lacked display at the plant control rooms, which is why the 02:15 detections had to be relayed through the CISF CCTV control room (verification at 02:18) before the Gas Terminal CR was telephoned at 02:18-02:20. The single-camera gap did not prevent detection — CISF Constable A heard and saw

CAUSAL EXPLANATION

The single intermittent camera meant the skid area was observed only when the pan cycle reached it, and no fixed or redundant camera could continue after the cable burned at 03:08:22. In the actual sequence, detection still came from perimeter staff at 02:15, but the verification relay through the CISF CCTV control room (02:18) added 3-5 minutes before the operating CR was notified at 02:18-02:20, and after ignition the incident commander lost the only remote visual feed. This Level-4 acquisition failure enabled the Level-2 findings PTE 2.6 system-redundancy gap (rating 3), PTE 2.8 workspace incompatibility (rating 3) and PTE 2.10 communications equipment inadequacy (ratings 3 and 2), and the Level-1 ASBE 1.6 visual-scan breakdown; it is rated 3 because the dominant 46-minute delay was driven by valve-actuation and authority failures rather than the monitoring design.

Linked Evidence: EVD-0210, EVD-2752

OIR 1.5 **Failure in Timely Acquisition/ Procurement/ Upgrade** **SUBSTANTIATED**

Three pressure transmitters shown on the drawing had no data in the DCS, leaving the metering lines without pressure-trend monitoring or a reconstructable pressure profile.

JUSTIFICATION

A further distinct equipment-upgrade failure concerns the pressure instrumentation at the gas terminal. [EVD-6470] records: 'Three Pressure transmitters 310PT4, 310PT5 & 310PT6 are shown in the drawing. It was informed that no pressure transmitter data are available in the system. No pressure data in Gas Terminal area is available/provided to check rise in pressure in the metering lines.' Three transmitters exist on the P&ID but the DCS was never configured to acquire, historize or trend their signals, so the metering system ran with zero pressure observability in both real time and forensic terms. The correct configuration would have brought these instruments into the DCS for live trend display and history, enabling pre-incident pressure profiling; the ACTUAL configuration left them connected nowhere that data could be checked. This acquisition/commissioning failure removed a potential early-anomaly indicator of the developing flange failure and denied investigators the pre-incident pressure profile needed to distinguish overpressure, installation error, or material defect as the failure mechanism. It is a secondary contributing factor because detection still occurred via CISF

CAUSAL EXPLANATION

Without any pressure data from 310PT4/5/6, the control room had no pressure-trend indication that could have surfaced the developing flange failure before the 02:15 perimeter detection, and no pressure history was available to confirm whether the 24-Sep-2020 failure was preceded by abnormal pressure behaviour after the 18-Sep-2020 gasket replacement. The gap compounded the annunciator-only F&G configuration (OIR 1.5 instance 2) and the DCS history gap [EVD-8657], and it enabled the Level-2 findings PTE 2.5 defective equipment in use (rating 2) and PTE 2.11 equipment not configured correctly (rating 3) plus the Level-3 finding SVF 3.2 on the uncorrected observability gap (rating 3). Its direct causal share is secondary because the loss of containment originated in the gasket joint, but it degraded both real-time diagnosis and post-incident certainty.

Linked Evidence: EVD-6470

OIR 1.6 **Failure in Maintenance/ Spares/ Support Programme** **SUBSTANTIATED**

Three gasket replacements in ~22 months on the 1986 meter never triggered root-cause investigation, OEM spares recovery or meter replacement; the same inadequate fix recurred until the fourth failure.

JUSTIFICATION

The maintenance-and-spares programme for meter 310FT01 failed to respond to a documented chronic sealing failure. [EVD-1494] records: 'Cleaning/servicing/calibration of meters 1 & 2 was carried out on 12th & 13th Nov 2018. Afterwards, meter no.1 was cleaned on 25/02/2019 & gasket was changed. Again on 18/09/2019, gasket was changed which leaked within one year' — with the contextual p.31 dating the most recent change as 18-Sep-2020, giving three gasket replacements in ~22 months (12-Nov-2018, 25-Feb-2019, 18-Sep-2020) on a 34-year-old meter. The evidence pack's own summary states the pattern 'did not trigger material/procedure review and the same locally-sourced O-ring continued in use.' The correct maintenance-organisation response to a second identical failure would have been root-cause analysis, re-establishment of the OEM gasket supply, or replacement of the meter itself; the ACTUAL programme treated each failure as a routine reseal and reordered the same non-conforming O-ring. This Level-4 maintenance-programme failure is a primary organizational cause — it guaranteed that the defective condition persisted

CAUSAL EXPLANATION

Because the maintenance organisation never converted the 12-Nov-2018 and 25-Feb-2019 failures into a root-cause investigation, an OEM seal supply chain, or a meter replacement decision, the 18-Sep-2020 job repeated the same unsuitable O-ring, returned the meter to service on 19-Sep-2020, and the joint failed at 02:12:50 on 24-Sep-2020, releasing vapour for 56 minutes before ignition at 03:08:15. This Level-4 failure is the organizational origin of the Level-3 findings SVI 1.6 (rating 4), SVI 1.8 (rating 4), SVF 3.2 recurrent-defect management (rating 4), SVF 3.4 (rating 4), SVV 4.5 (rating 4) and SVIO 2.7 inappropriate asset management (rating 3), and of the Level-1 AJDE 2.5 ignored-warning finding (rating 4) that allowed the initiating defect to persist.

Linked Evidence: EVD-1494

OIR 1.6 **Failure in Maintenance/ Spares/ Support Programme** **SUBSTANTIATED**

No hydrostatic test was performed after the 18-Sep-2020 gasket replacement and no SOP governed high-pressure meter return-to-service; the meter re-entered service on operating-pressure validation only.

JUSTIFICATION

A distinct maintenance-programme failure is the absence of the pressure-boundary verification step and its governing procedure. [EVD-2894] records: 'After calibration of the meter 310FE01, the gasket was replaced on 18.09.2020. No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated. Next day meter 310FE01 was taken on online and was in use at the time of Incident.' The same evidence quotes the committee's probable-cause finding: 'Probable cause of leak is due to use of nonstandard gasket and not testing the system to designed pressure. No standard operating procedure is in place for calibration and taking back in service of these meters which are operating at high pressures.' The correct programme would have required a hydrostatic test to design pressure after any pressure-boundary intervention, with a documented SOP governing calibration and return-to-service; the ACTUAL programme substituted an operating-pressure soak and had no governing procedure at all. This maintenance-programme gap removed the last technical barrier that could have

CAUSAL EXPLANATION

Because the maintenance programme neither required nor provided a procedure for a design-pressure hydrotest, the inadequate O-ring installation of 18-Sep-2020 was never verified at design load; a correct hydrotest would most likely have failed the joint under controlled water-pressure conditions. The meter instead entered live service on 19-Sep-2020 and the unvalidated seal failed at 02:12:50 on 24-Sep-2020, producing the 56-minute vapour accumulation that ignited at 03:08:15. This Level-4 programme failure is the organizational enabler of the Level-1 findings ASBE 1.2 checklist omission (rating 4), AJDE 2.7 delayed/omitted hydrotest (rating 4) and AJDE 2.11 misapplied testing policy (rating 4), the Level-2 findings PTE 2.12 (rating 4) and PTF 1.13 mission planning (rating 4), and the Level-3 findings SVI 1.4 (rating 4), SVI 1.10 (rating 4), SVF 3.4 (rating 4) and SVV 4.1 (rating 4).

Linked Evidence: EVD-2894

OIR 1.6 **Failure in Maintenance/ Spares/ Support Programme** **SUBSTANTIATED**

Electrical PM was breakdown-skewed; a Zone-1 HPMV flame-proof lamp with three missing bolts and a glass case open 5-6 cm remained in service and became the probable ignition source at 03:08:15.

JUSTIFICATION

The electrical maintenance programme failed to preserve flame-proof integrity in a Zone 1 hydrocarbon area, and that failure supplied the probable ignition source. [EVD-6916] records the physical state: 'three bolts on one of the flame proof lamps were missing resulting in it being open with its integrity compromised, glass case got detached from the main body and had opened out about 5 to 6 cms. The nearby fittings were found intact.' The systemic character of the failure is documented in [EVD-0496]: 'the electrical maintenance system is a lot more skewed towards breakdown maintenance and lays little emphasis on systematic preventive maintenance and inspection,' and the post-incident audit [EVD-9999] was 'too generic and short (One page)' and 'clubbed all nine 125 W HPMV FLP lamps together and reported cable burnt and fitting damaged due to heat as status for all of them.' The correct programme would have applied systematic preventive inspection of flame-proof enclosure integrity (bolt presence, glass-case security) in the hazardous-area lamp population; the ACTUAL programme was breakdown-reactive and never detected the missing bolts and open case.

CAUSAL EXPLANATION

Because no systematic preventive inspection of flame-proof fittings occurred in the metering blast radius, the HPMV lamp with three missing bolts and a 5-6 cm open glass case remained energised in a Zone 1 area. When the vapour cloud from the failed 310FT01 flange accumulated over 56 minutes (02:12:50 to 03:08:15), it found that exposed fitting and ignited at 03:08:15, only 4 minutes 15 seconds after the last isolation action — proving the already-released cloud alone was sufficient fuel. This Level-4 programme failure directly enabled the Level-3 findings SVI 1.1 (rating 3), SVV 4.1 flame-proof enforcement omission (rating 4), SVF 3.2 electrical PM management (rating 4), SVIO 2.7 inappropriate asset management (rating 3) and SVV 4.5 (rating 4), and the Level-2 findings PTE 2.12 (rating 4), PTE 2.15 (rating 4) and PTF 1.2 audit cross-monitoring (rating 3); it converted the hydrocarbon release into a major fire.

Linked Evidence: EVD-6916, EVD-0496, EVD-9999

OIR 1.6 **Failure in Maintenance/ Spares/ Support Programme** **SUBSTANTIATED**

The four-year thickness-measurement cycle was breached (2011-2017 gap of six years) and 2017 significant-thickness-loss findings had no corrective-action plan after 3+ years.

JUSTIFICATION

A distinct integrity-programme failure is documented for the gas-terminal pipeline fleet. [EVD-8307] records: 'The laid down schedule for thickness measurement is every four years... the last two thickness measurement exercise was carried out in 2011 and 2017 which suggests that the schedule of four years is not being adhered to strictly' — a six-year gap between the last two surveys and a further 3+ years elapsed after 2017. [EVD-5527] then records that the 2017 findings were never actioned: 'several pipes have been identified with significant thickness loss and need for corrective action. However, even after three years no plan has been developed for the corrective actions to be taken for those lines. Mere measurement of thickness without corrective action does not complete the process of maintaining the integrity of pipes.' The correct programme would have executed the four-year cadence and generated risk-based remediation plans for the identified wall losses; the ACTUAL programme ran six years between surveys and produced no remedial plan for three years following the findings. The trunk pipeline itself survived the fire intact [EVD-5780], so this

CAUSAL EXPLANATION

The unenforced four-year cadence and the three-year silence on the 2017 wall-loss findings normalised the organisational default of recording integrity warnings without remediation. This did not open the 24-Sep-2020 leak — the 36-inch trunk pipeline remained intact and undamaged [EVD-5780] — but it directly substantiates the systemic pattern that allowed the meter 310FT01 recurrence (three gasket failures) and the omitted hydrotest to go unactioned. This Level-4 instance is therefore the organizational counterpart of the Level-3 findings SVI 1.6 (rating 3), SVV 4.1 (rating 3), SVV 4.5 (rating 3), SVF 3.2 (rating 3) and SVF 3.4 (rating 4) on pipeline integrity, and of the Level-1 AJDE 2.5 ignored-warning (rating 3) and AJDE 2.7 delayed-action (rating 3) findings, reinforcing the culture in which the initiating gasket defect survived.

Linked Evidence: EVD-8307, EVD-5527

OIR 1.6 **Failure in Maintenance/ Spares/ Support Programme** **SUBSTANTIATED**

A 1.03 mm pitted wall-loss finding on a 23.83 mm zero-corrosion-allowance 36-inch segment received painting only, with no fitness-for-service evaluation.

JUSTIFICATION

An independently evidenced maintenance-programme failure concerns the adequacy of a corrective action on the 36-inch pipeline. [EVD-2397] records: 'a reduction of thickness of 1.03mm along with heavy pitting and painting peeling off was observed in one of the segments of 36 pipeline in gas terminal in July 2019. The corrective action of painting was completed in Dec. 2019. However, as per the available data, the design thickness of the pipeline is 23.83mm with zero corrosion allowance. It needs to be examined whether painting as a corrective action would suffice in such cases and the mechanism of deciding on a corrective action needs to be reviewed.' The correct treatment for 1.03 mm metal loss with active pitting on a pipe carrying zero corrosion allowance would be fitness-for-service evaluation (remaining-strength assessment) and, if warranted, weld repair or section replacement; the ACTUAL action was repainting, a coating-restoration measure that does nothing for lost wall thickness. That wrong-choice pattern — treating a structural integrity defect with a cosmetic remedy — is the same decision pattern that accepted locally-sourced O-rings for a

CAUSAL EXPLANATION

By accepting repainting as closure for measured wall loss with zero corrosion allowance, the maintenance programme institutionalised the norm that structural integrity deficiencies could be addressed symptomatically. This specific segment did not fail in the incident — the trunk line remained intact [EVD-5780] — but the same programme-level decision logic allowed the non-conforming O-ring to be repeatedly resealed on meter 310FT01 without engineering challenge, which did fail at 02:12:50 on 24-Sep-2020 and led to ignition at 03:08:15. This Level-4 instance underlies the Level-1 AJDE 2.8 wrong-choice-of-action finding (rating 3), the Level-3 SVI 1.6/SVF 3.4/SVV 4.5 painting-adequacy findings (ratings 2-3), and forms part of the organizational culture that enabled the initiating gasket failure.

Linked Evidence: EVD-2397

OIR 1.6 **Failure in Maintenance/ Spares/ Support Programme** **SUBSTANTIATED**

Dec-2019 calibration changed meter 310FT01 from 16000 to 2540 mmwc and back with no checklist and no record in the signed documents (recommendation D.1.i).

JUSTIFICATION

A distinct maintenance-programme gap is evidenced for the December 2019 calibration of the very meter that later became the leak source. [EVD-2506] records: 'During calibration from 17/12/2019 to 18/12/2019, calibration was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of this change in the calibration documents signed by the team. During next calibration for meter 310FT01 was again changed to 16000 MMWC. No checklist is followed while carrying such critical jobs of high pressure/risk.' A setpoint was moved by a factor of ~6.3 (16000 to 2540 mmwc) and later reversed, with neither change recorded in the signed calibration certificate and no checklist used for the high-pressure work. The committee's recommendation D.1.i (p.35) addresses exactly this gap: 'details of Master calibrator/action taken/replacement/deviation from procedure if any, is to be documented in the calibration certificate.' The correct programme would have required a checklist and documented, approved deviations for any setpoint change; the ACTUAL programme allowed silent excursions. This instance did not itself break the flange, but it entrenched

CAUSAL EXPLANATION

The undocumented 16000-to-2540 mmwc excursion and its silent reversal normalised checklist bypass and unrecorded modification of critical pressure-boundary instrumentation on meter 310FT01 nine months before the incident. That same programme discipline gap reappeared in the 18-Sep-2020 gasket replacement, where the design-pressure test was omitted with no checklist or supervisory challenge, and the unvalidated seal failed at 02:12:50 on 24-Sep-2020, leading to ignition at 03:08:15. This Level-4 instance enabled the Level-1 findings ASBE 1.2 calibration checklist-not-followed (rating 3), ASBE 1.5 under-controlled calibration (rating 3) and AJDE 2.10 calibration knowledge inadequacy (rating 3), and the Level-3 findings SVI 1.1/SVI 1.6/SVV 4.1 on the unrecorded setpoint change, all as a secondary contributor to the environment that produced the initiating failure.

Linked Evidence: EVD-2506

OIR 1.6 **Failure in Maintenance/ Spares/ Support Programme** **SUBSTANTIATED**

SAP notification 3800419840 raised 17-Sep-2020 was never closed or reviewed after the gasket job, removing the supervisory quality gate before return to service (recommendation D.2.ii).

JUSTIFICATION

A further distinct support-programme failure is the broken closure loop in the SAP maintenance workflow. [EVD-0591] records: 'Notification no. 3800419840 was raised on 17/09/2020 for malfunction of meter no.1' and that the notification 'was not closed after completion of the job' (p.27), while [EVD-0185] shows the meter was returned to service the next day (19-Sep-2020) and remained in use at the time of the incident. The committee's recommendation D.2.ii at p.35 — 'Sample checks/approval of closure of notifications by Senior Officers to be introduced' — confirms that no senior closure review existed as a mechanism. The correct programme would close each notification with a supervisor checking the job details (material used, tests performed, recurrence history); the ACTUAL programme left the notification open with no review, so the non-standard O-ring, the missing hydrotest and the third-failure-in-22-months recurrence all passed without management visibility. This closure-loop failure is a rated-3 secondary contributor: it removed a supervisory quality gate rather than creating the physical defect.

CAUSAL EXPLANATION

Because no senior officer closed or sample-checked notification 3800419840, the 18-Sep-2020 repair package — locally-sourced O-ring, no design-pressure hydrotest, third gasket failure in 22 months — entered service on 19-Sep-2020 without any management review. The unvalidated joint then failed at 02:12:50 on 24-Sep-2020, releasing vapour for 56 minutes until ignition at 03:08:15. This Level-4 closure-loop failure enabled the Level-1 findings AV 1.1 non-closure violation (rating 3) and AJDE 2.11 misapplied closure policy (rating 3), the Level-2 findings PTF 1.12 failed feedback/acknowledgement (rating 3) and PTE 2.12 missing supervisory gate (rating 3), and the Level-3 findings SVI 1.1/SVI 1.6/SVF 3.4/SVV 4.1 on the unclosed notification (ratings 3).

Linked Evidence: EVD-0591, EVD-0185

OIR 1.6 **Failure in Maintenance/ Spares/ Support Programme** **SUBSTANTIATED**

Original metering-system P&ID lost; vendor replacement never checked or approved by ONGC; field valves/meters configuration does not match the drawing.

JUSTIFICATION

A distinct configuration-documentation failure affected the very skid that failed. [EVD-0933] records: 'Reference drawings no. 02-310-AO-101 Rev.1, 2x6 lines & 1x2 lines for metering system are shown but valves & meters configuration are not matching. As old drawings were lost, new drawings are made by a vendor but the same has not been checked & approved by ONGC.' The controlled as-built drawing for the metering system therefore did not exist: the original was lost, the vendor replacement was never checked or approved by the operator, and the field valves/meters configuration did not match the unapproved drawing. The correct configuration-management programme would maintain approved as-built drawings against which maintenance could verify the correct gasket design and flange details; the ACTUAL programme provided no authoritative reference. This instance is rated 2 because the physical failure was driven by the incorrect seal and the missing hydrotest rather than the drawing mismatch alone, but the uncontrolled drawing is part of the documentation environment that let the OEM gasket specification substitution go unchallenged.

CAUSAL EXPLANATION

Without an approved, accurate as-built drawing of the metering skid, maintenance personnel had no controlled configuration baseline to confirm the correct rectangular-slot gasket design or to verify the substituted O-ring against the flange, compounding the lost OEM gasket specification [EVD-8645]. This Level-4 configuration-management failure is the organizational root of the Level-1 AV 1.1 configuration-management breakdown (rating 2), the Level-2 PTE 2.14 unfamiliar-equipment/non-custodial operations finding (rating 2) and PTE 2.7 unclear technical guidance on the P&ID (rating 3), and the Level-3 SVI 1.4 uncontrolled-drawing-policy finding (rating 2); it did not directly initiate the release, so it is rated 2.

Linked Evidence: EVD-0933

OIR 1.6 **Failure in Maintenance/ Spares/ Support Programme** **SUBSTANTIATED**

Missing permits went untracked, JSPs were absent from SAP, permit closure went unreviewed, and notification-driven work blurred into permit-controlled work (recommendation B.5).

JUSTIFICATION

The work-control support system had a further, independently evidenced failure at permit level. [EVD-5105] records: 'Work Permits are issued in SAP and hard copy of work permits are at sites during execution of work. After completion of work, the originals are filed. On checking the same, it was found that few of the work permits were missing. No data is kept on such missing permits. Job safety Plan is not attached/given in SAP document. Closing of permits & actions indicated in work needs review.' The same evidence cites committee recommendation B.5 at p.34: 'Maintenance notifications are not be assumed as permit to work' — implying maintenance was at times executed under notification without a separate permit authorisation. The correct programme would track every permit, attach a JSP for non-routine/hazardous work in SAP, and review permit closure; the ACTUAL programme had untracked missing permits, no JSP attachments, and no reviewed closure. This instance is rated 2 because the evidence does not tie a specific missing permit to the 18-Sep-2020 gasket job, but the systemic permit gap removed the task-hazard gate that should

CAUSAL EXPLANATION

Without a mandatory JSP and auditable permit closure, the 18-Sep-2020 high-risk gasket replacement could proceed without a documented task hazard analysis identifying the required design-pressure hydrotest and gasket material-conformance checks, so the omissions passed into service on 19-Sep-2020 and the joint failed on 24-Sep-2020 at 02:12:50. This Level-4 permit-system failure is the organizational enabler of the Level-2 finding PTE 2.7 incomplete permit guidance (rating 2), the Level-3 findings SVI 1.4/SVV 4.1 on permit enforcement (ratings 2-3) and SVF 3.4 (rating 2), and it contributed secondarily to the Level-1 procedural omissions (ASBE 1.2 rating 4), though the available evidence does not tie a specific missing permit to the failed joint, hence its rating of 2.

Linked Evidence: EVD-5105

OIR 1.5 **Failure in Timely Acquisition/ Procurement/ Upgrade** **SUBSTANTIATED**

OEM gasket specification lost and never re-established; locally-purchased unspecified O-rings were substituted as regular practice with no purchase spec or QC, per committee root causes #2 and #3 (p.33).

JUSTIFICATION

The organization's material-acquisition process delivered a technically unsuitable seal for the meter 310FT01 flange. [EVD-8645] records, verbatim: 'The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket.' Six discrete acquisition failures are documented in those two sentences: the specification was lost and never re-established, OEM procurement was abandoned, a market O-ring was substituted, no purchase specification was applied, no incoming quality control was performed, and no dated or authorized decision for the change exists. The official inquiry reached the same conclusion, naming committee root cause #2 'Use of nonstandard O ring' and #3 'Lack of control over material sourcing' (p.33). The recurrence record [EVD-1494] shows the

CAUSAL EXPLANATION

Because the acquisition chain never produced the correct rectangular-slot gasket, the round O-ring installed on 18-Sep-2020 could not fill the machined flange groove and left a leak path from installation. The meter returned to service on 19-Sep-2020, operated for six days, and the joint failed at 02:12:50 on 24-Sep-2020 [EVD-1284], releasing vapour that accumulated for 56 minutes before igniting at 03:08:15 [EVD-4507]. This Level-4 procurement failure created the conditions for, and directly enabled, the lower-level findings AJDE 2.8 wrong choice of action (rating 4), ASBE 1.3 procedure not followed correctly (rating 4), AV 1.1 routine/widespread substitution violation (rating 4), PTE 2.12 defective maintenance output (rating 4), and the supervisory failures SVIO 2.5 authorized unnecessary hazard and SVV 4.2 allowing unwritten policies to become standard. It was itself enabled by the dormant MOC process (SVI 1.4, rating 3) that should have screened the substitution as an engineering change.

Linked Evidence: EVD-8645, EVD-1494

OIR 1.7 **Phase-Out Policies (Failure to Remove Inadequate, Worn Out Equipm** SUBSTANTIATED

A 1986 meter with three gasket failures in 22 months and an open malfunction notification was not retired or quarantined; it failed a fourth time and became the leak source.

JUSTIFICATION

The 1986-vintage meter 310FT01 was allowed to remain in service despite an objectively measurable pattern of sealing failures. [EVD-1494] records: 'Cleaning/servicing/calibration of meters 1 & 2 was carried out on 12th & 13th Nov 2018. Afterwards, meter no.1 was cleaned on 25/02/2019 & gasket was changed. Again on 18/09/2019, gasket was changed which leaked within one year' - with the narrative fixing the latest replacement as 18-Sep-2020, that is three gasket replacements in approximately 22 months on a meter installed 34 years earlier. [EVD-0591] shows the final warning signal: 'Notification no. 3800419840 was raised on 17/09/2020 for malfunction of meter no.1' and the notification 'was not closed after completion of the job' (p.27). Instead of quarantining, redesigning, replacing, or otherwise removing the chronic-failure pressure boundary from service, the organisation repeated the same non-OEM O-ring repair on 18-Sep-2020 and returned the meter to line on 19-Sep-2020; [EVD-0064] records the result: '6 meter with ID 310FE1 was extensively damaged with all bolts out of the flange arrangement' and 'The condensate might have leaked from the

CAUSAL EXPLANATION

Because the obsolete, repeatedly failing meter remained in service, the inadequate O-ring fitted on 18-Sep-2020 was subjected to six days of live condensate pressure and failed at 02:12:50 on 24-Sep-2020, initiating the 56-minute vapour accumulation and the 03:08:15 ignition. This Level-4 phase-out failure created the environment for the lower-level AJDE 2.5 ignored-warning finding (rating 4), the PTE 2.12 recurring-maintenance finding (rating 3), the SVI 1.6 lack-of-supervisory-response pattern, and the SVF 3.4 no-mitigation finding, which together allowed the initiating equipment failure to occur.

Linked Evidence: EVD-1494, EVD-0591, EVD-0064

OIR 1.7 **Phase-Out Policies (Failure to Remove Inadequate, Worn Out Equipm** SUBSTANTIATED

A Zone 1 flame-proof lamp with three missing bolts and a glass case opened 5-6 cm was not removed or repaired; it supplied the ignition energy at 03:08:15.

JUSTIFICATION

The probable ignition source was a Zone 1 flame-proof lamp whose explosion-protection integrity had already been lost, and the organisation had no effective removal/repair process for such defective electrical equipment. [EVD-6916] records 'three bolts on one of the flame proof lamps were missing resulting in it being open with its integrity compromised, glass case got detached from the main body and had opened out about 5 to 6 cms. The nearby fittings were found intact' (p.32), bounding the failure to that single fitting. [EVD-0496] shows the post-incident electrical audit 'has clubbed all nine 125 W HPMV FLP lamps together and reported cable burnt and fitting damaged due to heat as status for all of them', and the unique pre-existing condition of the one lamp 'was not mentioned in the report' (p.28), while [EVD-9999] states 'the electrical maintenance system is a lot more skewed towards breakdown maintenance and lays little emphasis on systematic preventive maintenance and inspection' (p.28). A flame-proof enclosure with missing bolts and an open glass case in a Zone 1 hydrocarbon area should have been removed or repaired; the phase-out/inspection process left it

CAUSAL EXPLANATION

The defective flame-proof enclosure was an unnecessary ignition source that remained available in the metering blast radius because no removal or repair trigger existed. After the failed gasket released vapour from 02:12:50, the cloud accumulated for 56 minutes and, at 03:08:15, after both isolation actions had already occurred (MOV 54 at 03:01, 36-inch PCV at 03:04), the already-released cloud contacted the exposed lamp contacts and ignited. This Level-4 phase-out failure directly enabled the lower-level PTE 2.15 inadequate-protection finding, the SVI 1.8 failed-to-correct-risky-practices finding, and the SVV 4.1 failure-to-enforce-flame-proof-rules finding, converting a vapour release into a major fire.

Linked Evidence: EVD-6916, EVD-0496, EVD-9999

OIR 1.7 **Phase-Out Policies (Failure to Remove Inadequate, Worn Out Equipm** SUBSTANTIATED

MOV 54 and the gas-terminal MOVs were left local-manual with no emergency remote operation, forcing a 46-minute physical-access delay before first isolation.

JUSTIFICATION

The gas-terminal emergency isolation valves, including MOV 54, remained configured for local manual operation only, an obsolete emergency-response configuration for a major-hydrocarbon-release scenario. [EVD-3932] states 'MOVs are provided in the gas terminal station for local operations but they are not having remote operation from control room during emergency' (p.26), confirming the field-level limitation. The consequence is visible in [EVD-9612], where the GSP Shift In-charge 'tried to enter the Gas Control room area but could not enter due to heavy gas smell... [he] closed the MOV 54 inlet condensate line from slug catcher to 36 metering area at about 0301 hrs' - the first isolation occurred only after the senior officer physically reached the gate, 46 minutes after first physical detection at 02:15 and 48 minutes after CCTV-confirmed leak onset at 02:12:50. The organisation did not phase out or modernise these manual-only MOVs despite the known consequence that a dense vapour cloud would prevent physical approach; this equipment-removal/modernisation failure created the unsafe field condition that delayed source isolation.

CAUSAL EXPLANATION

Because the obsolete manual-only MOV configuration was not removed or upgraded, the control-room operators who received the annunciator alarm at 02:20 could not execute immediate isolation themselves, and the senior officer had to travel to the gate and act at 03:01. The 46-minute unisolated release window that resulted filled the area with the vapour cloud documented at 03:00:44 as 'vapour all around even at height' [EVD-8213], which ignited at 03:08:15. This Level-4 phase-out failure enabled the lower-level PTE 2.4 controls-inadequate finding, the PTE 2.8 workspace-incompatible finding, the AJDE 2.7 delayed-necessary-action finding, and the PTF 1.13 planning failure that produced the response delay.

Linked Evidence: EVD-3932, EVD-9612

OIR 1.7 **Phase-Out Policies (Failure to Remove Inadequate, Worn Out Equipm** SUBSTANTIATED

A 1.03 mm pitted, zero-corrosion-allowance pipeline segment was retained in service after painting only; it did not cause this incident but evidences the phase-out gap.

JUSTIFICATION

A pipe segment on the 36-inch pipeline with measured metal loss and heavy pitting remained in service after only a painting corrective action, with no removal, repair, or fitness-for-service retirement decision. [EVD-2397] records 'a reduction of thickness of 1.03mm along with heavy pitting and painting peeling off was observed in one of the segments of 36 pipeline in gas terminal in July 2019. The corrective action of painting was completed in Dec. 2019. However, as per the available data, the design thickness of the pipeline is 23.83mm with zero corrosion allowance. It needs to be examined whether painting as a corrective action would suffice' (p.30). The phase-out process for degraded pressure equipment was therefore inadequate: known wall loss on a zero-corrosion-allowance design was not the trigger for removal or replacement. This instance is rated 2 because [EVD-5780] confirms the 36-inch trunk pipeline itself was intact after the fire, so this equipment-retention failure did not directly drive the 24-Sep-2020 outcome, but it substantiates the same systemic phase-out deficiency.

CAUSAL EXPLANATION

This instance did not initiate the 24-Sep-2020 release, so it is rated present without direct outcome impact. Its causal significance is systemic: the same failure to retire or remediate degraded pressure equipment normalised the retention of known integrity defects and supported the lower-level AJDE 2.8 wrong-choice finding on painting as corrective action and the SVI 1.1/SVI 1.6 supervisory patterns that also allowed the meter 310FT01 sealing defect to persist until it failed at 02:12:50 on 24-Sep-2020.

Linked Evidence: EVD-2397, EVD-5780

OIR 1.7 **Phase-Out Policies (Failure to Remove Inadequate, Worn Out Equipm** SUBSTANTIATED

Three of four diesel fire-water pumps were unreliable or under maintenance and were not repaired or retired; electrical pumps masked the gap during this incident.

JUSTIFICATION

Three of the four diesel fire-water pumps were unavailable or unreliable, and the equipment was neither repaired, retired, nor replaced. [EVD-2432] records 'Out of 4 diesel pumps 410 m3/hr, 2 pumps are under maintenance, one operated for 30 minutes & fourth pumps is tripping after 2 minutes. The reason for tripping is not identified or actions have not been recorded' (p.27). The same record shows the consequence was latent rather than actual: 'Both the Tank level and the Fire water header pressure were healthy throughout the incident. Three electrical Fire water pumps namely 47PE003 A, C, D were operated during the cooling activity' (p.11). The phase-out/removal process left known-unreliable emergency equipment in the backup inventory without remediation, but because the electrical pumps carried the entire fire-water load and the fire was extinguished by approximately 06:00, this instance is rated present without impact on the incident outcome.

CAUSAL EXPLANATION

This degraded-backup condition did not change the 24-Sep-2020 outcome because the electrical fire-water pumps maintained tank levels and header pressure throughout the 02:15-to-07:30 emergency. It is nonetheless a substantiated phase-out/repair failure that created a latent resilience gap and exemplifies the same supervisory pattern of accepting known defective equipment that elsewhere enabled the meter gasket and flame-proof lamp defects to persist into the incident.

Linked Evidence: EVD-2432, EVD-2432

OIR 1.8 **Failure to Provide Adequate Operational Information Resources/ Support** **SUBSTANTIATED**

No OEM gasket specification, substitute specification, or purchase QC information was available, so a locally procured O-ring was fitted to a rectangular-slot flange.

JUSTIFICATION

The maintenance organisation lacked the single most important operational information resource for the 18-Sep-2020 repair: the OEM gasket specification for the 1986 rectangular-slot flange. [EVD-8645] documents six information omissions in two sentences: 'The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket' (p.31). Without the correct geometry, material, and pressure-rating information, the maintainer could not verify that a round O-ring was incompatible with a rectangular-slot seal design, and without a substitute-specification resource the procurement process had no acceptance basis. The committee's root cause #2 'Use of nonstandard O ring' and root cause #3 'Lack of control over material sourcing' (p.33) flow directly from this

CAUSAL EXPLANATION

The absent specification information directly produced the lower-level PTE 2.7 technical-guidance-unavailable finding (rating 4), the AJDE 2.8 wrong-choice finding, the ASBE 1.3 wrong-gasket procedure finding, and the AV 1.1 routine-substitution violation. The round O-ring installed without any specification failed under operating pressure on 24-Sep-2020 at 02:12:50, initiating the 56-minute vapour accumulation and the 03:08:15 ignition, so this Level-4 information failure is a primary cause of the loss of containment.

Linked Evidence: EVD-8645, EVD-2894

OIR 1.8 **Failure to Provide Adequate Operational Information Resources/ Support** **SUBSTANTIATED**

310PT4/PT5/PT6 produced no retained data, eliminating pressure-trend monitoring in the metering area.

JUSTIFICATION

The metering area had three pressure transmitters shown on the drawing but no pressure data available in the DCS, depriving operators and investigators of a basic operational information resource. [EVD-6470] records 'Three Pressure transmitters 310PT4, 310PT5 & 310PT6 are shown in the drawing. It was informed that no pressure transmitter data are available in the system. No pressure data in Gas Terminal area is available/provided to check rise in pressure in the metering lines' (p.26). This absence removed the ability to trend pressure before and during the developing release, so the machine-recorded leak at 02:12:50 could not be confirmed or anticipated from pressure data, and post-incident reconstruction was impaired. The information gap was secondary rather than primary because human perimeter detection at 02:15 and the annunciator at 02:20 still occurred, but it compounded the delayed detection and diagnosis environment.

CAUSAL EXPLANATION

Without pressure-trend information, operators had no instrument-based early indication of the developing flange failure before the 02:15 perimeter detection or the 02:20 annunciator, and investigators could not anchor the pre-incident pressure history. This Level-4 information-resource failure supported the lower-level PTE 2.5 defective-equipment observation and PTE 2.11 equipment-not-configured-correctly finding on the missing transmitter data, contributing secondarily to the detection and diagnostic limitations within the 46-minute response window.

Linked Evidence: EVD-6470

OIR 1.8 **Failure to Provide Adequate Operational Information Resources/ Support** **SUBSTANTIATED**

No 1986 baseline thickness data, original drawings, or pipe schedules were available, leaving only 2011 and 2017 measurements for trend analysis.

JUSTIFICATION

For the 1986-vintage gas-terminal pipelines, the organisation lacked the baseline information required to trend integrity degradation. [EVD-8287] records 'It is observed that for a large number of pipes, especially in gas terminal area baseline data of pipe thickness in not available. The original drawings and the schedules of the pipes are also not available. In most cases only two thickness details of 2011 and 2017 is available for a pipe which is commissioned in 1986' (p.29). With only two data points across 34 years of service, wall-loss trends could not be rigorously established or prioritised, which reinforced the three-year non-action on the 2017 thickness findings. This instance is rated present without direct impact because the 36-inch pipeline itself remained intact during the incident, but it substantiates the missing operational information resources for pipeline integrity management.

CAUSAL EXPLANATION

The absent baseline information prevented rigorous trending of the 2017 thickness-loss findings and supported the lower-level AJDE 2.5 ignored-warning and AJDE 2.7 delayed-corrective-action findings. This compounded the integrity-governance weakness recorded in EVD-5527 and EVD-2397, but because the 36-inch line did not fail, it is rated as a systemic enabler rather than an outcome-causing instance.

Linked Evidence: EVD-8287

OIR 1.8 **Failure to Provide Adequate Operational Information Resources/ Support** **SUBSTANTIATED**

The self-safety audit checklist was not plant-specific and approved manuals/SOPs were unavailable, so plant-specific defects were not sampled.

JUSTIFICATION

The self-safety audit instrument did not provide plant-specific assurance information, and approved manuals/SOPs were not available to auditors. [EVD-9692] records 'The checklist is comprehensive but not related to specific plants and in the process, lot of observations are missing... Approved manuals/SOPs were not available' (p.26), and the committee's recommendation B.1 at p.34 states 'Checklist for Self-Safety Audit for individual plants are to be developed.' Because the checklist was generic, it did not direct auditors to verify the Zone 1 flame-proof lamp integrity or the metering gasket substitution practice, so the defective ignition-source lamp remained unreported. This is a secondary but substantive failure to provide adequate operational assurance information, directly connected to the survival of the compromised HPMV lamp that ignited the accumulated cloud at 03:08:15.

CAUSAL EXPLANATION

The generic audit information resource failed to force inspection of the plant-specific ignition-source and gasket-substitution hazards, enabling the lower-level ASBE 1.2 audit-checklist failure, the PTF 1.2 cross-monitoring failure, and the PPI 5.6 inaccurate-expectancy condition. As a result, the compromised HPMV lamp with three missing bolts remained undetected and supplied the ignition at 03:08:15, converting the release into a major fire.

Linked Evidence: EVD-9692, EVD-9692

OIR 1.8 **Failure to Provide Adequate Operational Information Resources/ Support** **SUBSTANTIATED**

Internal safety audits produced observations with no filled checklists, making audit completeness unverifiable.

JUSTIFICATION

The 2018 and 2019 internal safety audits did not preserve the checklist information needed to verify audit coverage. [EVD-7681] records 'Internal Safety Audit was conducted in 2018 & 2019 for 4 & 3 days respectively by a 3-member committee and the observations are given without the filled checklist. The checklist format is not available to check completeness of audit areas' (p.26). Without a filled checklist trail, managers and later investigators could not establish whether the metering-skid pressure boundary work or the flame-proof lamp enclosures had ever been within audit scope, so latent defects could persist. This instance had no direct outcome impact because it was an assurance-traceability gap rather than the initiating cause, but it is a substantiated missing information resource.

CAUSAL EXPLANATION

The absent checklist trail removed the information needed to challenge the assurance claims and supported the lower-level ASBE 1.2 audit-checklist failure and the PTF 1.2 cross-monitoring failure. This indirectly allowed the compromised HPMV lamp and the meter gasket practice to persist undetected, warranting a present-without-direct-impact rating of 2.

Linked Evidence: EVD-7681

OIR 1.8 **Failure to Provide Adequate Operational Information Resources/ Support** **SUBSTANTIATED**

Fire drill records in SAP lacked actions and observations, with a separate non-conforming dataset, making readiness unmeasurable.

JUSTIFICATION

The fire drill records were an inadequate information resource for measuring emergency-readiness. [EVD-2130] records 'Fire drill reports are available in SAP but without any data on actions taken/recommendations. Separate data is maintained which is not in line with the approved guidelines' (p.26). Because the SAP records contained no actions or observations, drill effectiveness could not be evidenced, and any deficiencies rehearsed in drills could not be tracked to closure. This instance is rated present without direct outcome impact because the fire team performed adequately on the night, but the missing readiness information left the organisation unable to validate or correct the escalate-first response pattern that later produced the 46-minute isolation delay.

CAUSAL EXPLANATION

The missing drill actions data removed the information loop that could have exposed and corrected the isolation-response deficiency before 24-Sep-2020. It did not directly cause the release or ignition, but it contributed to the organisation's unverified emergency-readiness and supported the SVI 1.7 lack-of-standardisation observation, hence rating 2.

Linked Evidence: EVD-2130

OIR 1.8 **Failure to Provide Adequate Operational Information Resources/ Support** **SUBSTANTIATED**

Work permits were missing untracked and JSPs were absent from SAP, leaving high-risk maintenance without task-specific hazard information.

JUSTIFICATION

The work-permit system did not reliably supply task-specific hazard information for high-risk maintenance. [EVD-5105] records 'few of the work permits were missing. No data is kept on such missing permits. Job safety Plan is not attached/given in SAP document. Closing of permits & actions indicated in work needs review' (p.26). Without a Job Safety Plan attached in SAP, the 18-Sep-2020 high-pressure gasket replacement could proceed without a documented hazard analysis identifying the required design-pressure hydrotest and material-conformance checks. The evidence does not tie a specific missing permit directly to the failed joint, so this instance is rated present without direct outcome impact, but it removed an information barrier that might have exposed the procedural omissions before return to service.

CAUSAL EXPLANATION

The absent JSP information meant no document forced the maintenance crew to specify the hydrostatic test and gasket-conformance steps before the 18-Sep-2020 job, supporting the lower-level ASBE 1.2 omissions and the SVI 1.4 permit-policy gap. Because the direct physical cause was the non-conforming seal itself, this remains a secondary systemic information failure rated 2.

Linked Evidence: EVD-5105

OIR 1.5 **Failure in Timely Acquisition/ Procurement/ Upgrade** **SUBSTANTIATED**

DCS timestamp drift of 2-12 minutes across control rooms was never synchronisation-corrected, complicating alarm correlation and event reconstruction.

JUSTIFICATION

A sixth, lower-severity acquisition/configurational failure is the unremedied DCS clock drift. [EVD-5532] records: 'Time stamping in DCS systems provided in various control rooms is varying from 2 minutes to 12 minutes and data matching is difficult.' Multi-control-room timestamps differed by up to 12 minutes, forcing the investigation to treat CCTV timestamps as canonical and making cross-room alarm correlation unreliable. The correct configuration for a distributed control system is a single time-synchronisation source (e.g. NTP) across all control rooms; the ACTUAL configuration allowed each system to drift independently, and the drift was observed but never corrected. This acquisition/configurational gap did not change the 24-Sep-2020 physical outcome, because the primary sequence was independently reconstructed from CCTV, witness statements and operator logs, but it materially hampered accurate sequence-of-events analysis and could mislead operators correlating simultaneous alarms across control rooms in real time.

CAUSAL EXPLANATION

Because no time-synchronisation upgrade was procured or configured, simultaneous alarms in different control rooms could appear separated by 2-12 minutes in the logs, degrading operators' ability to correlate events during an emergency and forcing the investigation to discard DCS timestamps in favour of CCTV. This Level-4 configuration lapse maps to the Level-2 finding PTE 2.6 clock-drift instance (rating 2) and PTE 2.11 configuration gap (rating 2), and is rated 2 because it had no direct impact on the incident outcome, only on forensic and real-time correlation reliability.

Linked Evidence: EVD-5532

OIR 1.6 Failure in Maintenance/ Spares/ Support Programme

SUBSTANTIATED

Three of four 410 m3/hr diesel fire-water pumps were under maintenance or unreliable, with a tripping fault undiagnosed and unrecorded; electrical pumps carried the incident.

JUSTIFICATION

A distinct maintenance-support failure affected the fire-water diesel backup. [EVD-2432] records: 'Out of 4 diesel pumps 410 m3/hr, 2 pumps are under maintenance, one operated for 30 minutes & fourth pumps is tripping after 2 minutes. The reason for tripping is not identified or actions have not been recorded.' The correct support programme would have kept a defined fraction of the diesel fire-water capacity available and would have diagnosed and recorded the 2-minute tripping fault; the ACTUAL state was that three of four diesel pumps were unavailable or unreliable with the trip cause unresolved and unrecorded. The firefighting outcome was not degraded because 'Both the Tank level and the Fire water header pressure were healthy throughout the incident' and 'Three electrical Fire water pumps namely 47PE003 A, C, D were operated during the cooling activity.' This instance is therefore a substantiated latent resilience gap with no direct impact on the 24-Sep-2020 outcome, rated 2, but it is part of the same maintenance-programme governance failure pattern seen elsewhere.

CAUSAL EXPLANATION

The degraded diesel pump train did not change the outcome — the three running electrical pumps kept tank level and header pressure in normal bands, active cooling began at ~03:50 and the fire was extinguished by ~06:00 with all-clear at 07:30 — but had electrical supply failed or cooling demand exceeded electric capacity, the unreliable diesel backup would have constrained the response. The unrepaired, undocumented tripping fault is the same programme characteristic that allowed the meter gasket recurrence and the compromised lamp to persist, linking this Level-4 instance to the lower-level latent findings PTE 2.5 diesel-pump instance (rating 2), SVIO 2.7 (rating 2), SVF 3.4 and SVV 4.5 on the fire-water system, all rated present-without-direct-impact.

Linked Evidence: EVD-2432

OIC 2.1 Organizational Culture Allows for Unsafe Task/Mission

SUBSTANTIATED

Locally purchased O-rings were used as a 'regular practice' for a rectangular-slot flange without specification, QC, or recorded authorization, with three replacements in 22 months.

JUSTIFICATION

The evidence shows this unsafe maintenance practice had become an organizational norm, not an isolated error. [EVD-8645] states: "The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket" (p.31). The phrase "as a regular practice" and the absence of any recorded authorization show leadership accepted a material change with no engineering review. The committee's root cause #2 ("Use of nonstandard O ring") and #3 ("Lack of control over material sourcing") flow directly from this finding. [EVD-1494] documents the recurrence: "Cleaning/servicing/calibration of meters 1 & 2 was carried out on 12th & 13th Nov 2018. Afterwards, meter no.1 was cleaned on 25/02/2019 & gasket was changed. Again on 18/09/2019, gasket was changed which leaked within

CAUSAL EXPLANATION

This organizational climate directly produced the initiating equipment failure: the round O-ring could not fill the rectangular slot, leaving a leak path from installation. Because the culture treated this as acceptable routine work, the same wrong material was reinstalled on 18-Sep-2020, the meter returned to service on 19-Sep-2020, and the joint failed at 02:12:50 on 24-Sep-2020. The release accumulated for 56 minutes and ignited at 03:08:15. This Level-4 factor created the environment for lower-level findings AJDE 2.8 (wrong choice of action), ASBE 1.3 (procedure not followed correctly), AV 1.1 (routine/widespread violation) and PTE 2.7 (technical guidance unavailable), and was enabled by the supervisory failure SVI 1.4 and the dormant MOC process.

Linked Evidence: EVD-8645, EVD-1494

OIC 2.1 **Organizational Culture Allows for Unsafe Task/Mission** **SUBSTANTIATED**

Three gasket replacements in ~22 months did not trigger root-cause analysis or engineering correction; the same inadequate relining was repeated until the fourth failure.

JUSTIFICATION

The repeated failure pattern was known but not escalated to root-cause investigation, which is a direct expression of an organizational culture that tolerates chronic safety-critical defects. [EVD-1494] records: "Cleaning/servicing/calibration of meters 1 & 2 was carried out on 12th & 13th Nov 2018. Afterwards, meter no.1 was cleaned on 25/02/2019 & gasket was changed. Again on 18/09/2019, gasket was changed which leaked within one year" (p.27). [EVD-0591] shows the latest malfunction notification (SAP 3800419840 of 17-Sep-2020) existed precisely seven days before the incident but was not closed. The committee's recommendation D.2.ii (p.35), "Sample checks/approval of closure of notifications by Senior Officers to be introduced", is an explicit recognition that this recurrence was not being reviewed at the leadership level.

CAUSAL EXPLANATION

Each replacement generated a clear signal that the O-ring substitution was inadequate, but the organization did not act. The 18-Sep-2020 repair therefore repeated the same wrong material and the same no-hydrotest shortcut, returned the meter to service on 19-Sep-2020, and the fourth failure occurred at 02:12:50 on 24-Sep-2020, leading to ignition at 03:08:15. This Level-4 climate directly enabled lower-level AJDE 2.5 (ignored warning, rating 4), AV 1.1 (routine violation), SVI 1.8 (failure to correct unsafe practices) and PTF 1.13 (mission planning without analysis).

Linked Evidence: EVD-1494, EVD-0591

OIC 2.1 **Organizational Culture Allows for Unsafe Task/Mission** **SUBSTANTIATED**

The MOC SOP existed since Aug-2016 but was not applied at all after 2018, so the O-ring material substitution was never subjected to change review.

JUSTIFICATION

[EVD-2986] documents that the plant's formal change-management control had been effectively dead for years: "Management of Change (MOC) philosophy/SOP was published in August 2016. However, the procedure indicated in the document is not implemented while carrying out change of Low-Pressure Gas to Kribhico from Medium Pressure Gas headers in LPG Plant in 2013 & 2018. It is informed no more MOCs are implemented in the plant since 2018" (p.30). This is significant because the switch from OEM rectangular-slot gaskets to locally purchased O-rings was exactly the kind of material change that should have required MOC. The absence of any MOC activity from 2018 onward is consistent with [EVD-8645], which notes "There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket". The culture thereby allowed an unsafe mission to continue without any engineering evaluation or documented approval.

CAUSAL EXPLANATION

Because no MOC was applied, the O-ring substitution bypassed the only formal gate that could have challenged geometry, material compatibility, and pressure rating. This allowed the wrong material to be installed on 18-Sep-2020 and to fail at 02:12:50 on 24-Sep-2020, producing the release and fire. This Level-4 factor is the organizational root of lower-level AJDE 2.11 (misapplication/unawareness of MOC policy), AV 1.1 (routine violation), SVI 1.4 (failure to provide policy guidance) and SVIO 2.4 (no formal risk assessment of the substitution).

Linked Evidence: EVD-2986, EVD-8645

OIC 2.1 **Organizational Culture Allows for Unsafe Task/Mission** **SUBSTANTIATED**

The required four-year UT thickness schedule was not followed (2011 to 2017 = six years), with no enforcement.

JUSTIFICATION

[EVD-8307] records the formal monitoring requirement was not being met: "The laid down schedule for thickness measurement is every four years... As per the records, the last two thickness measurement exercise was carried out in 2011 and 2017 which suggests that the schedule of four years is not being adhered to strictly" (p.29). This is a distinct manifestation of the culture of allowing mandatory preventive tasks to lapse, separate from the failure to act on the 2017 findings. It is rated 2 because the missed inspection schedule did not directly produce the fire (the 36-inch pipeline was intact), but it demonstrates leadership tolerance of known schedule violations.

CAUSAL EXPLANATION

The unenforced inspection cadence meant pipe wall-loss trends were not refreshed as required, leaving degradation uncharacterized. Although this instance did not directly cause the 24-Sep-2020 fire, it reinforced the organizational environment in which mandatory integrity tasks, such as the post-gasket hydrostatic test, were treated as optional. It supports lower-level AJDE 2.7 and SVV 4.1/SVF 3.2 pipeline-integrity findings.

Linked Evidence: EVD-8307

OIC 2.1 **Organizational Culture Allows for Unsafe Task/Mission** **SUBSTANTIATED**

MOV 54 and the Gas Terminal MOVs had no remote emergency operation, so isolation required physical approach that the dense vapor cloud denied until 03:01.

JUSTIFICATION

[EVD-3932] records the committee's structural finding: "MOVs are provided in the gas terminal station for local operations but they are not having remote operation from control room during emergency" (p.26). The consequence was demonstrated in real time: [EVD-9612] shows the GSP Shift In-charge could only close MOV 54 after physically reaching the Gas Terminal Gate at approximately 02:35, executing the closure at 03:01, 46 minutes after first detection at 02:15. The dense vapor cloud prevented earlier physical approach, and because the valve was not remote-operable, the control room operators could not isolate it from the blast-proof building at 02:20. This design left the emergency response unable to stop the source for nearly three-quarters of an hour.

CAUSAL EXPLANATION

The manual-only MOV configuration converted a potentially rapid control-room response into a field-transport and escalation sequence: leak onset 02:12:50, human detection 02:15, alarm 02:20, supervisor arrival ~02:35, MOV 54 closure 03:01. The unisolated line fed the release for 46 minutes, producing the cloud that ignited at 03:08:15. This organizational climate/equipment posture is the Level-4 root of lower-level PTE 2.4/PTE 2.11/PTE 2.8, PTF 1.4 and AJDE 2.7 findings.

Linked Evidence: EVD-3932, EVD-9612

OIC 2.1 **Organizational Culture Allows for Unsafe Task/Mission** **SUBSTANTIATED**

Self-safety and internal audits used generic checklists, reported compliance without site verification, and retained no filled checklists to demonstrate coverage.

JUSTIFICATION

[EVD-9692] records the self-safety audit instrument was not fit for purpose: "The checklist is comprehensive but not related to specific plants and in the process, lot of observations are missing. The duration of audit is not indicated and in most of the items, compliances are indicated, but during our inspection, the data is found to be incomplete & not in line with OISD standards/manuals. Approved manuals/SOPs were not available" (p.26). [EVD-7681] shows the internal audits had no auditable checklist trail: "Internal Safety Audit was conducted in 2018 & 2019 for 4 & 3 days respectively by a 3-member committee and the observations are given without the filled checklist" (p.26). This is a culture of audit theater, in which written compliance was accepted without verification.

CAUSAL EXPLANATION

The audit layer that should have caught the compromised HPMV flame-proof lamp and the uncontrolled gasket practice instead generated false assurance. The lamp with three missing bolts and glass case opened 5-6 cm ([EVD-6916]) remained in Zone 1 service and became the probable ignition source at 03:08:15. This Level-4 culture factor enabled lower-level ASBE 1.2 (audit checklist not followed), PPI 5.6 (inaccurate expectancy), SVI 1.1/1.4 and PTF 1.2 findings.

Linked Evidence: EVD-9692, EVD-7681

OIC 2.1 **Organizational Culture Allows for Unsafe Task/Mission** **SUBSTANTIATED**

Calibration of 310FT01 was changed from 16000 to 2540 mmwc and back in Dec-2019 with no documentation, no checklist, and no approval record.

JUSTIFICATION

[EVD-2506] documents a separate unsafe maintenance task on the same meter: "During calibration from 17/12/2019 to 18/12/2019, calibration was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of this change in the calibration documents signed by the team. During next calibration for meter 310FT01 was again changed to 16000 MMWC. No checklist is followed while carrying such critical jobs of high pressure/risk" (p.27). The setting was changed by a factor of ~6.3 and then reverted with no documented justification, no checklist, and no recorded authorization. The committee recommendation D.1.i (p.35) explicitly requires that "details of Master calibrator/action taken/replacement/deviation from procedure if any, is to be documented in the calibration certificate", indicating this was a systemic documentation omission.

CAUSAL EXPLANATION

The undocumented calibration changes did not by themselves open the leak, but they normalized checklist-free, unrecorded modifications to the same meter that later failed. That cultural precedent carried into the 18-Sep-2020 gasket replacement, where the hydrostatic test was omitted without challenge. It therefore enabled lower-level ASBE 1.2/1.5, AJDE 2.10 and AV 1.1 findings, and contributes to the integrity-management environment that culminated in the 02:12:50 release on 24-Sep-2020.

Linked Evidence: EVD-2506

OIC 2.1 Organizational Culture Allows for Unsafe Task/Mission

SUBSTANTIATED

Missing work permits were untracked, JSPs were absent from SAP, and permit closure was not reviewed, allowing hazardous tasks without documented hazard analysis.

JUSTIFICATION

[EVD-5105] records: "Work Permits are issued in SAP and hard copy of work permits are at sites during execution of work. After completion of work, the originals are filed. On checking the same, it was found that few of the work permits were missing. No data is kept on such missing permits. Job safety Plan is not attached/given in SAP document. Closing of permits & actions indicated in work needs review" (p.26). The committee also noted (p.34) that "Maintenance notifications are not be assumed as permit to work", indicating that notification-driven work was being executed without separate task-specific hazard authorization. This is a distinct manifestation of a culture in which paperwork controls were allowed to be incomplete or absent.

CAUSAL EXPLANATION

The incomplete permit/JSA system meant the 18-Sep-2020 high-risk gasket replacement could proceed without a JSP forcing identification of the required design-pressure hydrotest and material-conformance checks. This is rated 2 because the evidence does not tie a specific missing permit directly to the failed joint, but it contributed to the procedural environment that allowed the hydrotest omission to pass into service; it supports lower-level PTE 2.7, SVI 1.4 and SVV 4.1 findings.

Linked Evidence: EVD-5105

OIC 2.3 Organisational Over confidence or Under confidence in Equipment

SUBSTANTIATED

The repaired 310FT01 flange was accepted as pressure-boundary-sound after only an operating-pressure soak, with no design-pressure hydrotest.

JUSTIFICATION

[EVD-2894] records the organization treated an operating-pressure soak as sufficient proof of joint integrity: "No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated. Next day meter 310FE01 was taken on online and was in use at the time of Incident" (p.31). This is a specific instance of overconfidence in the repaired equipment: the joint was declared fit without being tested to design pressure, and the organization accepted that confidence by placing the meter in hydrocarbon service. The committee's probable-cause statement, "Probable cause of leak is due to use of nonstandard gasket and not testing the system to designed pressure" (p.32), ties this overconfidence directly to the loss of containment.

CAUSAL EXPLANATION

The organization's confidence in the repaired flange was based on a test that could not reveal the geometry mismatch between the round O-ring and the rectangular slot. The meter entered service on 19-Sep-2020 and withstood operating pressure for about six days before the joint failed at 02:12:50 on 24-Sep-2020. This is the Level-4 origin of lower-level ASBE 1.2, AJDE 2.7 and AJDE 2.11, and the 56-minute release that followed ended in ignition at 03:08:15.

Linked Evidence: EVD-2894

OIC 2.3 Organisational Over confidence or Under confidence in Equipment

SUBSTANTIATED

A 1.03 mm pitted zero-corrosion-allowance pipeline segment was considered acceptable after painting only, without evaluating remaining strength.

JUSTIFICATION

[EVD-2397] shows management treated a cosmetic coating as sufficient closure for a structural integrity finding: "a reduction of thickness of 1.03mm along with heavy pitting and painting peeling off was observed in one of the segments of 36 pipeline in gas terminal in July 2019. The corrective action of painting was completed in Dec. 2019. However, as per the available data, the design thickness of the pipeline is 23.83mm with zero corrosion allowance. It needs to be examined whether painting as a corrective action would suffice in such cases" (p.30). The organization thereby expressed confidence in the continued fitness of a degraded zero-allowance pipe without any fitness-for-service or remaining-strength evaluation. This is a substantiated instance of organizational overconfidence in equipment, though it did not itself fail during the incident.

CAUSAL EXPLANATION

The overconfidence in the pitted pipe segment did not directly produce the fire, since the 36-inch line remained intact and the release occurred at the metering skid. However, it is the same organizational confidence pattern that accepted the nonconforming O-ring and the omitted hydrostatic test on the meter: a known defect was treated as resolved by a superficial action. It therefore substantiates lower-level AJDE 2.8 (wrong choice of painting as corrective action) and the supervisory culture findings SVI 1.6/SVIO 2.5, contributing to the environment in which the 24-Sep-2020 release could occur.

Linked Evidence: EVD-2397

OIR 1.6 Failure in Maintenance/ Spares/ Support Programme

SUBSTANTIATED

For pipes commissioned in 1986, no baseline thickness data and no original drawings/schedules exist; only 2011 and 2017 measurements are on file.

JUSTIFICATION

The pipeline integrity support programme also failed to preserve baseline records. [EVD-8287] records: 'It is observed that for a large number of pipes, especially in gas terminal area baseline data of pipe thickness in not available. The original drawings and the schedules of the pipes are also not available. In most cases only two thickness details of 2011 and 2017 is available for a pipe which is commissioned in 1986.' For an asset commissioned in 1986 and still in service in 2020, only two measurement data points (2011, 2017) exist, with no commissioning baseline, no original drawings and no pipe schedules — so genuine thickness trend analysis across the 34-year asset life is impossible. The correct records-management programme would have preserved or re-established these engineering baselines; the ACTUAL programme allowed them to be lost without replacement. This instance did not contribute directly to the 24-Sep-2020 failure — the leak was at the metering flange and the trunk line survived intact — but it degraded the integrity-management data foundation and is therefore rated 2.

CAUSAL EXPLANATION

Without a 1986 baseline and original pipe schedules, the 2011/2017 measurements could not be trended against design starting points, weakening prioritisation of the 2017 significant-thickness-loss findings and reinforcing the three-year non-action on corrective plans. This Level-4 records-management failure is the organizational origin of the Level-2 finding PTE 2.7 baseline-data availability (rating 3) and the Level-3 finding SVI 1.4 policy gap on baseline preservation (rating 2), and it is part of the broader loss-of-technical-records pattern (lost OEM gasket spec, lost P&ID) that left maintainers without authoritative references. Its direct causal share in the 24-Sep-2020 incident is limited, so it is rated 2.

Linked Evidence: EVD-8287

OIR 1.7 Phase-Out Policies (Failure to Remove Inadequate) Work Out Equipm

SUBSTANTIATED

The F&G system remained annunciator-only with no auto-shutdown or suppression interlock, converting a machine-detected 02:12:50 leak into a 46-minute manual response.

JUSTIFICATION

The plant's Flame & Gas detection system remained in service as a display-only annunciator system with no automatic protective action, and no phase-out or upgrade corrected this known inadequacy before the incident. [EVD-8657] records 'Hydrocarbon/H2S detectors are configured to Annunciator Panel for display but not in DCS for history' and 'F&G detection system does not have provision for auto shut down of plant or activation of fire suppression system' (p.26). [EVD-3452] confirms the fix was only proposed after the event: the committee's recommendation A.1 at p.34 calls for 'F&G detection system needs to be interlocked to automatic plant shut down and activation of fire suppression system.' A safety system whose machine detection at 02:12:50 produced no automatic isolation was inadequate for the major-release scenario it was installed to protect against; the failure to remove, replace, or upgrade this obsolete configuration left the plant dependent on a manual human response chain that took 46 minutes to close the first valve.

CAUSAL EXPLANATION

With no automatic shutdown interlock, the leak witnessed by CCTV at 02:12:50 continued unisolated through the entire human chain: CISF detection 02:15, CCTV verification 02:18, control-room notification 02:18-02:20, senior call-out 02:35, MOV 54 closure 03:01, and 36-inch PCV closure 03:04. The 46-minute release produced the large flammable cloud that ignited at 03:08:15. This Level-4 phase-out failure enabled the lower-level PTE 2.11 equipment-misconfiguration finding, the PTF 1.4 over-reliance-on-human-response finding, the AV 1.1 F&G-configuration violation, and the SVI 1.4 failure-to-require-interlocks finding.

Linked Evidence: EVD-8657, EVD-3452

OIR 1.8 **Failure to Provide Adequate Operational Information Resources/ Support** **SUBSTANTIATED**

The DMP/ERDMP covered the leak scenario but provided no isolation/stoppage decision criteria or authority, leaving responders without a pre-planned protective action.

JUSTIFICATION

The emergency planning information available to shift personnel defined the hydrocarbon-leak scenario but omitted the decision rule that mattered most: when and by whom to isolate. [EVD-4052] records 'As per DMP On site Hazira Plant, guidelines to be followed in case of hydrocarbon leakage are included but does not include isolation/stoppage of facilities/pipelines' (p.32), reinforced at p.27: 'In ERDMP, scenario for Level 3 emergencies are taken from the plant failures. Heavy product/gas leakages from pipelines & tanks are considered but decision on isolation/closure of plant/pipelines is not mentioned.' The committee's root cause #6 'Non isolation/stoppage of 36 pipeline operation' flows from this missing planning information. Without a pre-communicated isolation criterion and authority, operators and supervisors at 02:15-02:20 had to improvise a serial escalation path rather than execute a pre-planned isolation action, which is precisely the information-resource failure defined by OIR 1.8.

CAUSAL EXPLANATION

The missing DMP isolation information forced the night-shift response into physical verification and serial management escalation: leak onset 02:12:50, first detection 02:15, annunciator 02:20, senior call-out 02:35, MOV 54 closure 03:01, 36-inch PCV closure 03:04, ignition 03:08:15. This Level-4 planning-information failure directly enabled the lower-level AJDE 2.9, AJDE 2.10, and AJDE 2.11 decision failures, the ASBE 1.2 emergency-checklist omission, the PTF 1.13 planning failure, and the SVI 1.4 policy-guidance failure, producing the 46-minute unisolated release that created the ignitable cloud.

Linked Evidence: EVD-4052

OIR 1.8 **Failure to Provide Adequate Operational Information Resources/ Support** **SUBSTANTIATED**

The metering-system P&ID was a lost-original, unapproved vendor document that mismatched the field, leaving no accurate as-built information.

JUSTIFICATION

The metering system's controlled engineering drawing resource was missing: the original was lost, the vendor replacement was never approved by ONGC, and the field configuration did not match even that replacement. [EVD-0933] records 'Reference drawings no. 02-310-AO-101 Rev.1, 2x6 lines & 1x2 lines for metering system are shown but valves & meters configuration are not matching. As old drawings were lost, new drawings are made by a vendor but the same has not been checked & approved by ONGC' (p.26). Without an accurate as-built reference, maintenance personnel had no authoritative configuration baseline to verify the correct rectangular-slot gasket design, flange details, or valve arrangement, which reinforced the undocumented O-ring substitution and the no-hydrotest return to service. This information-resource failure is secondary in the causal chain because the physical failure was driven by the wrong seal and omitted test, but it removed a documentation control that could have challenged the substitution.

CAUSAL EXPLANATION

The missing controlled drawing removed the configuration baseline that maintenance needed to identify the correct OEM rectangular-slot gasket, enabling the lower-level PTE 2.7 technical-guidance gap, the PTE 2.14 unfamiliar-equipment condition, and the AV 1.1 configuration-management breakdown. This contributed secondarily to the installation of the non-conforming O-ring that failed at 02:12:50 on 24-Sep-2020.

Linked Evidence: EVD-0933

OIC 2.4 **Unit Mission/ Aircraft/ Vehicle/ Equipment Change or Unit Deactivation** **SUBSTANTIATED**

The OEM rectangular-slot gasket was replaced by a locally procured round O-ring with no specification, quality control, recorded authority or MOC, repeated across three repairs in 22 months.

JUSTIFICATION

A material-equipment change was made on a pressure boundary without any managed change process, and the official committee traced two of its root causes directly to that change. [EVD-8645] records 'Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket', and the same evidence notes committee root cause #2 'Use of nonstandard O ring' and #3 'Lack of control over material sourcing'. The correct configuration is the 1986 rectangular-slot gasket that beds into the machined groove of meter 310FE1/310FT01; the actual substitute was a round locally bought O-ring whose cross-section cannot fill that rectangular slot and can only make line contact, leaving a leak path at the joint. The change was also repeated, not isolated: [EVD-1494] records gasket replacements on 12-Nov-2018, 25-Feb-2019 and 18-Sep-2020, so the same

CAUSAL EXPLANATION

The unmanaged change from a controlled OEM rectangular-slot gasket to a locally purchased O-ring created a flange joint that was geometrically unable to seal the 1986 meter's rectangular slot. Because the change carried no recorded authority, specification, or quality control, the same deficient substitution was reinstalled on 12-Nov-2018, 25-Feb-2019 and 18-Sep-2020; after the last repair the meter returned to service on 19-Sep-2020 and the joint failed at 02:12:50 on 24-Sep-2020, initiating a 56-minute vapour accumulation that ignited at 03:08:15. This Level 4 change-management failure created the conditions for the lower-level wrong choice of action (AJDE 2.8), routine/widespread violation (AV 1.1), incorrect gasket procedure (ASBE 1.3) and the maintenance-not-conducted-correctly precondition (PTE 2.12), and it corresponds directly to committee root causes #2 and #3.

Linked Evidence: EVD-8645, EVD-1494

OIC 2.4 **Unit Mission/ Aircraft/ Vehicle/ Equipment Change or Unit Deactivation** **SUBSTANTIATED**

The 18-Sep-2020 gasket replacement was returned to service on 19-Sep-2020 after only an operating-pressure pressurisation, with no hydrostatic test to design pressure, despite this being a high-pressure pressure-boundary change.

JUSTIFICATION

The equipment-change back-to-service transition was executed through a rushed validation path rather than the mandated design-pressure hydrostatic test, and the committee explicitly linked this to the probable cause. [EVD-2894] records 'After calibration of the meter 310FE01, the gasket was replaced on 18.09.2020. No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated. Next day meter 310FE01 was taken on online and was in use at the time of Incident', and it also records the committee's conclusion 'Probable cause of leak is due to use of nonstandard gasket and not testing the system to designed pressure. No standard operating procedure is in place for calibration and taking back in service of these meters which are operating at high pressures'. The correct transition after a pressure-boundary change is a hydrostatic test to design pressure before hydrocarbon service; the actual transition applied only an operating-pressure soak with both meters in line and isolated, so the new O-ring joint was never loaded to its design requirement. [EVD-0185] confirms the

CAUSAL EXPLANATION

Because the equipment change was not validated to design pressure, the non-conforming O-ring joint was accepted into live hydrocarbon service on 19-Sep-2020 with an unknown pressure capability. The meter then operated for approximately six days under condensate pressure before the seal failed at 02:12:50 on 24-Sep-2020, producing the vapour cloud that accumulated for 56 minutes and ignited at 03:08:15. This Level 4 change-control omission directly enabled the lower-level ASBE 1.2 checklist-not-followed finding (rating 4), AJDE 2.7 delayed/omitted necessary action (rating 4), AJDE 2.11 misapplied testing policy (rating 4) and the PTF 1.13 mission-planning failure, and it is the reason the defective seal was not discovered in a controlled test before operation.

Linked Evidence: EVD-2894, EVD-0185

OIC 2.4 **Unit Mission/ Aircraft/ Vehicle/ Equipment Change or Unit Deactivation** **SUBSTANTIATED**

The 310FT01 calibration setpoint was changed 16000 to 2540 mmwc and later back to 16000 mmwc with no entry in the signed calibration documents, indicating uncontrolled equipment parameter changes on the failed meter.

JUSTIFICATION

A critical equipment parameter was changed and then reversed without being managed through the calibration record, on the same meter that later became the leak source. [EVD-2506] records 'During calibration from 17/12/2019 to 18/12/2019, calibration was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of this change in the calibration documents signed by the team. During next calibration for meter 310FT01 was again changed to 16000 MMWC. No checklist is followed while carrying such critical jobs of high pressure/risk'. The correct process would record the setpoint change, the reason, and the approval before signing the calibration certificate; the actual process changed the range by a factor of about 6.3 and then changed it back, with neither change documented in the signed record and no checklist used. This is a separate manifestation of unmanaged equipment change from the gasket substitution because it occurred at a different time window on the calibration system of the same meter.

CAUSAL EXPLANATION

The undocumented 16000-to-2540 mmwc change and its reversal created measurement uncertainty and, more importantly, normalized uncontrolled changes to critical pressure-boundary instrumentation on the same meter. Nine months later, the same maintenance organization installed an unspecified O-ring and returned the meter to service without a hydrotest, suggesting the calibration change precedent helped weaken configuration control on 310FT01. The instance therefore supports the lower-level ASBE 1.2 calibration-checklist failure (rating 3), ASBE 1.5 under-controlled calibration (rating 3), AJDE 2.10 inadequate calibration knowledge (rating 3) and SVF 3.1 personnel-management findings that fed the 24-Sep-2020 failure at 02:12:50.

Linked Evidence: EVD-2506

OIR 1.8 **Failure to Provide Adequate Operational Information Resources/ Support** **SUBSTANTIATED**

DCS clocks across control rooms drifted by 2-12 minutes, making event correlation and data matching difficult.

JUSTIFICATION

The DCS systems across the multiple control rooms did not share a synchronised clock, degrading the event-time information needed for both real-time correlation and reconstruction. [EVD-5532] records 'Time stamping in DCS systems provided in various control rooms is varying from 2 minutes to 12 minutes and data matching is difficult' (p.26). The incident narrative confirms operators had to adopt CCTV timestamps as canonical because of this drift. This information-resource deficiency did not alter the incident outcome, since the sequence was independently reconstructed from CCTV, logs, and witness statements, but it removed a trustworthy cross-control-room timing resource that could have supported coordinated emergency decision-making and forensic analysis.

CAUSAL EXPLANATION

The asynchronous DCS clocks did not cause the release or ignition, but they removed a reliable timing information resource for correlating alarms across control rooms and forced the investigation to depend on CCTV timestamps. This supported the lower-level PTE 2.6/PTE 2.11 configuration findings and marginally compounded forensic difficulty, hence the present-without-impact rating of 2.

Linked Evidence: EVD-5532

OIR 1.8 **Failure to Provide Adequate Operational Information Resources/ Support** **SUBSTANTIATED**

Coverage relied on one 90-second-pan PTZ camera with no redundant display; its cable burned out 7 seconds after ignition, ending visual evidence.

JUSTIFICATION

The only video monitoring resource for the metering skid was a single PELCO PTZ D5230D camera that panned 360 degrees every 90 seconds, with limited visibility and no separate display in the plant control rooms. [EVD-0210] records 'PELCO make PTZ cameras (Model D5230D) are provided at few places and has turn around (360 degree) at the time of about 90 seconds. Visibility from this camera is limited' (p.30). The information loss became complete at ignition: [EVD-2752] records 'major fire continued, CCTV capturing discontinued due to burnt of CCTV cable' (p.25), seven seconds after ignition at 03:08:15. The intermittent coverage and the CCTV-controlled verification step added latency between the 02:15 perimeter detections and the 02:18-02:20 control-room notification, and the cable loss ended the visual evidence chain for the post-ignition phase. This is therefore a secondary but real operational information-resource deficiency.

CAUSAL EXPLANATION

The single intermittent camera meant the leak onset at 02:12:50 did not produce continuous operator alerting, and the CISF-internal verification step via the CCTV control room consumed the 02:15-02:18 interval before the GT CR was called. The cable loss at 03:08:22 then removed the only remote visual feed at the fire transition. This Level-4 information-resource failure enabled the lower-level PTE 2.10 communications-equipment findings, the PTF 1.16 miscommunication findings, and the committee's recommendation for per-plant CCTV with master-control display.

Linked Evidence: EVD-0210, EVD-2752

OIC 2.4 Unit Mission/ Aircraft/ Vehicle/ Equipment Change or Unit Deactivation

SUBSTANTIATED

The newly built Gas Terminal Control Room was delivered with blast-proof walls but non-blast-proof doors, leaving an incomplete protective envelope in an area that became vapour-filled during the incident.

JUSTIFICATION

The Gas Terminal Control Room was a new facility transition whose blast protection was incomplete at the doors, a discrepancy logged in the facility equipment references. [EVD-1460] lists the linked equipment 'Gas Terminal Control Room (new blast-proof; doors not blast-proof)', and [EVD-9941] repeats the same configuration reference. The correct change outcome for a control room in a gas-processing area would be a complete blast-proof envelope including the doors; the actual delivered state left the doors non-blast-proof. The vapour cloud reached the road and control-room area within approximately 40 seconds of first detection, as [EVD-9941] documents, so personnel inside were operating in a facility whose protective envelope had a known gap. The deficiency did not cause the release or the ignition and no occupant was injured, so it is rated present without direct outcome impact, but it is a further example of an equipment/facility change completed in an unsafe, unverified condition.

CAUSAL EXPLANATION

This instance was not part of the causal chain to the gasket failure or ignition, and no injury resulted, but it shows that the organization accepted a partially compliant blast-protection change. The observed incomplete transition is the same failure pattern identified elsewhere: changes were closed out without verifying the final protective state, a culture that also permitted the unvalidated gasket change to enter service on 19-Sep-2020.

Linked Evidence: EVD-1460, EVD-9941

OIC 2.5 Organisational Structure

SUBSTANTIATED

The DMP/ERDMP omitted isolation criteria and no standing authority existed below Plant Manager to close MOV 54, forcing a 46-minute escalation instead of immediate isolation.

JUSTIFICATION

The organization's emergency decision structure left no role below the senior management chain with standing authority to isolate the 36-inch supply, and [EVD-3932] records the structural consequence: 'NO role between Shift In-charge and Plant Manager has standing authority to close MOV 54 (or any other emergency-isolation valve) without telephoning up the chain — confirmed by the committee finding (p.26) that "MOVs are provided in the gas terminal station for local operations but they are not having remote operation from control room during emergency."' The governing plan itself omitted the decision rule: [EVD-4052] records 'As per DMP On site Hazira Plant, guidelines to be followed in case of hydrocarbon leakage are included but does not include isolation/stoppage of facilities/pipelines', and the committee's root cause #6 is 'Non isolation/stoppage of 36 pipeline operation'. The correct structure would have pre-authorized a named on-shift role to order isolation immediately on confirmed release and defined the criteria for doing so; the actual structure required serial escalation and physical presence, which consumed the 46 minutes between first detection at 02:15 and

CAUSAL EXPLANATION

Because the organizational structure did not delegate isolation authority or define isolation criteria, the 02:15 detection and 02:20 annunciator alarm could not trigger pre-authorized action; the response passed serially through the shift in-charge, management chain and physical arrival at the gate before MOV 54 was closed at 03:01, 46 minutes after first detection. The vapour released during that window was already 'all around even at height' at 03:00:44 and ignited at 03:08:15. This Level 4 structural failure caused the lower-level AJDE 2.7 delayed necessary action (rating 4), AJDE 2.9 decision-making during operation, AJDE 2.10/2.11 knowledge and policy failures, ASBE 1.3 emergency-isolation procedural failure, and the PTF 1.5/PTF 1.6 role and authority-gradient preconditions, and it is the organizational root beneath committee root cause #6.

Linked Evidence: EVD-4052, EVD-3932

OIC 2.5 **Organisational Structure** **SUBSTANTIATED**

CISF perimeter detections could not reach the Gas Terminal Control Room directly and had to relay through the CCTV Control Room, adding a 3-5 minute notification lag at first detection.

JUSTIFICATION

The detection organization required every perimeter alarm to pass through a verification node rather than reaching the operating control room directly. [EVD-8871] records 'At 02:18 he visually verified "heavy leakage and formation of gas cloud through CCTV" (timeline p.12) and notified the Gas Terminal Control Room and CISF Main Gate. The CISF-internal verification step adds a ~3 minute latency between physical detection (02:15) and operating-CR notification (02:18-02:20). The CCTV CR is treated as an obligatory verification node, but the same architecture means perimeter detection cannot reach the operating CR directly.' [EVD-6234] independently documents the same routing: 'At 0216. He informed CCTV Control room and CISF main gate control room.' This communication architecture was structurally inadequate for time-critical warnings because the two simultaneous 02:15 detections by Constable A and Head Constable B both bypassed the Gas Terminal Control Room, so the crew responsible for the area learned of the event only at 02:18-02:20.

CAUSAL EXPLANATION

Because the communication map placed the CCTV Control Room as an obligatory intermediary, the 02:15 physical detections reached the Gas Terminal CR only at 02:18-02:20, five minutes before the annunciator. The operating crew therefore started their assessment after the cloud was already dense, contributing to the lost early minutes and to the later 46-minute cumulative response delay ending with MOV 54 closure at 03:01 and ignition at 03:08:15. This structural routing gap enabled the lower-level PTF 1.10 critical-information-not-communicated findings, PTF 1.16 miscommunication, ASBE 1.6 visual-scan failure, and AJDE 2.4 misinterpretation.

Linked Evidence: EVD-8871, EVD-6234

OIC 2.5 **Organisational Structure** **SUBSTANTIATED**

SAP notification 3800419840 for meter 310FT01 remained open and unreviewed after the gasket replacement, with no assigned owner for closure or quality verification.

JUSTIFICATION

The maintenance workflow assigned no one accountable for closing SAP notification 3800419840 after the gasket replacement, removing the supervisory review gate before the meter returned to service. [EVD-0591] records 'SAP Notification 3800419840 dt. 17.09.2020 was raised for checking meter 1', and the description adds the notification 'was not closed after completion of the job' (p.27) — no supervisory closure check. The correct workflow would name a responsible senior officer to review the completed work, confirm the required tests were performed, and approve closure; the actual workflow left the notification open and the job entered service with no formal hand-back. Because no one owned that step, the non-conforming gasket, the missing hydrotest, and the repeated failure pattern all passed into operation unchallenged.

CAUSAL EXPLANATION

The unclosed and unreviewed SAP notification allowed the 18-Sep-2020 repair to escape any management quality gate, so the non-standard O-ring and omitted hydrotest were not detected before the meter returned to service on 19-Sep-2020. The defective joint then failed at 02:12:50 on 24-Sep-2020. This structural accountability gap enabled lower-level PTF 1.12 failure-to-communicate feedback, SVI 1.1/1.6 supervisory non-response, AV 1.1 routine non-closure violation, AJDE 2.11 policy misapplication, and PTE 2.12 missing supervisory gate.

Linked Evidence: EVD-0591

OIC 2.5 **Organisational Structure** **SUBSTANTIATED**

Control-room operators had only a four-day general training course with no defined qualification in HR/Operations Policy, leaving emergency competence structurally unassigned.

JUSTIFICATION

The organization had no defined qualification standard for control-room roles, leaving night-shift emergency competence unowned and unverified. [EVD-5634] records 'control room operations are carried out by persons inadequately trained in safety, operations, control room handling, logbook records and actions during emergency. It is informed that four days training programme is conducted for new recruitment/transfer staff which is general in nature and later the job training is carried out. Qualification of this staff is also not defined in HR/Operations Policy. Most of the staff have attended only One Basic Course since 2018'. The correct structure would define a qualification and recurrent competency standard in HR/Operations Policy, with simulator-based emergency training before assignment to control-room duties; the actual structure assigned the two night-shift engineers on the basis of a four-day general course and no defined competency gate. This structural gap manifested at 02:20 when the control-room staff attempted physical source verification and telephone escalation rather than trained isolation action.

CAUSAL EXPLANATION

Because no organizational role owned or enforced control-room qualification, the crew on duty at 02:20 had no drilled emergency-isolation response, contributing to the 46-minute gap between first detection and MOV 54 closure at 03:01 and to the 03:08:15 ignition of the accumulated cloud. This structural gap enabled lower-level SVV 4.4 authorizing unqualified individuals, SVI 1.3/1.4/1.11/1.14 supervisory training failures, PTF 1.3 task delegation, PTF 1.17 unfamiliarity with workspace, and AJDE 2.10 inadequate knowledge for isolation decisions.

Linked Evidence: EVD-5634

OIC 2.1 **Organizational Culture Allows for Unsafe Task/Mission** **SUBSTANTIATED**

The 18-Sep-2020 gasket replacement was accepted with only an operating-pressure soak and no design-pressure hydrotest, with no SOP governing high-pressure meter return-to-service.

JUSTIFICATION

A second distinct unsafe task was permitted by the same climate: pressure-boundary work was accepted with no hydrostatic test. [EVD-2894] records: "After calibration of the meter 310FE01, the gasket was replaced on 18.09.2020. No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated. Next day meter 310FE01 was taken on online and was in use at the time of Incident" (p.31). [EVD-0185] confirms the meter "was taken on online" the next day. The committee directly linked this to the cause: "Probable cause of leak is due to use of nonstandard gasket and not testing the system to designed pressure. No standard operating procedure is in place for calibration and taking back in service of these meters which are operating at high pressures" (p.32). The organization thus formally acknowledged that it had no SOP and allowed an untested pressure boundary into live hydrocarbon service.

CAUSAL EXPLANATION

Because senior leaders accepted the no-hydrotest practice, the defective O-ring joint was never subjected to design pressure before hydrocarbon service. Meter 310FT01 operated for approximately six days (19-24 Sep 2020) and failed at 02:12:50; the resulting vapor accumulated for 56 minutes and ignited at 03:08:15. This organizational cause enabled the lower-level unsafe acts AJDE 2.5/2.7 (ignoring or delaying the hydrotest requirement), ASBE 1.2 (checklist not followed) and AJDE 2.11 (misapplied testing policy), and is the Level-4 origin of supervisory failures SVI 1.10, SVIO 2.5 and SVF 3.4.

Linked Evidence: EVD-2894, EVD-0185

OIC 2.1 **Organizational Culture Allows for Unsafe Task/Mission** **SUBSTANTIATED**

Known pipeline wall-loss findings from 2017 had no corrective action plan after 3+ years, showing a cultural acceptance of deferred integrity actions.

JUSTIFICATION

[EVD-5527] documents a separate manifestation of the same culture on the pipeline system: "As per the report of 2017, several pipes have been identified with significant thickness loss and need for corrective action. However, even after three years no plan has been developed for the corrective actions to be taken for those lines. Mere measurement of thickness without corrective action does not complete the process of maintaining the integrity of pipes" (p.30). This is an official committee observation that known integrity deficiencies were measured but never remediated, a pattern also observed in the 2017-to-2020 metering recurrence. The finding is substantiated because the report's existence and the absence of action are documented; it is rated secondary rather than primary because the 36-inch pipeline itself remained intact during the fire.

CAUSAL EXPLANATION

This instance did not directly cause the metering-skid fire, but it created and normalized the norm that identified risks could be recorded without being resolved. That same organizational default was applied to the meter 310FT01 recurrence and the skipped hydrostatic test, which did directly cause the 24-Sep-2020 release. It is the Level-4 climate behind lower-level findings AJDE 2.5 (ignored warning on pipeline thickness) and AJDE 2.7 (delayed necessary action), and the supervisory non-response SVI 1.6.

Linked Evidence: EVD-5527

OIC 2.1 **Organizational Culture Allows for Unsafe Task/Mission** **SUBSTANTIATED**

A 1.03 mm wall-loss finding on a zero-corrosion-allowance 36-inch segment was treated with painting only, without remaining-strength evaluation.

JUSTIFICATION

[EVD-2397] documents a structurally inadequate remedy being accepted as closure: "a reduction of thickness of 1.03mm along with heavy pitting and painting peeling off was observed in one of the segments of 36 pipeline in gas terminal in July 2019. The corrective action of painting was completed in Dec. 2019. However, as per the available data, the design thickness of the pipeline is 23.83mm with zero corrosion allowance. It needs to be examined whether painting as a corrective action would suffice in such cases and the mechanism of deciding on a corrective action needs to be reviewed" (p.30). The official wording explicitly questions whether the chosen action was adequate. This demonstrates a culture in which cosmetic measures were used to close engineering integrity deficiencies without fitness-for-service analysis.

CAUSAL EXPLANATION

The painting-only decision did not cause the metering failure, but it is the same organizational pattern as accepting the O-ring substitution without engineering review: an identified structural risk was addressed by a superficial action. This Level-4 climate enabled the lower-level wrong-choice finding AJDE 2.8 (painting instead of analysis/replacement) and the supervisory acceptance SVI 1.6/SVIO 2.5, reinforcing the environment in which the meter's inadequate gasket was repeatedly accepted until it failed at 02:12:50 on 24-Sep-2020.

Linked Evidence: EVD-2397

OIC 2.7 **Inconsistent Risk Management Practices** **SUBSTANTIATED**

A formal MOC SOP existed on paper but no MOCs were implemented since 2018, so materials changes such as the gasket substitution bypassed review.

JUSTIFICATION

EVD-2986 records that 'Management of Change (MOC) philosophy/SOP was published in August 2016. However, the procedure indicated in the document is not implemented while carrying out change of Low-Pressure Gas to Kribhico from Medium Pressure Gas headers in LPG Plant in 2013 & 2018. It is informed no more MOCs are implemented in the plant since 2018' (p.30). This creates a direct mixed-message condition within OIC 2.7: senior management issued a formal MOC procedure, but the operating site stopped applying it entirely after 2018, so the workforce received the signal that process-safety change controls were optional. The same governance gap is then connected in EVD-8645 to the undocumented switch from OEM gaskets to locally purchased O-rings, which was never subjected to MOC review.

CAUSAL EXPLANATION

This Level-4 non-implementation enabled lower-level findings AJDE 2.11 (misapplied or unaware of MOC policy, rating 3), SVI 1.4 (failed to provide working change-control guidance, rating 3), SVIO 2.4 (substitution never formally risk-assessed, rating 4), and SVI 1.6 (lack of supervisory response to critical information, rating 3). Because no MOC gate was applied, the locally purchased O-ring substitution was treated as routine maintenance rather than as a change requiring engineering evaluation, allowing the deficient seal installed on 18-Sep-2020 to enter service and fail at 02:12:50 on 24-Sep-2020.

Linked Evidence: EVD-2986

OIC 2.7 **Inconsistent Risk Management Practices** **SUBSTANTIATED**

The emergency plan named the hydrocarbon-leak scenario but omitted isolation/stoppage decision criteria and authority, leaving responders without clear guidance.

JUSTIFICATION

EVD-4052 records that 'As per DMP On site Hazira Plant, guidelines to be followed in case of hydrocarbon leakage are included but does not include isolation/stoppage of facilities/pipelines' (p.32), and reinforces this at p.27 with the ERDMP observation that heavy gas leakage scenarios are considered but 'decision on isolation/closure of plant/pipelines is not mentioned'. The committee's root cause #6 ('Non isolation/stoppage of 36 pipeline operation') flows directly from this gap, confirming that the written emergency guidance named the hazard but provided no decision criteria or delegated authority for the protective action. This is a textbook OIC 2.7 manifestation of unclear guidance and mixed messages: the DMP purported to cover hydrocarbon leakage, yet personnel were left without any pre-authorised isolation rule.

CAUSAL EXPLANATION

This instance is the Level-4 cause of the 46-minute response delay findings: lower-level AJDE 2.9 (decision-making during operation, rating 3/4), AJDE 2.10 (inadequate knowledge of isolation criteria, rating 3), AJDE 2.11 (unawareness of the DMP gap, rating 4), PTF 1.13 (mission planning failure, rating 4), and SVI 1.4 (supervision failed to provide policy, rating 4). With no pre-authorised isolation action, the night shift from 02:15 to 03:01 verified, escalated, and waited for authority while the release continued; MOV 54 was closed only at 03:01, the 36-inch PCV at 03:04, and the accumulated cloud ignited at 03:08:15.

Linked Evidence: EVD-4052

OIC 2.7 Inconsistent Risk Management Practices

SUBSTANTIATED

A flame-proof lamp with three missing bolts and its glass case open 5-6 cm remained energised in Zone 1 and became the probable ignition source.

JUSTIFICATION

EVD-6916 records that 'three bolts on one of the flame proof lamps were missing resulting in it being open with its integrity compromised, glass case got detached from the main body and had opened out about 5 to 6 cms. The nearby fittings were found intact' (p.32). A flame-proof enclosure in a Zone 1 hydrocarbon area is a safety-critical risk barrier, and its loss of integrity was not identified or corrected under a regime that EVD-0496 describes as 'a lot more skewed towards breakdown maintenance and lays little emphasis on systematic preventive maintenance and inspection' (p.28). The risk was therefore neither raised through any effective inspection channel nor addressed before the incident, making this a substantiated OIC 2.7 instance in the electrical-integrity domain.

CAUSAL EXPLANATION

This instance converted the vapour release into a major fire: lower-level findings PTE 2.12 (maintenance not conducted correctly on FLP lamp, rating 4), PTE 2.15 (inadequate protection equipment, rating 4), and SVI 1.8 (supervision failed to correct the unsafe condition, rating 4) are all rooted in this organisational failure to maintain and enforce flame-proof integrity. When the vapour cloud accumulated over 56 minutes from leak onset at 02:12:50, the exposed contacts supplied the ignition energy at 03:08:15, only 4 minutes 15 seconds after the final isolation action at 03:04.

Linked Evidence: EVD-6916

OIC 2.7 Inconsistent Risk Management Practices

SUBSTANTIATED

The 2017 UT report identified significant pipe-wall loss requiring corrective action, but no plan existed more than three years later.

JUSTIFICATION

EVD-5527 records that 'As per the report of 2017, several pipes have been identified with significant thickness loss and need for corrective action. However, even after three years no plan has been developed for the corrective actions to be taken for those lines. Mere measurement of thickness without corrective action does not complete the process of maintaining the integrity of pipes' (p.30). This is direct evidence that a formal integrity measurement programme generated a risk warning requiring corrective action, but no plan was developed from 2017 to the incident on 24-Sep-2020. The organisation thus demonstrated to maintainers and engineers that risk findings could be acknowledged and then left unresolved indefinitely, which is the cultural signature of OIC 2.7.

CAUSAL EXPLANATION

Although the 36-inch trunk pipeline itself remained intact on 24-Sep-2020 (EVD-5780), this instance is the organisational root of the lower-level ignored-warning pattern: AJDE 2.5 (2017 warning ignored, rating 3), AJDE 2.7 (corrective action delayed 3+ years, rating 3), SVI 1.6 (supervisory non-response to critical information, rating 3), and SVF 3.4 (ineffective mitigation of identified risk, rating 3). It normalised the perception that raising an integrity risk would not be met with management action, which is the same environment in which the meter 310FT01 recurrence was allowed to persist until the 02:12:50 failure on 24-Sep-2020.

Linked Evidence: EVD-5527

OIC 2.7 **Inconsistent Risk Management Practices** **SUBSTANTIATED**

The mandated four-year UT thickness cycle was breached, with a six-year gap from 2011 to 2017 and the next cycle overdue by September 2020.

JUSTIFICATION

EVD-8307 records that 'The laid down schedule for thickness measurement is every four years... As per the records, the last two thickness measurement exercise was carried out in 2011 and 2017 which suggests that the schedule of four years is not being adhered to strictly' (p.29). The interval between 2011 and 2017 was six years rather than the mandated four years, and by the incident date the next cycle was already overdue. This schedule breach was a known control failure that management did not enforce, demonstrating that formally mandated risk-monitoring cadences were treated as flexible rather than mandatory.

CAUSAL EXPLANATION

This schedule failure did not directly initiate the incident, because the 36-inch trunk line remained intact (EVD-5780), but it is the systemic enabler of the lower-level AJDE 2.7 pattern of deferring required integrity actions. The unenforced cadence meant degraded wall loss could run uncharacterised for extended periods; more broadly, it taught the organisation that a formally defined risk-control cadence could be missed without consequence, contributing to the environment in which the meter hydrotest and notification-closure requirements were also treated as optional before the 24-Sep-2020 release.

Linked Evidence: EVD-8307

OIC 2.7 **Inconsistent Risk Management Practices** **SUBSTANTIATED**

A 16000-to-2540 mmwc setpoint change on meter 310FT01 in Dec-2019 was made and reversed without signed record or checklist use.

JUSTIFICATION

EVD-2506 records that 'During calibration from 17/12/2019 to 18/12/2019, calibration was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of this change in the calibration documents signed by the team. During next calibration for meter 310FT01 was again changed to 16000 MMWC. No checklist is followed while carrying such critical jobs of high pressure/risk' (p.27). The committee recommendation D.1.i at p.35, requiring documentation of master calibrator, action taken, replacement, and any deviation from procedure in the calibration certificate, directly addresses this gap. A change of approximately 6.3 times the setpoint was made and then reversed on the same meter that later became the leak source, with neither the change nor its reversal reflected in the signed records, demonstrating that critical deviations were not being surfaced or challenged.

CAUSAL EXPLANATION

This calibration instance did not itself open the 24-Sep-2020 leak, but it is the same OIC 2.7 reporting failure that later allowed the fatal gasket repair to escape control: it normalised undocumented, checklist-free changes on meter 310FT01 and enabled lower-level findings ASBE 1.2 (calibration checklist not followed, rating 3), ASBE 1.5 (under-controlled system, rating 3), AJDE 2.10 (inadequate calibration knowledge, rating 3), and SVI 1.6 (supervisory non-response to the unrecorded change, rating 3). The procedural-discipline decay established in December 2019 reappeared in the 18-Sep-2020 gasket replacement on the same meter, which omitted the hydrotest and produced the seal that failed at 02:12:50 on 24-Sep-2020.

Linked Evidence: EVD-2506

OIC 2.7 **Inconsistent Risk Management Practices** **SUBSTANTIATED**

Twelve external-audit recommendations were extended to 30-Apr-2021 without the procedurally required Director Offshore approval.

JUSTIFICATION

EVD-6852 records that 'External safety audit was conducted in February 2018, and 296 out of 308 recommendations are complied with. Balance are in various stages and dates are extended till 30/04/2021. Document indicating approval from Director Offshore for extending the target date was not available as per the procedure' (p.26). Of 308 formal external-audit recommendations, 12 remained open with target dates extended to 30-Apr-2021, but the extension lacked the required competent-authority approval. The existence of a formally mandated approval step that was bypassed is a clear signal to the organisation that control deadlines and approval requirements could be stretched without consequence, a substantiated OIC 2.7 mixed-message condition.

CAUSAL EXPLANATION

This instance had no direct mechanical link to the 24-Sep-2020 fire, so it is rated 2, but it is part of the organisational causal field: lower-level SVI 1.6 and SVF 3.4 findings on overdue recommendation management show the same non-closure behaviour that left the meter gasket recurrence unactioned. By normalising unapproved extensions of identified risks, the organisation weakened the expectation that reported hazards must be resolved, which contributed to the environment in which the 310FT01 repair was not closed out and the recurring gasket failure was not investigated.

Linked Evidence: EVD-6852

OIC 2.7 Inconsistent Risk Management Practices

SUBSTANTIATED

The permit-to-work system had untracked missing permits, JSPs absent from SAP, and unreviewed permit closure.

JUSTIFICATION

EVD-5105 records that 'Work Permits are issued in SAP and hard copy of work permits are at sites during execution of work. After completion of work, the originals are filed. On checking the same, it was found that few of the work permits were missing. No data is kept on such missing permits. Job safety Plan is not attached/given in SAP document. Closing of permits & actions indicated in work needs review' (p.26). The permit system therefore had missing originals with no log of which ones were missing, JSPs were not attached in SAP, and permit closure went unreviewed; the committee's recommendation B.5 reinforces that maintenance notifications were not to be assumed to be permits to work. This is a substantiated OIC 2.7 instance where formal risk-control documentation could disappear or be incompletely populated without detection.

CAUSAL EXPLANATION

This permit-control gap contributed secondarily to the lower-level ASBE 1.2 hydrotest omission and AJDE 2.11 policy misapplication findings by allowing high-risk maintenance such as the 18-Sep-2020 gasket replacement to proceed without a documented Job Safety Plan that would have identified the design-pressure test and material-conformance requirements. Because the evidence does not tie a specific missing permit to the failed joint, it is rated 2 rather than a direct cause, but it shows the same risk-documentation culture that let the defective seal return to service and fail at 02:12:50 on 24-Sep-2020.

Linked Evidence: EVD-5105

OIC 2.7 Inconsistent Risk Management Practices

SUBSTANTIATED

Control-room operators received only four days of general training with no defined qualification and no simulator emergency practice.

JUSTIFICATION

EVD-5634 records that 'control room operations are carried out by persons inadequately trained in safety, operations, control room handling, logbook records and actions during emergency. It is informed that four days training programme is conducted for new recruitment/transfer staff which is general in nature and later the job training is carried out. Qualification of this staff is also not defined in HR/Operations Policy. Most of the staff have attended only One Basic Course since 2018' (p.30). The committee's own benchmark at p.36 recommends at least one month of simulator-based training before assigning control-room duties, an approximately eightfold increase over the actual four days. This known competency risk was not remediated through HR policy, recruitment checks, or training investment, so the organisation accepted a control-room workforce that was not certified for emergency isolation decisions.

CAUSAL EXPLANATION

This instance is a Level-4 driver of the response delay: it enabled lower-level findings AJDE 2.10 (inadequate knowledge to make isolation decisions, rating 3/4), AJDE 2.11 (unawareness of existing policy, rating 3/4), PTF 1.3 (task delegation to under-trained staff, rating 3), PTF 1.17 (team unfamiliar with workspace, rating 3), and SVIO 2.1/2.3 (mission beyond capability and selection of inexperienced staff, rating 3). When the 02:20 annunciator alarm sounded, the night-shift engineers had no practised isolation script and defaulted to on-foot verification followed by telephone escalation; MOV 54 was not closed until 03:01, 46 minutes after first detection, and the accumulated cloud ignited at 03:08:15.

Linked Evidence: EVD-5634

OIC 2.7 **Inconsistent Risk Management Practices** SUBSTANTIATED

The post-incident electrical audit produced a one-page report that clubbed all nine HPMV lamps into a single generic damage status.

JUSTIFICATION

EVD-9999 records that 'Post the incident, an electrical audit of the affected area was carried out by the plant maintenance team. Considering that the audit was carried out to identify the damage to the equipment, the report is too generic and short (One page) and does not cover details of damages... the report has clubbed all nine 125 W HPMV FLP lamps together and reported cable burnt and fitting damaged due to heat as status for all of them' (p.28). EVD-0496 adds that the unique condition of one lamp, with three missing bolts and an open glass case, was not mentioned in the report, and that the electrical maintenance system is skewed toward breakdown maintenance. Even after a major fire, the risk-characterisation exercise failed to record per-equipment detail, showing the same weak reporting discipline under OIC 2.7.

CAUSAL EXPLANATION

This post-incident audit did not affect the incident outcome, because the fire had already occurred, so it is rated 2. Its significance is diagnostic: the same weak risk-reporting culture that failed to characterise lamp-specific damage after the event is the culture under whose authority the compromised Zone-1 HPMV lamp with three missing bolts went uninspected before the event and later supplied the ignition at 03:08:15. It corroborates the lower-level ASBE 1.2 audit-checklist failures rather than independently causing the fire.

Linked Evidence: EVD-9999, EVD-0496

OIC 2.1 **Organizational Culture Allows for Unsafe Task/Mission** SUBSTANTIATED

The F&G system was deliberately configured display-only with no auto-shutdown or suppression interlock, leaving the plant dependent on slow manual response.

JUSTIFICATION

[EVD-3452] records the accepted design posture: "F&G detection system does not have provision for auto shut down of plant or activation of fire suppression system" (p.26), and the committee's recommendation A.1 (p.34) explicitly requires that the system "needs to be interlocked to automatic plant shut down and activation of fire suppression system." [EVD-8657] adds: "Hydrocarbon/H2S detectors are configured to Annunciator Panel for display but not in DCS for history" (p.26). The consequence is visible in the timeline: leak onset at 02:12:50, first annunciator alarm at ~02:20, and first isolation MOV 54 only at 03:01. The organization had accepted a safety architecture in which a machine-detected leak could not stop the plant automatically.

CAUSAL EXPLANATION

Because the culture tolerated a non-interlocked F&G design, the leak witnessed by CCTV at 02:12:50 produced no automatic protective action. The full human detection-verification-escalation chain consumed 46 minutes until MOV 54 closure at 03:01, allowing the large flammable cloud documented at 03:00:44 to accumulate and ignite at 03:08:15. This Level-4 factor is the organizational root of lower-level PTE 2.11 (equipment not configured correctly), PTE 2.2/PTE 2.8 (inadequate instrument/workspace), AV 1.1 and SVF 3.2 findings.

Linked Evidence: EVD-3452, EVD-8657

OIC 2.1 **Organizational Culture Allows for Unsafe Task/Mission** SUBSTANTIATED

The DMP/ERDMP named major hydrocarbon leaks as a Level-3 scenario but contained no criteria or authority for isolation/stoppage, leaving responders without a planned action.

JUSTIFICATION

[EVD-4052] directly documents that the emergency planning documents defined the hazard scenario but not the required response: "As per DMP On site Hazira Plant, guidelines to be followed in case of hydrocarbon leakage are included but does not include isolation/stoppage of facilities/pipelines" (p.32). The report reinforces this at p.27: "Heavy product/gas leakages from pipelines & tanks are considered but decision on isolation/closure of plant/pipelines is not mentioned." This is a clear organizational omission in a formal emergency planning document, not a single individual error. The committee's root cause #6 ("Non isolation/stoppage of 36 pipeline operation") flows from this gap. The culture had allowed a critical emergency scenario to remain without documented decision criteria despite being explicitly named.

CAUSAL EXPLANATION

With no pre-planned isolation decision rule or delegated authority, the night-shift response defaulted to physical verification and serial escalation: detection 02:15, alarm 02:20, shift-in-charge called ~02:35, MOV 54 closure 03:01, 36-inch PCV closure 03:04. The 46-minute unisolated release produced the mature cloud seen at 03:00:44 and ignited at 03:08:15. This Level-4 culture factor enabled lower-level AJDE 2.9/2.10/2.11, PTF 1.5, PTE 2.7, SVI 1.4 and SVIO 2.4 findings.

Linked Evidence: EVD-4052

OIC 2.1 **Organizational Culture Allows for Unsafe Task/Mission** **SUBSTANTIATED**

Night-shift control-room personnel were assigned emergency duties with only a four-day general course, no defined qualification and no simulator emergency training.

JUSTIFICATION

[EVD-5634] documents the accepted training posture: "control room operations are carried out by persons inadequately trained in safety, operations, control room handling, logbook records and actions during emergency. It is informed that four days training programme is conducted for new recruitment/transfer staff which is general in nature and later the job training is carried out. Qualification of this staff is also not defined in HR/Operations Policy. Most of the staff have attended only One Basic Course since 2018" (p.30). The committee recommended at p.36 that operating staff be trained "for at least 1 month (preferably on simulators)" before control-room duties, an almost eightfold expansion of the current initial training. This demonstrates that a known competency gap was allowed to persist as organizational policy.

CAUSAL EXPLANATION

Because no competency standard existed, the two engineers who received the alarm at 02:20 treated the major release as a locate-and-inspect task rather than an isolation trigger; they attempted a physical approach, were repelled by vapor, and then escalated by telephone. This contributed to the 46-minute gap from 02:15 to MOV 54 closure at 03:01 and to ignition at 03:08:15. This Level-4 factor created the conditions for lower-level AJDE 2.10/2.11, PTF 1.3/1.17, SVI 1.9/SVIO 2.1 and PSE 3.1 findings.

Linked Evidence: EVD-5634

OIC 2.1 **Organizational Culture Allows for Unsafe Task/Mission** **SUBSTANTIATED**

Preventive maintenance for Zone 1 flame-proof electrical fittings was breakdown-oriented and generic, leaving a compromised HPMV lamp with three missing bolts and an open glass case in service.

JUSTIFICATION

[EVD-9999] records the electrical maintenance posture explicitly: "the report is too generic and short (One page) and does not cover details of damages... the report has clubbed all nine 125 W HPMV FLP lamps together and reported 'cable burnt and fitting damaged due to heat' as status for all of them" (p.28), and "the electrical maintenance system is a lot more skewed towards breakdown maintenance and lays little emphasis on systematic preventive maintenance and inspection" (p.28). The consequence was a Zone 1 lamp with its explosion protection already defeated: [EVD-6916] states "three bolts on one of the flame proof lamps were missing resulting in it being open with its integrity compromised, glass case got detached from the main body and had opened out about 5 to 6 cms. The nearby fittings were found intact" (p.32). This is a culture of accepting unsafe electrical equipment in a hazardous area.

CAUSAL EXPLANATION

The culture that accepted generic, breakdown-based electrical maintenance left the flawed flame-proof lamp with exposed contacts in the metering blast radius. After 56 minutes of vapor accumulation from the failed flange at 02:12:50, the cloud found this ready ignition source at 03:08:15, converting the release into a major fire. This Level-4 factor is the root of lower-level PTE 2.12/2.15, ASBE 1.2 (audit checklist missed the defect), SVI 1.8/SVF 3.1 and SVIO 2.7 findings.

Linked Evidence: EVD-9999, EVD-6916

OIP 3.2 **Organisational Program/ Policy Risks not Adequately Assessed** **SUBSTANTIATED**

Locally purchased O-rings replaced lost OEM rectangular-slot gaskets as regular practice without specification, QC, MOC, or any recorded authority.

JUSTIFICATION

The organization's material-sourcing program contained an unassessed technical risk: [EVD-8645] records 'The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket' (p.31). [EVD-2986] shows the MOC SOP published in August 2016 was not implemented after 2018, so this material substitution was never processed through the formal change-risk review that should have compared the round O-ring cross-section with the rectangular-slot sealing geometry and hydrocarbon service requirements. The committee's root cause #2 (use of nonstandard O ring) and #3 (lack of control over material sourcing) flow directly from this unassessed program risk. This is the Level-4 origin of lower-level SVIO 2.4 MOC failure, SVV 4.2 unwritten-policy standard, AJDE

CAUSAL EXPLANATION

Because the program risk was never assessed, the round O-ring was authorized repeatedly for a flange designed for a rectangular-slot gasket. The 18-Sep-2020 installation returned the meter to service on 19-Sep-2020 and the defective seal failed at 02:12:50 on 24-Sep-2020; the vapour accumulated for 56 minutes and ignited at 03:08:15. This Level-4 gap is the enabling cause of the initiating loss of containment.

Linked Evidence: EVD-8645, EVD-2986

OIP 3.2 **Organisational Program/ Policy Risks not Adequately Assessed** **SUBSTANTIATED**

The DMP/ERDMP named the hydrocarbon-leak scenario but never assessed or defined isolation and stoppage criteria, leaving no pre-authorized protective action.

JUSTIFICATION

The site's emergency-response program identified the hydrocarbon-leak scenario but never assessed the associated isolation decision risk. [EVD-4052] records 'As per DMP On site Hazira Plant, guidelines to be followed in case of hydrocarbon leakage are included but does not include isolation/stoppage of facilities/pipelines' (p.32), and further that in ERDMP Level 3 scenarios 'Heavy product/gas leakages from pipelines & tanks are considered but decision on isolation/closure of plant/pipelines is not mentioned' (p.27). A program risk assessment for a major vapour release should have defined pre-authorized isolation criteria and authority; instead the plan named the scenario but omitted the protective decision entirely. This program gap directly produced the lower-level SVI 1.4 DMP policy failure, AJDE 2.9/2.10/2.11 operational decision failures, and the committee root cause #6 regarding non-isolation/stoppage of the 36-inch pipeline operation.

CAUSAL EXPLANATION

Because the isolation-decision risk was never assessed or documented, responders at 02:15 and 02:20 had no pre-authorized action and defaulted to physical verification and serial management escalation. MOV 54 was closed only at 03:01 and the 36-inch PCV at 03:04, after 46 minutes of continuous release; the mature cloud was visible at 03:00:44 and ignited at 03:08:15. This Level-4 gap is the root of the response delay that determined incident severity.

Linked Evidence: EVD-4052, EVD-6414

OIP 3.2 **Organisational Program/ Policy Risks not Adequately Assessed** **SUBSTANTIATED**

The risk that emergency MOVs would be inaccessible during a vapour release was never assessed; they remained local-manual only.

JUSTIFICATION

The isolation architecture program accepted a risk that was never assessed for the case of an unapproachable vapour cloud. [EVD-3932] records 'MOVs are provided in the gas terminal station for local operations but they are not having remote operation from control room during emergency' (p.26), and [EVD-9612] shows the consequence: the GSP Shift In-charge had to physically reach the Gas Terminal Gate before closing MOV 54 at 03:01, 46 minutes after first detection at 02:15. A program risk assessment for major-hydrocarbon-release scenarios would have identified that dense vapour could prevent local access precisely when isolation is most urgent, and would have required remote or automatically operated isolation. This Level-4 gap directly enabled lower-level PTE 2.4/PTE 2.8 manual control inadequacy, AJDE 2.7 delayed necessary action and PTF 1.4 over-reliance on manual access.

CAUSAL EXPLANATION

Because the isolation valves had no remote operation, the control room could not close MOV 54 at 02:20 when the alarm sounded. The DGM had to be called at 02:35 and then physically reach the gate before closing the valve at 03:01, while the release continued and the cloud grew to the state documented at 03:00:44. This program-level risk gap is a primary enabler of the 46-minute unisolated release that fuelled the fire.

Linked Evidence: EVD-3932, EVD-9612

OIP 3.2 **Organisational Program/ Policy Risks not Adequately Assessed** **SUBSTANTIATED**

Breakdown-skewed electrical PM left Zone-1 flame-proof lamp integrity unassessed, allowing a three-bolt-missing lamp to become the probable ignition source.

JUSTIFICATION

The program risk from degraded flame-proof electrical fittings in Zone 1 was not assessed in the maintenance system. [EVD-0496] records the post-incident audit clubbed all nine 125 W HPMV FLP lamps together as 'cable burnt and fitting damaged due to heat' and that 'the electrical maintenance system is a lot more skewed towards breakdown maintenance and lays little emphasis on systematic preventive maintenance and inspection' (p.28); [EVD-9999] notes the one-page report did not cover per-equipment damage details. The result is documented in [EVD-6916]: 'three bolts on one of the flame proof lamps were missing resulting in it being open with its integrity compromised, glass case got detached from the main body and had opened out about 5 to 6 cms' (p.32). Had the Zone-1 electrical-integrity risk been assessed with preventive inspection, this fitting would not have remained as the probable ignition source.

CAUSAL EXPLANATION

The unassessed Zone-1 electrical risk allowed the compromised lamp to remain energised and exposed in the metering blast radius. When the vapour cloud accumulated for 56 minutes after the 02:12:50 release, it reached the fitting and ignited at 03:08:15, converting a major release into a major fire. This Level-4 gap therefore enabled the ignition transition rather than the initiating leak.

Linked Evidence: EVD-0496, EVD-6916

OIP 3.2 **Organisational Program/ Policy Risks not Adequately Assessed** **SUBSTANTIATED**

The hazard of assigning emergency control-room duties to operators with only a four-day general course and no defined qualification was never programmatically assessed.

JUSTIFICATION

The program risk of staffing a major-hazard control room with broadly trained operators was not assessed. [EVD-5634] records 'control room operations are carried out by persons inadequately trained in safety, operations, control room handling, logbook records and actions during emergency. It is informed that four days training programme is conducted for new recruitment/transfer staff which is general in nature... Qualification of this staff is also not defined in HR/Operations Policy. Most of the staff have attended only One Basic Course since 2018' (p.30). The committee's recommendation to train operating staff for at least one month, preferably on simulators, before control-room duties is effectively an admission that the prior training-program risk was unassessed. This gap most directly enabled lower-level AJDE 2.10/2.11 decision failures, SVIO 2.1 leading an untrained crew, and PTF 1.3/PTF 1.17 training and unfamiliarity preconditions during the emergency response.

CAUSAL EXPLANATION

The unassessed training risk left the night-shift crew without practised emergency-isolation competence. At 02:20 on 24-Sep-2020 they attempted on-foot source verification and then telephone escalation rather than initiating isolation, contributing to the 46-minute delay before MOV 54 closed at 03:01 and the subsequent 03:08:15 ignition. This Level-4 gap is a secondary but real enabler of the response delay.

Linked Evidence: EVD-5634

OIP 3.2 **Organisational Program/ Policy Risks not Adequately Assessed** **SUBSTANTIATED**

A 1.03 mm pitted thickness loss on a zero-corrosion-allowance 36-inch segment was treated with painting without a program risk assessment.

JUSTIFICATION

A measured integrity risk on a zero-corrosion-allowance pipe segment was accepted with a coating action, without any program-level fitness-for-service assessment. [EVD-2397] records 'a reduction of thickness of 1.03mm along with heavy pitting and painting peeling off was observed in one of the segments of 36 pipeline in gas terminal in July 2019. The corrective action of painting was completed in Dec. 2019. However... the design thickness of the pipeline is 23.83mm with zero corrosion allowance. It needs to be examined whether painting as a corrective action would suffice' (p.30). The risk that painting was not an adequate mitigation for metal loss on a zero-allowance line was therefore identified after the fact but not assessed at the time. This instance directly underlies lower-level AJDE 2.8 wrong choice of painting, SVI 1.6 acceptance of inadequate remedy, and SVIO 2.7 inappropriate asset management, though the segment survived the fire, so rating is 2.

CAUSAL EXPLANATION

The accepted painting-only remedy normalised the selection of superficial fixes for structural-integrity risks. This did not directly fail on 24-Sep-2020, but it substantiates the same management decision pattern that accepted the O-ring substitution for the meter pressure boundary, which did fail at 02:12:50 and led to ignition at 03:08:15.

Linked Evidence: EVD-2397

OIP 3.2 **Organisational Program/ Policy Risks not Adequately Assessed** **SUBSTANTIATED**

The sole metering-area CCTV camera had a 90-second pan and limited visibility, and this coverage risk was never assessed or mitigated.

JUSTIFICATION

The coverage risk of the sole PELCO PTZ camera was not assessed despite known limitations. [EVD-0210] records 'PELCO make PTZ cameras (Model D5230D) are provided at few places and has turn around (360 degree) at the time of about 90 seconds. Visibility from this camera is limited' (p.30). [EVD-2752] shows the single camera cable burned at 03:08:22, seven seconds after ignition, terminating the visual record. The committee's recommendation for per-plant CCTV with displays at plant control rooms and master control/CISF station confirms the prior coverage-risk assessment was inadequate. This instance contributed to the CISF-verification hop and loss of post-ignition situational evidence but did not initiate the release, so it is rated 2 rather than higher.

CAUSAL EXPLANATION

The lack of redundancy and continuous fixed coverage added uncertainty during initial notification and left no independent post-ignition visual record. The camera still recorded leak onset and ignition, so this gap is secondary, but it illustrates a broader pattern of accepting single points of failure in critical monitoring without formal risk assessment.

Linked Evidence: EVD-0210, EVD-2752

OIP 3.2 **Organisational Program/ Policy Risks not Adequately Assessed** **SUBSTANTIATED**

Three metering-line pressure transmitters retained no data and DCS clocks drifted 2-12 minutes; neither program risk was assessed.

JUSTIFICATION

The DCS program failed to retain pressure data and did not synchronise clocks, and neither gap was risk-assessed. [EVD-6470] records 'Three Pressure transmitters 310PT4, 310PT5 & 310PT6 are shown in the drawing. It was informed that no pressure transmitter data are available in the system. No pressure data in Gas Terminal area is available/provided to check rise in pressure in the metering lines' (p.26). [EVD-5532] records 'Time stamping in DCS systems provided in various control rooms is varying from 2 minutes to 12 minutes and data matching is difficult' (p.26). The missing pressure-trend data denied operators a direct early indicator and impaired the investigation, while drift forced reliance on CCTV timestamps. These program gaps are rated 2 because the release was ultimately detected and the sequence established through other channels, but they are substantiated configuration-program deficiencies.

CAUSAL EXPLANATION

The missing pressure data removed a diagnostic layer that could have shown pre-incident anomalies after the 18-Sep-2020 gasket replacement, and the time drift complicated alarm correlation across control rooms. These gaps did not initiate the fire, but they weakened both real-time detection and forensic reconstruction around the 02:12:50 leak and 03:08:15 ignition.

Linked Evidence: EVD-6470, EVD-5532

OIP 3.3 **Provided Inadequate Procedural Guidance/ Publications** **SUBSTANTIATED**

Maintenance was performed without an SOP for meter calibration/return-to-service and without the OEM rectangular-slot gasket specification.

JUSTIFICATION

The published procedural guidance for the failed meter was either absent or inadequate. [EVD-2894] records that after the 18-Sep-2020 gasket replacement 'No hydro test was done... The system was only pressurised to operating pressure (Both meters were in line) and isolated', and quotes the committee: 'No standard operating procedure is in place for calibration and taking back in service of these meters which are operating at high pressures' (p.32). [EVD-8645] adds that the original gasket specifications were not available and no specification or quality control existed for the substitute O-rings. Correct written guidance would have prescribed the OEM rectangular-slot gasket geometry, specified material conformance, and required a design-pressure hydrostatic test before return to service; the actual documentation contained none of these. This inadequate procedural guidance directly enabled lower-level PTE 2.7 technical guidance unavailable, ASBE 1.2 hydrotest checklist omission and AJDE 2.7 delayed necessary action.

CAUSAL EXPLANATION

Without the SOP and OEM specification, the 18-Sep-2020 crew fitted an undocumented round O-ring to the rectangular-slot flange and returned the meter to service on 19-Sep-2020 without design-pressure validation. The unvalidated joint failed at 02:12:50 on 24-Sep-2020, causing the 56-minute vapour accumulation and the 03:08:15 ignition. This procedural publication gap is the direct Level-4 enabler of the initiating equipment failure.

Linked Evidence: EVD-2894, EVD-8645

OIP 3.3 **Provided Inadequate Procedural Guidance/ Publications** **SUBSTANTIATED**

The published DMP/ERDMP had no isolation/stoppage decision criteria or authority for hydrocarbon leaks.

JUSTIFICATION

The site's formal emergency publications were inadequate for the actual leak scenario. [EVD-4052] states 'As per DMP On site Hazira Plant, guidelines to be followed in case of hydrocarbon leakage are included but does not include isolation/stoppage of facilities/pipelines' (p.32), and the ERDMP section is similarly missing a decision rule for isolation/closure of plant/pipelines (p.27). A complete procedure would specify when a confirmed vapour release requires immediate isolation, who is authorized to order it, and which valves are to be closed in what sequence. The actual DMP/ERDMP text supplied none of these, which left every responder from 02:15 to 03:01 improvising through serial escalation. This publication gap directly enabled lower-level PTE 2.7 guidance unavailable, AJDE 2.9/2.11 decision failures, ASBE 1.3 delayed isolation, and committee root cause #6.

CAUSAL EXPLANATION

Because the published guidance lacked isolation criteria, the night-shift response had no documented trigger for immediate valve closure. The resulting serial escalation stretched the response to 46 minutes before MOV 54 closed at 03:01, while the release accumulated the cloud that ignited at 03:08:15. This is a primary procedural-publication cause of the incident's severity.

Linked Evidence: EVD-4052

OIP 3.3 **Provided Inadequate Procedural Guidance/ Publications** **SUBSTANTIATED**

Critical high-pressure calibration work on 310FT01 had no checklist, allowing unrecorded 16000-to-2540-mmwc setpoint changes.

JUSTIFICATION

The calibration documentation for meter 310FT01 was inadequate because no checklist governed the work. [EVD-2506] records 'During calibration from 17/12/2019 to 18/12/2019, calibration was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of this change in the calibration documents signed by the team... No checklist is followed while carrying such critical jobs of high pressure/risk' (p.27), and the setting was later changed back to 16000 mmwc, also unrecorded. Correct published guidance would have included a mandatory checklist requiring documentation and approval of any setpoint deviation; the actual guidance left the change invisible. Committee recommendation D.1.i at p.35 explicitly requires documenting master calibrator, actions, replacements and deviations. This inadequate procedural guidance enabled lower-level ASBE 1.2 calibration checklist failure, ASBE 1.5 under-controlled calibration, and the normalisation of undocumented changes on the same meter that later failed.

CAUSAL EXPLANATION

The missing checklist allowed a 6.3x setpoint change to pass without documented approval, degrading confidence in the meter and normalising checklist-free high-pressure work. This procedural gap contributed to the weak maintenance-governance environment on meter 310FT01 that culminated in the 18-Sep-2020 gasket substitution and 24-Sep-2020 failure, though it did not itself open the leak.

Linked Evidence: EVD-2506

OIP 3.3 **Provided Inadequate Procedural Guidance/ Publications** **SUBSTANTIATED**

Baseline thickness data, original drawings and schedules for the 1986 pipelines were not available, leaving only 2011 and 2017 data points.

JUSTIFICATION

The published integrity records for the 1986 gas-terminal pipelines were incomplete. [EVD-8287] records 'for a large number of pipes, especially in gas terminal area baseline data of pipe thickness in not available. The original drawings and the schedules of the pipes are also not available. In most cases only two thickness details of 2011 and 2017 is available for a pipe which is commissioned in 1986' (p.29). Correct integrity documentation would preserve the 1986 baseline and original schedules so that 2011 and 2017 measurements could be trended against design thickness. The actual records had only two data points across 34 years, making degradation assessment unreliable. This publication gap supported lower-level PTE 2.7 baseline-data absence and SVI 1.4 failure to maintain configuration documentation; it is rated 2 because the 36-inch pipeline itself remained intact [EVD-5780].

CAUSAL EXPLANATION

The incomplete published baseline prevented rigorous trending of the 2017 wall-loss findings, contributing to the three-year absence of corrective action. This did not directly cause the fire, but it formed part of the integrity-documentation weakness that paralleled the lost meter gasket specification, which was directly causal.

Linked Evidence: EVD-8287

OIP 3.3 **Provided Inadequate Procedural Guidance/ Publications** **SUBSTANTIATED**

The Self Safety Audit used a generic, non-plant-specific checklist that recorded unverified compliance and missed plant-specific hazards.

JUSTIFICATION

The Self Safety Audit publication was inadequate for the plant-specific risks it was supposed to sample. [EVD-9692] records 'the checklist is comprehensive but not related to specific plants and in the process, lot of observations are missing. The duration of audit is not indicated and in most of the items, compliances are indicated, but during our inspection, the data is found to be incomplete & not in line with OISD standards/manuals. Approved manuals/SOPs were not available' (p.26). Committee recommendation B.1 at p.34 calls for individual plant-specific SSA checklists, confirming the existing instrument was not fit for purpose. Because the checklist was generic, plant-specific defects including the Zone-1 HPMV lamp integrity and the gasket substitution practice were not selected for verification. This inadequate guidance enabled lower-level ASBE 1.2 audit checklist failure, PTF 1.2 audit cross-monitoring failure and PTE 2.7 generic guidance gap.

CAUSAL EXPLANATION

The generic audit document did not force verification of the meter skid or the flame-proof fittings, so the latent ignition-source lamp and the uncontrolled gasket practice survived to the incident. When the vapour cloud formed after the 02:12:50 release, the compromised lamp became the probable ignition source at 03:08:15, making this publication gap a secondary enabler of the transition to fire.

Linked Evidence: EVD-9692

OIP 3.3 **Provided Inadequate Procedural Guidance/ Publications** **SUBSTANTIATED**

2018 and 2019 internal safety audits produced observations without filled checklists, leaving audit coverage unverifiable.

JUSTIFICATION

The internal audit publications were not auditable because the checklists themselves were missing. [EVD-7681] records 'Internal Safety Audit was conducted in 2018 & 2019 for 4 & 3 days respectively by a 3-member committee and the observations are given without the filled checklist. The checklist format is not available to check completeness of audit areas' (p.26). Correct audit documentation would retain the filled checklist showing which areas were inspected and which were skipped. The actual records contained only observations, so coverage of the metering skid and hazardous-area electrical fittings could not be confirmed. This inadequate publication supported lower-level ASBE 1.2 audit checklist failure and SVI 1.1 unverifiable audit oversight, but it is rated 2 because its causal effect was limited to lost assurance rather than a direct physical failure.

CAUSAL EXPLANATION

The absence of an auditable checklist trail meant management could not demonstrate that the Zone-1 lamp integrity or the meter maintenance practice had ever been within audit scope, allowing those defects to persist undetected until the 03:08:15 ignition. This is an assurance-process publication gap rather than a direct cause of the release.

Linked Evidence: EVD-7681

OIP 3.3 **Provided Inadequate Procedural Guidance/ Publications** **SUBSTANTIATED**

Permit documentation was incomplete: missing permits untracked, JSPs absent from SAP, and no reviewed closure.

JUSTIFICATION

The published permit/JSA system did not provide adequate task-specific guidance. [EVD-5105] records 'few of the work permits were missing. No data is kept on such missing permits. Job safety Plan is not attached/given in SAP document. Closing of permits & actions indicated in work needs review' (p.26). Recommendation B.5 at p.34 states 'Maintenance notifications are not be assumed as permit to work', confirming the procedural documentation was inadequate. Because JSPs were not attached in SAP, a high-risk job such as the 18-Sep-2020 gasket replacement had no documented hazard analysis specifying the hydrostatic test and gasket-conformance requirements. This gap supported lower-level PTE 2.7 permit guidance incomplete and ASBE 1.2 hydrotest omission, but is rated 2 because no direct evidence ties a missing permit specifically to the failed joint.

CAUSAL EXPLANATION

The inadequate permit/JSP documentation meant the 18-Sep-2020 pressure-boundary job was not forced to identify its required design-pressure test or material-control steps before execution. This contributed to the procedural environment in which the unvalidated O-ring seal entered service on 19-Sep-2020 and failed on 24-Sep-2020, though the primary causes were the missing SOP and uncontrolled material substitution.

Linked Evidence: EVD-5105

OIP 3.3 **Provided Inadequate Procedural Guidance/ Publications** **SUBSTANTIATED**

Monthly fire drill records lacked actions and observations, and the parallel informal dataset did not comply with approved guidelines.

JUSTIFICATION

The published fire-drill records were inadequate to verify emergency-response learning. [EVD-2130] records 'Fire drill reports are available in SAP but without any data on actions taken/recommendations. Separate data is maintained which is not in line with the approved guidelines' (p.26). Correct drill documentation would capture observations and action items so that deficiencies such as slow isolation or escalation could be tracked to closure. The actual records were split across SAP and an informal non-conforming dataset, making drill effectiveness unmeasurable. This gap is rated 2 because the fire team's performance on the night was adequate and no specific drill finding is tied to the release, but it removed a feedback mechanism that might otherwise have surfaced and corrected the response deficiencies that contributed to the 46-minute delay.

CAUSAL EXPLANATION

The incomplete drill records did not directly cause the release or ignition, but they prevented the organization from evidencing follow-up on emergency-response learning. This publication gap left the escalate-first and slow-isolation pattern uncorrected through training, supporting secondary involvement in the response-phase delay.

Linked Evidence: EVD-2130

OIP 3.3 **Provided Inadequate Procedural Guidance/ Publications** SUBSTANTIATED

Twelve Feb-2018 external audit recommendations were extended to 30-Apr-2021 without the required Director Offshore approval document.

JUSTIFICATION

The procedural documentation for overdue external-audit recommendations was inadequate. [EVD-6852] records 'External safety audit was conducted in February 2018, and 296 out of 308 recommendations are complied with. Balance are in various stages and dates are extended till 30/04/2021. Document indicating approval from Director Offshore for extending the target date was not available as per the procedure' (p.26). Correct program documentation would have recorded competent-authority approval for the 12 extended recommendations. The actual record lacked the required approval document, evidencing a publication/approval gap in the recommendation-tracking process. This instance is rated 2 because no specific overdue recommendation is linked to the 24-Sep-2020 fire, but it substantiates the systemic documentation weakness also visible in the unclosed SAP notification and the unrecorded MOC decision.

CAUSAL EXPLANATION

The missing approval documentation did not directly cause the fire, but it shows the organization's corrective-action documentation lacked procedural control. This pattern of unresolved and undocumented risk items is the same pattern that allowed the meter gasket recurrence and missing hydrotest to evade review until the 24-Sep-2020 loss of containment.

Linked Evidence: EVD-6852

OIC 2.3 **Organisational Over confidence or Under confidence in Equipment** SUBSTANTIATED

The site repeatedly trusted a locally bought O-ring as equivalent to the OEM rectangular-slot gasket, with no spec, QC, or post-failure revalidation.

JUSTIFICATION

The organization repeatedly accepted a round O-ring for a flange designed for a rectangular-slot gasket, evidencing overconfidence in equipment it had never validated. [EVD-8645] states: "The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available... use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings" (p.31). This is not a one-off misjudgment; [EVD-1494] shows the same substitute was used after the 12-Nov-2018 and 25-Feb-2019 failures. The organization acted as though the substitute seal was equivalent despite three failures in 22 months, indicating an unjustified confidence in the equipment rather than a simple procurement error.

CAUSAL EXPLANATION

The unjustified confidence in the O-ring substitute meant the organization did not seek the correct rectangular-slot gasket, did not impose material QC, and did not revalidate the seal after repeated failures. The 18-Sep-2020 installation therefore replicated the same defective equipment, which failed at 02:12:50 on 24-Sep-2020, releasing vapor for 56 minutes until ignition at 03:08:15. This Level-4 overconfidence is the direct enabler of lower-level AJDE 2.8, ASBE 1.3, PTE 2.5, SVI 1.7 and SVV 4.1 findings.

Linked Evidence: EVD-8645, EVD-1494

OIC 2.3 **Organisational Over confidence or Under confidence in Equipment** SUBSTANTIATED

The organization trusted a non-interlocked, annunciator-only F&G system and manual isolation chain to handle a major hydrocarbon release, a confidence contradicted by the 46-minute delay.

JUSTIFICATION

[EVD-3452] documents that the safety architecture was accepted with no automatic action: "F&G detection system does not have provision for auto shut down of plant or activation of fire suppression system" (p.26). [EVD-8657] records the resulting performance: the annunciator alarmed at approximately 02:20, five minutes after human detection and more than seven minutes after CCTV-confirmed leak onset at 02:12:50. The organization had for years relied on a display-and-human-response system despite knowing that a dense vapor cloud could deny access to the area. The committee's recommendation A.1 (p.34) necessarily implies that the existing confidence in the non-interlocked F&G system was misplaced.

CAUSAL EXPLANATION

The overconfidence in the F&G/manual-response posture meant the leak that was machine-visible at 02:12:50 produced no automatic isolation or suppression. The entire human chain consumed 46 minutes before MOV 54 closed at 03:01, allowing the cloud seen at 03:00:44 ('vapour all around even at height') to form and ignite at 03:08:15. This Level-4 factor created the environment for lower-level PTE 2.11, PTE 2.2, PTF 1.4 and AV 1.1 findings.

Linked Evidence: EVD-3452, EVD-8657

OIP 3.4 Organisational Formal Training Issues/ Programs are Inadequate or Unavailable

SUBSTANTIATED

Control-room staff received only a 4-day general course with no defined qualification and no simulator training, versus the committee's recommended minimum of one month.

JUSTIFICATION

The formal training programme that prepares control-room operators for emergency duties was structurally inadequate rather than individually failed. [EVD-5634] records that "control room operations are carried out by persons inadequately trained in safety, operations, control room handling, logbook records and actions during emergency", that "four days training programme is conducted for new recruitment/transfer staff which is general in nature", that "Qualification of this staff is also not defined in HR/Operations Policy", and that "Most of the staff have attended only One Basic Course since 2018" (p.30). The committee's own remedial benchmark at p.36, "operating staffs of control room needs to be trained for at least 1 month (preferably on simulators) before being assigning Control Room duties enabling them to handle operations & emergencies effectively", is nearly eight times the current four days and confirms the deficiency was programme-level. This is the Level-4 origin of the lower-level SVI 1.3 and SVI 1.14 training-governance findings, the SVIO 2.3/SVV 4.4 selection and authorisation failures for the night-shift crew, and the Level-2 PTF 1.3/PTF 1.17

CAUSAL EXPLANATION

Because the only formal initial training was a four-day general course with no defined qualification and no simulator-practised emergency isolation, the two Gas Terminal control-room engineers who received the annunciator alarm and the CCTV phone call at 02:20 on 24-Sep-2020 treated the event as a locate-and-inspect task rather than an immediate-isolation trigger. They attempted an on-foot approach, were forced back by the dense vapour, and then used serial telephone escalation; the GSP Shift In-charge reached the gate only at approximately 02:35 and closed MOV 54 at 03:01, 46 minutes after first physical detection at 02:15 and 48 minutes 10 seconds after CCTV-confirmed leak onset at 02:12:50, with the 36-inch PCV closing at 03:04. The vapour released during that window was already "vapour all around even at height" at 03:00:44 [EVD-8213] and ignited at 03:08:15. This programme gap therefore causally enabled the lower-level AJDE 2.10/AJDE 2.11 decision-making failures and the PTF 1.3/PTF 1.17 preconditions that extended the pre-isolation release window.

Linked Evidence: EVD-5634

OIP 3.5 Flawed Doctrine/ Philosophy

SUBSTANTIATED

An organisation-wide preference for breakdown repair over preventive inspection left a Zone 1 flame-proof lamp with three missing bolts and an open glass case in service as the probable ignition source.

JUSTIFICATION

The plant's electrical maintenance doctrine was explicitly breakdown-oriented rather than preventive, and this organisational philosophy is documented in the enquiry report itself. [EVD-0496] records that "the electrical maintenance system is a lot more skewed towards breakdown maintenance and lays little emphasis on systematic preventive maintenance and inspection" (p.28), and the post-incident audit described in [EVD-9999] was "too generic and short (One page)", clubbing "all nine 125 W HPMV FLP lamps" into a single status line of "cable burnt and fitting damaged due to heat". The consequence of that philosophy is the defect at [EVD-6916]: "three bolts on one of the flame proof lamps were missing resulting in it being open with its integrity compromised, glass case got detached from the main body and had opened out about 5 to 6 cms", while "[t]he nearby fittings were found intact" (p.32), showing a specific pre-existing loss of flame-proof integrity in a Zone 1 hydrocarbon area that no preventive inspection had detected. This is the Level-4 origin of the Level-3 SVI 1.11 and SVV 4.1 electrical oversight failures and the Level-2 PTE 2.12/PTE 2.15

CAUSAL EXPLANATION

Because prevention was not the operating doctrine, the 125 W HPMV FLP lamp with three missing bolts and a glass case open 5-6 cm remained energised in the metering blast radius. When the condensate vapour released from the failed 310FT01 gasket accumulated for 56 minutes from 02:12:50 to 03:08:15, the compromised enclosure supplied exposed electrical contacts and ignited the cloud at 03:08:15, 4 minutes 15 seconds after the 36-inch PCV was closed at 03:04 and 7 minutes after MOV 54 closure at 03:01, proving the already-released cloud was sufficient fuel. This organisational doctrine therefore converted a vapour release into the major fire and directly enabled the lower-level ignition-source findings.

Linked Evidence: EVD-9999, EVD-6916

OIP 3.5 **Flawed Doctrine/ Philosophy** **SUBSTANTIATED**

The site's philosophy accepted locally bought, unspecified O-rings as a 'regular practice' for a rectangular-slot 1986 meter flange, repeated across three replacements in 22 months.

JUSTIFICATION

The organisation's maintenance doctrine accepted a known non-conforming material substitution as a normal, unregulated practice rather than treating it as an engineering change requiring specification control. [EVD-8645] records that "[t]he meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket" (p.31). The recurrence data in [EVD-1494] shows this philosophy was exercised repeatedly, with gasket replacements on 12-Nov-2018, 25-Feb-2019 and 18-Sep-2020, and the committee's official root causes #2 "Use of nonstandard O ring" and #3 "Lack of control over material sourcing" (p.33) name exactly this accepted-risk doctrine. This instance is the Level-4 origin of the lower-level AJDE 2.8 wrong choice, AV 1.1 routine violation, ASBE 1.3 incorrect procedure and SVI

CAUSAL EXPLANATION

Because the accepted doctrine did not require OEM specification recovery, material conformance or engineering approval for seal substitution, the round O-ring installed on 18-Sep-2020 could not fill the rectangular slot and left a leak path from installation. The meter returned to service on 19-Sep-2020 and ran for six days before the joint failed at 02:12:50 on 24-Sep-2020 [EVD-1284], releasing condensate vapour for 56 minutes until ignition at 03:08:15 [EVD-4507]. The philosophy thus directly produced the lower-level wrong-choice and routine-violation unsafe acts that initiated the loss of containment.

Linked Evidence: EVD-8645, EVD-1494

OIP 3.5 **Flawed Doctrine/ Philosophy** **SUBSTANTIATED**

The DMP/ERDMP doctrine defined the major-leak scenario but contained no isolation/stoppage criteria or authority, forcing serial escalation and a 46-minute unisolated release.

JUSTIFICATION

The plant's emergency doctrine named the hydrocarbon-leak scenario but deliberately or negligently omitted the doctrine required to stop it, leaving no pre-authorized isolation rule. [EVD-4052] records that "[a]s per DMP On site Hazira Plant, guidelines to be followed in case of hydrocarbon leakage are included but does not include isolation/stoppage of facilities/pipelines" (p.32), reinforced at p.27: "In ERDMP, scenario for Level 3 emergencies are taken from the plant failures. Heavy product/gas leakages from pipelines & tanks are considered but decision on isolation/closure of plant/pipelines is not mentioned." The committee's root cause #6 is "Non isolation/stoppage of 36 pipeline operation". [EVD-3932] further shows the resulting authority design: no role between shift in-charge and plant manager had standing authority to close emergency isolation valves without telephoning up the chain. This flawed concept of operations converted a machine-detected 02:12:50 leak into a 46-minute manual escalation before first isolation at 03:01 [EVD-6414], and is the Level-4 origin of the lower-level DMP/role-allocation and 46-minute delay findings.

CAUSAL EXPLANATION

With no documented criterion for when a hydrocarbon leak warranted pipeline isolation and no delegated authority below the senior chain, the night-shift response became a discovery and approval process: leak onset at 02:12:50 [EVD-1284], first human detection at 02:15, annunciator alarm at 02:20, shift in-charge called at approximately 02:35 and arriving at the gate, and only then MOV 54 closure at 03:01, 46 minutes after first detection, followed by the 36-inch PCV at 03:04. Every minute of that window released live inventory; the cloud was already "vapour all around even at height" at 03:00:44 [EVD-8213] and ignited at 03:08:15, after isolation, proving the already-released inventory was sufficient fuel. Had the doctrine specified immediate isolation and pre-authorized the control room, the first closure could have occurred within minutes of 02:15, substantially reducing the flammable envelope.

Linked Evidence: EVD-4052, EVD-6414

OIP 3.5 **Flawed Doctrine/ Philosophy** **SUBSTANTIATED**

A 1.03 mm pitted wall-loss finding on a zero-corrosion-allowance 36-inch line was closed with painting only, institutionalising symptomatic treatment of structural integrity risks.

JUSTIFICATION

The organisation's philosophy treated a protective coating as an adequate corrective action for a structural integrity defect, an acceptance of risk that the committee itself questioned. [EVD-2397] records that "a reduction of thickness of 1.03mm along with heavy pitting and painting peeling off was observed in one of the segments of 36 pipeline in gas terminal in July 2019", that "the corrective action of painting was completed in Dec. 2019", and that "the design thickness of the pipeline is 23.83mm with zero corrosion allowance. It needs to be examined whether painting as a corrective action would suffice in such cases and the mechanism of deciding on a corrective action needs to be reviewed" (p.30). Painting restores coating, not wall thickness; accepting it closed the finding without any fitness-for-service evaluation. This philosophical choice to treat symptoms rather than root causes is the same doctrine that accepted unspecified O-rings for a rectangular-slot flange, and it is the Level-4 origin of the lower-level AJDE 2.8 wrong-choice finding on the pipeline and part of the culture that permitted the initiating gasket substitution.

CAUSAL EXPLANATION

Although the 36-inch trunk pipeline remained intact on 24-Sep-2020 [EVD-5780] and the loss of containment occurred at the metering flange, this philosophy is a substantiated organisational risk-acceptance pattern. It directly enabled the lower-level AJDE 2.8 wrong-choice-of-corrective-action finding and the SVI 1.8/SVV 4.5 supervisory acceptance of inadequate remedies. The same symptom-based decision logic allowed the locally purchased O-ring, which addressed the visible leak symptom without restoring the engineering basis of the seal, to persist across replacements on 12-Nov-2018, 25-Feb-2019 and 18-Sep-2020, culminating in failure at 02:12:50 on 24-Sep-2020 and ignition at 03:08:15.

Linked Evidence: EVD-2397

OIP 3.6 **Inadequate Program Oversight/ Program Management** **SUBSTANTIATED**

The 2018-2019 internal audits produced observations without filled checklists or a checklist format, leaving audit coverage unverifiable.

JUSTIFICATION

The internal safety audit programme produced findings without an auditable checklist trail, so audit completeness could not be verified after the fact. [EVD-7681] records that "Internal Safety Audit was conducted in 2018 & 2019 for 4 & 3 days respectively by a 3-member committee and the observations are given without the filled checklist. The checklist format is not available to check completeness of audit areas" (p.26). Without the completed checklists, neither the auditors nor later managers could establish whether the metering skid's pressure-boundary work or the HPMV lamp enclosures had ever been within audit scope, allowing latent integrity defects to persist unnoticed into 2020. This programme oversight contributed secondarily to the lower-level ASBE 1.2 audit-checklist findings and to the persistence of the compromised flame-proof lamp that became the probable ignition source at 03:08:15.

CAUSAL EXPLANATION

The absent checklist trail weakened the assurance layer most likely to surface the Zone 1 lamp defect and the non-standard gasket practice. This did not directly initiate the 02:12:50 release, but it enabled the lower-level SVI 1.1/SVI 1.4 audit oversight failures and the PPI 5.6 inaccurate-expectancy condition, allowing the lamp with three missing bolts and glass case open 5-6 cm [EVD-6916] to remain in service until it ignited the accumulated vapour at 03:08:15.

Linked Evidence: EVD-7681

OIP 3.6 **Inadequate Program Oversight/ Program Management** **SUBSTANTIATED**

The self-safety audit used a generic, non-plant-specific checklist and recorded compliance without verifying site data, creating false assurance.

JUSTIFICATION

The self-safety audit programme was built on a checklist that was comprehensive in name but not specific to any plant, producing compliance assertions that site inspection disproved. [EVD-9692] records that "Safety self-audit is conducted by a multi-disciplinary team based on the checklist prepared by the location. The checklist is comprehensive but not related to specific plants and in the process, lot of observations are missing. The duration of audit is not indicated and in most of the items, compliances are indicated, but during our inspection, the data is found to be incomplete & not in line with OISD standards/manuals. Approved manuals/SOPs were not available" (p.26). The committee's recommendation B.1, "Checklist for Self-Safety Audit for individual plants are to be developed" (p.34), confirms the instrument itself was unfit. Because the instrument was not plant-specific, it never forced examination of the metering-skid flame-proof fittings or the gasket substitution practice, allowing the compromised HPMV lamp [EVD-6916] and the non-standard seal to survive to the incident.

CAUSAL EXPLANATION

The generic instrument prevented the audit programme from directing auditors toward plant-specific hazards, including the Zone 1 HPMV lamp with three missing bolts and a 5-6 cm open glass case and the uncontrolled O-ring practice. This is why the latent ignition source and the material-substitution hazard survived to participate in the event: when the vapour released from the failed 310FT01 gasket accumulated for 56 minutes, the uncompromised-assurance assumption was falsified by ignition at 03:08:15. The programme failure directly enabled the lower-level ASBE 1.2 audit-checklist finding, PTF 1.2 cross-monitoring failure and PPI 5.6 inaccurate expectancy.

Linked Evidence: EVD-9692

OIP 3.6 **Inadequate Program Oversight/ Program Management** **SUBSTANTIATED**

12 of 308 external-audit recommendations were overdue on extensions lacking the required Director Offshore approval.

JUSTIFICATION

The external audit follow-up programme lacked competent-authority control over overdue actions. [EVD-6852] records that "External safety audit was conducted in February 2018, and 296 out of 308 recommendations are complied with. Balance are in various stages and dates are extended till 30/04/2021. Document indicating approval from Director Offshore for extending the target date was not available as per the procedure" (p.26). The programme had achieved 96% closure, but the remaining 12 items carried extensions without the Director Offshore approval the procedure required, meaning schedule relief was granted outside the management system. This instance did not directly cause the 24-Sep-2020 release, but it substantiates the programme-oversight pattern that allowed identified actions to age without controlled closure, the same pattern that left the meter 310FT01 recurrence and 2017 pipe findings unactioned.

CAUSAL EXPLANATION

This programme-control failure had no direct outcome impact on the 24-Sep-2020 fire, so it is rated present without direct causal effect. Its systemic significance is that the extension of 12 safety recommendations without procedural approval normalised the deferral of known findings, the same deferral logic that allowed the three gasket replacements on 310FT01 (12-Nov-2018, 25-Feb-2019, 18-Sep-2020) to proceed without root-cause review and permitted the unvalidated joint to fail at 02:12:50 on 24-Sep-2020.

Linked Evidence: EVD-6852

OIP 3.6 **Inadequate Program Oversight/ Program Management** **SUBSTANTIATED**

The 2017 thickness report flagged pipes needing corrective action, but no plan existed over three years later - measurement without remediation.

JUSTIFICATION

The integrity programme collected critical data but had no working mechanism to convert findings into corrective action. [EVD-5527] records that "[a]s per the report of 2017, several pipes have been identified with significant thickness loss and need for corrective action. However, even after three years no plan has been developed for the corrective actions to be taken for those lines. Mere measurement of thickness without corrective action does not complete the process of maintaining the integrity of pipes" (p.30). From 2017 to the 24-Sep-2020 incident, zero remediation plan existed. The committee's phrase "Mere measurement of thickness without corrective action does not complete the process" is an explicit programme-management finding. This failure normalised the pattern of acknowledging integrity warnings without acting on them, and it is the Level-4 origin of the lower-level AJDE 2.5 and AJDE 2.7 pipeline findings and the SVI 1.6/SVV 4.5 no-action findings.

CAUSAL EXPLANATION

Although the 36-inch trunk pipeline itself remained intact on 24-Sep-2020, this three-year non-action is the clearest organisational evidence of a findings-to-action closure failure. That same closure failure was replicated on meter 310FT01, where three gasket replacements in 22 months [EVD-1494] were recorded but never escalated to root-cause review. The resulting fourth failure occurred at 02:12:50 on 24-Sep-2020, releasing vapour for 56 minutes until ignition at 03:08:15, so this programme gap contributed systemically to the initiating defect.

Linked Evidence: EVD-5527

OIP 3.6 **Inadequate Program Oversight/ Program Management** **SUBSTANTIATED**

SAP notification 3800419840 was never closed or senior-reviewed after the 18-Sep-2020 repair, removing the last administrative gate before return to service.

JUSTIFICATION

The maintenance work-management programme's close-out gate was not executed for the very notification that triggered the defective repair. [EVD-0591] records that "Notification no. 3800419840 was raised on 17/09/2020 for malfunction of meter no.1" and that it "was not closed after completion of the job" (p.27), and [EVD-0185] confirms the meter was returned to service without post-maintenance validation while the notification remained open. The committee's recommendation D.2.ii, "Sample checks/approval of closure of notifications by Senior Officers to be introduced" (p.35), shows the programme lacked a routine senior review gate. Because no closure review occurred, the substitute O-ring, the omitted hydrostatic test and the third-in-22-months recurrence all passed without supervisory challenge. This is the Level-4 origin of the lower-level AV 1.1/SVV 4.1 notification non-closure findings and the PTF 1.12 feedback failure.

CAUSAL EXPLANATION

Because the notification remained open and unreviewed, the 18-Sep-2020 repair package escaped the supervisory review that could have caught the non-standard O-ring, the missing design-pressure test and the recurrence pattern. The meter returned to service on 19-Sep-2020 and the unvalidated joint failed at 02:12:50 on 24-Sep-2020 [EVD-1284], releasing vapour for 56 minutes until ignition at 03:08:15 [EVD-4507]. This programme gap therefore enabled the lower-level AV 1.1 and AJDE 2.11 findings and allowed the initiating defect to pass unchecked.

Linked Evidence: EVD-0591, EVD-0185

OIP 3.6 **Inadequate Program Oversight/ Program Management** **SUBSTANTIATED**

Fire drill records in SAP lacked action/observation data, and a separate informal dataset did not follow approved guidelines, making drill effectiveness unmeasurable.

JUSTIFICATION

The emergency drill programme's documented record could not demonstrate that drill findings were followed up. [EVD-2130] records that "Fire drill reports are available in SAP but without any data on actions taken/recommendations. Separate data is maintained which is not in line with the approved guidelines" (p.26). Two parallel records existed: an SAP dataset containing only dates and no actions or observations, and an informal dataset that did not comply with approved guidelines. Drill effectiveness could therefore not be measured and lessons related to emergency response, including the escalate-first isolation behaviour observed on 24-Sep-2020, could not be identified or corrected through the drill programme. The fire team itself performed adequately on the night, so this programme gap did not change the firefighting outcome.

CAUSAL EXPLANATION

This programme-control failure did not directly cause the release or ignition; the fire was extinguished by approximately 06:00 and all-clear declared at 07:30 [EVD-0368]. Its significance is that the drill programme could not evidence testing of the rapid-isolation and escalation actions that failed between 02:15 and 03:01 on 24-Sep-2020, so emergency readiness for a major hydrocarbon release remained unverified before the incident.

Linked Evidence: EVD-2130

OIP 3.6 **Inadequate Program Oversight/ Program Management** **SUBSTANTIATED**

Two of four diesel fire-water pumps were under maintenance, one tripped after two minutes with unrecorded cause, and no actions were documented.

JUSTIFICATION

The maintenance programme for the diesel fire-water backup left the majority of its pumps unavailable without documented diagnosis. [EVD-2432] records that during the incident "[b]oth the Tank level and the Fire water header pressure were healthy throughout" and that three electrical pumps operated, while the committee noted (p.27): "Out of 4 diesel pumps 410 m3/hr, 2 pumps are under maintenance, one operated for 30 minutes & fourth pumps is tripping after 2 minutes. The reason for tripping is not identified or actions have not been recorded." Three of four diesel pumps therefore had maintenance or reliability issues, and the unexplained two-minute tripping fault remained unactioned. Because three electrical pumps maintained adequate fire water, this programme gap did not change the outcome and is rated 2, but it left the response dependent on electrical supply and exposed the same known-defects-accepted-without-action programme pattern seen on the meter and lamp systems.

CAUSAL EXPLANATION

This latent reliability gap had no direct impact on the 24-Sep-2020 outcome because the electrical fire-water pumps maintained tank level and header pressure throughout cooling, with the fire extinguished by approximately 06:00. Had electrical supply failed or cooling demand exceeded electric capacity, the degraded diesel train would have constrained suppression, but on the incident timeline it remained a standby weakness rather than a causal factor. It substantiates the same programme-oversight culture that left known equipment defects without documented corrective action.

Linked Evidence: EVD-2432

OIP 3.6 Inadequate Program Oversight/ Program Management

SUBSTANTIATED

Three metering-line pressure transmitters shown on the drawing had no data available in the DCS, eliminating pressure-trend monitoring for the Gas Terminal area.

JUSTIFICATION

The programme for pressure-data retention and DCS configuration failed for the transmitters monitoring the very lines that failed. [EVD-6470] records that "[t]hree Pressure transmitters 310PT4, 310PT5 & 310PT6 are shown in the drawing. It was informed that no pressure transmitter data are available in the system. No pressure data in Gas Terminal area is available/provided to check rise in pressure in the metering lines" (p.26). Three transmitters appear on the P&ID but zero pressure data were retained for the pre-incident window, removing both real-time pressure-trend monitoring and forensic reconstruction. This programme gap did not begin the flange failure and human detection still occurred at 02:15, so it is rated present without direct outcome impact, but it compounded the detection and diagnosis limitations during the 46-minute pre-isolation window.

CAUSAL EXPLANATION

The absence of pressure data removed a potential early indicator of abnormal conditions before the 02:15 perimeter detection and the 02:20 annunciator alarm, and denied investigators the pressure profile needed to determine whether the 18-Sep-2020 gasket failed from overpressure, installation error or material defect. This supported the lower-level PTE 2.5/PTE 2.11 configuration findings and limited diagnosis of the release that ran from 02:12:50 until MOV 54 closure at 03:01 and ignited at 03:08:15, but the primary delay drivers were the manual MOV and missing DMP criteria, so the rating is 2.

Linked Evidence: EVD-6470

OIP 3.2 Organisational Program/ Policy Risks not Adequately Assessed

SUBSTANTIATED

The F&G system was designed as display-only with no auto-shutdown/suppression interlock because the protective-action risk was never assessed.

JUSTIFICATION

The program risk that gas detection would need automatic protective action was not assessed in the F&G system design. [EVD-8657] records 'Hydrocarbon/H2S detectors are configured to Annunciator Panel for display but not in DCS for history' (p.26), and [EVD-3452] records 'F&G detection system does not have provision for auto shut down of plant or activation of fire suppression system' (p.26). Consequently, the leak machine-witnessed at 02:12:50 produced no automatic isolation or deluge; the response depended on the full human chain, with first detection at 02:15, annunciator alarm at 02:20, and first isolation only at 03:01. The committee's recommendation A.1, calling for F&G interlock to automatic plant shutdown and fire suppression, confirms this was a recognized but previously unassessed program risk. It is the Level-4 origin of lower-level PTE 2.11 equipment misconfiguration, AV 1.1 F&G violation, and the 46-minute release window.

CAUSAL EXPLANATION

The unassessed program risk meant the 02:12:50 leak could not trigger automatic protection, leaving a manual response that consumed 46 minutes before MOV 54 closure at 03:01. The vapour released during that period formed the flammable cloud that ignited at 03:08:15. This gap converted a machine-detectable leak into a prolonged release and major fire.

Linked Evidence: EVD-3452, EVD-8657

OIC 2.4 Unit Mission/ Aircraft/ Vehicle/ Equipment Change or Unit Deactivation

SUBSTANTIATED

The original metering-system P&ID was lost and replaced by an unapproved vendor drawing that failed to match the field configuration, leaving no verifiable as-built reference during the gasket replacement.

JUSTIFICATION

A controlled configuration document was lost and replaced outside the change-management system, leaving the field configuration without an approved as-built reference. [EVD-0933] records 'P&ID Drawing 02-310-AO-101 Rev.1 covering the 2x6" lines and 1x2" line metering system... Reference drawings no. 02-310-AO-101 Rev.1, 2x6 lines & 1x2 lines for metering system are shown but valves & meters configuration are not matching. As old drawings were lost, new drawings are made by a vendor but the same has not been checked & approved by ONGC'. The correct process is to maintain an as-built drawing verified and approved by ONGC engineering against the field; the actual process produced an unapproved vendor drawing that did not match the installed valves and meters. This change-based documentation gap was not the direct leak initiator, but it deprived maintainers and supervisors of the authoritative configuration needed to challenge the non-conforming O-ring substitution.

CAUSAL EXPLANATION

Without an approved as-built drawing, no one could independently verify the correct gasket geometry, valve arrangement, or meter details for the 1986 skid. This allowed the undocumented O-ring substitution to proceed without a configuration-based challenge, compounding the lost OEM specification recorded in EVD-8645. It therefore operated as a secondary enabler of the wrong-material choice that failed on 24-Sep-2020 at 02:12:50, supporting the lower-level AV 1.1 configuration-management breakdown (rating 2), ASBE 1.3 wrong-gasket procedure and PTE 2.14 unfamiliar-equipment precondition.

Linked Evidence: EVD-0933

OIP 3.7 Purchasing or Providing Poorly Designed or Unsuitable Equipment (Acquisition Policies)

SUBSTANTIATED

The acquisition process supplied a locally bought, unspecified round O-ring for a 1986 flange machined for a rectangular-slot gasket, without specification or QC, repeated across three replacements in 22 months.

JUSTIFICATION

The acquisition process for the meter 310FT01 seal allowed a geometry-incompatible, locally purchased part to become the regular repair material. [EVD-8645] records: "The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket." The committee's formal findings as cited in [EVD-8645] are root cause #2 ("Use of nonstandard O ring") and root cause #3 ("Lack of control over material sourcing"), which name exactly this acquisition-control failure. The recurrence record in [EVD-1494] shows the same inadequate seal was fitted on 12-Nov-2018, 25-Feb-2019 and 18-Sep-2020 — three replacements in approximately 22 months — so this was a sustained procurement practice, not a one-off. The consequence is physical: [EVD-0064] records "6 meter with ID

CAUSAL EXPLANATION

The round-cross-section O-ring procured for the 1986 rectangular-slot flange could not fill the machined groove, so the joint had a continuous leak path from installation. The meter was returned to service on 19-Sep-2020 and operated under live condensate pressure for approximately five to six days before the seal failed at 02:12:50 on 24-Sep-2020, releasing vapour that accumulated for 56 minutes and ignited at 03:08:15. This Level-4 acquisition-policy failure is the organizational origin of the lower-level unsafe acts that directly caused the loss of containment: AJDE 2.8 (wrong choice of substitute O-ring, rating 4), ASBE 1.3 (incorrect gasket procedure, rating 4), AV 1.1 (routine/widespread substitution violation, rating 4), and the Level-2 precondition PTE 2.5/PTE 2.7 (defective equipment with lost technical guidance). It also explains the supervisory findings SVI 1.4 (failed to provide procurement/material policy), SVI 1.7 (lack of standardization), SVIO 2.5 (authorised unnecessary hazard), SVV 4.2 (unwritten policy became standard), and SVF 3.4/SVV 4.5 (no mitigation of the identified nonstandard O-ring risk). Had the acquisition process required

Linked Evidence: EVD-8645, EVD-1494, EVD-0064

OIP 3.7 **Purchasing or Providing Poorly Designed or Unsuitable Equipment (Acquisition Policies)** [REJECTED]

MOV 54 and the Gas Terminal emergency MOVs were provided as local-manual valves without control-room remote operation, forcing a 46-minute physical-access and escalation delay before the first isolation at 03:01.

JUSTIFICATION

The emergency isolation valves provided for the Gas Terminal station were acquired without remote operation capability, leaving the control-room operators unable to isolate the release from a safe location. [EVD-3932] records the committee finding (p.26): "MOVs are provided in the gas terminal station for local operations but they are not having remote operation from control room during emergency." The consequence is documented in [EVD-9612], where the GSP Shift In-charge "closed the MOV 54 inlet condensate line from slug catcher to 36 metering area at about 0301 hrs. Thereafter fire occurred." That closure came 46 minutes after first physical detection at 02:15 and 48 minutes after CCTV-confirmed leak onset at 02:12:50, because the senior officer had to be called from quarters at ~02:35 and physically reach the Gas Terminal Gate before the local valve could be operated. The committee's response analysis in [EVD-6414] reinforces the causal link: "Decision of closure of valve was taken after about 40 minutes resulting the spread of vapour cloud in increased area." Correct emergency isolation design for a hydrocarbon release would have provided control-room

CAUSAL EXPLANATION

Because MOV 54 and the surrounding MOVs had no remote operation, the control-room engineers who received the annunciator alarm and CCTV phone call at approximately 02:20 could not initiate source isolation. The DGM was telephoned at ~02:35, had to travel to the Gas Terminal Gate, and only then closed MOV 54 at 03:01 — 46 minutes after first detection at 02:15 and 48 minutes 10 seconds after leak onset at 02:12:50. During that unisolated window the release continued, and by 03:00:44 the cloud was recorded as "vapour all around even at height" [EVD-8213]; the accumulated inventory ignited at 03:08:15 [EVD-4507], after both isolation actions (MOV 54 at 03:01 and 36-inch PCV at 03:04) were complete. This Level-4 acquisition/design deficiency directly enabled the lower-level findings PTE 2.4 and PTE 2.8 (workspace/controls incompatible with operations, manual-only MOVs), PTE 2.11 (equipment not configured for remote emergency operation), PTF 1.4 (over-reliance on individual local action), and the Level-1 response failures AJDE 2.7 (delayed necessary action, rating 4) and AJDE 2.9/ASBE 1.3 (failure to execute immediate isolation). It also

Linked Evidence: EVD-3932, EVD-9612, EVD-6414

OIP 3.7 **Purchasing or Providing Poorly Designed or Unsuitable Equipment (Acquisition Policies)** [REJECTED]

The F&G system was supplied as annunciator-only with no auto-shutdown or fire-suppression interlock, so the machine-detected leak at 02:12:50 was left to a 46-minute manual response before first isolation.

JUSTIFICATION

The flame and gas detection equipment at the Gas Terminal was acquired and configured without the automatic protective actions required for time-critical hydrocarbon detection. [EVD-8657] records: "Hydrocarbon/H2S detectors are configured to Annunciator Panel for display but not in DCS for history" and "F&G detection system does not have provision for auto shut down of plant or activation of fire suppression system." The consequence is quantified in the same evidence: leak onset at 02:12:50 (CCTV), first CISF human detection at 02:15, and the first annunciator alarm only at approximately 02:20 — roughly 5 minutes after human detection and 7 minutes 10 seconds after leak onset — with no automatic isolation or suppression action produced at any point. The committee's recommendation A.1 quoted in [EVD-3452] confirms the acquisition/design inadequacy: "F&G detection system needs to be interlocked to automatic plant shut down and activation of fire suppression system." Correct design for a Zone-1 hydrocarbon metering area would have linked confirmed gas detection to automatic MOV/PCV closure or deluge initiation within seconds; the actual

CAUSAL EXPLANATION

With no automatic shutdown or suppression interlock, the leak that began at 02:12:50 and was machine-visible on CCTV could not trigger any protective action. The response instead required the full human chain: CISF detection at 02:15, CCTV verification and GT CR notification at 02:18-02:20, annunciator alarm at 02:20, shift-in-charge call-out at ~02:35, and manual MOV 54 closure at 03:01, with the 36-inch PCV closed only at 03:04. Every minute of that 46-minute interval released additional inventory, and the cloud documented at 03:00:44 as "vapour all around even at height" [EVD-8213] ignited at 03:08:15. This Level-4 acquisition/design failure is the enabling condition for the lower-level findings PTE 2.6 and PTE 2.11 (system automation/configuration deficiencies, rating 4), PTE 2.15 and PTF 1.4 (over-reliance on human response to a display-only system), AV 1.1 (alarm-only configuration as a systemic violation), and the Level-1 delayed-action findings AJDE 2.7 and AJDE 2.9. Had the supplied F&G system been interlocked to automatic isolation on confirmed gas detection, the release window would have been reduced from 46 minutes to seconds, substantially

Linked Evidence: EVD-8657, EVD-3452, EVD-1284

OIP 3.7 **Purchasing or Providing Poorly Designed or Unsuitable Equipment (Acquisition Policies)** SUBSTANTIATED

The metering area was provided with only one PELCO PTZ D5230D CCTV camera with a ~90-second pan and limited visibility, and its cable burned through seven seconds after ignition, ending all visual coverage.

JUSTIFICATION

The monitoring equipment provided for the metering skid was a single PELCO PTZ D5230D camera with limited visibility, which was inadequate for continuous coverage of a safety-critical operating area. [EVD-0210] records: "PELCO make PTZ cameras (Model D5230D) are provided at few places and has turn around (360 degree) at the time of about 90 seconds. Visibility from this camera is limited" (p.30). The acquisition/design limitation became materially relevant when the camera's cable burned through seven seconds after ignition: [EVD-2752] records "major fire continued, CCTV capturing discontinued due to burnt of CCTV cable" (03:08:22) and cites the committee recommendation that "CCTV arrangement shall be provided for each plant including separately with display at plant control rooms as well as master control/CISF station." The single intermittent-view camera was therefore the sole remote visual channel for the metering area, with no fixed redundant coverage and no independent plant-control-room display. This instance is rated present with no direct impact on the outcome because the camera still recorded the 02:11:43 baseline and the 02:12:50 leak

CAUSAL EXPLANATION

The single pan-tilt-zoom camera could observe the metering skid only intermittently, with up to approximately 90 seconds between returned views, and there was no redundant fixed camera or plant-control-room display. This did not initiate the release or the ignition — the leak was detected by CISF perimeter personnel at 02:15 and the camera did capture leak onset at 02:12:50 and ignition at 03:08:15 — but it contributed to the monitoring and verification architecture that routed perimeter detection through the CCTV Control Room before reaching the Gas Terminal Control Room at 02:18-02:20. When the cable burned at 03:08:22, the sole remote visual channel was lost, preventing any post-ignition visual evidence and leaving the incident commander without camera coverage during firefighting. This Level-4 acquisition deficiency therefore supports the lower-level findings PTE 2.6 (single-camera monitoring gap), PTE 2.8 and PTE 2.10 (workspace/communications equipment inadequacy), and PTF 1.4 (over-reliance on one monitoring channel), but its causal contribution is secondary and did not change the physical outcome of the 24-Sep-2020 fire.

Linked Evidence: EVD-0210, EVD-2752

OIS 4.1 **Personnel Recruiting and Selection Policies** SUBSTANTIATED

HR/Operations Policy defined no qualification for control-room operators, and the selected night-shift crew had only a four-day general course with no simulator training before facing the release.

JUSTIFICATION

The organizational recruiting/selection policy for control-room personnel is directly evidenced as inadequate by [EVD-5634], which records that "control room operations are carried out by persons inadequately trained in safety, operations, control room handling, logbook records and actions during emergency. It is informed that four days training programme is conducted for new recruitment/transfer staff which is general in nature and later the job training is carried out. Qualification of this staff is also not defined in HR/Operations Policy. Most of the staff have attended only One Basic Course since 2018" (p.30). This is a selection-and-placement deficiency, not merely a training shortfall: HR/Operations Policy contained no defined qualification gate for assigning individuals to safety-critical control-room duties, so the two night-shift engineers on 24-Sep-2020 had been selected into those positions without any verified emergency-response competence. The consequence is visible in [EVD-1460], which states that at around 02:20 the Chief Engineer heard the annunciator alarm and the CCTV phone call, went outside, saw the heavy cloud but could not approach, and then

CAUSAL EXPLANATION

Because the organizational policy placed no qualification gate on control-room staffing, the two engineers on night duty had not been screened for, or trained in, the emergency-isolation actions the release demanded. When the annunciator alarmed at 02:20 — roughly 7 minutes after leak onset at 02:12:50 and 5 minutes after CISF perimeter detection at 02:15 — both engineers left the control room to locate the source on foot, were forced back by the dense vapour cloud, and then spent the 02:20–02:35 window telephoning senior managers rather than initiating available isolation. Senior responders then had to be called from quarters and physically reach the Gas Terminal Gate before MOV 54 was closed at 03:01, 46 minutes after first detection, with the 36-inch PCV closed at 03:04 and ignition following at 03:08:15. This Organizational Influences gap is the Level-4 origin of the lower-level SVIO 2.3 selected-individual-without-current-experience findings, PTF 1.3 task delegation to under-trained staff, and AJDE 2.10 inadequate knowledge to make isolation decisions, all of which extended the vapour accumulation window that made the

Linked Evidence: EVD-5634, EVD-1460

OIS 4.2 Personnel Resources Selection & Staffing

SUBSTANTIATED

The 18-Sep-2020 gasket job was resourced without any independent QC role, hydrotest witness, or closing supervisor, allowing the unspecified O-ring and untested joint to pass directly into service.

JUSTIFICATION

The personnel-resource allocation for the 18-Sep-2020 meter 310FT01 gasket replacement was objectively inadequate: the job was resourced without any independent QC role, without a hydrotest witness or test authority, and without a supervisor assigned to close and review the maintenance notification. [EVD-2894] records that "After calibration of the meter 310FE01, the gasket was replaced on 18.09.2020. No hydro test was done after replacement of the gasket. The system was only pressurised to operating pressure (Both meters were in line) and isolated" (p.31) — meaning no personnel resource was assigned or required to perform or witness the design-pressure test. [EVD-8645] records that the substitute O-ring was sourced with "no specification or quality control during the purchase of the O rings" and with "no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket" (p.31), establishing that no quality-control or material-inspection function was staffed onto the job. [EVD-0591] further records that SAP Notification 3800419840, raised on 17-Sep-2020 for the meter malfunction, "was not closed after

CAUSAL EXPLANATION

Because the organization allocated no quality inspector, no hydrotest witness, and no closing supervisor to the 18-Sep-2020 job, the non-conforming round O-ring installed into the 1986 rectangular-slot flange was accepted without challenge and without a design-pressure validation. Meter 310FT01 returned to service on 19-Sep-2020 and operated under condensate pressure for approximately five days until the joint failed at 02:12:50 on 24-Sep-2020, releasing vapour that accumulated for 56 minutes and ignited at 03:08:15. This Level-4 manning-resource failure is the direct organizational enabler of the lower-level SVIO 2.6 insufficient-manning finding, the PTF 1.4 over-reliance-on-individual precondition, the PTF 1.5 unallocated-quality-roles precondition, and the Level-1 unsafe acts AJDE 2.8 wrong choice of substitute gasket, ASBE 1.3 procedure not followed correctly, and ASBE 1.2 omitted hydrotest checklist — collectively the initiating loss of containment.

Linked Evidence: EVD-2894, EVD-8645, EVD-0591

OIC 2.4 Unit Mission/ Aircraft/ Vehicle/ Equipment Change or Unit Deactivation

SUBSTANTIATED

The 2016 MOC procedure was not used for the 2013 and 2018 LP-gas-to-Kribhico header changes and no MOCs were implemented since 2018, demonstrating plant-wide loss of equipment-change control.

JUSTIFICATION

The plant's formally published MOC process was allowed to fall into disuse, as demonstrated by two named process-equipment changes that were executed without it. [EVD-2986] records 'Management of Change (MOC) philosophy/SOP was published in August 2016. However, the procedure indicated in the document is not implemented while carrying out change of Low-Pressure Gas to Kribhico from Medium Pressure Gas headers in LPG Plant in 2013 & 2018. It is informed no more MOCs are implemented in the plant since 2018'. This is a separate unmanaged equipment/process change from the metering gasket substitution (different plant system and years) and is direct evidence that the MOC control was inactive across the organization, not merely missed once. This LP-to-Kribhico change did not itself affect the 24-Sep-2020 gas-terminal fire, so it is rated as present without direct outcome impact, but it explains why the gasket substitution also passed without MOC review.

CAUSAL EXPLANATION

The documented non-implementation of MOC since 2018 is the organizational condition that allowed the OEM-to-O-ring substitution on meter 310FT01 to become 'regular practice' without engineering review. Although the LP-to-Kribhico change itself did not initiate the 24-Sep-2020 release, it demonstrates that the equipment-change control failure was systemic rather than local to one maintenance crew, reinforcing the lower-level AJDE 2.11 MOC-policy misapplication and AV 1.1 routine-substitution findings that produced the initiating flange failure.

Linked Evidence: EVD-2986

OIC 2.5 Organisational Structure

SUBSTANTIATED

Emergency escalation was organized as a serial management chain with a mid-incident SIC-to-CIC hand-off, adding delay at the moment rapid notification and action were needed.

JUSTIFICATION

The intended escalation structure was a serial cascade rather than parallel notification, and it included a mid-emergency hand-off of the Site Incident Controller role. [EVD-7340] records that the SIC 'instructed to close the 36 line pressure control valve and get out of the control room. He informed LPG C/R and asked immediately informed to Resident Engineer, Head operations, Plant Manager and all other concerned', and the evidence description adds 'The escalation cascaded serially up the management chain (Resident Engineer -> Head Operations -> Plant Manager) rather than via parallel auto-paging. The SIC role itself was later transferred to Head Operations on his arrival at site (p.10) — a hand-off pattern that adds time but is encoded in plant procedure.' Although this structure was procedural, it was inadequate for a time-critical hydrocarbon containment because each hop added delay and the SIC hand-off inserted a role transition during the most critical phase. The 36-inch PCV was not closed until 03:04, three minutes after MOV 54 and only about four minutes before ignition, while notification duties continued serially.

CAUSAL EXPLANATION

The serial hierarchy meant that the initial alert reached senior managers one by one, and the SIC role hand-off occurred while the release was still uncontained. This structural inefficiency consumed part of the 02:35-to-03:04 window, leaving the 36-inch PCV closure until 03:04, only 4 minutes 15 seconds before ignition at 03:08:15. It enabled the lower-level PTF 1.1 crew leadership failure, PTF 1.15 failure to replan, SVI 1.1 supervisory oversight gap, and AJDE 2.9 operational decision-making failure, and it amplified the primary authority gap rather than acting as a prompt parallel alerting system.

Linked Evidence: EVD-7340

OIC 2.5 Organisational Structure

SUBSTANTIATED

No organizational role owned the meter gasket specification, procurement quality control, or the authority to approve the substitution from OEM rectangular-slot gasket to locally procured O-ring.

JUSTIFICATION

The organization created a formal accountability vacuum over the metering pressure-boundary material: no person or role owned the specification, the procurement control, or the decision to change the seal type. [EVD-8645] records 'There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket', and 'There was no specification or quality control during the purchase of the O rings'. Because ownership was unallocated, the locally purchased O-ring repeated through three replacements without any designated quality role challenging its conformity. The correct accountability structure would assign a single authority responsible for seal specification, procurement conformance, and MOC review for the 1986 meter; the actual structure left maintenance fitters with no controlled specification and no accountable gate to block the non-conforming part.

CAUSAL EXPLANATION

With no accountable owner, the switch from the OEM rectangular-slot gasket to a round O-ring was never challenged on geometry or material grounds, so the defective seal was installed on 18-Sep-2020 and returned to service on 19-Sep-2020. The joint failed at 02:12:50 on 24-Sep-2020, releasing vapour for 56 minutes until ignition at 03:08:15. This structural accountability gap directly enabled lower-level PTF 1.5 roles-not-allocated, PTF 1.4 over-reliance on individual execution, SVI 1.4/1.7 lack of policy and standardization, AJDE 2.8 wrong choice, and ASBE 1.3 wrong gasket procedure, and it is the organizational form of committee root cause #3 on lack of material-sourcing control.

Linked Evidence: EVD-8645

OIC 2.5 Organisational Structure

SUBSTANTIATED

Twelve external-audit recommendations remained overdue on extensions to 30-Apr-2021 for which no Director Offshore approval existed, breaking the formal accountability chain for closing findings.

JUSTIFICATION

The safety-assurance accountability chain failed to enforce its own approval rule for twelve overdue audit recommendations. [EVD-6852] records 'External safety audit was conducted in February 2018, and 296 out of 308 recommendations are complied with. Balance are in various stages and dates are extended till 30/04/2021. Document indicating approval from Director Offshore for extending the target date was not available as per the procedure'. The correct structure requires the competent authority, Director Offshore, to approve any target-date extension; the actual structure allowed extensions without that approval, meaning accountability for closing findings was no longer formally assigned. This instance did not directly affect the 24-Sep-2020 fire, and the 12 recommendations have no identified link to the metering skid, so it is rated present without direct outcome impact, but it documents the same broken formal-accountability structure that elsewhere left safety-critical items unowned.

CAUSAL EXPLANATION

The unapproved extensions show that the organization's formal accountability for safety findings could be bypassed without consequence, normalizing the deferral of known risks. This same non-closure pattern reappeared in the unclosed SAP notification on meter 310FT01 and in the three-year absence of a corrective plan for the 2017 thickness findings. It is therefore a systemic structural contributor to the environment that allowed the meter gasket defect to persist, supporting lower-level SVI 1.6 and SVF 3.4 findings, though it did not itself initiate the 24-Sep-2020 release.

Linked Evidence: EVD-6852

OIC 2.7 **Inconsistent Risk Management Practices** **SUBSTANTIATED**

Three gasket replacements in ~22 months on meter 310FT01 were never escalated for root-cause action, allowing the same deficient repair to recur.

JUSTIFICATION

EVD-1494 records the SAP maintenance history verbatim: 'Cleaning/servicing/calibration of meters 1 & 2 was carried out on 12th & 13th Nov 2018. Afterwards, meter no.1 was cleaned on 25/02/2019 & gasket was changed. Again on 18/09/2019, gasket was changed which leaked within one year' (p.27), while EVD-0591 documents that the final pre-incident warning was raised on 17-Sep-2020 as 'SAP Notification 3800419840 dt. 17.09.2020 was raised for checking meter 1' (p.33). Three gasket replacements in approximately 22 months on a hydrocarbon pressure boundary were therefore treated as isolated repair events rather than as a risk signal requiring root-cause analysis, material review, or meter replacement; the committee's later recommendation D.2.ii for senior-officer sample checks of notification closure (EVD-0185, p.35) is the formal recognition that the site's risk-tracking loop did not close. This is a substantiated manifestation of OIC 2.7: the organisation's own maintenance records repeatedly surfaced a hazard, but the absence of action told staff that such reports would not be addressed.

CAUSAL EXPLANATION

This Level-4 failure is the organisational root of lower-level finding AJDE 2.5 (three gasket failures ignored, rating 4), SVI 1.8 (risky practice not identified/corrected, rating 4), SVF 3.4 (ineffective mitigation of known risk, rating 4), and AV 1.1 (routine substitution violation, rating 4). Because the recurrence was not treated as a risk, the 18-Sep-2020 repair repeated the same local O-ring in the rectangular-slot flange and omitted the design-pressure test; the meter returned to service on 19-Sep-2020 and failed at 02:12:50 on 24-Sep-2020, releasing vapour for 56 minutes until ignition at 03:08:15.

Linked Evidence: EVD-1494, EVD-0591

OIC 2.7 **Inconsistent Risk Management Practices** **SUBSTANTIATED**

The OEM rectangular-slot gasket was replaced by unspecified locally purchased O-rings as a 'regular practice' with no specification, QC or documented decision.

JUSTIFICATION

EVD-8645 records that 'The meter is very old (1986) and having rectangular slot for the gasket. Original Specifications of gasket are not available, and the gaskets were not being procured from the OEMs. Instead, use of locally purchased O ring was done in its place as a regular practice. There was no specification or quality control during the purchase of the O rings. There is no record of date or authority of the decision to switch over from OEM supplied gasket to locally purchased gasket' (p.31). The evidence item then states explicitly that committee root cause #2 ('Use of nonstandard O ring') and root cause #3 ('Lack of control over material sourcing') flow from this finding, making it an authoritative confirmation of an organisational risk-management failure rather than a single procurement slip. A material change affecting a hydrocarbon pressure boundary was therefore allowed to operate outside specification, quality-control, and approval controls, demonstrating that the site's risk-management system did not catch or correct the substitution.

CAUSAL EXPLANATION

This instance enabled the directly causal lower-level unsafe acts AJDE 2.8 (wrong choice of substitute gasket, rating 4), AV 1.1 (routine substitution violation, rating 4), and ASBE 1.3 (wrong gasket procedure, rating 4), as well as the technical-guidance precondition PTE 2.7 (rating 4). The round O-ring could not fill the 1986 rectangular slot, so the joint was defective from installation on 18-Sep-2020; it ran six days unvalidated because no hydrostatic test was done and failed at 02:12:50 on 24-Sep-2020, producing the 56-minute vapour accumulation that ignited at 03:08:15.

Linked Evidence: EVD-8645

OIC 2.7 **Inconsistent Risk Management Practices** **SUBSTANTIATED**

The self-safety audit used a generic checklist that recorded compliances unsupported by data, allowing plant-specific integrity defects to remain hidden.

JUSTIFICATION

EVD-9692 records that 'Safety self-audit is conducted by a multi-disciplinary team based on the checklist prepared by the location. The checklist is comprehensive but not related to specific plants and in the process, lot of observations are missing. The duration of audit is not indicated and in most of the items, compliances are indicated, but during our inspection, the data is found to be incomplete & not in line with OISD standards/manuals. Approved manuals/SOPs were not available' (p.26). The committee's recommendation B.1 at p.34 requires plant-specific checklists, which is formal recognition that the audit instrument was not fit for purpose. By recording compliance without verified data on a generic checklist, the organisation manufactured an assurance signal that did not reflect actual plant risk, directly matching the OIC 2.7 poor-reporting-culture condition.

CAUSAL EXPLANATION

This instance enabled lower-level ASBE 1.2 (audit checklist not followed correctly, rating 3), PPI 5.6 (inaccurate expectancy, rating 3), and SVI 1.4 (supervision provided inadequate audit policy, rating 3). Because the audit did not force verification of plant-specific hazards such as flame-proof lamp integrity or the gasket substitution practice, the compromised Zone-1 lamp with three missing bolts and the uncontrolled O-ring repair survived undetected, allowing the release on 24-Sep-2020 to find a ready ignition source at 03:08:15.

Linked Evidence: EVD-9692

OIC 2.7 **Inconsistent Risk Management Practices** **SUBSTANTIATED**

A 1.03 mm pitted wall-loss finding on a zero-corrosion-allowance 36-inch segment was closed with painting only instead of fitness-for-service evaluation.

JUSTIFICATION

EVD-2397 records that 'a reduction of thickness of 1.03mm along with heavy pitting and painting peeling off was observed in one of the segments of 36 pipeline in gas terminal in July 2019. The corrective action of painting was completed in Dec. 2019. However, as per the available data, the design thickness of the pipeline is 23.83mm with zero corrosion allowance. It needs to be examined whether painting as a corrective action would suffice in such cases and the mechanism of deciding on a corrective action needs to be reviewed' (p.30). A measured loss of 1.03 mm with heavy pitting on a zero-corrosion-allowance line was therefore closed with a cosmetic coating, without fitness-for-service analysis or remaining-strength assessment. This shows an organisational habit of accepting inadequate corrective actions for identified structural-integrity risks, another substantiated OIC 2.7 manifestation.

CAUSAL EXPLANATION

This instance did not directly cause the 24-Sep-2020 fire because the 36-inch line survived intact (EVD-5780), but it is the organisational evidence for lower-level AJDE 2.5 (painting accepted for metal loss, rating 3), AJDE 2.8 (wrong choice of corrective action, rating 3), and SVI 1.6 (supervisory acceptance of cosmetic remedy, rating 3). It reinforced the same acceptance-of-inadequate-response culture that permitted the substitute O-ring to be treated as an adequate replacement for the OEM rectangular-slot gasket, which failed at 02:12:50 on 24-Sep-2020 and initiated the release.

Linked Evidence: EVD-2397

OIC 2.7 **Inconsistent Risk Management Practices** **SUBSTANTIATED**

The 2018 and 2019 internal safety audits produced observations but no filled checklists, making audited coverage unverifiable.

JUSTIFICATION

EVD-7681 records that 'Internal Safety Audit was conducted in 2018 & 2019 for 4 & 3 days respectively by a 3-member committee and the observations are given without the filled checklist. The checklist format is not available to check completeness of audit areas' (p.26). Because the completed checklists were not retained, the organisation could not demonstrate what had actually been inspected, and later reviewers could not verify whether coverage was complete or selective. This absence of an auditable trail is a substantiated OIC 2.7 manifestation in the assurance domain: the audit output existed, but its completeness could not be challenged or relied upon.

CAUSAL EXPLANATION

This instance is a secondary systemic enabler of the lower-level ASBE 1.2 audit checklist failure (rating 3) and SVI 1.4/1.6 supervisory oversight gaps. Without filled checklists, plant-specific risks such as the compromised HPMV flame-proof lamp and the gasket substitution practice could fall outside audit scope without detection, allowing those latent conditions to persist until the 24-Sep-2020 vapour release found the ready ignition source at 03:08:15. It did not directly initiate the loss of containment, so it is rated 2.

Linked Evidence: EVD-7681

OIC 2.7 **Inconsistent Risk Management Practices** **SUBSTANTIATED**

The malfunction notification for meter 310FT01 was raised on 17-Sep-2020 but never closed or reviewed after the gasket job.

JUSTIFICATION

EVD-0591 records that 'Notification no. 3800419840 was raised on 17/09/2020 for malfunction of meter no.1' (p.27) and confirms on p.33 that 'SAP Notification 3800419840 dt. 17.09.2020 was raised for checking meter 1', but states the notification 'was not closed after completion of the job' (p.27). EVD-0185 further notes the committee recommendation D.2.ii at p.35 that sample checks or approval of notification closure by senior officers should be introduced. The open notification removed the administrative gate at which a senior reviewer could have challenged the non-standard O-ring, the missing hydrotest, and the recurrence history, so the risk-reporting loop was severed at exactly the point where it should have produced supervisory scrutiny.

CAUSAL EXPLANATION

This instance enabled lower-level findings AV 1.1 (routine non-closure violation, rating 2-3), AJDE 2.11 (misapplied SAP closure policy, rating 3), PTF 1.12 (failure to acknowledge completion, rating 3), and PTE 2.12 (missing supervisory quality gate, rating 3). Without closure review, the defective 18-Sep-2020 gasket replacement passed unchecked and the meter returned to service on 19-Sep-2020; it failed at 02:12:50 on 24-Sep-2020, starting the 56-minute vapour accumulation that ended in ignition at 03:08:15.

Linked Evidence: EVD-0591, EVD-0185

OIC 2.7 **Inconsistent Risk Management Practices** **SUBSTANTIATED**

Fire drill records in SAP lacked action/observation data, and a separate informal dataset was not aligned with approved guidelines.

JUSTIFICATION

EVD-2130 records that 'Fire drill reports are available in SAP but without any data on actions taken/recommendations. Separate data is maintained which is not in line with the approved guidelines' (p.26). Two parallel record sets existed: SAP entries containing dates only, with no actions or observations, and an informal dataset that did not follow approved guidelines. Neither record allowed drill effectiveness or corrective-action closure to be measured, so the emergency-preparedness reporting system was generating activity records without learning value.

CAUSAL EXPLANATION

The incomplete drill records did not directly affect the 24-Sep-2020 fire outcome, because the fire team still extinguished the fire by approximately 06:00, but the evidence shows the organisation could not verify whether drills had ever tested rapid hydrocarbon-release isolation and escalation. This reporting weakness removed a feedback channel that could have surfaced and corrected the slow response behaviour observed between 02:15 and 03:01 on 24-Sep-2020, so it is a secondary OIC 2.7 contributor rated 2.

Linked Evidence: EVD-2130

OIC 2.7 **Inconsistent Risk Management Practices** **SUBSTANTIATED**

Three of four diesel fire-water pumps were unavailable or unreliable, with one tripping after 2 minutes and no recorded cause or action.

JUSTIFICATION

EVD-2432 records that 'Out of 4 diesel pumps 410 m3/hr, 2 pumps are under maintenance, one operated for 30 minutes & fourth pumps is tripping after 2 minutes. The reason for tripping is not identified or actions have not been recorded' (p.27). Three of four diesel fire-water pumps were unavailable or unreliable, and the tripping fault on the fourth had neither an identified reason nor a recorded corrective action. This known loss of fire-water redundancy was left open without diagnosis, fitting the OIC 2.7 pattern of identified equipment risks that were neither investigated nor actioned.

CAUSAL EXPLANATION

This latent gap did not change the incident outcome because three electrical fire-water pumps maintained tank level and header pressure throughout the fire, which was brought under control by approximately 06:00, so it is rated 2. It remains a substantiated organisational risk-management failure: a known, unrepaired redundancy defect was left without diagnosis or action, illustrating the same non-closure culture that allowed the meter gasket recurrence and missing hydrotest to persist until the 24-Sep-2020 release and 03:08:15 ignition.

Linked Evidence: EVD-2432

OIC 2.7 Inconsistent Risk Management Practices

SUBSTANTIATED

The metering-system P&ID was a lost-original, unapproved vendor drawing that did not match the field configuration.

JUSTIFICATION

EVD-0933 records that 'Reference drawings no. 02-310-AO-101 Rev.1, 2x6 lines & 1x2 lines for metering system are shown but valves & meters configuration are not matching. As old drawings were lost, new drawings are made by a vendor but the same has not been checked & approved by ONGC' (p.26). The original controlled drawings were lost, the vendor-prepared replacement was never checked or approved by the operator, and the field configuration did not match the drawing. This left maintenance and operations without an authoritative configuration baseline for the metering skid, a clear OIC 2.7 condition of unclear guidance and uncontrolled technical documentation.

CAUSAL EXPLANATION

This configuration-management failure contributed secondarily to the lower-level AV 1.1 configuration-management breakdown (rating 2), PTE 2.7 technical-guidance gap (rating 3), and the ASBE 1.3 wrong-gasket procedure. Without an accurate approved as-built drawing, maintenance personnel could not independently verify the correct 1986 rectangular-slot gasket design for meter 310FT01, so the locally sourced O-ring substitution could proceed unchallenged and the meter returned to service on 19-Sep-2020 before failing at 02:12:50 on 24-Sep-2020. It is rated 2 because the direct initiating defect was the physical seal, not the drawing alone.

Linked Evidence: EVD-0933

OIP 3.2 Organisational Program/ Policy Risks not Adequately Assessed

SUBSTANTIATED

Three gasket replacements in 22 months were treated as routine resealing with no program-level reliability or root-cause risk assessment.

JUSTIFICATION

The recurrence itself was a program risk signal that was never assessed: [EVD-1494] records 'Cleaning/servicing/calibration of meters 1 & 2 was carried out on 12th & 13th Nov 2018. Afterwards, meter no.1 was cleaned on 25/02/2019 & gasket was changed. Again on 18/09/2019, gasket was changed which leaked within one year' (p.27), i.e. three replacements in about 22 months, all using the same locally sourced O-ring [EVD-8645]. [EVD-0591] confirms SAP Notification 3800419840 for the latest malfunction was not closed after completion of the job (p.27), so no programmatic review ever questioned why the same pressure boundary kept failing. Correct reliability management would have escalated the second recurrence for root-cause analysis, OEM seal restoration, or meter replacement; the actual program treated each failure as routine resealing. This unassessed recurrence program risk enabled lower-level SVI 1.8 failure to correct unsafe practices, AJDE 2.5 ignored caution warning, PTF 1.13 planning without analysis and PTE 2.12 recurrent maintenance defect.

CAUSAL EXPLANATION

Without a program-level recurrence risk assessment, the same inadequate O-ring repair was repeated on 18-Sep-2020. The meter then operated from 19-Sep to 24-Sep-2020 and failed at 02:12:50, producing the 56-minute vapour accumulation and the 03:08:15 ignition. The recurrence-assessment gap is why a known chronic sealing defect remained uncorrected until it caused the incident.

Linked Evidence: EVD-1494, EVD-0591

OIP 3.2 **Organisational Program/ Policy Risks not Adequately Assessed** **SUBSTANTIATED**

Significant 2017 pipe-wall-loss findings sat for three years without a corrective-action plan or risk assessment.

JUSTIFICATION

The pipeline-integrity program collected a warning but never assessed the risk of leaving it open. [EVD-5527] records 'As per the report of 2017, several pipes have been identified with significant thickness loss and need for corrective action. However, even after three years no plan has been developed for the corrective actions to be taken for those lines. Mere measurement of thickness without corrective action does not complete the process of maintaining the integrity of pipes' (p.30). [EVD-8307] shows the mandatory four-year thickness-measurement schedule was already breached, with measurements in 2011 and 2017. This unassessed program risk did not directly fail the 36-inch pipeline, which remained intact [EVD-5780], but it institutionalised the practice of acknowledging findings without action—the same pattern that allowed the meter gasket recurrence to persist. It enabled lower-level AJDE 2.5, AJDE 2.7, SVI 1.6 and SVF 3.4 pipeline findings.

CAUSAL EXPLANATION

This instance is rated 2 because the 36-inch trunk line did not fail and the source was the metering flange, but it is substantiated evidence of the organizational pattern in which identified risks were measured and then left unresolved. That same pattern is why the three gasket failures and the missing hydrotest went unchallenged until the 24-Sep-2020 release.

Linked Evidence: EVD-5527, EVD-8307

OIP 3.2 **Organisational Program/ Policy Risks not Adequately Assessed** **SUBSTANTIATED**

The 1986 pipeline baseline thickness data, original drawings and schedules were absent, and the resulting trend-analysis risk was never assessed.

JUSTIFICATION

The pipeline-integrity program operated without the baseline data needed to assess degradation trends, and this gap was not itself risk-assessed. [EVD-8287] records 'for a large number of pipes, especially in gas terminal area baseline data of pipe thickness in not available. The original drawings and the schedules of the pipes are also not available. In most cases only two thickness details of 2011 and 2017 is available for a pipe which is commissioned in 1986' (p.29). Without a 1986 baseline, the 2017 thickness-loss findings could not be rigorously trended or prioritised, which reinforced the three-year non-action on corrective measures. This program gap supported lower-level PTE 2.7 baseline-data unavailability and SVI 1.4 failure to preserve configuration data, but it did not directly initiate the 24-Sep-2020 release, so it is rated 2.

CAUSAL EXPLANATION

The missing baseline prevented robust trending of known wall-loss findings and allowed corrective actions to be deferred indefinitely. This data gap did not directly fail the trunk line, but it contributed to the integrity-governance environment in which known hazards were neither quantified nor escalated, paralleling the lost OEM gasket specification that directly enabled the 24-Sep-2020 flange failure.

Linked Evidence: EVD-8287

OIP 3.2 **Organisational Program/ Policy Risks not Adequately Assessed** **SUBSTANTIATED**

Three of four diesel fire-water pumps were unavailable or unreliable with unassessed and unrecorded causes, though electric pumps carried the incident.

JUSTIFICATION

The fire-water program accepted known diesel-pump unreliability without recording diagnosis or actions, and the resulting risk was not assessed. [EVD-2432] records 'Out of 4 diesel pumps 410 m3/hr, 2 pumps are under maintenance, one operated for 30 minutes & fourth pumps is tripping after 2 minutes. The reason for tripping is not identified or actions have not been recorded' (p.27). During the incident the three electrical pumps 47PE003 A, C, D kept tank level and header pressure healthy, so this gap did not alter the outcome. It is nonetheless a substantiated program-level resilience risk: had electrical supply failed or cooling demand exceeded electric capacity, the degraded diesel backup would have constrained the response, paralleling the same weak follow-through seen on the meter and lamp defects.

CAUSAL EXPLANATION

This instance had no direct impact on the 24-Sep-2020 outcome because the electrical pumps maintained fire-water pressure throughout the 5-hour 15-minute emergency. It is retained as a substantiated program-risk finding because it demonstrates the same organizational pattern of accepting known equipment defects without assessment or corrective action that elsewhere produced the directly causal meter and lamp failures.

Linked Evidence: EVD-2432

OIP 3.2 **Organisational Program/ Policy Risks not Adequately Assessed** **SUBSTANTIATED**

Missing permits were untracked and JSPs were absent from SAP, leaving the permit-to-work program risk unassessed.

JUSTIFICATION

The permit-to-work program operated without tracking missing permits and without attaching job safety plans in SAP, and this control weakness was not assessed. [EVD-5105] records 'it was found that few of the work permits were missing. No data is kept on such missing permits. Job safety Plan is not attached/given in SAP document. Closing of permits & actions indicated in work needs review' (p.26), and recommendation B.5 states maintenance notifications are not to be assumed as permits to work. This meant a high-risk pressure-boundary job such as the 18-Sep-2020 gasket replacement could proceed without a formal JSA forcing identification of the hydrostatic test and material-conformance requirements. It enabled lower-level PTE 2.7 permit guidance gap, ASBE 1.2 testing omission and SVV 4.1 non-enforcement, but the evidence does not tie a specific missing permit to the failed joint, so rating is 2.

CAUSAL EXPLANATION

Because the permit/JSA program risk was not assessed, hazardous maintenance lacked documented task-specific hazard controls. For the 18-Sep-2020 job this contributed to the absence of a formal requirement to hydro-test the substitute O-ring seal, allowing the defective pressure boundary into service on 19-Sep-2020 until it failed on 24-Sep-2020.

Linked Evidence: EVD-5105

OIP 3.2 **Organisational Program/ Policy Risks not Adequately Assessed** **SUBSTANTIATED**

Perimeter leak detections could not be communicated directly to the Gas Terminal Control Room, and this notification delay risk was never assessed.

JUSTIFICATION

The program risk that perimeter detection could not reach the operating control room directly was not assessed. [EVD-6234] shows CISF Head Constable B detected the vapour at 02:15 but informed only the CCTV Control Room and CISF main gate, and [EVD-8871] shows Sub-Inspector C then had to visually verify via CCTV before calling the Gas Terminal Control Room at 02:18. The resulting CSIF-internal verification hop added approximately 3-5 minutes to initial notification, at a time when the cloud was already expanding rapidly across the GT CR, road and piping area within 40 seconds of first detection [EVD-9941]. This architecture gap enabled lower-level PTF 1.10 and PTF 1.16 notification delays, but the dominant 46-minute delay was driven by isolation authority and valve configuration, so rating is 2.

CAUSAL EXPLANATION

The indirect notification path consumed the earliest minutes of the response; the Gas Terminal CR operators began acting only at 02:20 rather than 02:15. This contributed a secondary delay to the already severe response gap, though the major driver remained the non-remote MOVs and absent DMP isolation criteria.

Linked Evidence: EVD-6234, EVD-8871

OIP 3.3 **Provided Inadequate Procedural Guidance/ Publications** **SUBSTANTIATED**

P&ID 02-310-AO-101 Rev.1 was vendor-produced, unapproved, and did not match the field configuration.

JUSTIFICATION

The engineering publication for the metering system was not a controlled, accurate document. [EVD-0933] records 'Reference drawings no. 02-310-AO-101 Rev.1, 2x6 lines & 1x2 lines for metering system are shown but valves & meters configuration are not matching. As old drawings were lost, new drawings are made by a vendor but the same has not been checked & approved by ONGC' (p.26). Correct document control would require an ONGC-approved as-built drawing reflecting the actual valve and meter configuration. The actual drawing was lost original, replaced by an unapproved vendor product, and still mismatched the field, so maintainers had no authoritative configuration baseline to verify the correct rectangular-slot gasket and meter details. This inadequate publication enabled lower-level AV 1.1 configuration-management breakdown, PTE 2.7 technical guidance absent and ASBE 1.3 wrong-gasket procedure.

CAUSAL EXPLANATION

The unapproved and inaccurate P&ID removed the controlled configuration reference that could have exposed the O-ring substitution and the 18-Sep-2020 no-hydrotest return to service. While the physical failure was the incorrect seal itself, the publication gap removed the technical documentation layer that should have challenged it before the 24-Sep-2020 release.

Linked Evidence: EVD-0933

OIP 3.6 **Inadequate Program Oversight/ Program Management** **SUBSTANTIATED**

The Aug-2016 MOC SOP was never applied after 2018, so the OEM-to-O-ring gasket substitution bypassed formal change review and became regular practice.

JUSTIFICATION

A formal Management of Change programme existed on paper but had no functional implementation, and the programme's lapse removed the engineering review gate that should have screened the gasket substitution. [EVD-2986] records that "Management of Change (MOC) philosophy/SOP was published in August 2016. However, the procedure indicated in the document is not implemented while carrying out change of Low-Pressure Gas to Kribhico from Medium Pressure Gas headers in LPG Plant in 2013 & 2018. It is informed no more MOCs are implemented in the plant since 2018" (p.30). With zero MOCs processed since 2018, the switch from the OEM rectangular-slot gasket to a locally purchased O-ring was treated as routine practice rather than a change requiring specification, material and geometric review, exactly the governance hole documented in [EVD-8645]. This programme-management failure is the Level-4 origin of the lower-level SVI 1.4/SVI 1.13 MOC policy failures, the AJDE 2.11 misapplication finding, and the SVF 3.2/SVF 3.4 MOC non-implementation findings.

CAUSAL EXPLANATION

Because the MOC programme was not operational after 2018, the substitution of locally purchased O-rings for the lost OEM rectangular-slot gasket was never subjected to a formal risk assessment of geometry, material compatibility or pressure rating. This allowed the non-conforming seal to be installed on 18-Sep-2020, returned to service on 19-Sep-2020, and to fail at 02:12:50 on 24-Sep-2020 [EVD-1284], releasing vapour for 56 minutes until ignition at 03:08:15 [EVD-4507]. The dormant programme therefore enabled the lower-level AJDE 2.11 and AV 1.1 findings by removing the change-management gate that should have blocked or documented the material substitution.

Linked Evidence: EVD-2986

OIP 3.6 **Inadequate Program Oversight/ Program Management** **SUBSTANTIATED**

The four-year thickness measurement schedule was breached (2011-to-2017 six-year gap, and the next cycle overdue by 2020).

JUSTIFICATION

The mandated four-year ultrasonic thickness measurement cycle was not managed to schedule. [EVD-8307] records that "[t]he laid down schedule for thickness measurement is every four years... As per the records, the last two thickness measurement exercise was carried out in 2011 and 2017 which suggests that the schedule of four years is not being adhered to strictly" (p.29). The 2011-to-2017 interval was six years, and the 2017-to-2020 follow-up was also overdue by the incident date. This programme-management lapse meant wall loss on the 1986-vintage gas-terminal pipelines was not being characterised at the prescribed cadence, reducing the site's ability to trend degradation and prioritise corrective action. It did not directly cause the 24-Sep-2020 fire because the 36-inch trunk pipeline remained intact [EVD-5780], so the rating is present without direct outcome impact, but it substantiates the same programme-oversight culture that allowed the meter gasket recurrence to escape scheduled review.

CAUSAL EXPLANATION

The lapsed schedule did not initiate the flange release, so it is rated 2. Its causal significance is systemic: the same programme-management failure to enforce a mandated control schedule is what allowed the mandatory hydrostatic test after the 18-Sep-2020 gasket replacement to be omitted without management consequence, permitting the defective joint to fail at 02:12:50 on 24-Sep-2020, 56 minutes before ignition at 03:08:15.

Linked Evidence: EVD-8307

OIP 3.6 **Inadequate Program Oversight/ Program Management** **SUBSTANTIATED**

Most 1986 gas-terminal pipes lacked baseline thickness data and original drawings, leaving only 2011 and 2017 measurements for trend analysis.

JUSTIFICATION

The programme for lifecycle asset data preservation failed to retain the 1986 baseline and original schedules for the gas-terminal pipelines. [EVD-8287] records that "for a large number of pipes, especially in gas terminal area baseline data of pipe thickness in not available. The original drawings and the schedules of the pipes are also not available. In most cases only two thickness details of 2011 and 2017 is available for a pipe which is commissioned in 1986" (p.29). For a 34-year-old asset, only two measurement points existed, and the baseline against which corrosion should have been trended was absent. This programme failure degraded the quality of the integrity decisions that depended on trending; it did not directly cause the 24-Sep-2020 release, so it is rated 2, but it is the data-level counterpart to the lost OEM gasket specification [EVD-8645], both showing the organisation did not maintain controlled technical baselines for its 1986 equipment.

CAUSAL EXPLANATION

The absent baseline prevented rigorous trending of the 2017 wall-loss findings and reinforced the three-year non-action on corrective measures. It is the same configuration-data management gap that left the OEM rectangular-slot gasket specification lost [EVD-8645], depriving the 18-Sep-2020 repair team of the reference needed to reject the round O-ring. The defective seal then returned to service on 19-Sep-2020 and failed at 02:12:50 on 24-Sep-2020, leading to the 56-minute release and ignition at 03:08:15. Its direct outcome impact is limited, hence rating 2.

Linked Evidence: EVD-8287

OIP 3.6 **Inadequate Program Oversight/ Program Management** **SUBSTANTIATED**

A 16000-to-2540 mmwc calibration change and its reversal on 310FT01 passed without checklist use or signed documentation, illustrating the calibration programme's missing oversight.

JUSTIFICATION

The calibration programme lacked the oversight controls that should have detected and recorded a major setpoint change on meter 310FT01. [EVD-2506] records that "[d]uring calibration from 17/12/2019 to 18/12/2019, calibration was changed from 16000 mmwc to 2540 mmwc for 310FT01 without any mention of this change in the calibration documents signed by the team. During next calibration for meter 310FT01 was again changed to 16000 MMWC. No checklist is followed while carrying such critical jobs of high pressure/risk" (p.27). The committee's recommendation D.1.i, "details of Master calibrator/action taken/replacement/deviation from procedure if any, is to be documented in the calibration certificate" (p.35), confirms the programme gap. This unmanaged change on the same meter normalised checklist-free high-pressure work and contributed to the procedural discipline failure that later allowed the 18-Sep-2020 gasket replacement without a hydrotest.

CAUSAL EXPLANATION

The unmanaged calibration swing did not directly open the leak, but it normalised undocumented deviations on the exact meter that later failed, supporting the lower-level ASBE 1.2/ASBE 1.5 calibration findings and the SVI 1.6/SVV 4.1 supervisory acceptance of incomplete records. The same procedural discipline gap reappeared in the 18-Sep-2020 replacement, where the required hydrostatic test was omitted; the resulting unvalidated joint failed at 02:12:50 on 24-Sep-2020, 56 minutes before ignition at 03:08:15.

Linked Evidence: EVD-2506

OIP 3.6 **Inadequate Program Oversight/ Program Management** **SUBSTANTIATED**

Missing work permits were untracked, Job Safety Plans were absent from SAP, and permit closure went unreviewed.

JUSTIFICATION

The permit and Job Safety Plan programme had untracked gaps that removed task-specific hazard guidance from high-risk maintenance. [EVD-5105] records that "Work Permits are issued in SAP and hard copy of work permits are at sites during execution of work. After completion of work, the originals are filed. On checking the same, it was found that few of the work permits were missing. No data is kept on such missing permits. Job safety Plan is not attached/given in SAP document. Closing of permits & actions indicated in work needs review" (p.26). The committee's recommendation B.5, "Maintenance notifications are not be assumed as permit to work" (p.34), shows maintenance work was being executed under SAP notifications without a separate permit control. This programme failure meant the 18-Sep-2020 gasket replacement was not forced to carry a JSP identifying the required hydrostatic test and material-conformance checks.

CAUSAL EXPLANATION

Without a mandatory JSP and an auditable permit closure, the high-risk gasket job could proceed without a documented hazard analysis naming the design-pressure test and seal-conformance verification. This contributed to the environment in which the lower-level ASBE 1.2 and AJDE 2.7 omissions occurred and the defective joint entered service on 19-Sep-2020, failing at 02:12:50 on 24-Sep-2020. Because the evidence does not tie a specific missing permit directly to the failed joint, this instance is rated present with secondary systemic impact.

Linked Evidence: EVD-5105

4. RECOMMENDATIONS

AJDE 2.5

Training levels and standards need review. Reinforce importance of responding to cautions and warnings. Address any issues with false alarms or alarm fatigue. Review alarm systems for reliability and relevance. Improve supervision and monitoring of alarm responses.

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AJDE 2.7

Training on timely decision-making. Simulation training for scenarios requiring prompt action. Reinforce SOPs on timing requirements. Improve confidence through experience and practice. Address any hesitation or decision-making barriers.

AJDE 2.8

Training on decision-making and problem diagnosis. Simulation training for complex scenarios. Reinforce SOPs through discussions and exercises. Raise experience levels for decision-making roles. Conduct briefings and brainstorming sessions. Improve information sharing within teams.

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AJDE 2.9

Training on emergency decision-making. Simulation training for time-constrained scenarios. Reinforce SOPs through emergency drills. Raise experience levels for handling emergencies. Conduct briefings and brainstorming on emergency response. Address cognitive task saturation issues.

AJDE 2.10

Training levels and standards need comprehensive review. Provide adequate initial and update training. Ensure proper orientation to work environment. Build experience through supervised practice. Improve knowledge of regulations and standards. Enhance technical and operational knowledge of systems. Simulation training for complex scenarios.

AV 1.1

Organisational head must reinforce the policy, procedure or regulation that was violated. Eliminate the need or motivation for violation. Punitive action must apply to everyone who violated and everyone who tolerated the violations irrespective of rank. Investigate for normalisation of deviance. Address systemic issues that led to widespread violation.

PTE 2.3

Review and modify designs to eliminate obstructions. Improve lighting systems. Address glare and reflection issues. Consider anti-glare treatments. Ensure critical visibility requirements are met. Relocate equipment that obstructs views.

PSE 3.1

Training levels and experience levels need review. Provide training for anticipated scenarios. Maintain proficiency through regular practice. Consider simulation training for rare events. Ensure adequate supervision for unfamiliar tasks. Develop procedures for novel situations.

PPE 1.2

Review operations planning for adverse weather. Ensure appropriate equipment for low visibility operations. Provide training for operations in limited visibility. Consider postponing operations when conditions are unsafe. Implement enhanced monitoring in poor visibility.

AJDE 2.4

Training on information interpretation and decision-making. Simulation training for decision scenarios. Reinforce SOPs through discussions and exercises. Improve communication and information presentation methods. Implement verification and cross-checking procedures.

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ASBE 1.3

Training levels and standards need review. Reinforce SOPs through discussions, briefings, and emergency drills before evolutions. Conduct simulation training. Increase frequency of drills and practice sessions. Improve quality and levels of supervision.

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ASBE 1.6

Technology and instrumentation improvements may be needed. Conduct simulation training. Reinforce proper scan patterns through training. Review and improve training levels and standards. Ensure SOPs clearly define required scan patterns and procedures.

ASBE 1.2

Revise/improve checklists by ensuring most critical items appear highlighted/bold or on top. Implement restart procedures if checklist interrupted. Use readback procedures to ensure critical items are completed. Require 'Checklist Completed' reports. Review organizational training on checklist discipline and importance.

ASBE 1.2

Revise/improve checklists by ensuring most critical items appear highlighted/bold or on top. Implement restart procedures if checklist interrupted. Use readback procedures to ensure critical items are completed. Require 'Checklist Completed' reports. Review organizational training on checklist discipline and importance.

AJDE 2.11

Ensure policies and procedures are effectively communicated. Provide training on current policies and SOPs. Implement system for tracking policy awareness. Regular updates and refresher training on policies. Improve accessibility of policy documents. Supervisory oversight of policy compliance.

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ASBE 1.5

Training levels and standards need review. Increase experience levels through practice sessions. Conduct recurrent training on proper control techniques. Improve levels and quality of supervision. Use simulation training to practice control in various conditions.

AJDE 2.10

Training levels and standards need comprehensive review. Provide adequate initial and update training. Ensure proper orientation to work environment. Build experience through supervised practice. Improve knowledge of regulations and standards. Enhance technical and operational knowledge of systems. Simulation training for complex scenarios.

APE 3.2

Technology and instrumentation to aid perception. Simulation training for various conditions. Reinforce proper scan patterns. Review training levels and standards. Ensure adequate instrumentation for altitude, height, and distance measurement.

AV 1.1

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PTF 1.4

Design missions with redundancy in mind. Train for contingency operations. Diversify team roles. Audit for single point failures in planning.

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PTF 1.5

Conduct explicit role briefings before missions. Use role matrices for clarity. Train on hierarchy and conflict resolution. Review role understanding in debriefs.

PTF 1.6

Balance team composition for optimal gradients. Train on effective authority dynamics. Encourage assertive communication regardless of rank. Monitor team interactions for gradient effects.

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PTE 2.4

Review and improve control and switch design. Ensure proper location and spacing. Improve labeling and identification. Standardize control design where possible. Ensure adequate lighting. Address reliability issues. Apply human factors engineering principles.

PTE 2.5

Implement strict policies against using defective equipment. Improve maintenance programs. Ensure adequate spares and replacement equipment. Enhance defect reporting and tracking. Provide adequate resources for equipment replacement. Strengthen supervisory oversight of equipment status.

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PTE 2.6

Review and improve automation design and interfaces. Ensure clear automation logic and modes. Test and verify system redundancy. Improve software usability. Simplify menu structures. Provide comprehensive automation training. Implement clear mode annunciation.

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PTE 2.7

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PTE 2.8

Review and redesign workspace as needed. Improve lighting. Optimize layout for tasks performed. Apply ergonomic principles. Ensure adequate access to equipment. Address hygiene issues. Consult operators in workspace design.

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ASBE 1.2

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PTE 2.10

Ensure adequate communication equipment availability. Upgrade equipment to meet mission demands. Address coverage gaps and dead zones. Improve system reliability. Provide redundant communication methods. Test communication systems before operations.

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PTE 2.11

Improve configuration procedures and checklists. Provide training on equipment setup. Implement pre-operation configuration checks. Simplify configuration processes where possible. Use automated configuration where feasible. Ensure clear configuration guidance.

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PTE 2.12

Strengthen maintenance programs and schedules. Ensure qualified maintenance personnel. Improve maintenance training. Implement strict maintenance tracking and documentation. Address resource constraints affecting maintenance. Enhance quality control of maintenance work.

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PTF 1.1

Enhance leadership training with CRM focus. Conduct regular team climate assessments. Implement debriefing sessions for mission alignment. Promote inclusive decision-making practices.

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PTF 1.2

Reinforce cross-monitoring in CRM training. Use simulations for backup scenarios. Establish clear roles for monitoring. Review team performance metrics for monitoring effectiveness.

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PTE 2.14

Provide thorough familiarization training before non-custodial operations. Ensure adequate orientation to equipment differences. Consider qualification requirements for non-custodial operations. Provide technical support and documentation. Assess whether non-custodial operations should be permitted.

PTE 2.15

Ensure adequate stocks of protective equipment. Maintain equipment quality through regular inspection and replacement. Make equipment easily accessible. Establish equipment issue and accountability systems. Provide training on proper use of protective equipment.

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PTF 1.3

Train on effective task delegation and load balancing. Assess team skills before assignment. Implement workload monitoring tools. Review delegation in after-action reports.

PTF 1.10

Implement standardized communication protocols. Use checklists for critical info. Train on timely information sharing. Audit communication logs.

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PTF 1.11

Enforce standard terminology in training. Update glossaries regularly. Practice standard communications in drills. Correct non-standard usage immediately.

PTF 1.12

Mandate read-back for critical communications. Train on effective feedback loops. Use confirmation tools like checklists. Analyze communication breakdowns in reviews.

PTF 1.10

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PTF 1.13

Standardize mission planning templates. Allocate sufficient planning time. Involve all stakeholders in planning. Review planning effectiveness post-mission.

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PTF 1.14

Structure briefings with mandatory contingency sections. Ensure interactive discussion. Record and review briefing content. Train briefers on comprehensive coverage.

PTF 1.15

Embed re-planning checkpoints in operations. Train on adaptive planning. Use real-time risk assessment tools. Debrief on re-planning decisions.

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PTF 1.16

Promote closed-loop communications. Train on interpretation under stress. Use visual aids for complex info. Investigate miscommunications systematically.

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PTF 1.17

Implement rapid onboarding for substitutes. Provide workspace familiarization training. Use checklists for policy briefings. Pair new members with experienced ones.

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PPI 5.4

Improve training on dynamic environmental assessment. Use augmented reality for perception training. Enhance situational awareness briefings. Implement redundant checks for critical parameters.

PPI 5.5

Review instrument display ergonomics. Provide training on instrument interpretation under stress. Implement read-back protocols for critical readings. Conduct audits on instrument maintenance and calibration.

PPI 5.6

Train on expectancy bias and confirmation bias. Encourage questioning assumptions in briefings. Use scenario-based training to challenge expectations. Foster a culture of double-checking perceptions.

PPI 5.7

Optimize audio alert systems for clarity. Train on auditory cue recognition in varied conditions. Implement visual backups for critical auditory signals. Review communication protocols for noise environments.

PPI 5.9

Enhance multi-sensory integration training. Practice recovery maneuvers in simulators. Update SOPs for recognized disorientation scenarios. Monitor for sensory conflicts in mission planning.

SVI 1.1

Define oversight requirements for tasks. Train supervisors on competency gaps. Implement supervisory rotation for freshness. Audit oversight effectiveness.

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SVI 1.2

Train leaders on positive role-modeling. Monitor and correct improper behaviors. Use mentorship programs with guidelines. Evaluate role-model impact in reviews.

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SVI 1.3

Regularly audit local training programs. Ensure availability of all required trainings. Tailor programs to operational needs. Track training completion and effectiveness.

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Establish mandatory response timelines for critical info. Track action closures. Train on information prioritization. Audit supervisory responses.

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SVI 1.4

Maintain current policy libraries. Communicate changes effectively. Train on policy application. Solicit feedback on guidance adequacy.

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SVI 1.7

Develop and enforce uniform standards. Train supervisors on consistency. Monitor for equity in application. Update standards based on feedback.

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SVI 1.8

Implement regular safety audits. Train on risk identification. Establish corrective action plans. Reward proactive hazard reporting.

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SVI 1.9

Use skill matrices for assignments. Require proficiency certifications. Provide shadowing for skill gaps. Review selection criteria regularly.

SVI 1.6

Establish mandatory response timelines for critical info. Track action closures. Train on information prioritization. Audit supervisory responses.

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SVI 1.4

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SVI 1.10

Mandate intent communication in orders. Use standardized templates. Verify understanding through acknowledgements. Include intent in handover protocols.

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Mandate intent communication in orders. Use standardized templates. Verify understanding through acknowledgements. Include intent in handover protocols.

SVI 1.11

Foster supportive supervisory culture. Train on mentoring skills. Establish anonymous reporting channels. Evaluate support in performance reviews.

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SVI 1.7

Develop and enforce uniform standards. Train supervisors on consistency. Monitor for equity in application. Update standards based on feedback.

SVI 1.14

Mandate supervisor recurrency training. Include new equipment in training plans. Pair untrained supervisors with experts. Track training currency.

SVIO 2.1

Conduct capability-risk matching in planning. Train on mission feasibility assessment. Establish veto rights for unsafe orders. Review assignment rationales.

SVIO 2.1

Conduct capability-risk matching in planning. Train on mission feasibility assessment. Establish veto rights for unsafe orders. Review assignment rationales.

SVIO 2.2

Use skill-based team assembly tools. Pre-mission composition audits. Balance experience levels. Solicit team input on composition.

SVIO 2.2

Use skill-based team assembly tools. Pre-mission composition audits. Balance experience levels. Solicit team input on composition.

SVIO 2.3

Enforce recency standards. Provide refresher for lapsed experience. Log experience levels. Adjust assignments based on currency.

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SVI 1.8

Implement regular safety audits. Train on risk identification. Establish corrective action plans. Reward proactive hazard reporting.

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SVI 1.10

Mandate intent communication in orders. Use standardized templates. Verify understanding through acknowledgements. Include intent in handover protocols.

SVIO 2.4

Standardize risk assessment processes. Train on tool usage. Integrate multiple assessment methods. Review assessments post-mission.

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SVIO 2.5

Require justification for high-risk authorizations. Implement qualification gates. Train on hazard minimization. Audit authorization decisions.

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SVIO 2.6

Adhere to manning minima strictly. Plan for contingencies. Cross-train for flexibility. Review manning post-incident.

SVI 1.10

Mandate intent communication in orders. Use standardized templates. Verify understanding through acknowledgements. Include intent in handover protocols.

SVIO 2.7

Optimize asset allocation planning. Maintain buffer stocks. Track usage and shortages. Coordinate asset sharing across units.

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SVI 1.11

Foster supportive supervisory culture. Train on mentoring skills. Establish anonymous reporting channels. Evaluate support in performance reviews.

SVI 1.13

Require confirmation of policy receipt. Schedule regular briefings. Use digital dissemination tools. Audit communication compliance.

SVIO 2.2

Use skill-based team assembly tools. Pre-mission composition audits. Balance experience levels. Solicit team input on composition.

SVIO 2.4

Standardize risk assessment processes. Train on tool usage. Integrate multiple assessment methods. Review assessments post-mission.

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SVF 3.1

Conduct behavioral risk assessments. Implement remedial programs. Train on cultural sensitivity. Document and track interventions.

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SVF 3.2

Establish hazard correction timelines. Prioritize operational safety fixes. Empower supervisors for quick actions. Follow up on correction efficacy.

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SVF 3.3

Regular workspace safety inspections. Foster positive environment initiatives. Address morale issues promptly. Involve team in workspace improvements.

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SVIO 2.5

Require justification for high-risk authorizations. Implement qualification gates. Train on hazard minimization. Audit authorization decisions.

SVIO 2.7

Optimize asset allocation planning. Maintain buffer stocks. Track usage and shortages. Coordinate asset sharing across units.

SVF 3.4

Require verification of mitigation effectiveness. Set escalation thresholds. Track risk closure status. Conduct periodic risk reviews.

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SVF 3.1

Conduct behavioral risk assessments. Implement remedial programs. Train on cultural sensitivity. Document and track interventions.

SVV 4.1

Strengthen rule enforcement training. Implement compliance audits. Hold supervisors accountable for omissions. Use reminders and alerts for rules.

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Strengthen rule enforcement training. Implement compliance audits. Hold supervisors accountable for omissions. Use reminders and alerts for rules.

SVV 4.2

Document all policies formally. Prohibit unofficial practices. Communicate policy boundaries clearly. Monitor for deviation creep.

SVF 3.2

Establish hazard correction timelines. Prioritize operational safety fixes. Empower supervisors for quick actions. Follow up on correction efficacy.

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SVV 4.4

Automate qualification verification. Enforce currency requirements strictly. Train on authorization responsibilities. Penalize improper authorizations.

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SVV 4.5

Require action plans for all identified risks. Allocate resources for mitigations. Escalate unaddressed risks. Track risk status dashboards.

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- **OIR 1.1**

  Conduct resource adequacy audits. Prioritize funding for critical infrastructure. Maintain upkeep schedules. Integrate resource status in planning.

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OIR 1.5

Streamline procurement processes. Prioritize safety-critical acquisitions. Forecast upgrade needs. Track procurement timelines.

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OIR 1.6

Optimize maintenance scheduling. Stock adequate spares. Evaluate support programs. Implement predictive maintenance tech.

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OIR 1.5

Streamline procurement processes. Prioritize safety-critical acquisitions. Forecast upgrade needs. Track procurement timelines.

OIR 1.7

Define clear phase-out timelines. Inspect for wear regularly. Plan replacements in advance. Dispose safely post-phase-out.

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OIR 1.8

Ensure reliable info sources. Redundant data feeds. Train on info acquisition. Integrate info in planning tools.

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OIC 2.1

Assess and reshape safety culture. Leadership to model safe behaviors. Safety culture surveys. Reward safety-aligned actions.

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OIC 2.3

Base confidence on data and testing. Regular equipment reviews. Train on realistic capabilities. Balance optimism with caution.

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OIR 1.6

Optimize maintenance scheduling. Stock adequate spares. Evaluate support programs. Implement predictive maintenance tech.

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OIC 2.4

Plan structured transition programs. Provide change management training. Support during deactivations. Communicate change rationales clearly.

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OIC 2.5

Clarify org charts and delegations. Standardize communication flows. Define accountability matrices. Periodically restructure for efficiency.

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OIC 2.7

Unify risk management frameworks. Promote just culture for reporting. Consistent messaging from leadership. Track and close risk reports visibly.

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- **OIP 3.2**

  Mandate comprehensive risk assessments for programs. Involve multi-disciplinary teams. Document risk mitigations. Review assessments at milestones.

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OIP 3.3

Regularly update and validate publications. User-test guidance materials. Simplify complex procedures. Gather feedback for improvements.

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OIP 3.4

Ensure training program availability. Quality assure external trainings. Schedule proactively. Evaluate program outcomes.

OIP 3.5

Periodically review and update doctrine. Risk-benefit analyses for philosophies. Consult stakeholders in revisions. Align with best practices.

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OIP 3.6

Assign dedicated program managers. Set oversight milestones. Provide necessary support resources. Evaluate program progress regularly.

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Mandate comprehensive risk assessments for programs. Involve multi-disciplinary teams. Document risk mitigations. Review assessments at milestones.

OIC 2.4

Plan structured transition programs. Provide change management training. Support during deactivations. Communicate change rationales clearly.

OIP 3.7

Incorporate human factors in acquisition. Rigorous design reviews. Test suitability pre-purchase. Post-acquisition evaluations.

OIP 3.7

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OIS 4.1

Enhance screening criteria. Include safety in selection. Regular policy reviews. Track long-term suitability.

OIS 4.2

Forecast staffing needs accurately. Standardize handover procedures. Optimize placement algorithms. Monitor staffing impacts on safety.

OIC 2.4

Plan structured transition programs. Provide change management training. Support during deactivations. Communicate change rationales clearly.

OIC 2.5

Clarify org charts and delegations. Standardize communication flows. Define accountability matrices. Periodically restructure for efficiency.

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OIC 2.7

Unify risk management frameworks. Promote just culture for reporting. Consistent messaging from leadership. Track and close risk reports visibly.

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OIP 3.2

Mandate comprehensive risk assessments for programs. Involve multi-disciplinary teams. Document risk mitigations. Review assessments at milestones.

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OIP 3.3

Regularly update and validate publications. User-test guidance materials. Simplify complex procedures. Gather feedback for improvements.

OIP 3.6

Assign dedicated program managers. Set oversight milestones. Provide necessary support resources. Evaluate program progress regularly.

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PPE 1.12

Provide training on operations in brownout/whiteout conditions. Implement operational techniques to minimize particulate disturbance. Ensure appropriate sensors and aids for low visibility. Consider alternative approaches or procedures. Aviation-specific: brownout landing training and equipment.

PPE 1.13

Improve ventilation systems. Provide appropriate respiratory protection. Implement atmospheric monitoring. Address contamination sources. Provide training on operating in contaminated atmospheres. Establish exposure limits.

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PTE 2.2

Review and improve instrument and display design. Ensure adequate lighting and visibility. Improve warning and alarm system effectiveness. Standardize symbology. Ensure instruments are reliable. Optimize location and layout. Provide training on instrument interpretation.

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PSE 3.1

Training levels and experience levels need review. Provide training for anticipated scenarios. Maintain proficiency through regular practice. Consider simulation training for rare events. Ensure adequate supervision for unfamiliar tasks. Develop procedures for novel situations.

PSE 3.3

Training levels and experience needed for emergency operations. Provide simulation training for time-critical scenarios. Develop procedures balancing speed and safety. Ensure adequate resources to meet time constraints. Consider whether time constraints were self-imposed or necessary.

ASBE 1.2

Revise/improve checklists by ensuring most critical items appear highlighted/bold or on top. Implement restart procedures if checklist interrupted. Use readback procedures to ensure critical items are completed. Require 'Checklist Completed' reports. Review organizational training on checklist discipline and importance.

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ASBE 1.3

Training levels and standards need review. Reinforce SOPs through discussions, briefings, and emergency drills before evolutions. Conduct simulation training. Increase frequency of drills and practice sessions. Improve quality and levels of supervision.

PTF 1.5

Conduct explicit role briefings before missions. Use role matrices for clarity. Train on hierarchy and conflict resolution. Review role understanding in debriefs.

PTF 1.4

Design missions with redundancy in mind. Train for contingency operations. Diversify team roles. Audit for single point failures in planning.

PTF 1.6

Balance team composition for optimal gradients. Train on effective authority dynamics. Encourage assertive communication regardless of rank. Monitor team interactions for gradient effects.